

ARCHITECTURAL CONCEPTS

BECOME ACQUAINTED WITH CLOUD CONCEPTS

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Cloud
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ARCHITECTURAL CONCEPTS - SECTION OUTLINE

Lecture One

We will ***define what cloud computing*** is and ***review its unique characteristics***.

Cloud characteristics

- What is cloud computing?
 - Common definition
 - NIST 800-145 definition
- Common cloud characteristics

ARCHITECTURAL CONCEPTS - SECTION OUTLINE

Lecture Two

We will examine how to **establish business requirements** before moving assets to the cloud

Business requirements

- Understanding the existing state
- Cost/benefit analysis

ARCHITECTURAL CONCEPTS - SECTION OUTLINE

Lecture Three

We will define the 3 general categories of cloud services

Cloud computing service categories

- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)

ARCHITECTURAL CONCEPTS - SECTION OUTLINE

Lecture Four

We will examine 4 different deployment models which describe ***the place where*** your assets are hosted and ***who has access*** to them.

Cloud deployment models

- Private cloud
- Public cloud
- Hybrid cloud
- Community cloud

ARCHITECTURAL CONCEPTS - SECTION OUTLINE

Lecture Five

We will define the concepts of multi-tenancy and oversubscription.

Multi-tenancy

- Sharing the same cloud computing resources
- Oversubscription

ARCHITECTURAL CONCEPTS - SECTION OUTLINE

Lecture Six

We will examine the different roles and responsibilities people and organisations play in the cloud computing world.

Cloud computing roles and responsibilities

- Cloud Service Providers (CSPs)
- Cloud Service Customers (CSCs)
- Cloud Service Partners (CSNs)
- Regulators
- Cloud Access Security Broker (CASB)

ARCHITECTURAL CONCEPTS - SECTION OUTLINE

Lecture Seven

We will define **virtualisation** as the transformative change which is the ***driving force behind cloud computing infrastructure.***

Virtualisation

- Hypervisors
 - Type 1
 - Type 2
- Virtualisation security

ARCHITECTURAL CONCEPTS - SECTION OUTLINE

Lecture Eight

As you decide the appropriate ways to use the cloud, there are a set of considerations to keep in mind. We will examine these factors that must be considered.

Cloud shared considerations

- Security and privacy considerations
- Operational considerations

ARCHITECTURAL CONCEPTS - SECTION OUTLINE

Lecture Nine

We will examine emerging technologies that are especially significant in the world of cloud computing.

Emerging technologies

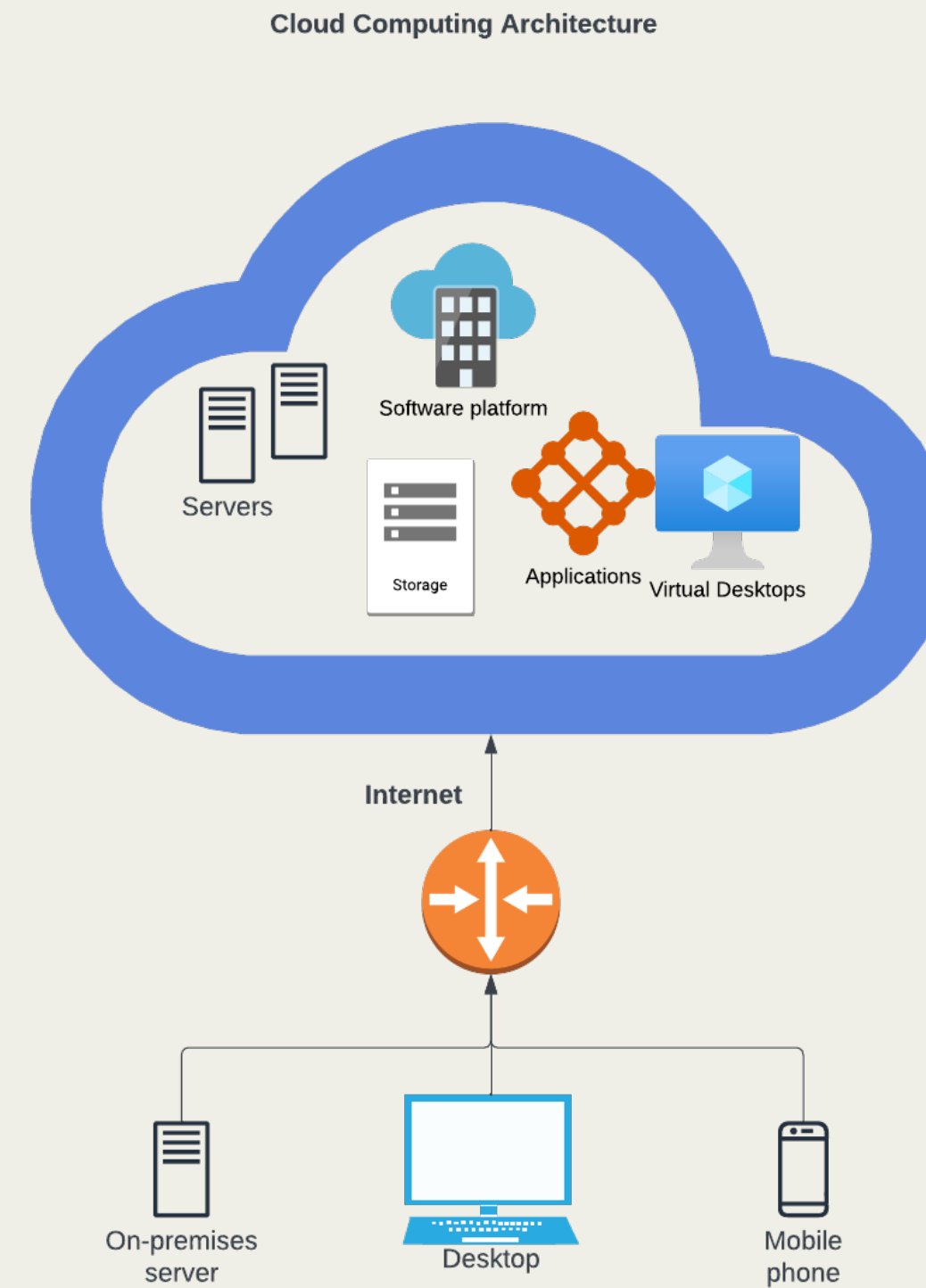
- ML and AI
- Blockchain
- IoT
- Containers
- Quantum computing
- Edge and fog computing
- Confidential computing
- DevOps and DevSecOps

CLOUD CHARACTERISTICS

CLOUD CHARACTERISTICS

What is cloud computing?

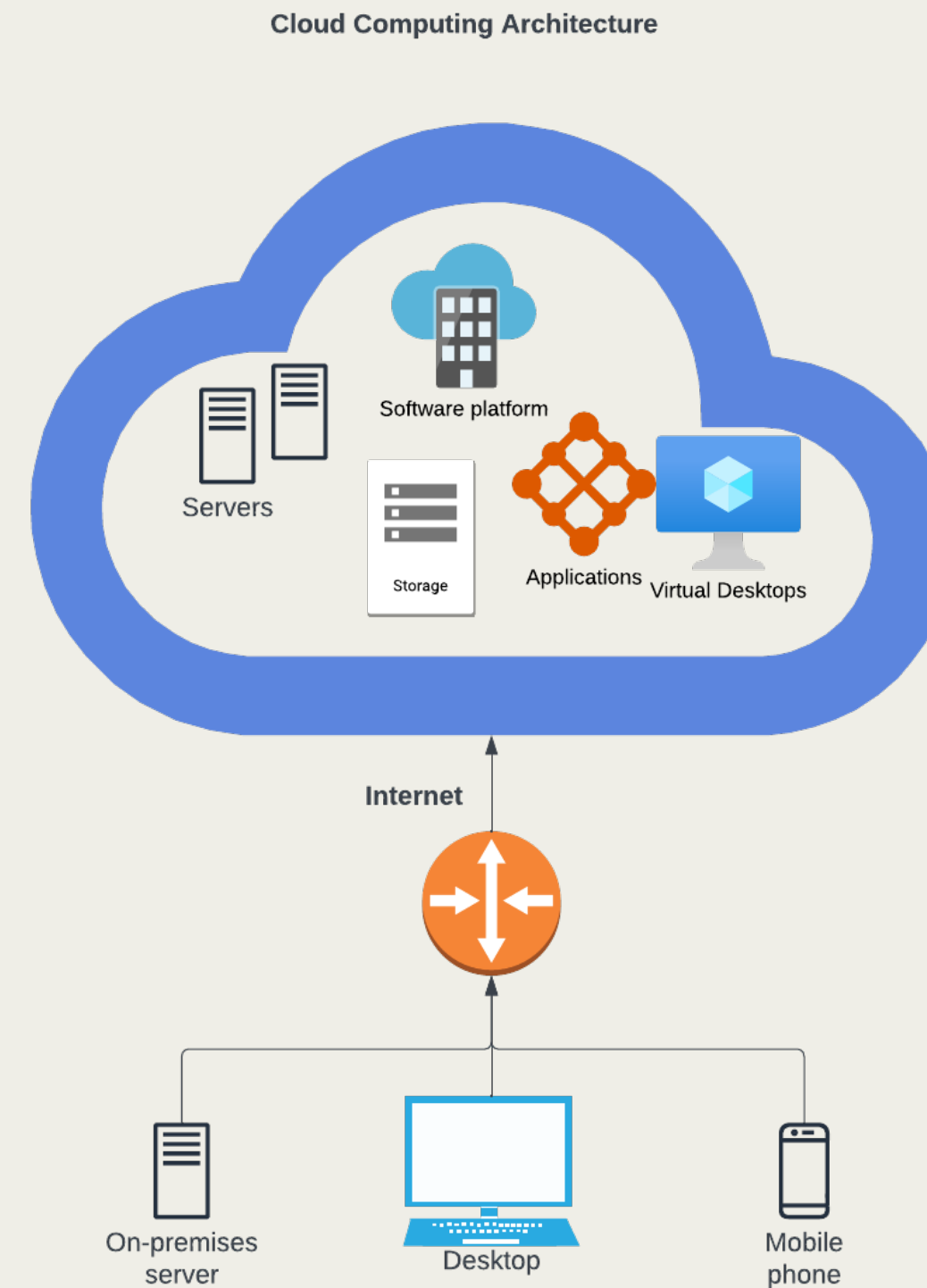
The CCSP study guide defines cloud computing as: *Any case where a provider is delivering computing to a customer at a remote location over a network.*



CLOUD CHARACTERISTICS

What is cloud computing?

The NIST definition of cloud computing says, "Cloud Computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction."



CLOUD CHARACTERISTICS

Imagine you wanted to try out a new idea. Before the cloud computing era, you'd:

- 1 Spec out the servers required to implement your idea
- 2 Obtain funding approval
- 3 Order the hardware
- 4 Wait for the hardware to arrive
- 5 Physically install the hardware
- 6 Configure the OS
- 7 Work on your idea

CLOUD CHARACTERISTICS

Imagine you wanted to try out a new idea. Today, in the cloud computing era...

- 1 Log into your cloud provider's console
- 2 Provision the server (VM)
- 3 Work on your idea

CLOUD CHARACTERISTICS

There are some common characteristics that we use to define cloud computing:

- Broad network access
- On-demand self-service
- Resource pooling
- Rapid elasticity and scalability
- Measured or metered service

CLOUD CHARACTERISTICS

Broad network access simply means that no matter where we or our users are physically located, we can access resources in the cloud.

- Broad network access
- On-demand self-service
- Resource pooling
- Rapid elasticity and scalability
- Measured or metered service

CLOUD CHARACTERISTICS

On-demand self-service refers to the model that allows customers to scale their compute and or storage needs with little or no intervention from or prior communication with the provider.

- Broad network access
- **On-demand self-service**
- Resource pooling
- Rapid elasticity and scalability
- Measured or metered service

CLOUD CHARACTERISTICS

Resource pooling is combining a large number of compute servers, storage devices, and network connections into a pool that can be shared by multiple users and or different organisations, with isolation ensured or provided by the cloud service provider.

- Broad network access
- On-demand self-service
- **Resource pooling**
- Rapid elasticity and scalability
- Measured or metered service

CLOUD CHARACTERISTICS

Rapid elasticity and scalability allows the customer to grow or shrink their compute, storage, or network needs as necessary to meet operational needs.

- Broad network access
- On-demand self-service
- Resource pooling
- Rapid elasticity and scalability
- Measured or metered service

CLOUD CHARACTERISTICS

Measured or metered service means that almost everything you do in the cloud is measured. This allows cloud service providers (CSPs) to charge you precisely for the services you use.

- Broad network access
- On-demand self-service
- Resource pooling
- Rapid elasticity and scalability
- **Measured or metered service**

CLOUD MIGRATION

CLOUD MIGRATION

Aligning specific security needs with broader business objectives is essential for achieving a comprehensive and effective cloud security posture.

Business objectives

- Cost optimisation
- Business continuity and resilience
- Competitive advantage

Security objectives

- Data protection and privacy
- Identity and Access Management
- Compliance & risk management

CLOUD MIGRATION

Business objectives

- Cost optimisation
- Business continuity and resilience
- Competitive advantage

Security objectives

- Data protection and privacy
- Identity and Access Management
- Compliance & risk management

By integrating security considerations into strategic planning processes and decision-making frameworks, organisations can enhance resilience, and drive business success in an increasingly interconnected and digital world.

CLOUD MIGRATION

Proactively addressing security risks

Enhances operational efficiency and reduces the total cost of ownership (TCO) associated with cloud services.

Business objectives



Cost optimisation

CLOUD MIGRATION

Business objectives

Proactively addressing security risks

Enables organisations to minimise the financial impact of security incidents, regulatory fines, and reputational damage.



Cost optimisation

CLOUD MIGRATION

Business objectives

- Cost optimisation
- Business continuity and resilience

Protecting critical assets and maintaining service availability are essential for sustaining business operations and customer satisfaction.

CLOUD MIGRATION

Business objectives

- Cost optimisation
- Business continuity and resilience

Investing in disaster recovery solutions, redundant infrastructure, and incident response plans mitigates the impact of disruptions and ensures continuity during unforeseen events.

CLOUD MIGRATION

Business objectives

- Cost optimisation
- Business continuity and resilience
- Competitive advantage

Demonstrating a commitment to security and compliance enhances the trust and credibility of the organisation among customers, partners, and stakeholders.

CLOUD MIGRATION

Business objectives

- Cost optimisation
- Business continuity and resilience
- Competitive advantage

By prioritising security as a competitive differentiator, businesses can gain a strategic edge in the market and attract new opportunities for growth and expansion.

CLOUD MIGRATION

Before moving your assets to the cloud, take three key steps to guarantee a long-term **return on investment (ROI)**.

CLOUD MIGRATION

1

Identify your business needs

Clearly define what problems cloud computing can solve for your company.

CLOUD MIGRATION

Evaluate how cloud migration will affect your current workflows and processes.

1

Identify your business needs

2

Assess the impact on operations (BIA)

CLOUD MIGRATION

Compare the potential cost savings and agility benefits of the cloud with migration expenses.

- 1 Identify your business needs
- 2 Assess the impact on operations (BIA)
- 3 Weigh the costs and benefits

CLOUD MIGRATION

1

Identify your business needs

A successful organisation will gather as much information about operational business requirements as possible.

CLOUD MIGRATION

1

Identify your business needs

Business requirements can be used for many purposes, including several functions in the security realm.

CLOUD MIGRATION

1

Identify your business needs

A network security administrator has to know what type of traffic to expect based on the business of the organisation.

CLOUD MIGRATION

The intrusion detection analyst has to understand

- What the organisation is doing
- How business activities occur
- Where the business is operating

In order to understand the nature and intensity of potential external attacks and how to adjust baselines accordingly.

1

Identify your business needs

CLOUD MIGRATION

1

Identify your business needs

The **security architect** has to understand the various needs of the organisational departments to enhance their operation without compromising their security profile.

CLOUD MIGRATION

Security leaders must

1

Identify your business needs

- Understand the technologies used by the organisation
- Understand the associated risks
- Know how to appropriately manage risks

CLOUD MIGRATION

The **business impact analysis (BIA)** is an assessment of the priorities given to each asset and process within the organisation.

2

Assess the impact on operations (BIA)

CLOUD MIGRATION

2.1

Determine the core activities and processes that are essential for the organisation's operations and continuity.

- Customer service
- Financial transactions
- Production channels
- Communication channels

2

Assess the impact on operations (BIA)

CLOUD MIGRATION

2.2 Identify the dependencies between different

- Business functions
- Applications
- Systems
- Resources

2 Assess the impact on operations (BIA)

CLOUD MIGRATION

Understand how these elements

2.2

interact with each other, thus you will be able to evaluate the potential impact of changes, such as cloud migration.

- Business functions
- Applications
- Systems
- Resources

2

Assess the impact on operations (BIA)

CLOUD MIGRATION

2.3

Evaluate the potential consequences of disruptions or changes to critical business functions

2.3.1

Consider scenarios such as service downtime, data loss, performance degradation and regulatory compliance.

2.3.2

Quantify the impact in terms of financial losses, reputational damage, productivity decline and customer satisfaction.

2

Assess the impact on operations (BIA)



CLOUD MIGRATION

2.4

Determine the acceptable downtime (RTO) and data loss (RPO) thresholds for each critical business function.

- Set realistic recovery targets
- Prioritise recovery efforts during and after the migration process

2

Assess the impact on operations (BIA)

CLOUD MIGRATION

2.5 Identify potential risks and vulnerabilities associated with cloud migration, such as

- Data breaches
- Service outages
- Compliance gaps

2 Assess the impact on operations (BIA)

CLOUD MIGRATION

2.5 Develop mitigation strategies to address identified risks, including

- Data encryption
- Redundant infrastructure
- Disaster recovery plans
- Security controls

2 Assess the impact on operations (BIA)

CLOUD MIGRATION

2.6

Collaborate with key stakeholders
(business leaders, IT, security experts,
vendors)



Gather insights, feedback, different
perspectives



Ensures a comprehensive
understanding of business priorities
and concerns

2

Assess the impact on operations
(BIA)



CLOUD MIGRATION

2.7

Document the results of the BIA,
including identified

- Critical business functions
- Dependencies
- Impact scenarios
- Recovery objectives
- Risk mitigation strategies
- Stakeholder inputs

2

Assess the impact on operations
(BIA)

CLOUD MIGRATION

2.7

Summarise key findings and recommendations to

- Inform decision-making

Guide the planning and

- implementation of the cloud migration strategy

2

Assess the impact on operations (BIA)



CLOUD MIGRATION

Organisations perceive cost savings through:

- Reduction in capital expenditure
- Reduction in operational costs
- Reduction in personnel costs
- Transfer of some regulatory costs to the CSP
- Reduction in costs for data archival and backup services

3

Weigh the costs and benefits

CLOUD MIGRATION

The BIA gives you a clear picture of what your organisation does in terms of lines of business and processes.

3

Weigh the costs and benefits



CLOUD MIGRATION

The BIA exercises gives you a better understanding of what the organisation might derive from cloud migration as well as the associated costs with the move.

3

Weigh the costs and benefits

CLOUD MIGRATION

Conducting a **cost/benefit analysis** helps you understand this trade-off in clear financial terms.

3

Weigh the costs and benefits

CLOUD MIGRATION

It is this cost-benefit calculation, driven by business needs but informed by security concerns, that would allow senior management to decide whether cloud migration of the organisation's operational environment makes sense.

3

Weigh the costs and benefits

CLOUD MIGRATION

3.1

Clearly outline what you aim to achieve with cloud migration. Is it

- Cost reduction
- Improved scalability
- Enhanced collaboration

3

Weigh the costs and benefits



CLOUD MIGRATION

3.2

List all expenses associated with cloud migration. This includes

● **One-time costs** - software licences,

data transfer fees, potential consulting expenses

● **Ongoing costs** - monthly cloud

service subscriptions, potential support charges

3

Weigh the costs and benefits



CLOUD MIGRATION

3.3

Two metrics help you perform more complex cost-benefit analysis:

● **Net Present Value (NPV)** - Considers the time value of money. Costs and benefits happening later are discounted to reflect their present-day value, providing a more nuanced picture.

3

Weigh the costs and benefits

CLOUD MIGRATION

3.3

Two metrics help you perform more complex cost-benefit analysis:

● **Benefit Cost Ratio (BCR)** - Divides the total benefits by the total costs, giving a ratio that indicates the return on investment (ROI) per dollar spent

3

Weigh the costs and benefits

CLOUD MIGRATION

3.4

Make a decision

- If benefits outweigh costs, the cloud migration might be worthwhile.

- Consider the NPV or BCR (if calculated) for a more informed decision

3

Weigh the costs and benefits

CLOUD MIGRATION

3.4

Remember, cost/benefit analysis is a tool to guide your decision, not a guarantee. To finalise your decision, factor in other considerations like

- Security
- Compliance
- Vendor lock-in

3

Weigh the costs and benefits

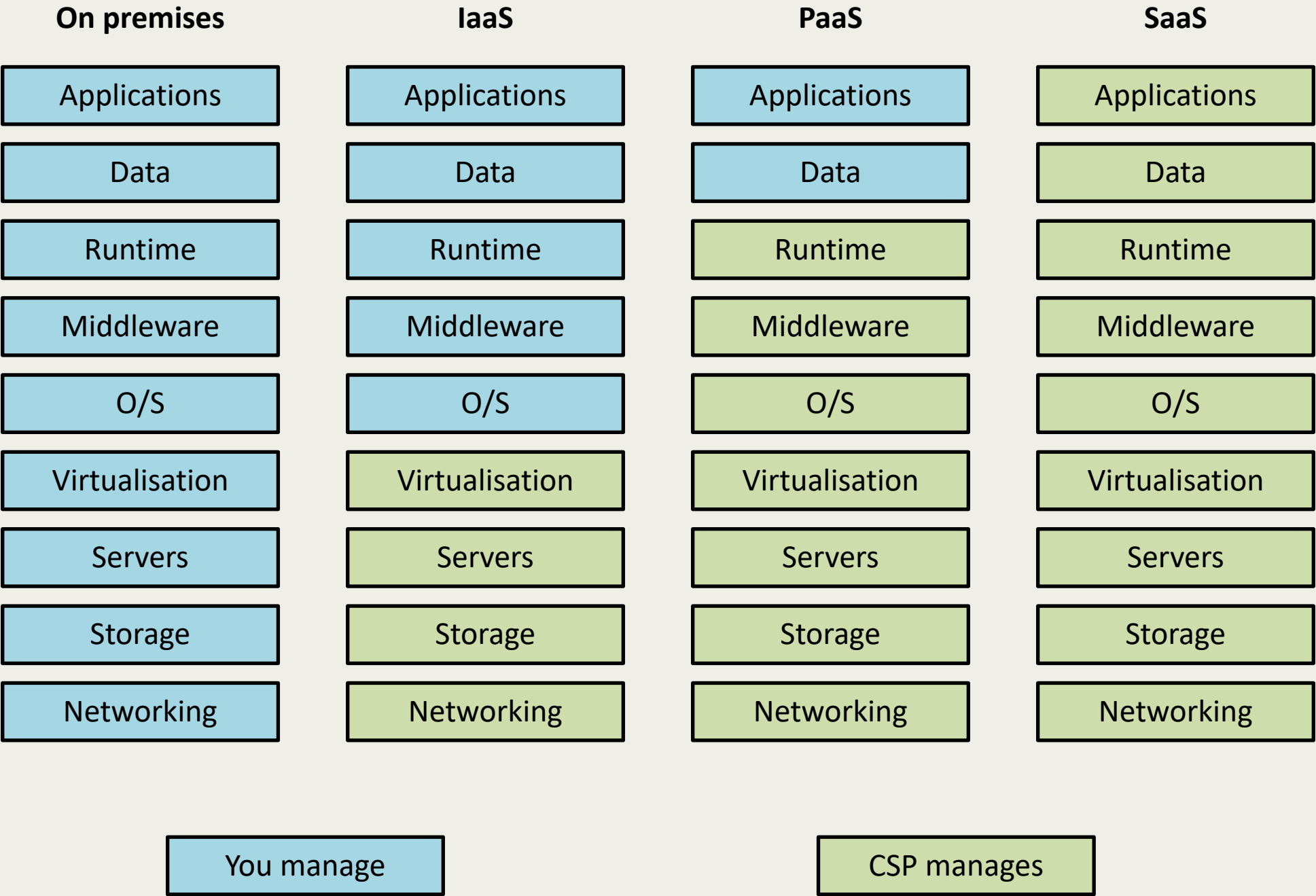


CLOUD SERVICE MODELS

CLOUD SERVICE MODELS

A cloud computing service model defines the type of service a CSP offers and the level of control you, the CSC, has over the resources.

- How much you (CSC) manage
- How much the CSP manages

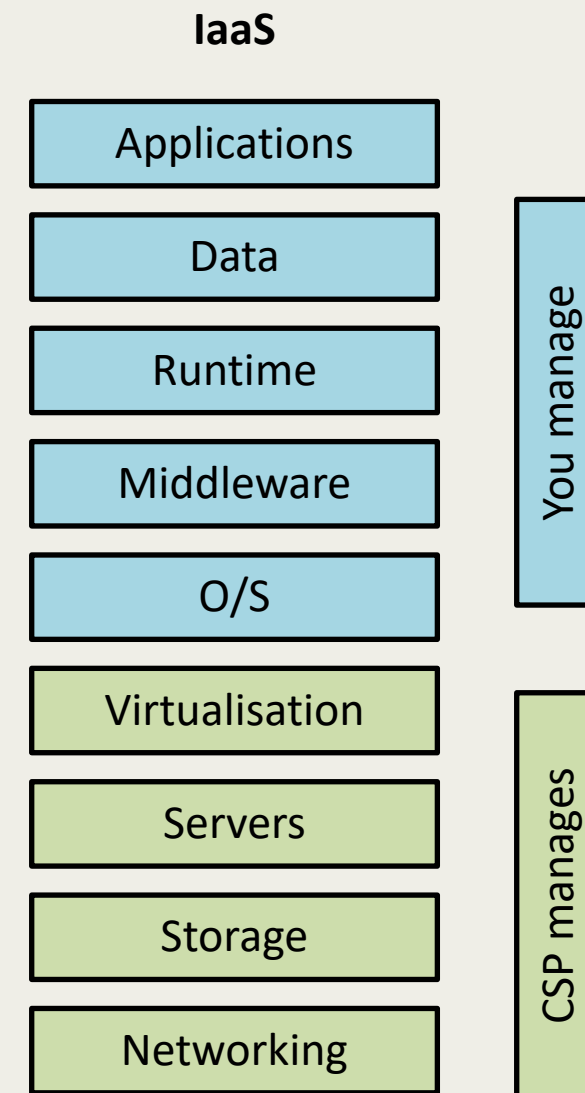


CLOUD SERVICE MODELS

Infrastructure as a Service (IaaS)
provides the fundamental building
blocks for cloud computing.

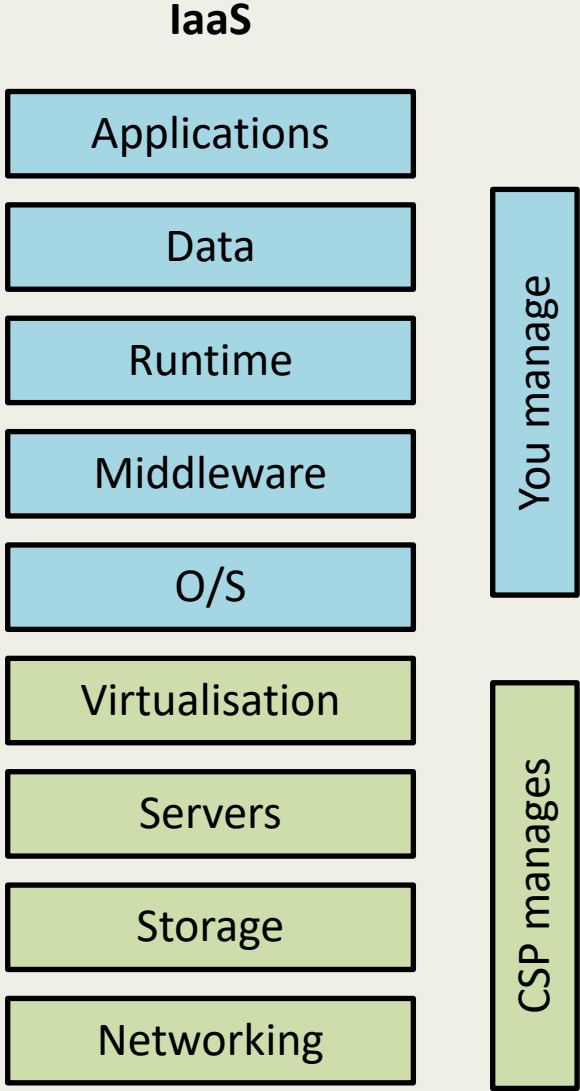
- Virtual servers
- Storage
- Networking

Think of it as renting pre-configured
servers and storage space,
eliminating the need to manage
physical hardware.



CLOUD SERVICE MODELS

- Amazon EC2 (virtual machines)
- Amazon S3 (storage)
- Amazon VPC (networking)

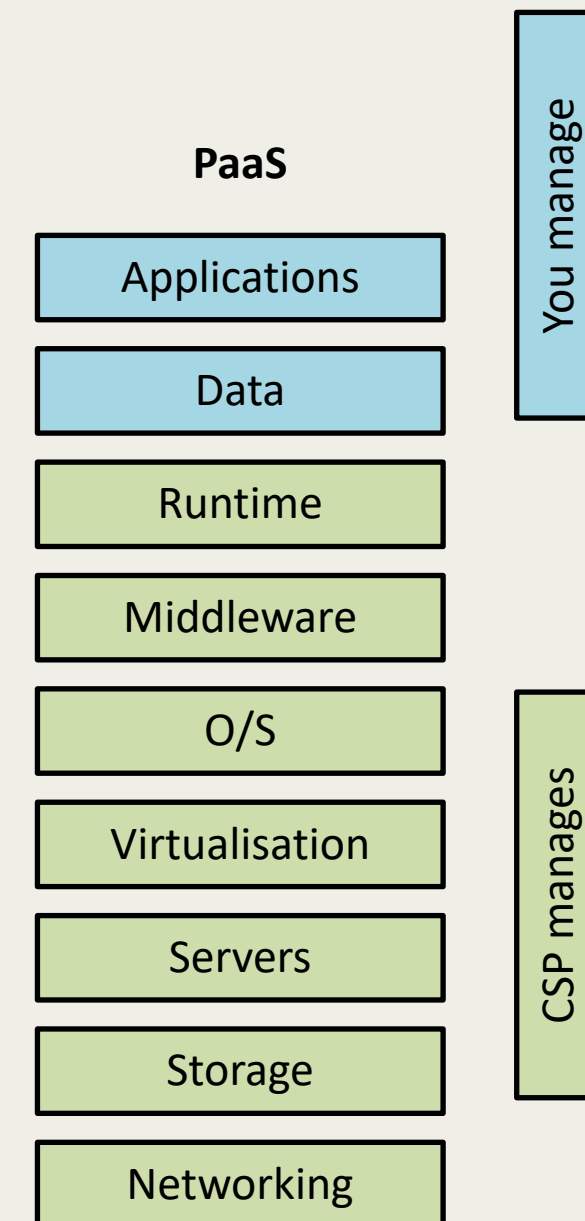


CLOUD SERVICE MODELS

Platform as a Service (PaaS) provides a complete development and deployment environment, including

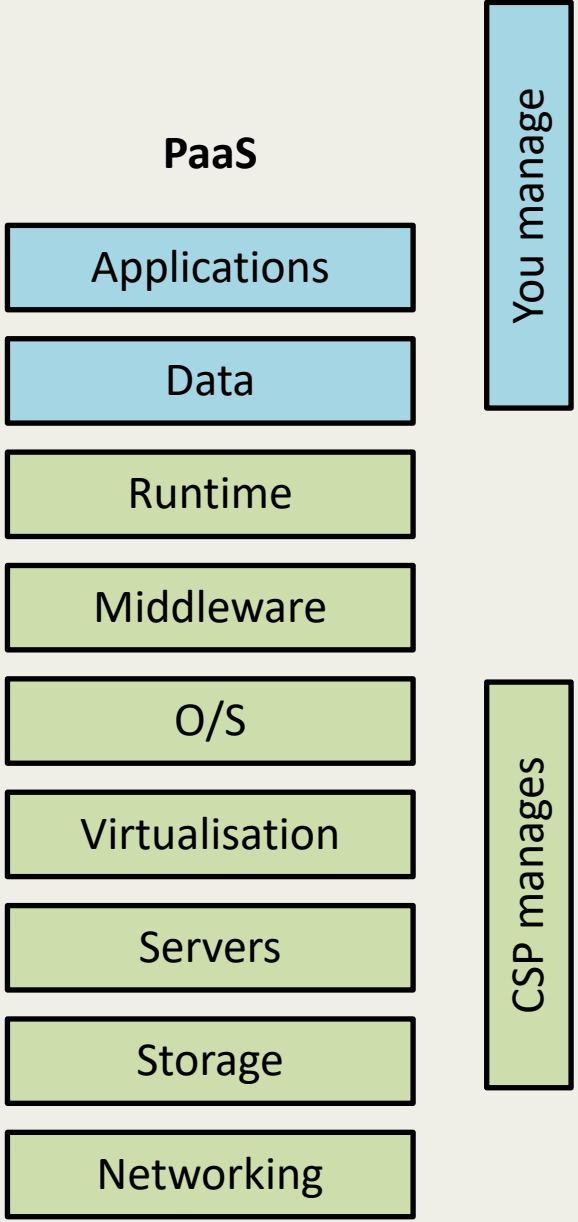
- Operating Systems
- Databases
- Development tools

It allows developers to focus on building applications without worrying about managing the underlying infrastructure.



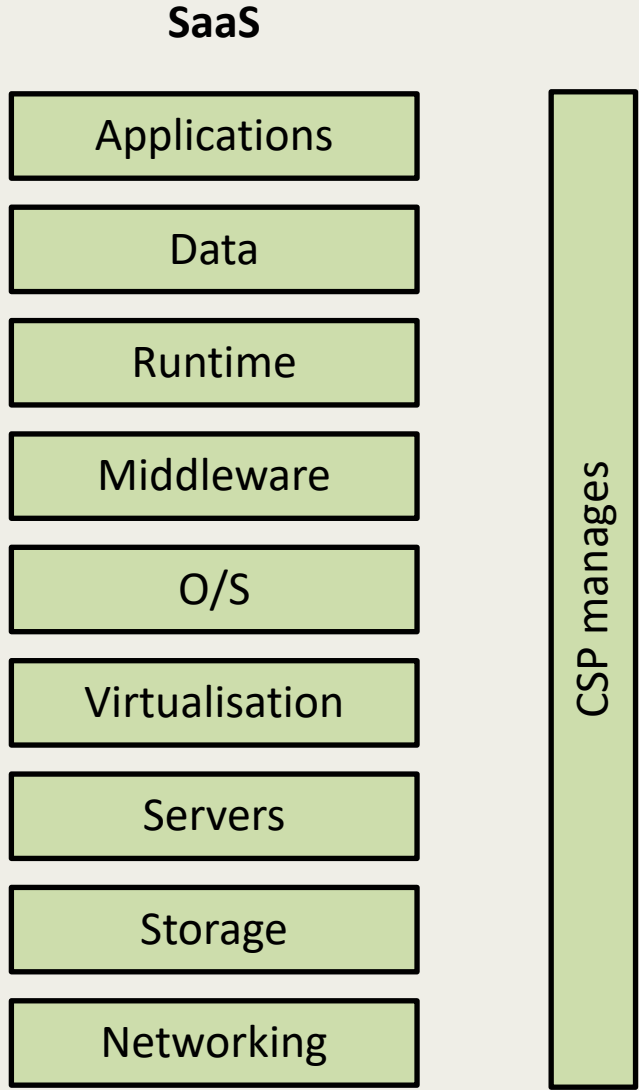
CLOUD SERVICE MODELS

- AWS Elastic Beanstalk (application deployment)
- AWS Lambda (serverless computing)
- Amazon RDS (relational database service)



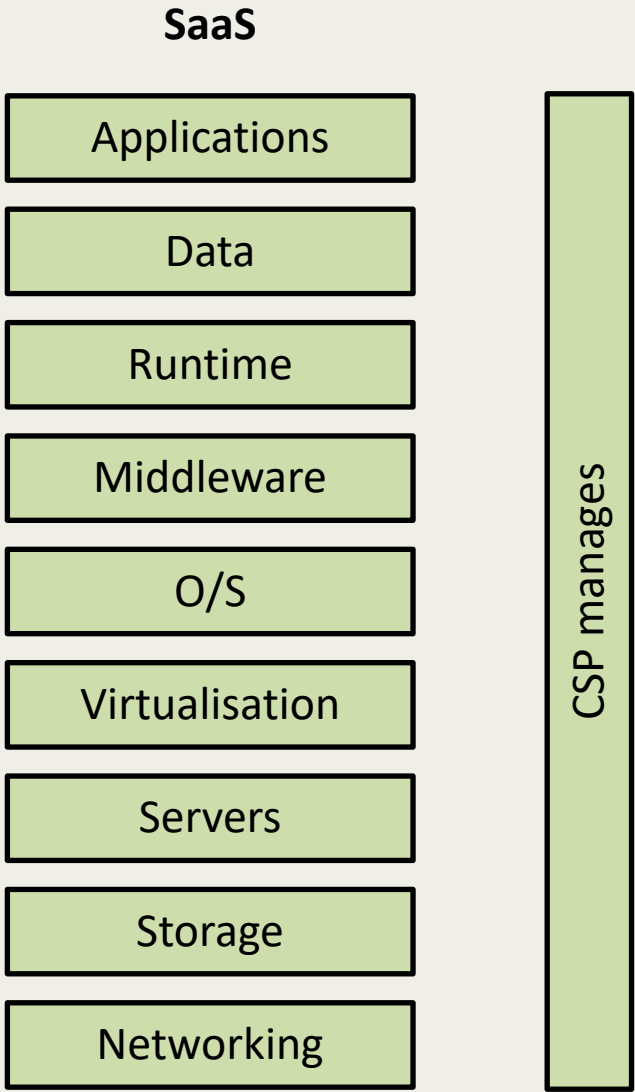
CLOUD SERVICE MODELS

Software as a Service (SaaS) provides ready-to-use applications delivered directly over the Internet.



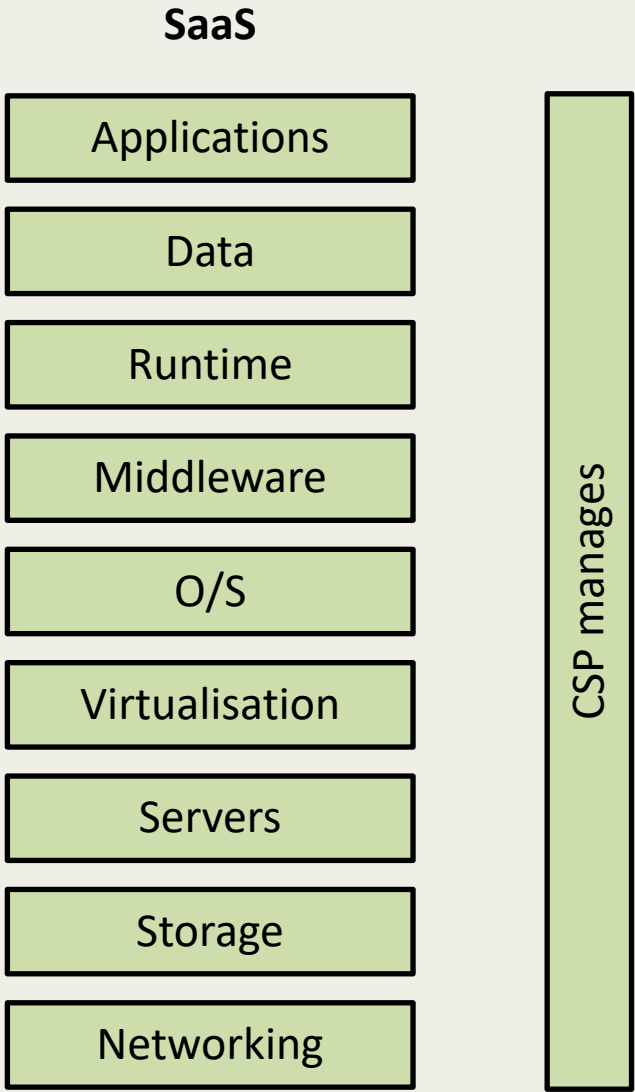
CLOUD SERVICE MODELS

Users simply access the application through a web browser or mobile app eliminating the need for installation or maintenance.



CLOUD SERVICE MODELS

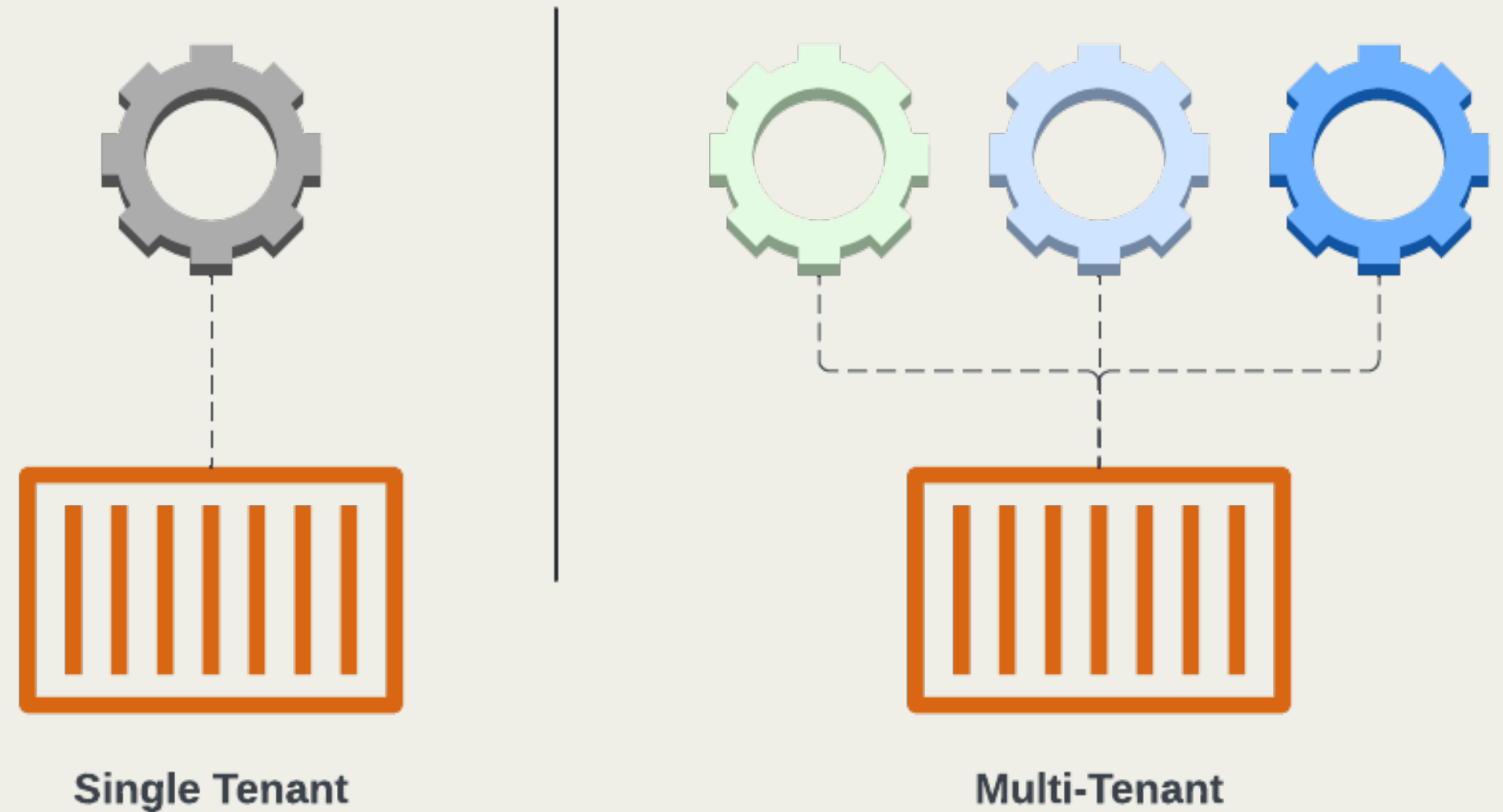
- Amazon WorkMail (email, calendar)
- Amazon WorkDocs (file storage, collaboration)
- Amazon Connect (Contact Center)



TENANCY MODELS IN CLOUD COMPUTING

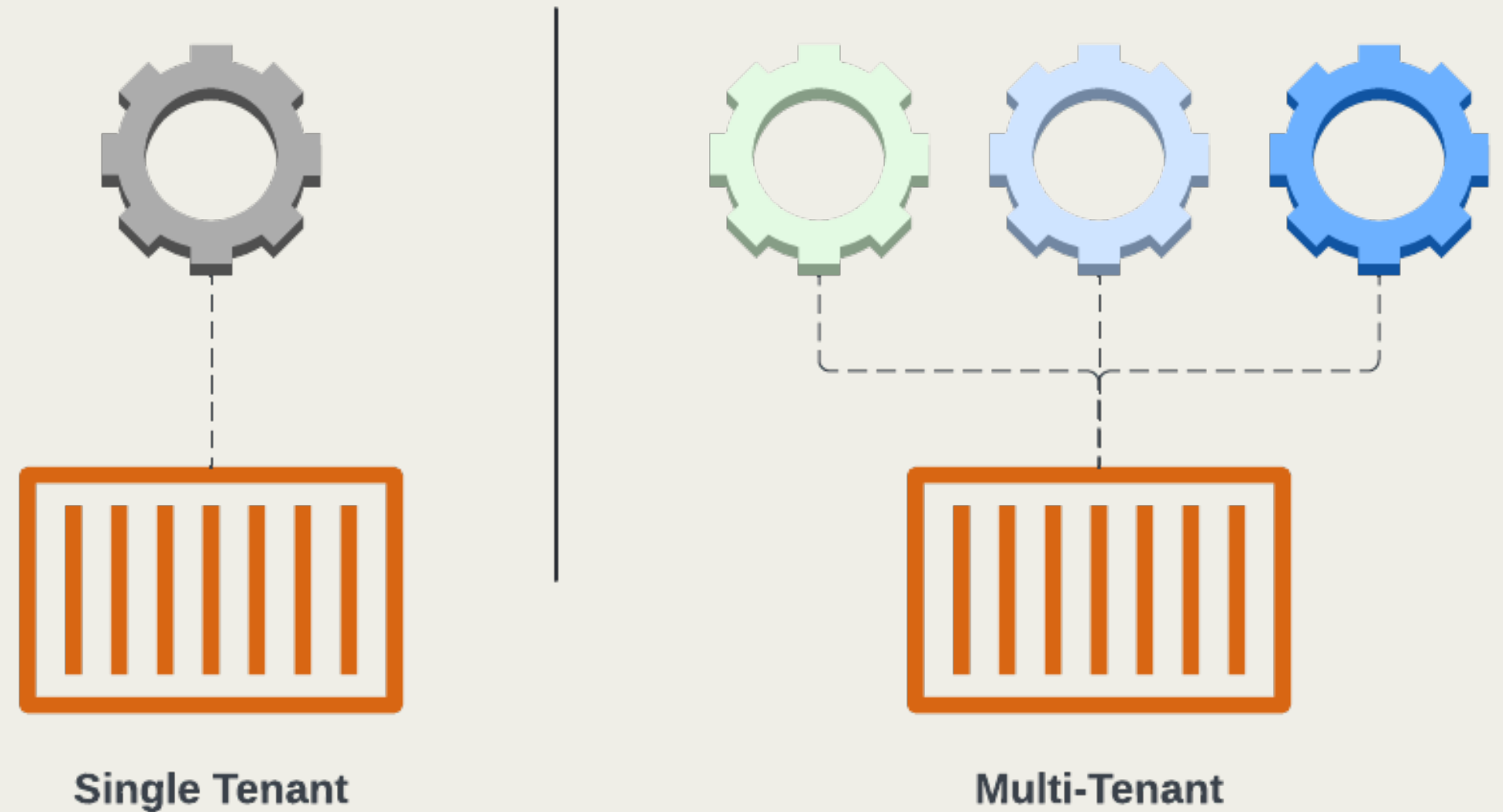
TENANCY MODELS IN CLOUD COMPUTING

Tenancy in cloud computing refers to the way computing resources are shared among different users or organisations within a cloud environment.



TENANCY MODELS IN CLOUD COMPUTING

Tenancy defines the isolation and access control mechanisms that separate user data and workloads.

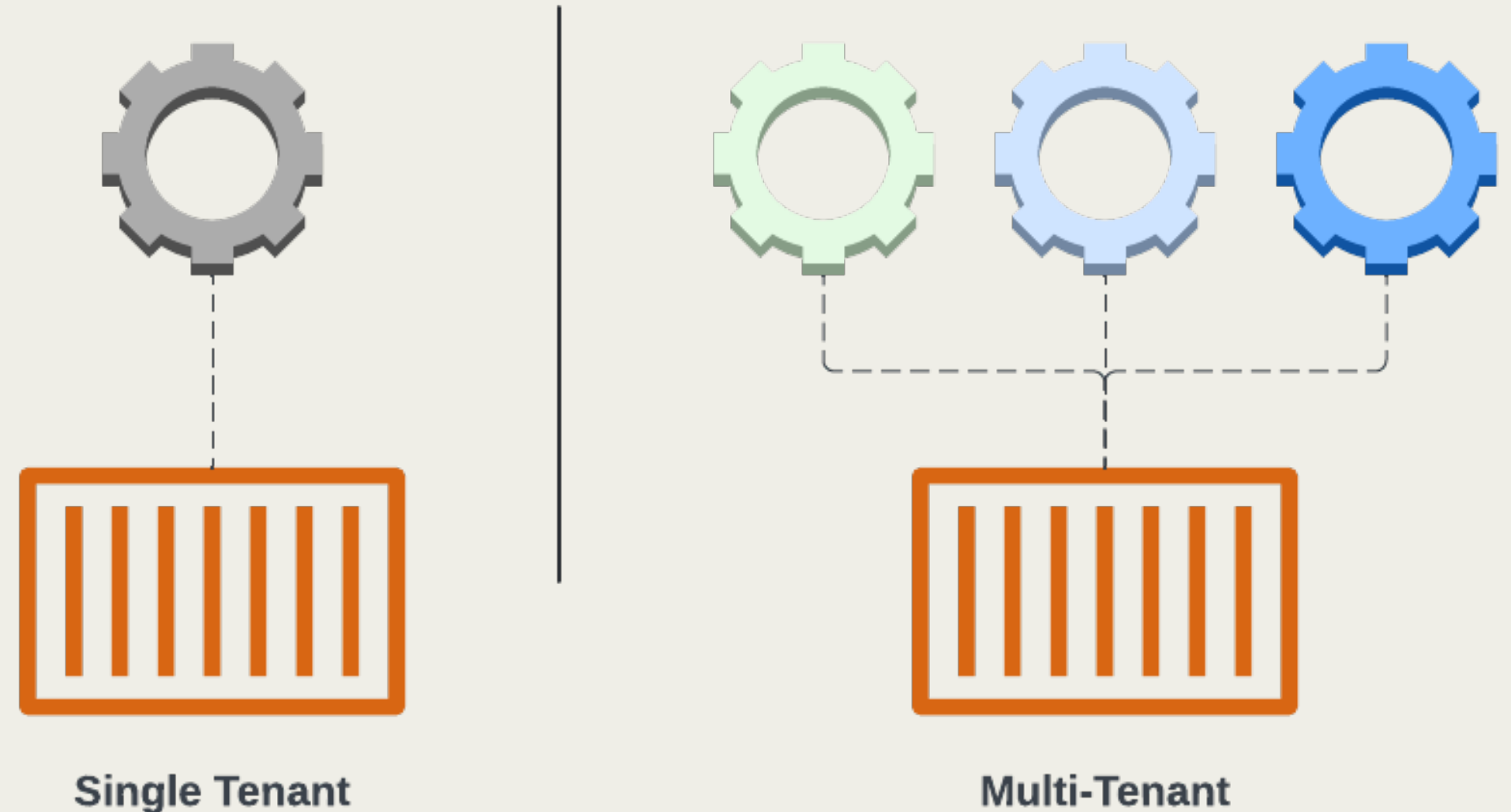


TENANCY MODELS IN CLOUD COMPUTING

Single tenant model

Provides dedicated resources (infrastructure, applications) to a single customer

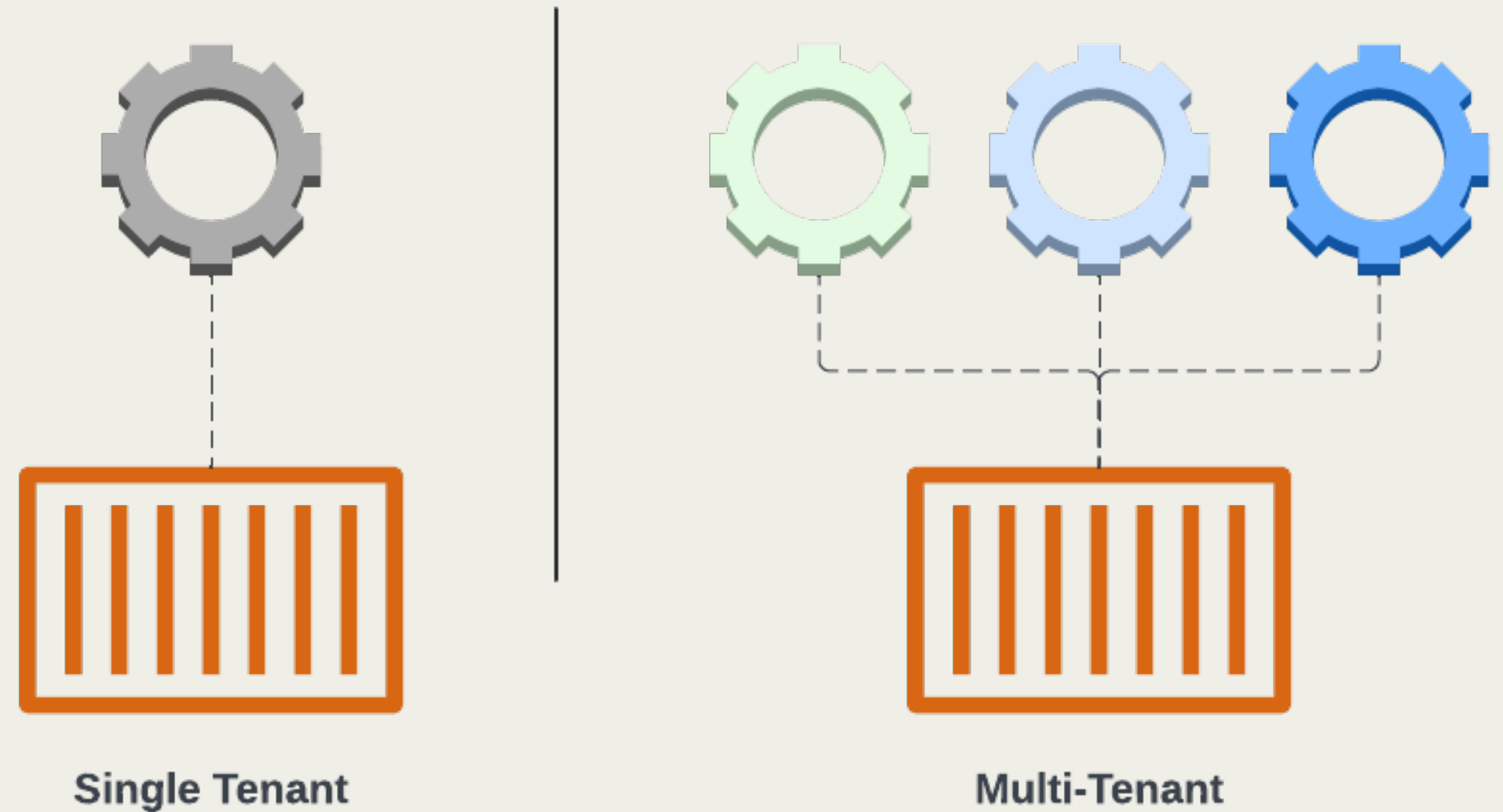
- The customer has complete control and isolation
- It offers the highest level of security and performance
- It can be less cost-effective for smaller workloads



TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

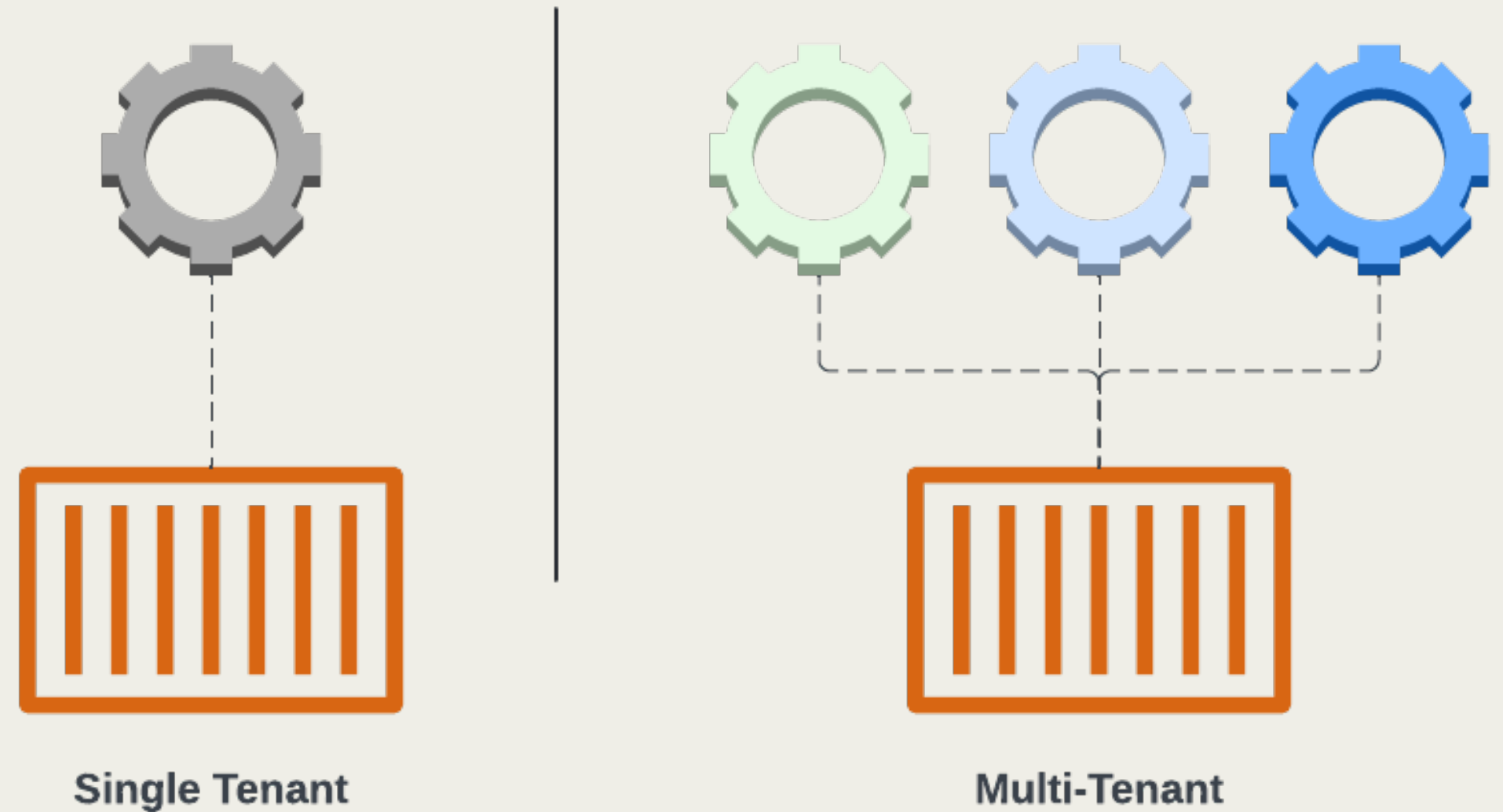
Shares the underlying infrastructure (hardware, storage, and networking) among multiple customers (tenants).



TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

The CSP manages the virtualisation and isolation between tenants to ensure security and performance.

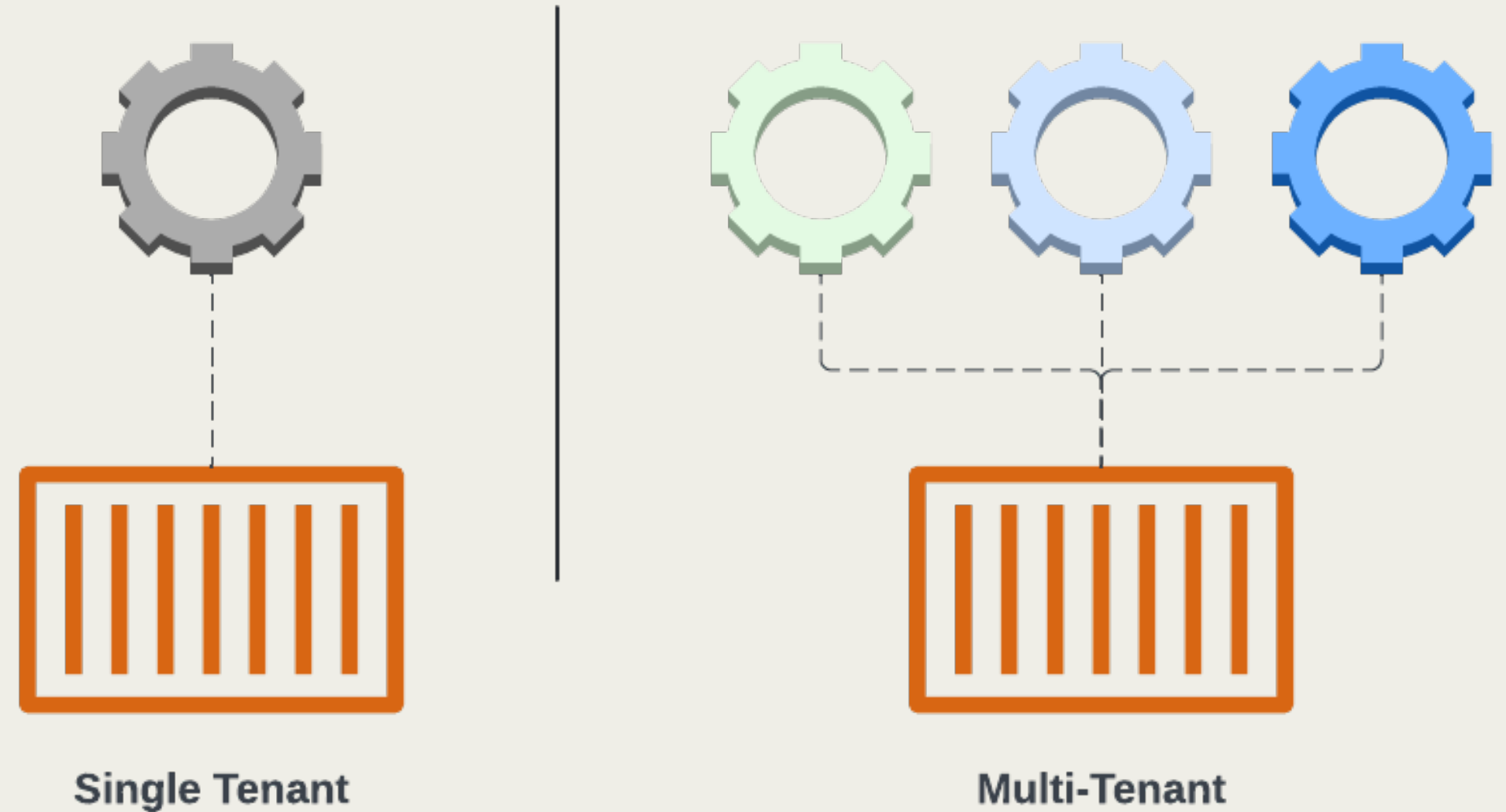


TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

Its sub-models offer scalability, and cost benefits.

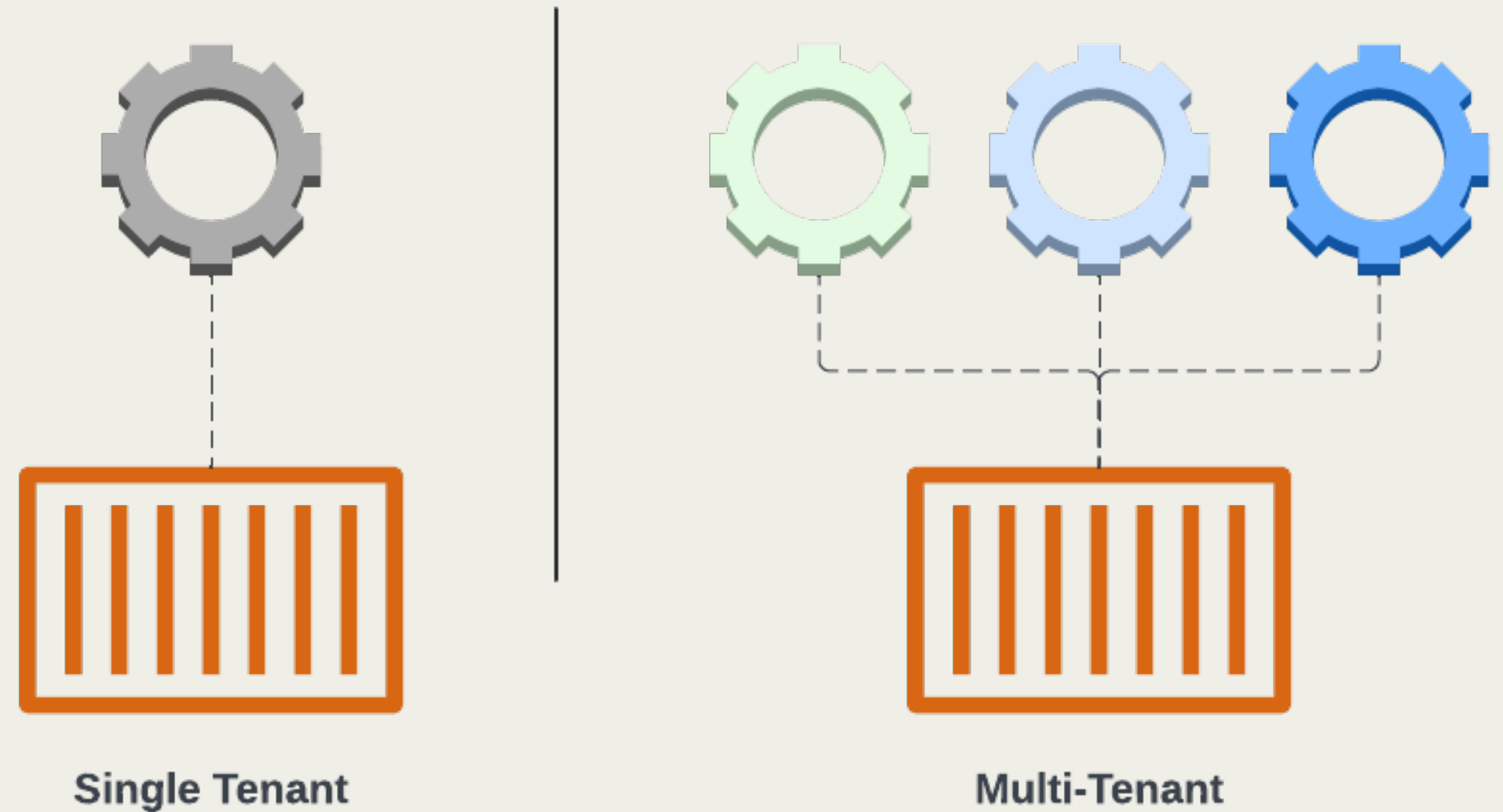
It may have limitations on control and customisation compared to the single tenant model.



TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

- Resource-level isolation
- Container-level isolation
- Serverless computing

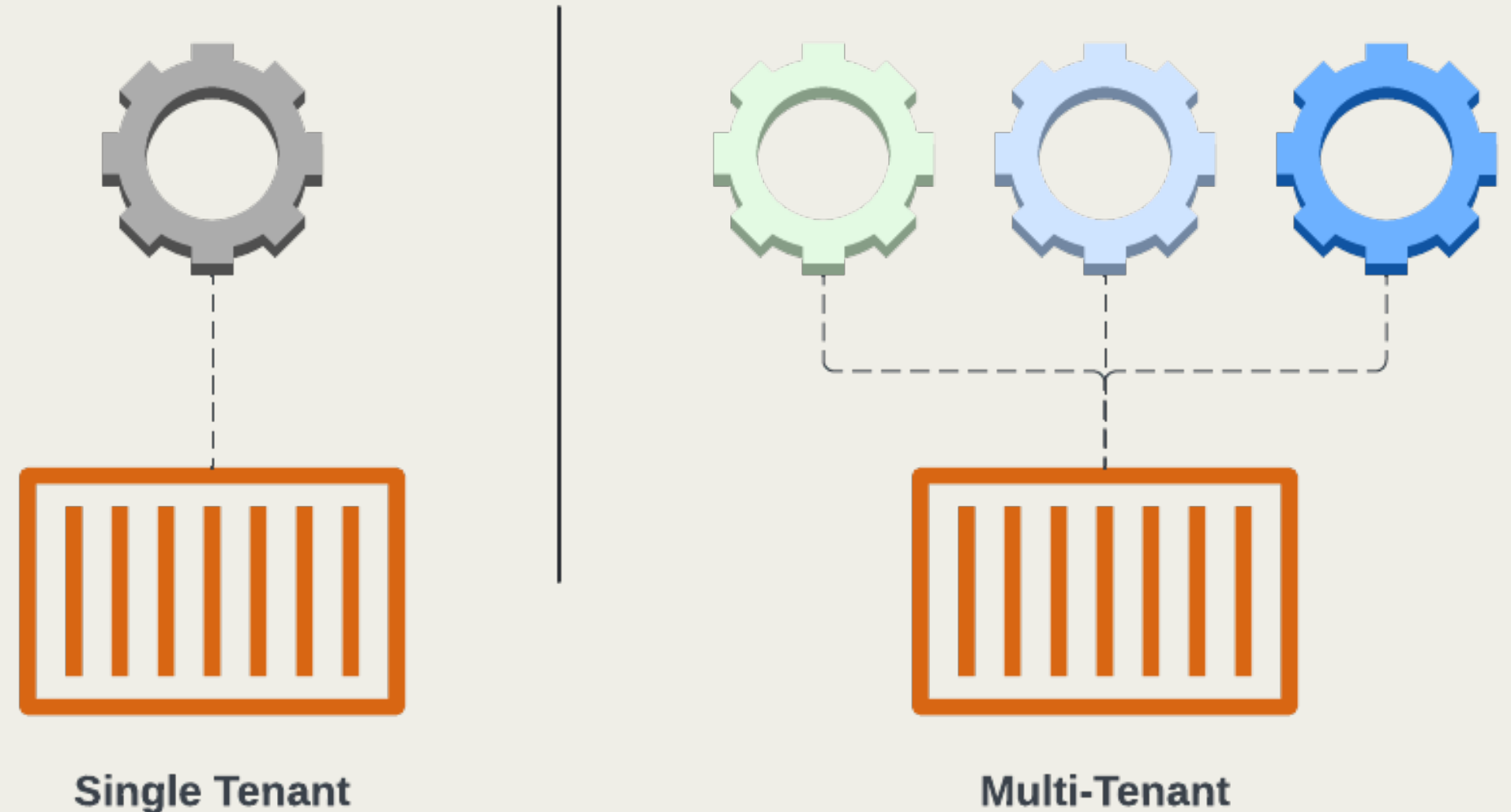


TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

Resource-level isolation

- Tenants share physical resources but are isolated at the virtual machine level.
- Each tenant gets their own virtualised environment with an operating system and applications.

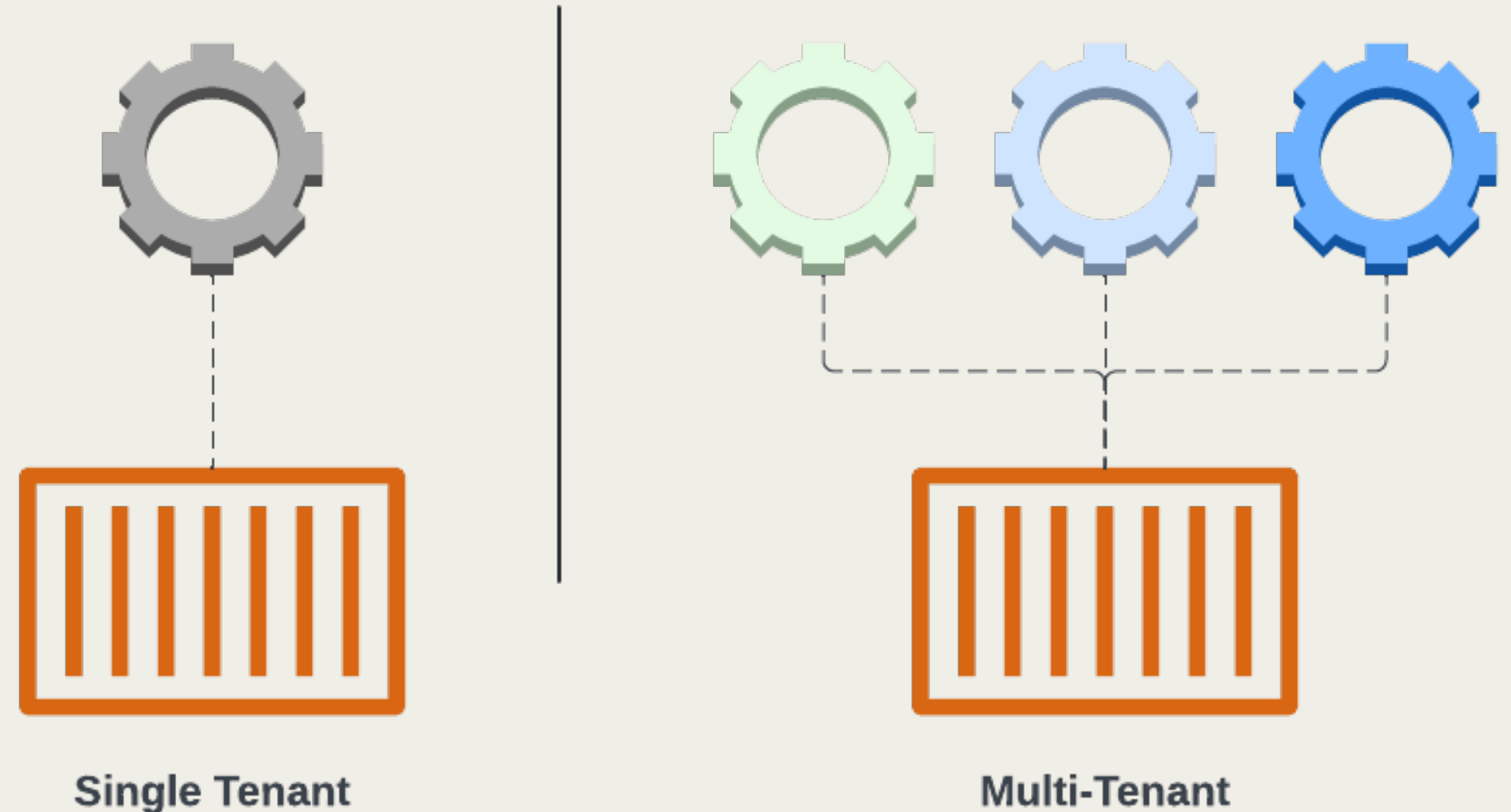


TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

Container-level isolation

- Tenants share the underlying infrastructure but are isolated at the container level.
- Containers are lightweight and share the OS kernel, making them more efficient than virtual machines.

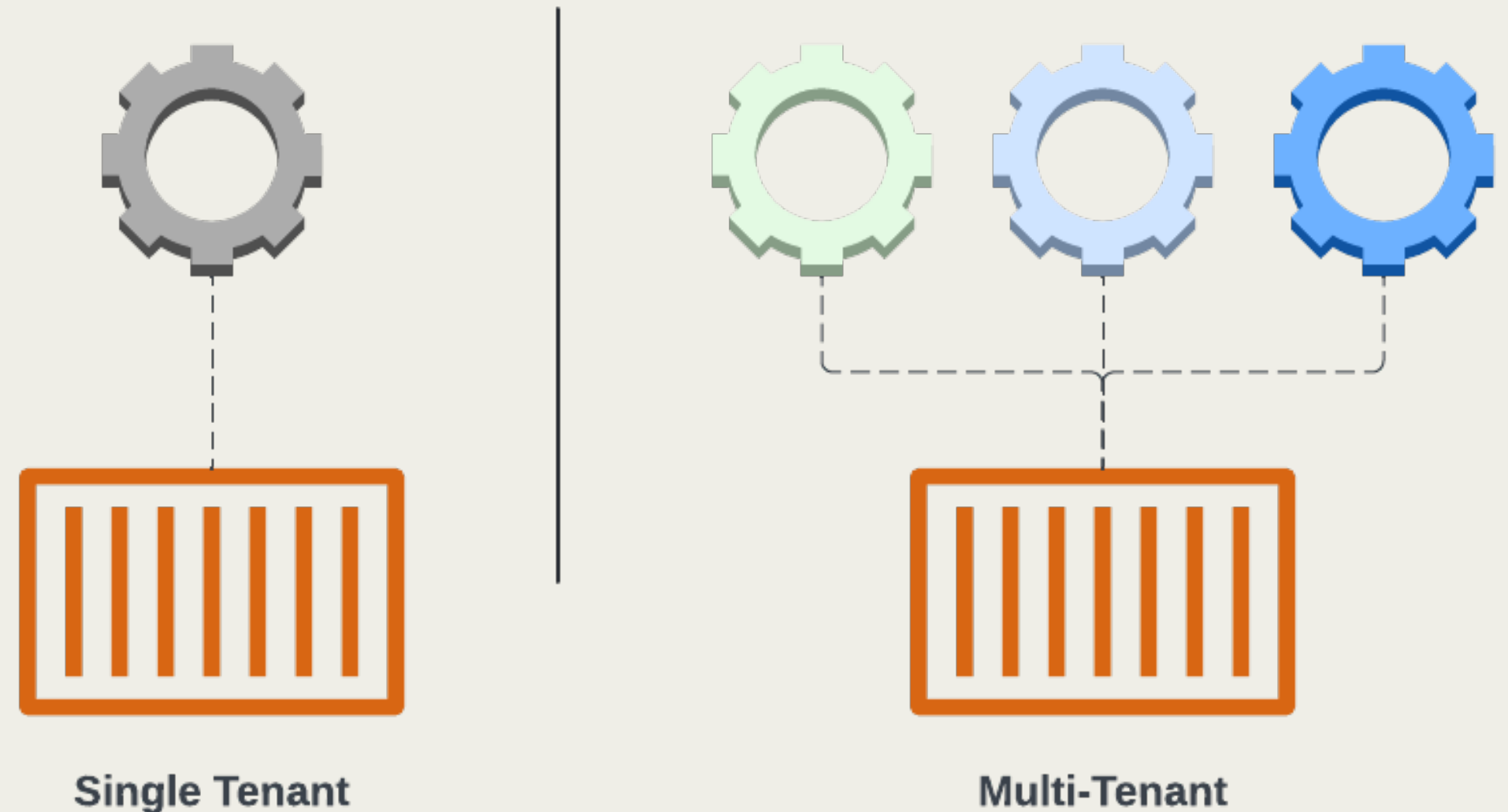


TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

Serverless computing

- The CSP manages the entire infrastructure and users pay based on usage.
- Tenants have no control over the underlying infrastructure but benefit from extreme scalability and cost efficiency.

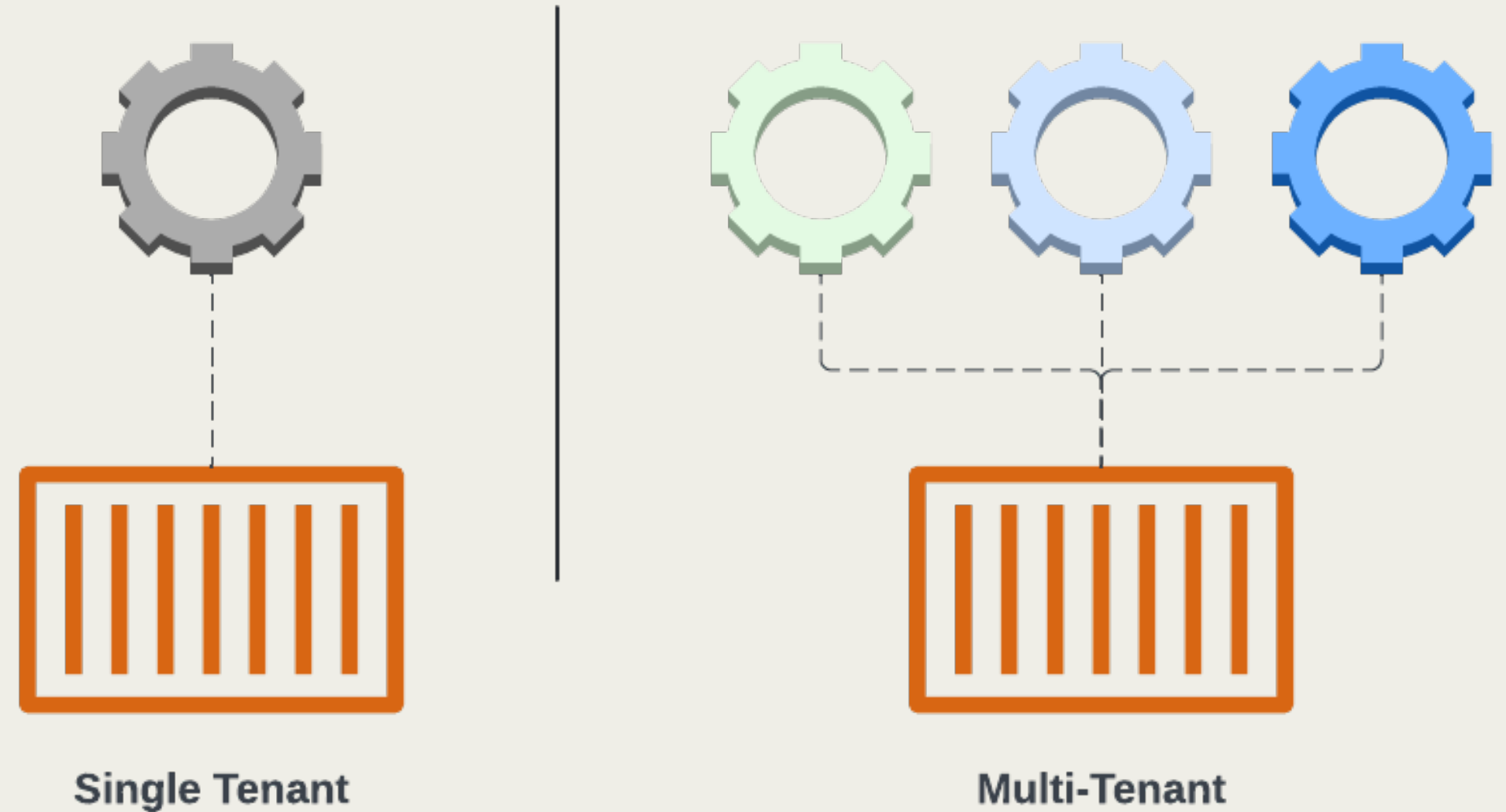


TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

Imagine a scenario where,

- Application A has high CPU utilisation for a few hours in the morning.
- Application B uses small peaks of maximum CPU throughout the day.

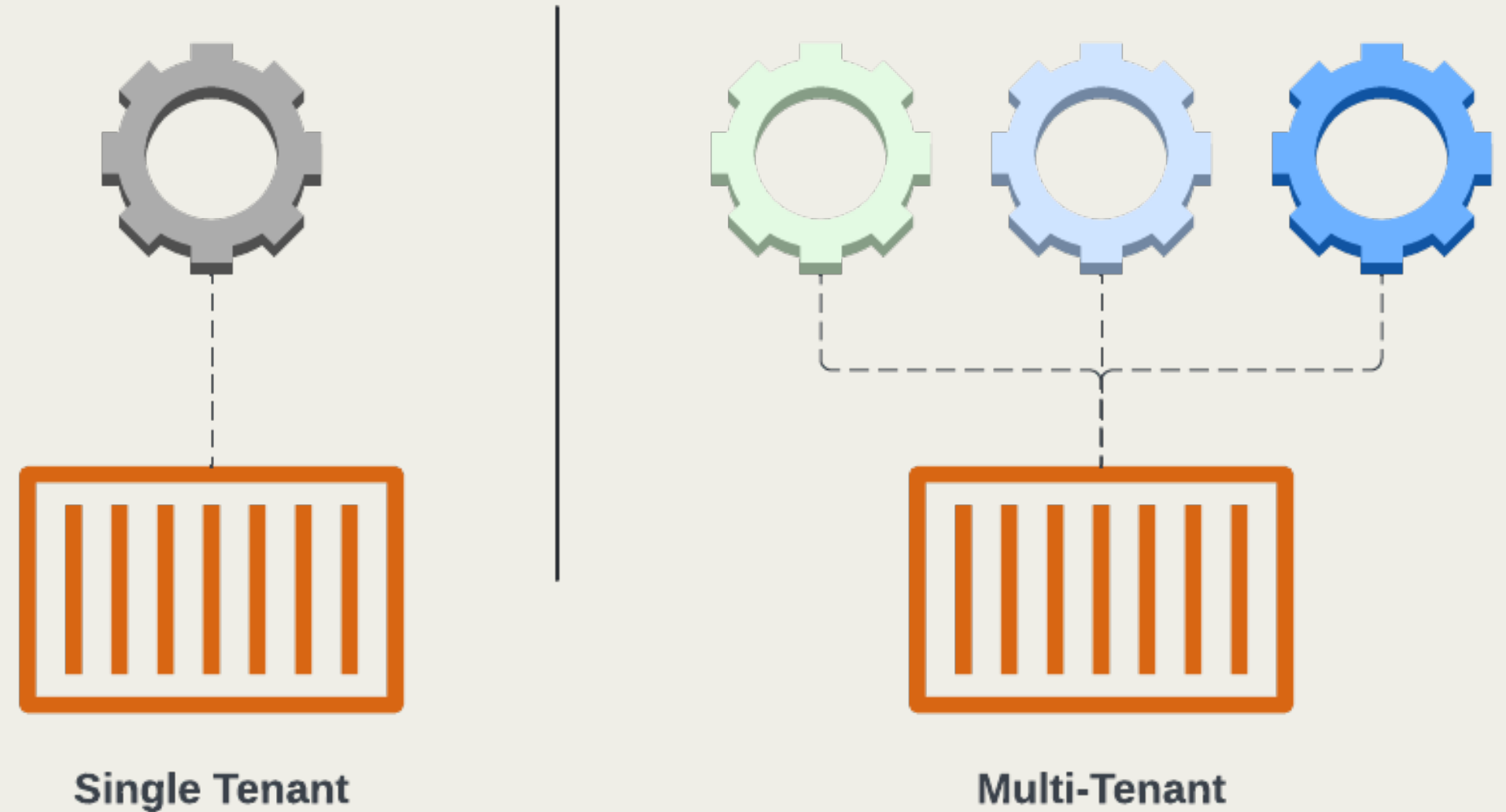


TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

Almost all computing workloads vary in their work needs over time.

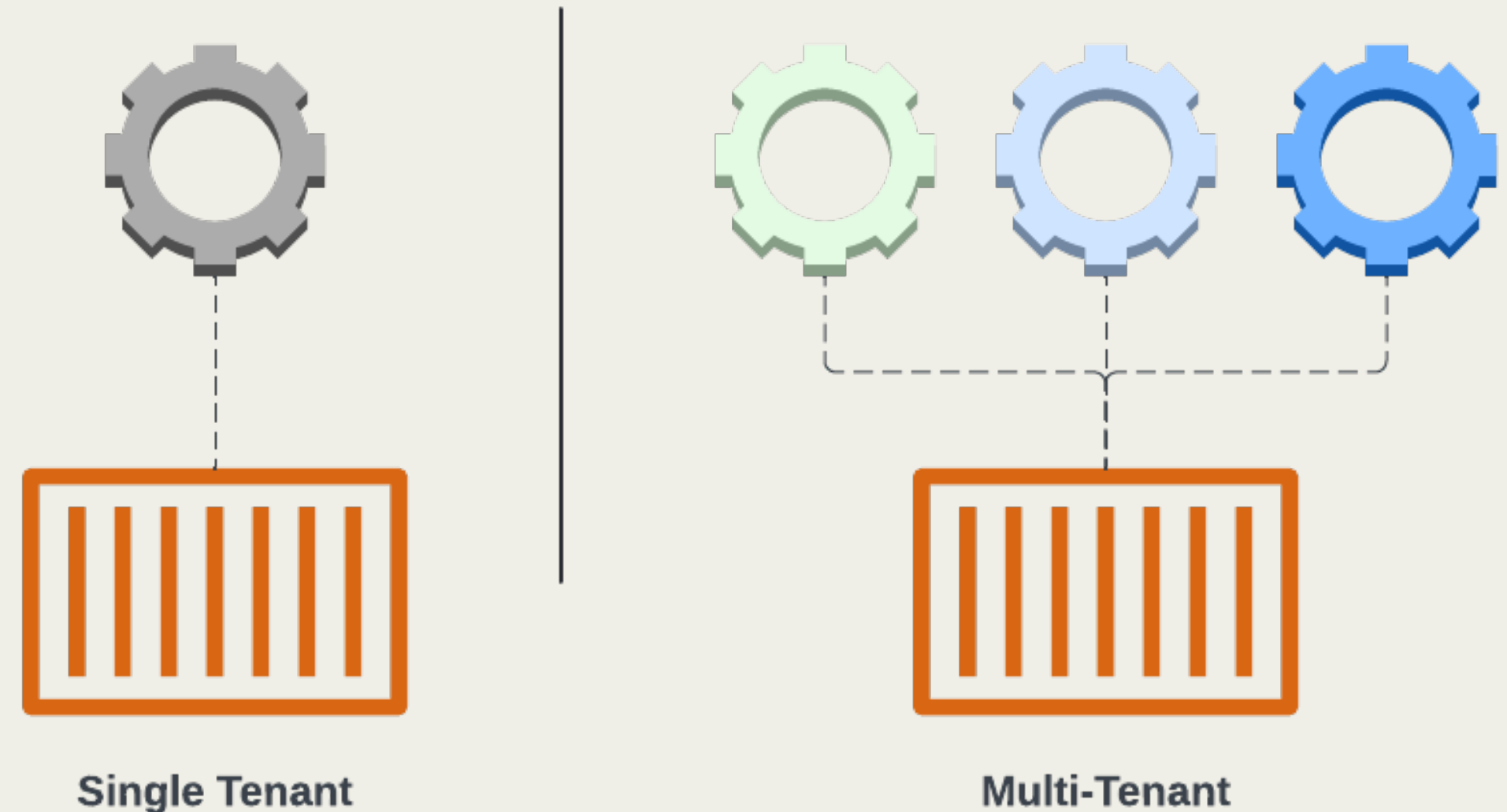
- Multi-tenancy allows CSPs to oversubscribe their resources.



TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

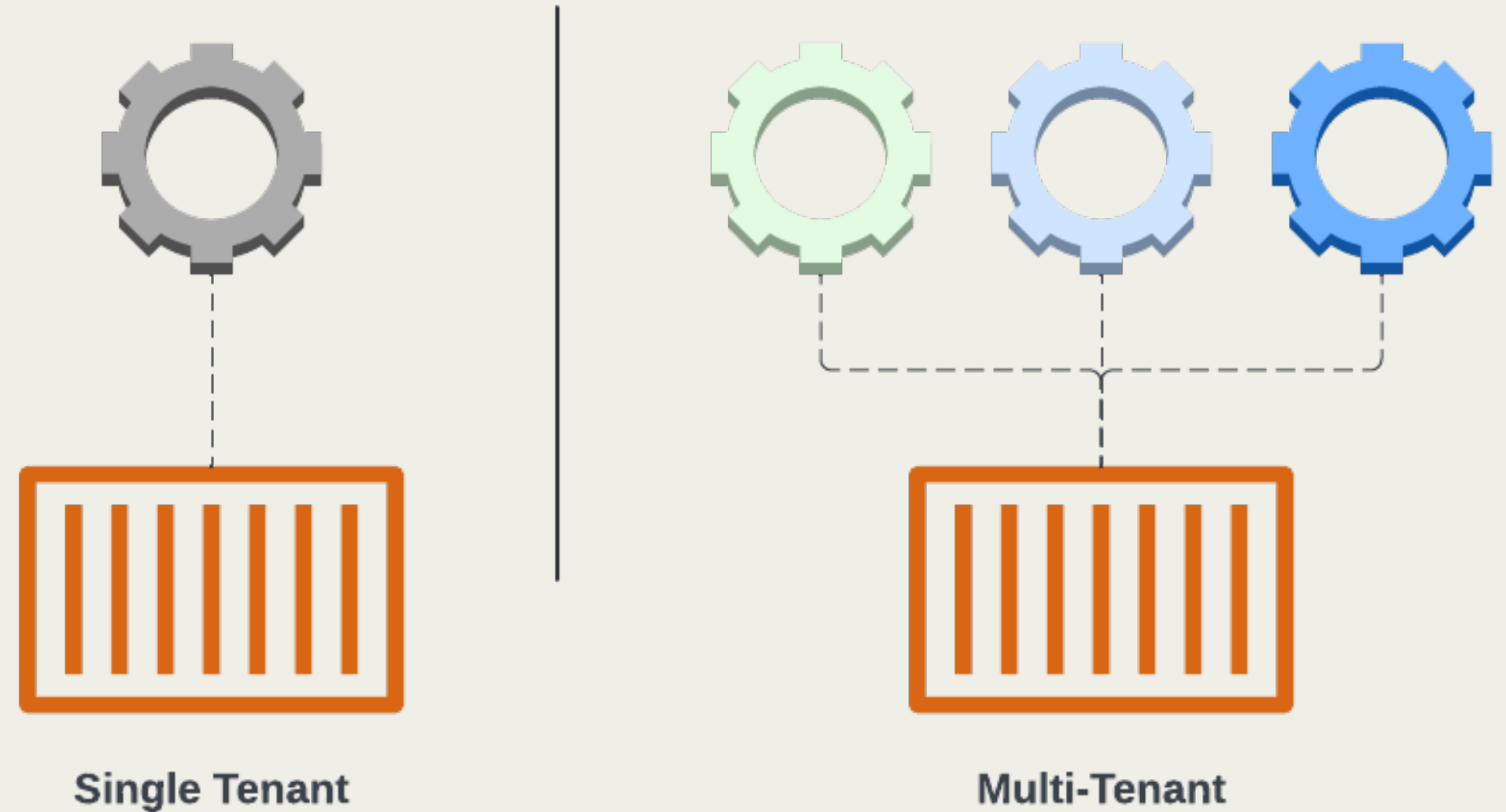
Oversubscription means that CSPs can sell customers a total capacity that exceeds the actual physical capacity of their infrastructure, because customers can never use all of their capacity simultaneously.



TENANCY MODELS IN CLOUD COMPUTING

Multi-tenant model

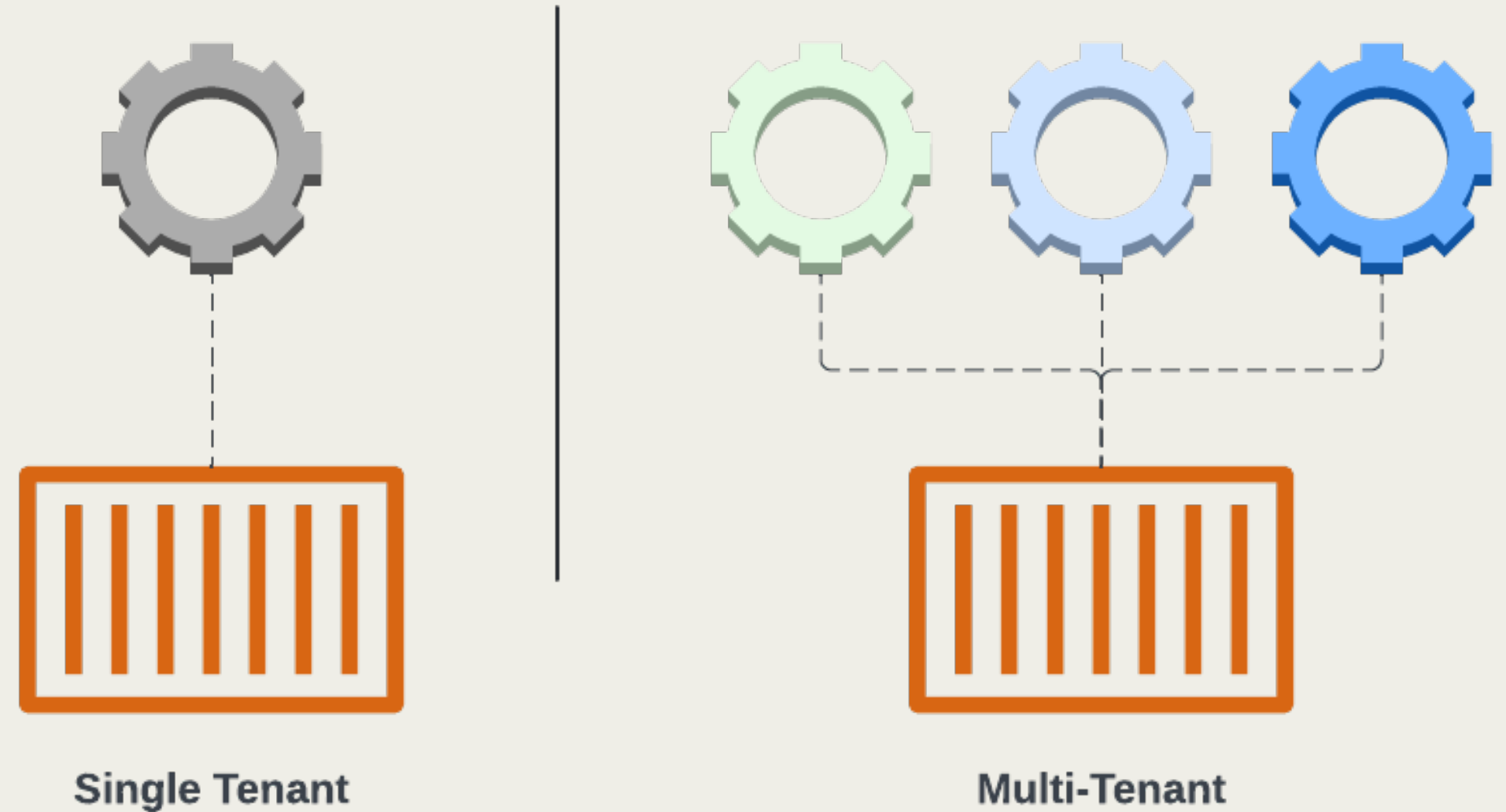
With oversubscription, the memory and CPU capacity of the physical environment are shared among many different users (resource pooling) and can be reassigned as needed.



TENANCY MODELS IN CLOUD COMPUTING

What are some of the drawbacks of multi-tenancy?

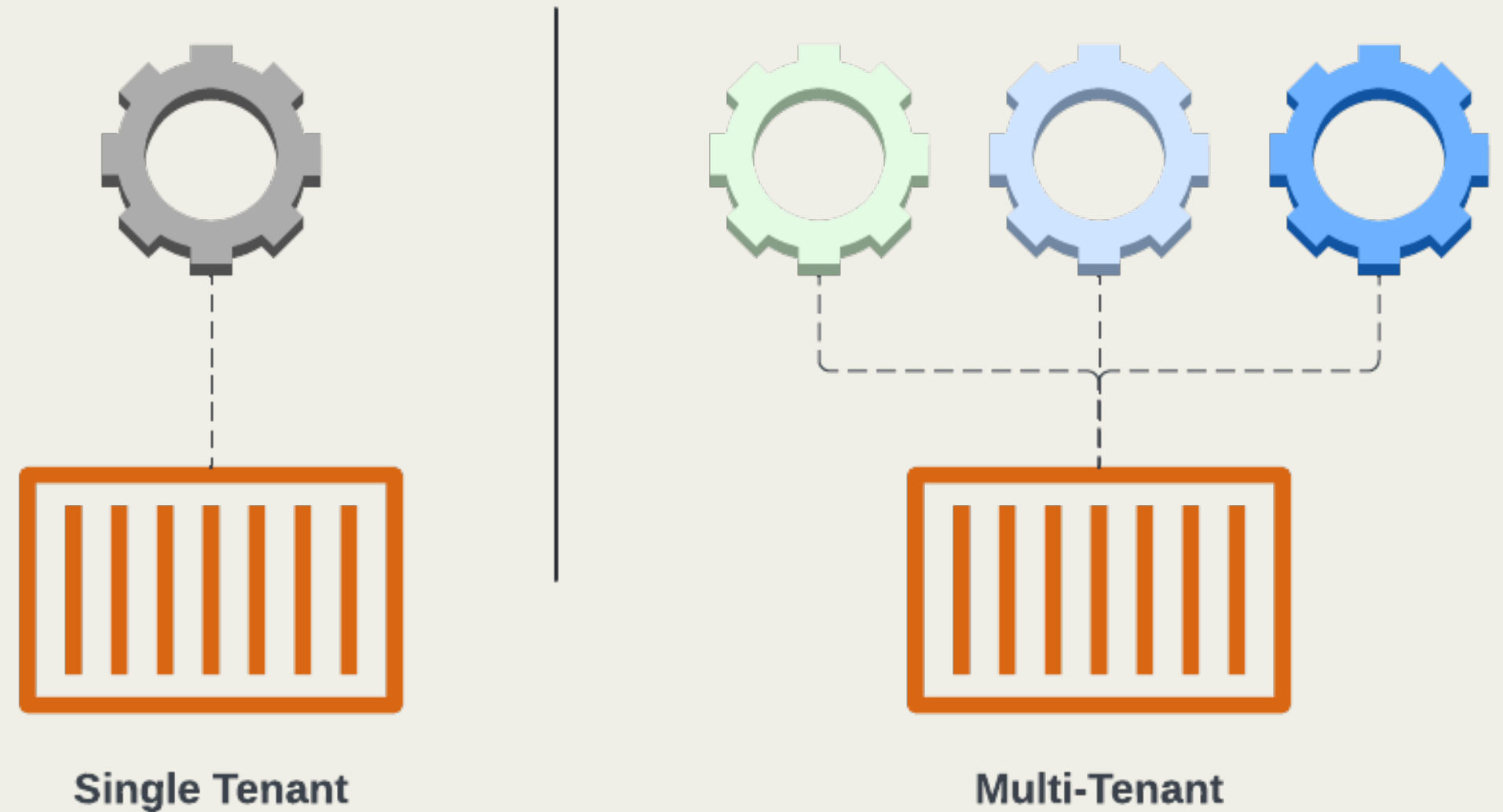
Some companies may not be able to store data within shared infrastructure, no matter how secure, due to regulatory requirements.



TENANCY MODELS IN CLOUD COMPUTING

What are some of the drawbacks of multi-tenancy?

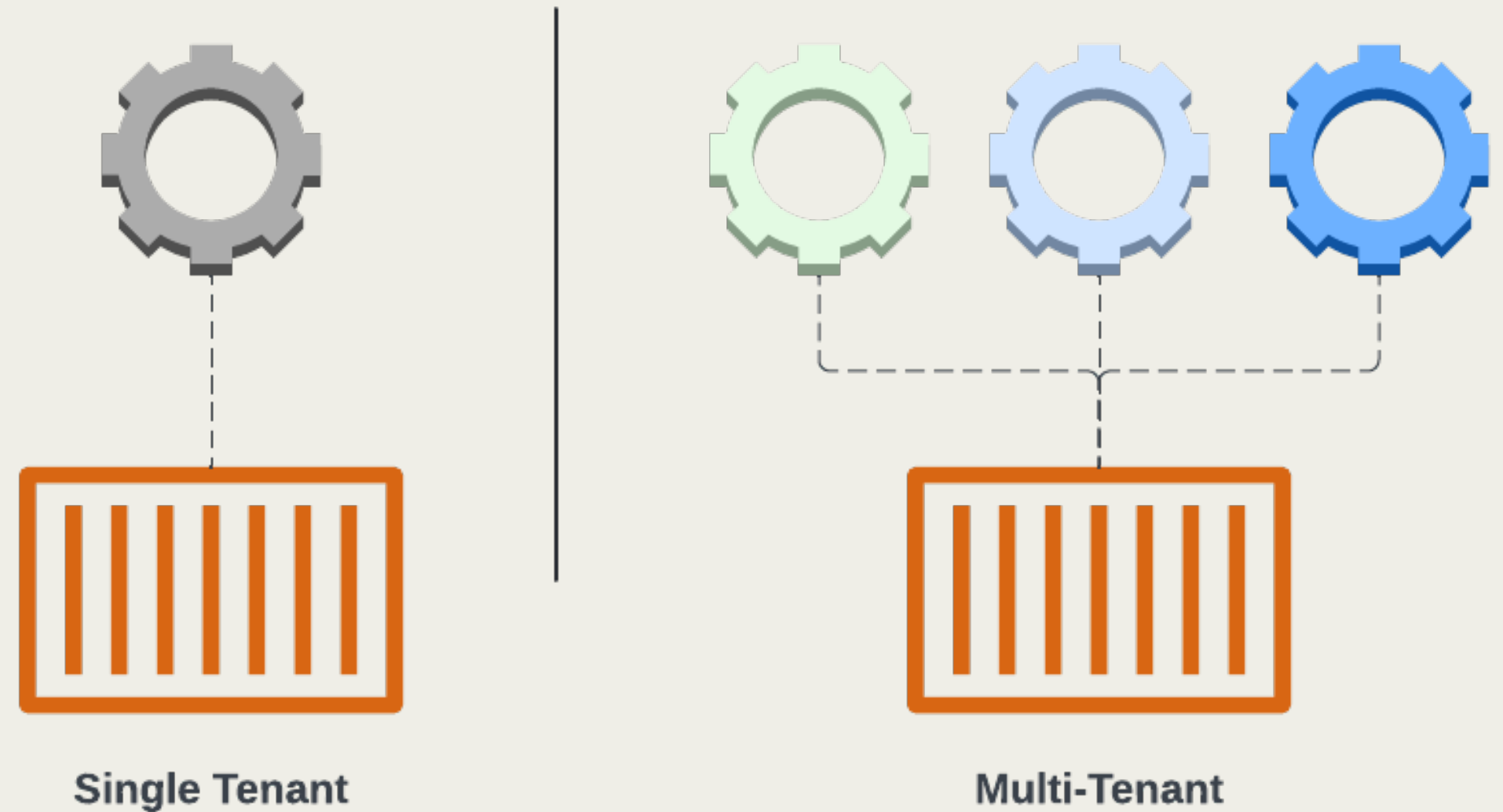
Security problems or corrupted data from one tenant could spread to other tenants on the same machine.



TENANCY MODELS IN CLOUD COMPUTING

What are some of the drawbacks of multi-tenancy?

"Noisy neighbour effect" - If one tenant is using an inordinate amount of computing power, this could slow down performance for the other tenants.



CLOUD DEPLOYMENT MODELS

DEPLOYMENT MODELS IN CLOUD COMPUTING

Cloud deployment models describe different **approaches** to the way organisations might implement cloud services.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

Cloud deployment models identify the specific type of cloud environment based on

- Ownership
- Scale
- Access
- Nature
- Purpose

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
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<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

Cloud deployment models define

- The location of the servers you are using
- Who controls access to the servers you are using

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
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<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The cloud deployment model specifies

- How your infrastructure will look like
- What you can change
- Whether you will be provided services or will have to create everything yourself

Factors	Public	Private	Community	Hybrid
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<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The cloud deployment model defines the **relationship** between the infrastructure and your users.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The public cloud

- Resources are owned and operated by a third-party CSP
- Resources are delivered over the Internet
- Resources are shared among multiple users (tenants)
- Resources are provided through a pay-as-you-go pricing model

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The public cloud

Security in the public cloud is a shared responsibility.

- The CSP is responsible for securing the underlying infrastructure
- Users are responsible for securing their data and applications

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The public cloud

Public clouds offer *high scalability*.

- Users can easily increase or decrease their resource usage based on demand
- Scalability eliminates the need to invest in and maintain hardware

Factors	Public	Private	Community	Hybrid
<i>Initial setup</i>	Easy	Complex	Complex	Complex
<i>Scalability and flexibility</i>	High	Less flexible	Fixed	High
<i>Cost-comparison</i>	Cost-effective	Costly	Distributed	Between public and private cloud
<i>Data security</i>	Shared responsibility	High	High	High
<i>Data privacy</i>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The public cloud

Public clouds offer various cloud service models

- IaaS - Provides raw compute like servers, storage and networking
- PaaS - Provides a platform for developing, deploying and managing applications
- SaaS - Delivers applications over the Internet

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The public cloud

Examples of public cloud providers include

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Factors	Public	Private	Community	Hybrid
<i>Initial setup</i>	Easy	Complex	Complex	Complex
<i>Scalability and flexibility</i>	High	Less flexible	Fixed	High
<i>Cost-comparison</i>	Cost-effective	Costly	Distributed	Between public and private cloud
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<i>Data privacy</i>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The private cloud

- Resources are dedicated to a single organisation
- Infrastructure can be on-premises in the organisation's data centre
- Infrastructure can be off-premises in dedicated environment managed by a service provider

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The private cloud

- Only accessible to authorised users within the organisation

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The private cloud

Private clouds offer a high level of security and control as resources are not shared with other organisations.

Organisations have full control over their data and applications.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
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<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The private cloud

- They offer scalability but are less flexible than public clouds.
- Scaling resources may require additional investment in hardware or software within the private cloud environment.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
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DEPLOYMENT MODELS IN CLOUD COMPUTING

The private cloud

They can be more expensive than public clouds to the cost of managing and maintaining the infrastructure.

Some organisations find the increased security and control justify the cost.

Factors	Public	Private	Community	Hybrid
<i>Initial setup</i>	Easy	Complex	Complex	Complex
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<i>Data privacy</i>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The private cloud

Examples of private cloud solutions include

- VMware Cloud Foundation
- OpenStack
- Microsoft Azure Stack

Factors	Public	Private	Community	Hybrid
<i>Initial setup</i>	Easy	Complex	Complex	Complex
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<i>Data privacy</i>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The community cloud

A deployment model where infrastructure and resources are shared among a specific group of organisations with common

- Interests
- Regulatory requirements
- Security needs

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
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<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The community cloud

This deployment model can be managed by

- One or more member organisations
- A third-party provider
- A combination of both

Factors	Public	Private	Community	Hybrid
<i>Initial setup</i>	Easy	Complex	Complex	Complex
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<i>Data privacy</i>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The community cloud

Accessibility is restricted to pre-approved organisations within the community.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The community cloud

Compared to the public cloud, security is enhanced due to the limited user base with shared concerns.

- However, it may not provide the same level of control as a private cloud.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
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<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The community cloud

Community clouds offer *scalability* based on the combined needs of member organisations.

Factors	Public	Private	Community	Hybrid
<i>Initial setup</i>	Easy	Complex	Complex	Complex
<i>Scalability and flexibility</i>	High	Less flexible	Fixed	High
<i>Cost-comparison</i>	Cost-effective	Costly	Distributed	Between public and private cloud
<i>Data security</i>	Shared responsibility	High	High	High
<i>Data privacy</i>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The community cloud

Costs are typically shared among member organisations, leading to potential cost savings compared to individual private cloud deployments.

- It can be more cost-effective than public clouds for specific workloads requiring high security and control.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The community cloud

Uses cases for community clouds could include

- A consortium of healthcare providers sharing a cloud for patient data management
- A group of universities collaborating on a research cloud environment.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
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<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The hybrid cloud

A hybrid cloud is a deployment model that combines public cloud, private cloud and on-premises infrastructure to create a single, flexible, IT environment.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The hybrid cloud

Resources can be allocated across these environments (public, private, on-premises) based on specific needs

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The hybrid cloud

Accessibility varies depending on the resources being accessed.

- Public cloud resources are accessed through the Internet.
- Private cloud and on-premises resources are restricted to authorised users.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The hybrid cloud

Security requires careful management due to the integration of different environments.

- Organisations need to establish consistent security policies and procedures across all platforms.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The hybrid cloud

Hybrid clouds offer high scalability.



Organisations can easily scale public cloud resources up or down to meet fluctuating demands.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The hybrid cloud

The cost of a hybrid cloud depends on the usage of each environment.

- Organisations can optimise costs by allocating workloads to the most cost-effective platform.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The hybrid cloud

Possible use cases for the hybrid cloud

- Using a public cloud for development and testing.
- Keeping customer data secure in a private cloud.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The multi cloud

Is a deployment strategy where an organisation uses a combination of two or more different CSPs.

- Public clouds
- Private clouds
- On-premises infrastructure

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The multi cloud

Accessibility varies depending on the specific cloud environment.

- Public clouds resources available over the Internet
- Private cloud/on-premises resources restricted to authorised users.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The multi cloud

Security requires a strong multi-cloud management strategy.

- Organisations need to establish consistent security policies and procedures
- Organisations need to consider the unique security posture of each CSP.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The multi cloud

Multi-cloud offers high scalability.

- Leverage on-demand resources of multiple public clouds to meet peak demands.
- Utilise on-premises or private infrastructure for stable workloads.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
<u>Data security</u>	Shared responsibility	High	High	High
<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The multi cloud

Multi-cloud use-cases:

- An e-commerce company uses CSP-A for its main storefront website and CSP-B for HPC tasks related to product recommendations.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
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<u>Data privacy</u>	Shared responsibility	High	High	High

DEPLOYMENT MODELS IN CLOUD COMPUTING

The multi cloud

Multi-cloud use-cases:

- A financial services firm utilising a private cloud for core banking applications and a public cloud for customer-facing mobile applications.

Factors	Public	Private	Community	Hybrid
<u>Initial setup</u>	Easy	Complex	Complex	Complex
<u>Scalability and flexibility</u>	High	Less flexible	Fixed	High
<u>Cost-comparison</u>	Cost-effective	Costly	Distributed	Between public and private cloud
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

CLOUD COMPUTING ROLES & RESPONSIBILITIES

In the world of cloud computing, people and organisations take on different *roles and responsibilities*.

As a cloud professional, it's important that you understand these different roles.

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
<u>Cloud Service Customer (CSC)</u>	Uses cloud services	Selects services, manages data security, pays for services	Company using a SaaS CRM platform
<u>Regulators</u>	Oversees cloud practices	Enforces regulations, sets security standards	General Data Protection Regulation (GDPR)
<u>Cloud Access Security Broker (CASB)</u>	Provides additional security for cloud	Monitors access, enforces security policies	Organisation using a CASB for cloud security
<u>Cloud Service Partner</u>	Provides services related to cloud	Cloud migration, managed services, app development	Consulting firm for cloud migration

CLOUD COMPUTING ROLES & RESPONSIBILITIES

The Cloud Service Provider (CSP)

The CSP delivers cloud services to customers.

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

The Cloud Service Provider (CSP)

- Manages cloud infrastructure
- Provides security and reliability
- Manages service uptime and performance
- Offers customer support
- Ensures data privacy and compliance

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

The Cloud Service Provider (CSP)

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

The Cloud Service Customer (CSC)

The CSC uses cloud services for various purposes like

- Application development
- Data storage
- Running business applications

Role	Focus	Responsibilities	Example
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

The Cloud Service Customer (CSC)

- Selects services based on its needs
- Manages data security and access within the cloud environment
- Pays for cloud services used

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

The Cloud Service Customer (CSC)

- Example: A company using a SaaS platform for customer relationship management (CRM)

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

Regulators

A regulator oversees cloud computing practices to ensure

- Data privacy
- Data security
- Consumer protection

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

Regulators

A regulator enforces regulations and compliance standards by

- 1 Defining data privacy laws
- 2 Enforcing data privacy laws
- 3 Developing cloud security standards
- 4 Investigating potential security breaches

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

Regulators

Example: The General Data Protection Regulation (GDPR) was drafted and adopted by the European Parliament and the Council of the European Union (EU)

● It is enforced by independent national Data Protection Authorities (DPAs) within each EU member state.

Role	Focus	Responsibilities	Example
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

Regulators

- GDPR
 - Empowers individuals with control over their personal data
 - It requires organisations to handle personal data transparently and securely
 - The DPAs can issue fines reaching up to 20 million Euros or 4% of a company's global turnover for non-compliance

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
<u>Cloud Service Customer (CSC)</u>	Uses cloud services	Selects services, manages data security, pays for services	Company using a SaaS CRM platform
<u>Regulators</u>	Oversees cloud practices	Enforces regulations, sets security standards	General Data Protection Regulation (GDPR)
<u>Cloud Access Security Broker (CASB)</u>	Provides additional security for cloud	Monitors access, enforces security policies	Organisation using a CASB for cloud security
<u>Cloud Service Partner</u>	Provides services related to cloud	Cloud migration, managed services, app development	Consulting firm for cloud migration

CLOUD COMPUTING ROLES & RESPONSIBILITIES

Cloud Access Security Broker (CASB)

A CASB acts as an intermediary between a CSC and the CSP.

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security Selects services, manages data security, pays for services	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

Cloud Access Security Broker (CASB)

It provides additional security controls and access management

- 1 Monitoring and enforcing security policies for cloud access
- 2 Identifying and mitigating security risks
- 3 User access control and authorisation for cloud resources

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security Selects services, manages data security, pays for services	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

Cloud Service Partner

Provides various services and solutions related to cloud computing.

- Helping customers migrate to the cloud
- Managing and maintaining cloud infrastructure for customers
- Developing cloud-based apps tailored to specific needs

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
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CLOUD COMPUTING ROLES & RESPONSIBILITIES

Cloud Service Partner

Examples:

- A consulting firm specialising in cloud migration
- Companies offering cloud-based accounting software

Role	Focus	Responsibilities	Example
<u>Cloud Service Provider (CSP)</u>	Delivers cloud services	Manages infrastructure, provides services, ensures security	Amazon Web Services (AWS)
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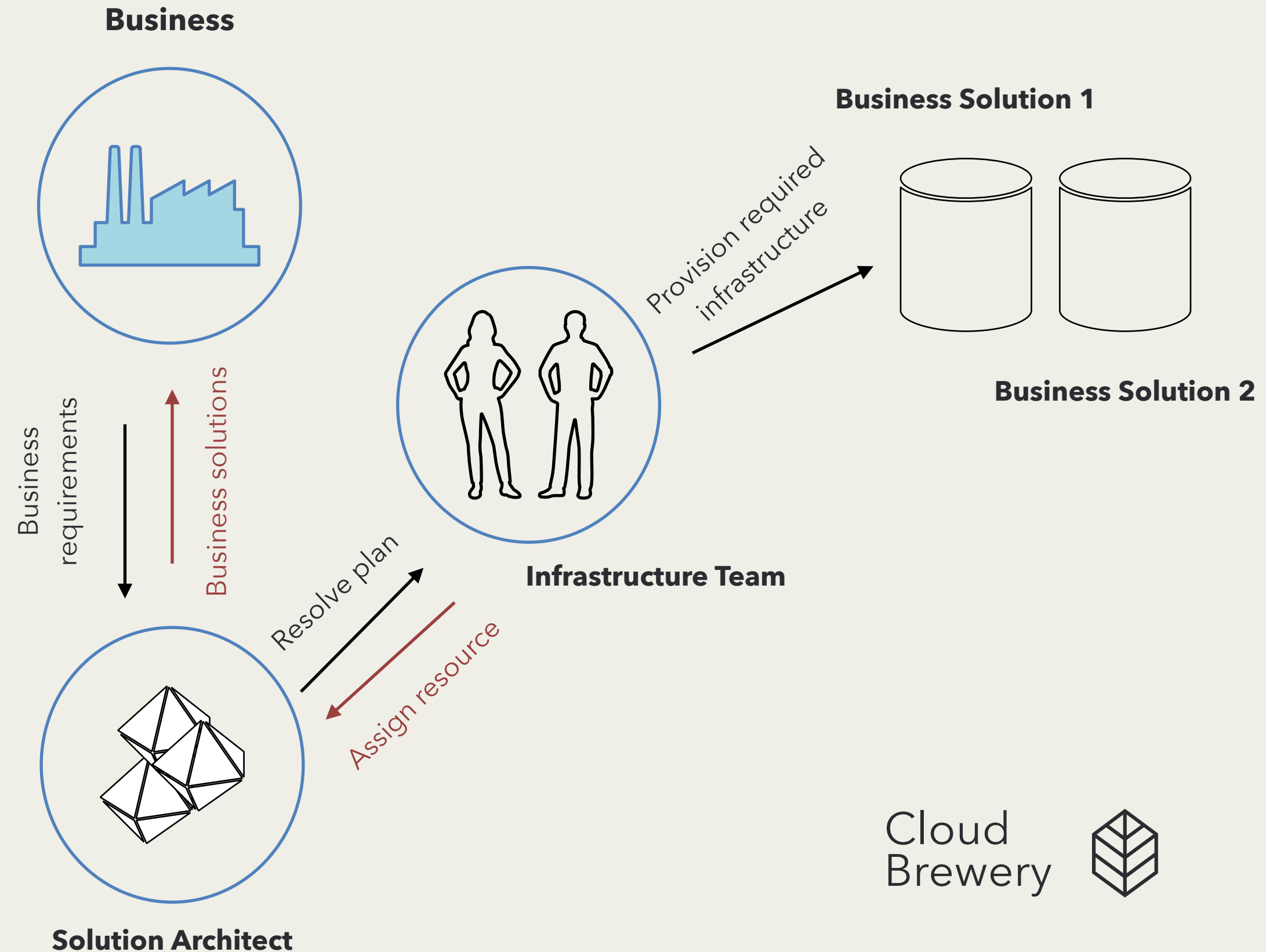
VIRTUALISATION & CLOUD COMPUTING

VIRTUALISATION & CLOUD COMPUTING

Challenges of traditional IT

Traditional IT provisioning often involves physical hardware setup and configuration - a slow and resource-intensive process, which can lead to

- Delays in deployment
- Delays in scaling
- Delays in response to changing market/customer needs
- Complex/expensive management of physical servers

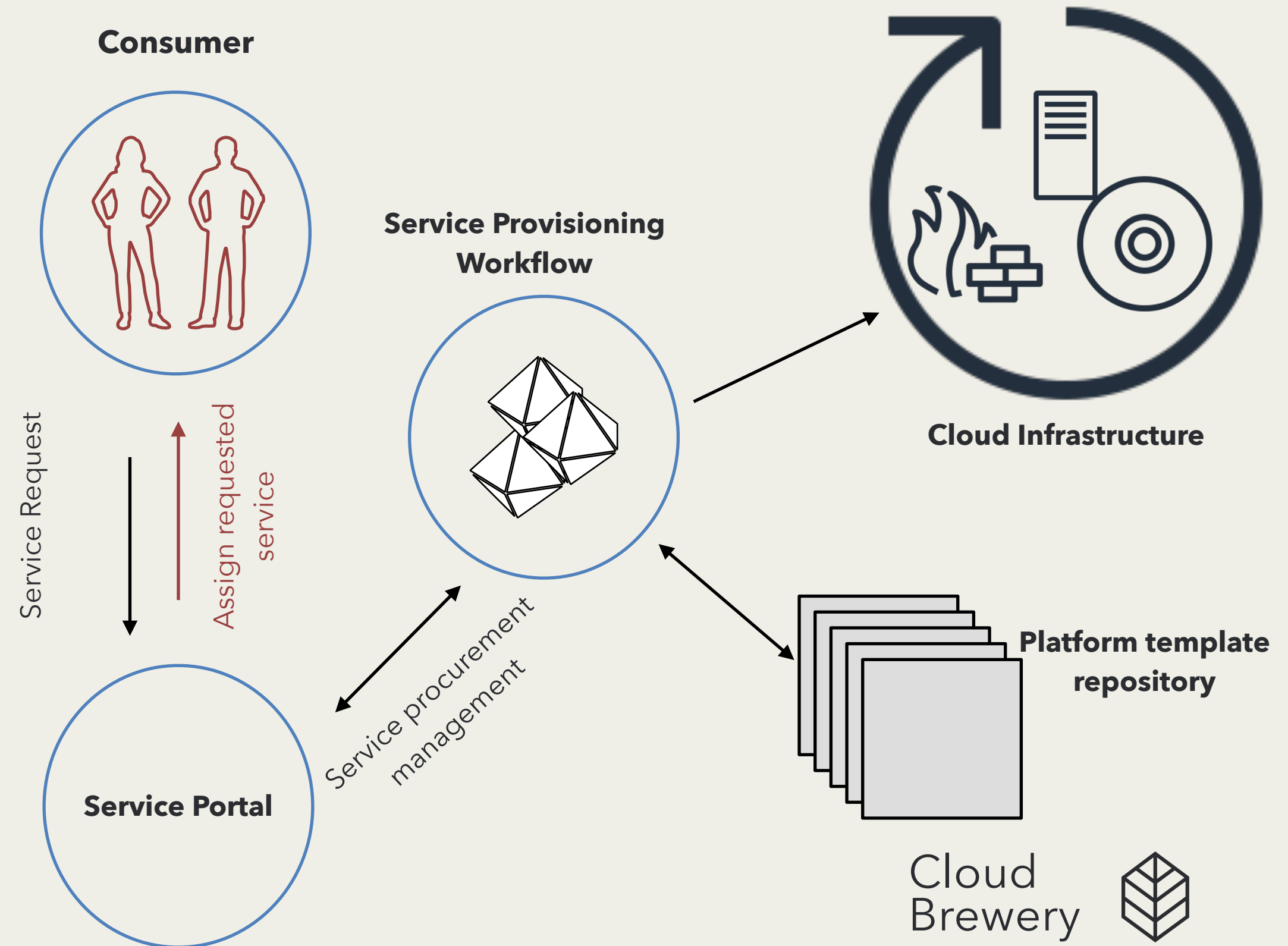


VIRTUALISATION & CLOUD COMPUTING

Needs of modern businesses

Today's business landscape is dynamic. Organisations need to be agile and adaptable to seize new opportunities and stay competitive.

- Readily available and easily scalable IT resources
- Seamless digital experiences
- that traditional IT infrastructure struggles to deliver

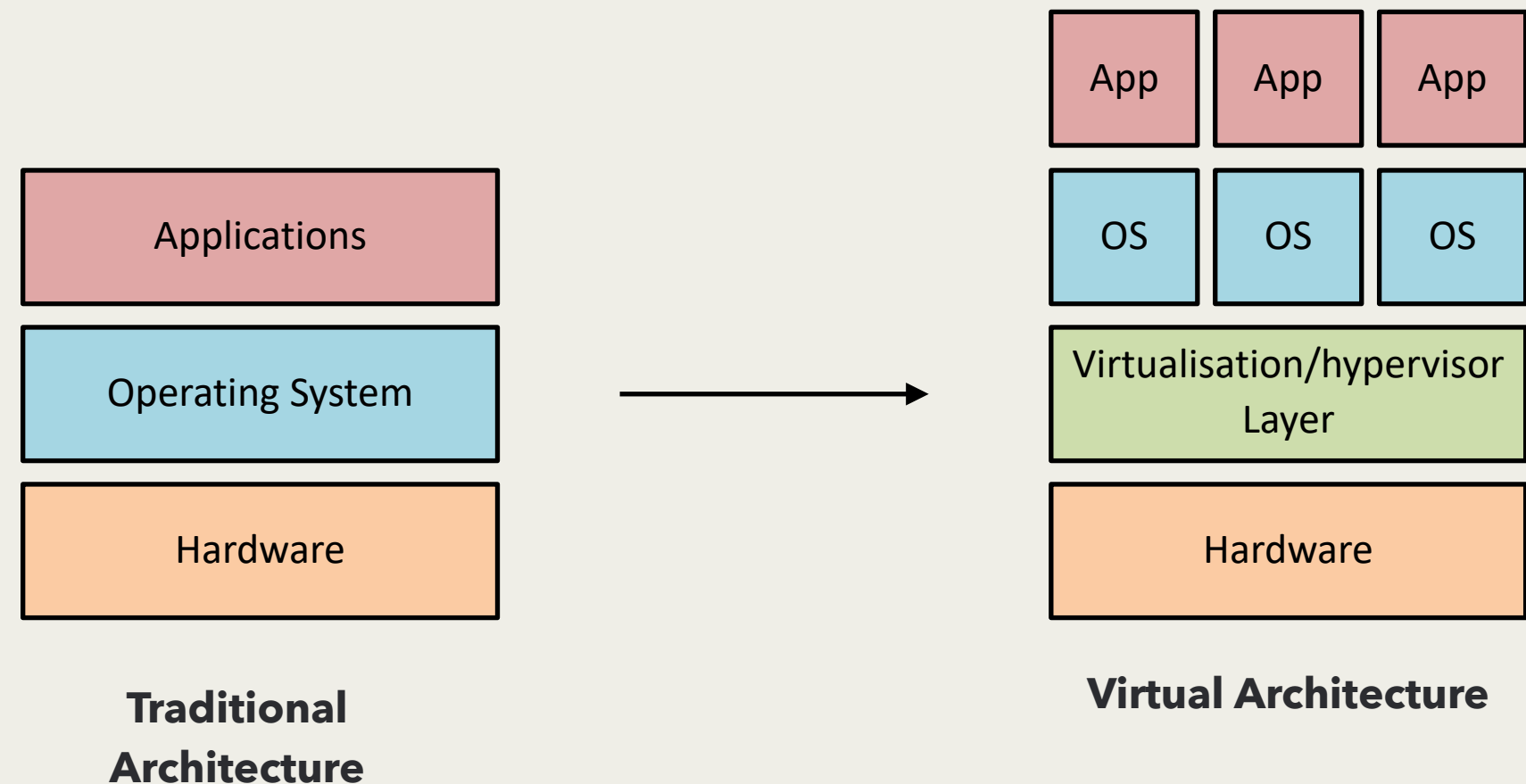


VIRTUALISATION & CLOUD COMPUTING

How virtualisation addresses modern business needs

Virtualisation solves the challenges presented by traditional IT and addresses the needs of modern businesses

- Faster resource provisioning
- Scalability on demand
- Improved digital experiences

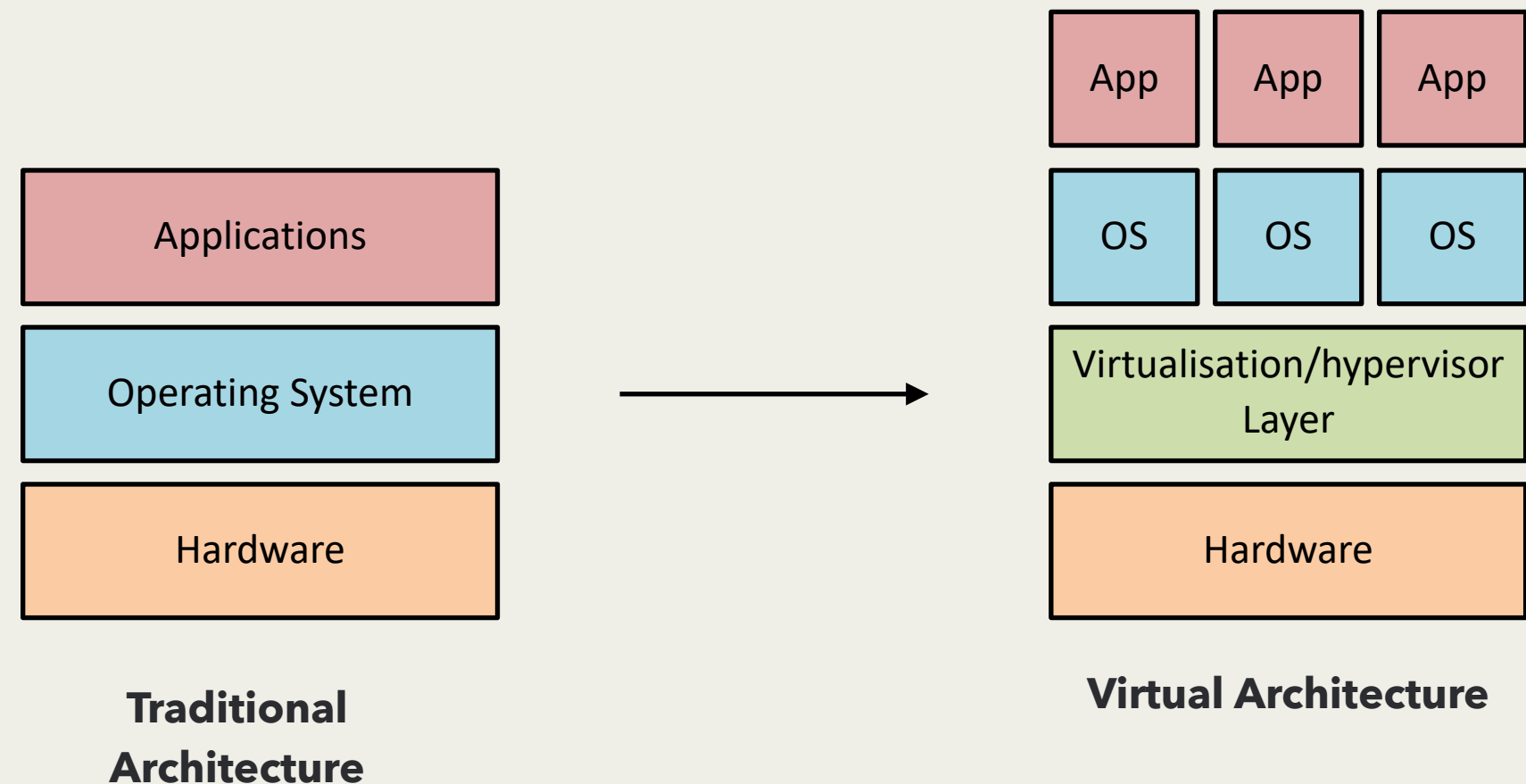


VIRTUALISATION & CLOUD COMPUTING

How virtualisation addresses modern business needs

Faster resource provisioning

Virtualisation software allows for the creation and deployment of virtual machines (VMs) in a manner of minutes, compared to the days or weeks it might take to setup a new physical server.

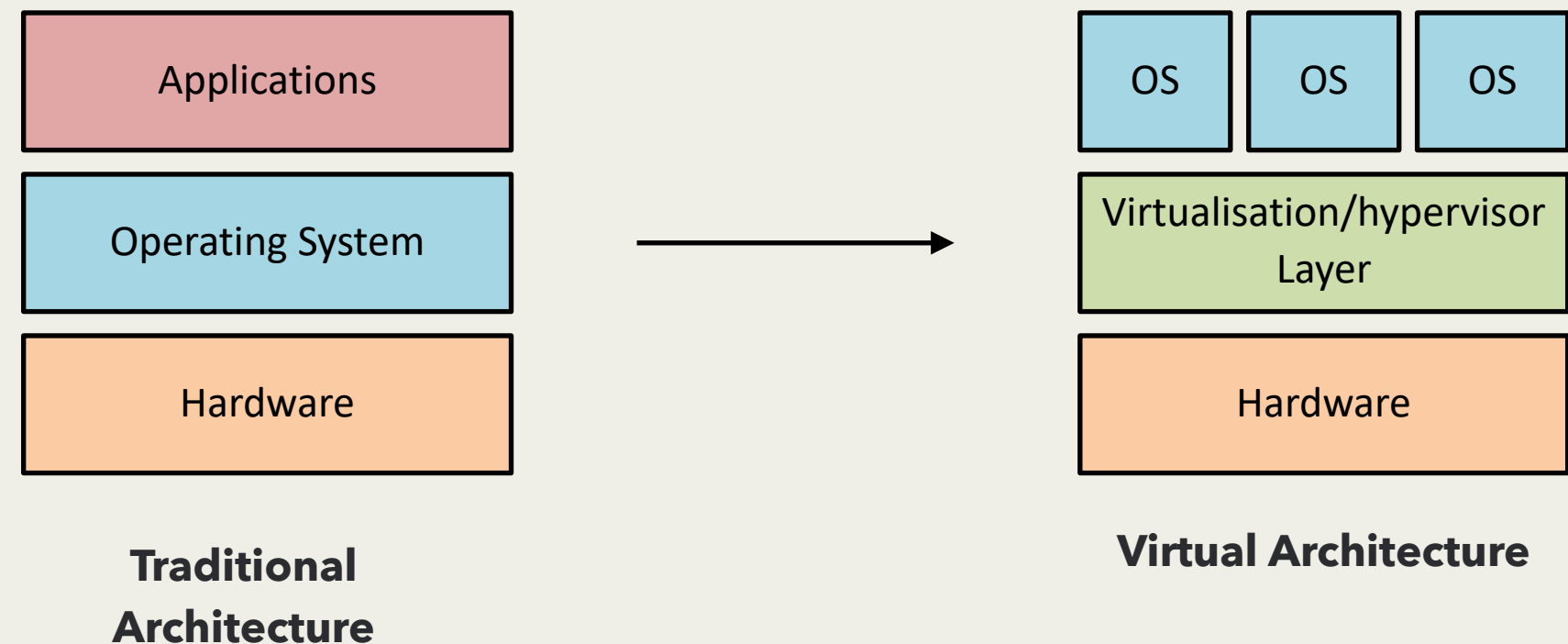


VIRTUALISATION & CLOUD COMPUTING

How virtualisation addresses modern business needs

Faster resource provisioning

This rapid provisioning enables organisations to respond quickly to changing market conditions and deploy new applications or features to meet evolving customer needs.

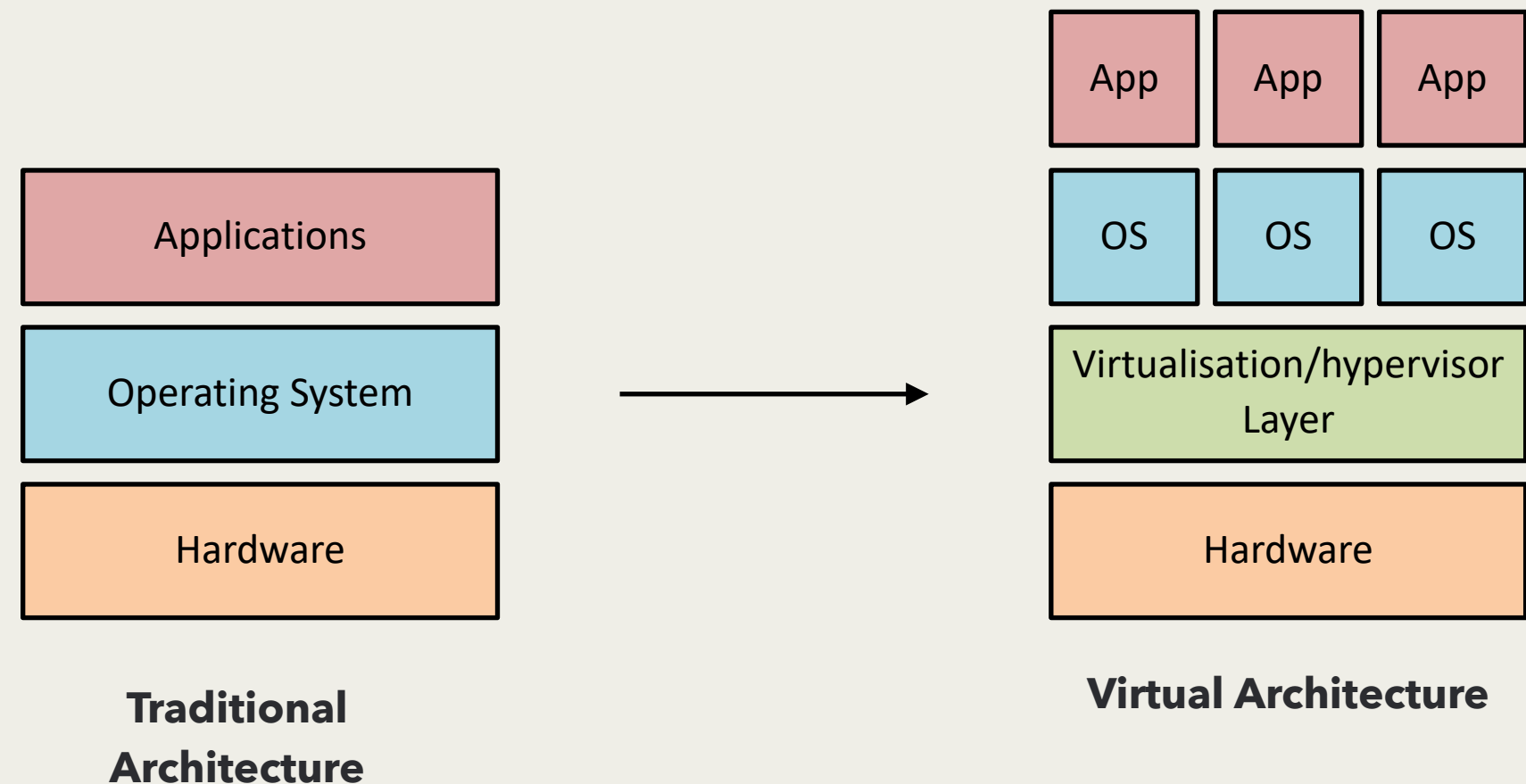


VIRTUALISATION & CLOUD COMPUTING

How virtualisation addresses modern business needs

Scalability on demand

Virtual resources like VMs are not limited to the physical constraints of a single server. They can be easily scaled up or down based on changing workloads.

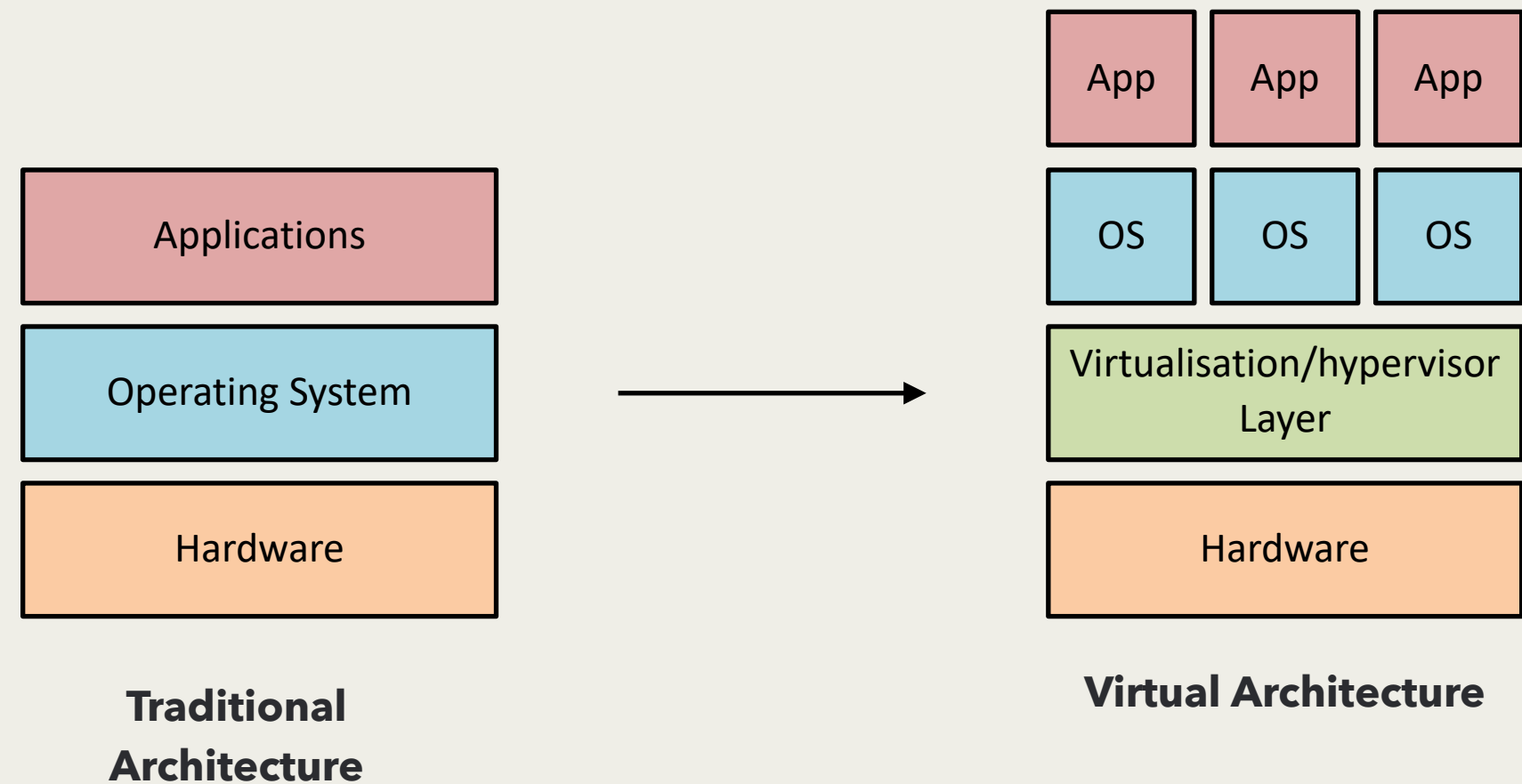


VIRTUALISATION & CLOUD COMPUTING

How virtualisation addresses modern business needs

Scalability on demand

This allows organisations to allocate resources efficiently, paying only for what they use.

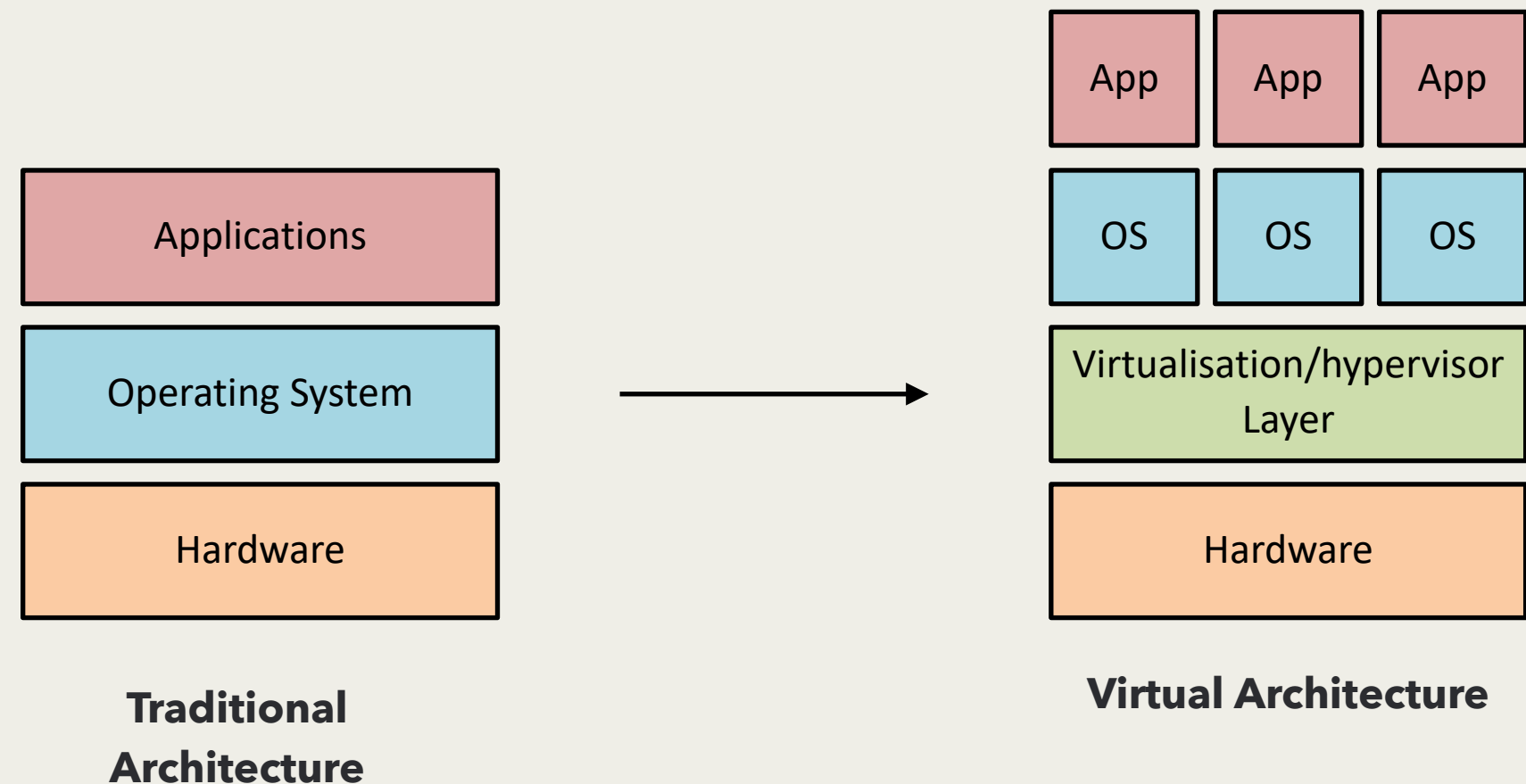


VIRTUALISATION & CLOUD COMPUTING

How virtualisation addresses modern business needs

Improved digital experiences

Virtualisation provides a foundation for delivering seamless digital experiences. It enables the creation of highly available and reliable IT infrastructure that can support modern applications and services.

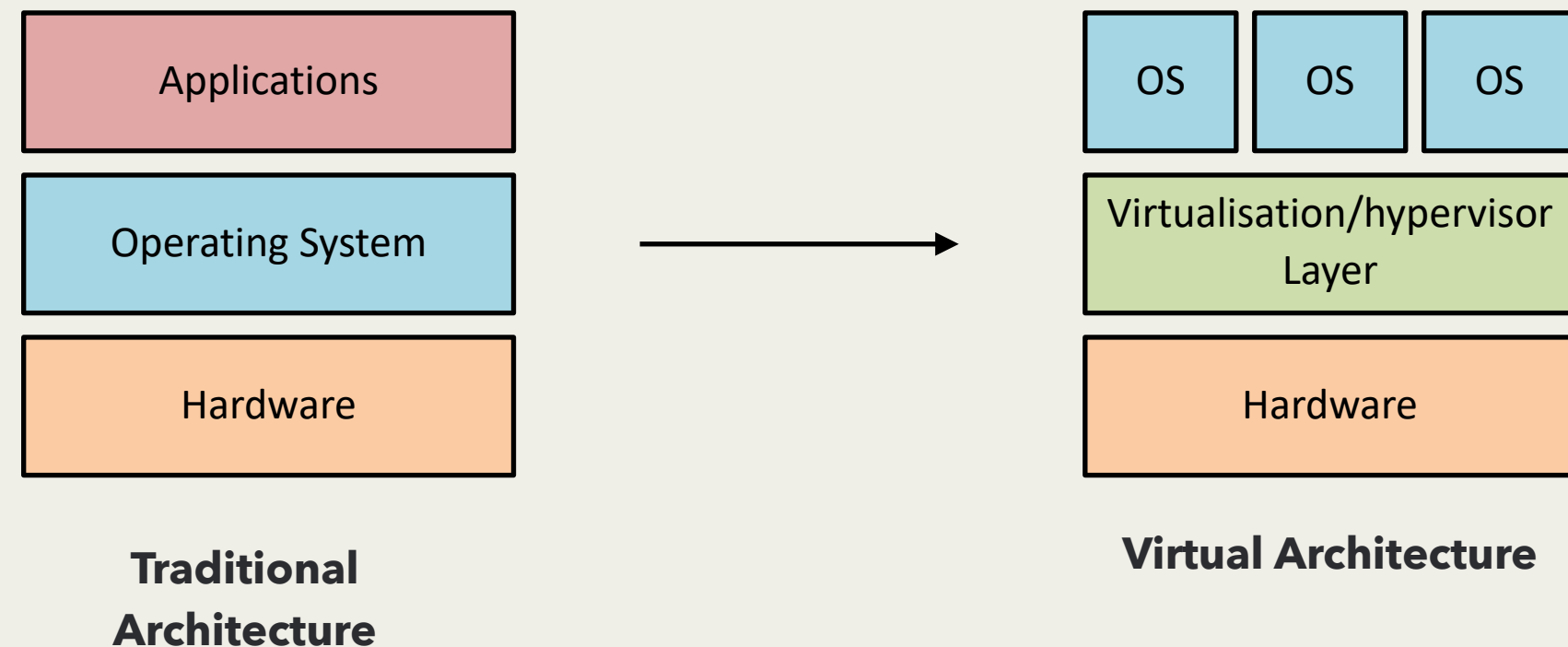


VIRTUALISATION & CLOUD COMPUTING

How virtualisation addresses modern business needs

Improved digital experiences

Virtualisation can be used to isolate development and testing environments, allowing faster development cycles and quicker delivery of new digital features.

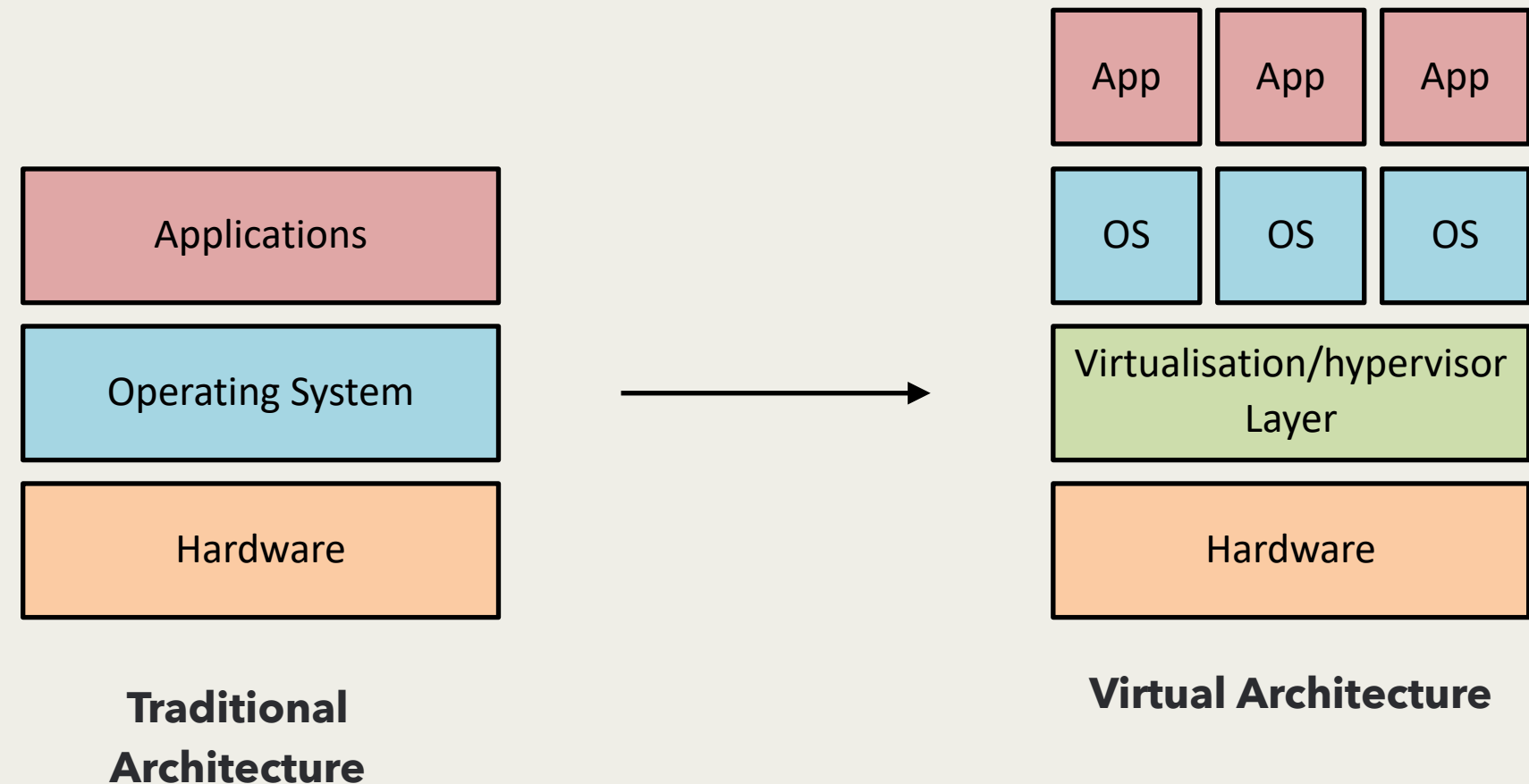


VIRTUALISATION & CLOUD COMPUTING

Virtualisation as the foundation for cloud computing

Virtualisation is the foundation that makes cloud computing possible, by enabling

- Resource pooling
- Multi-tenancy
- Elasticity
- Self-service provisioning



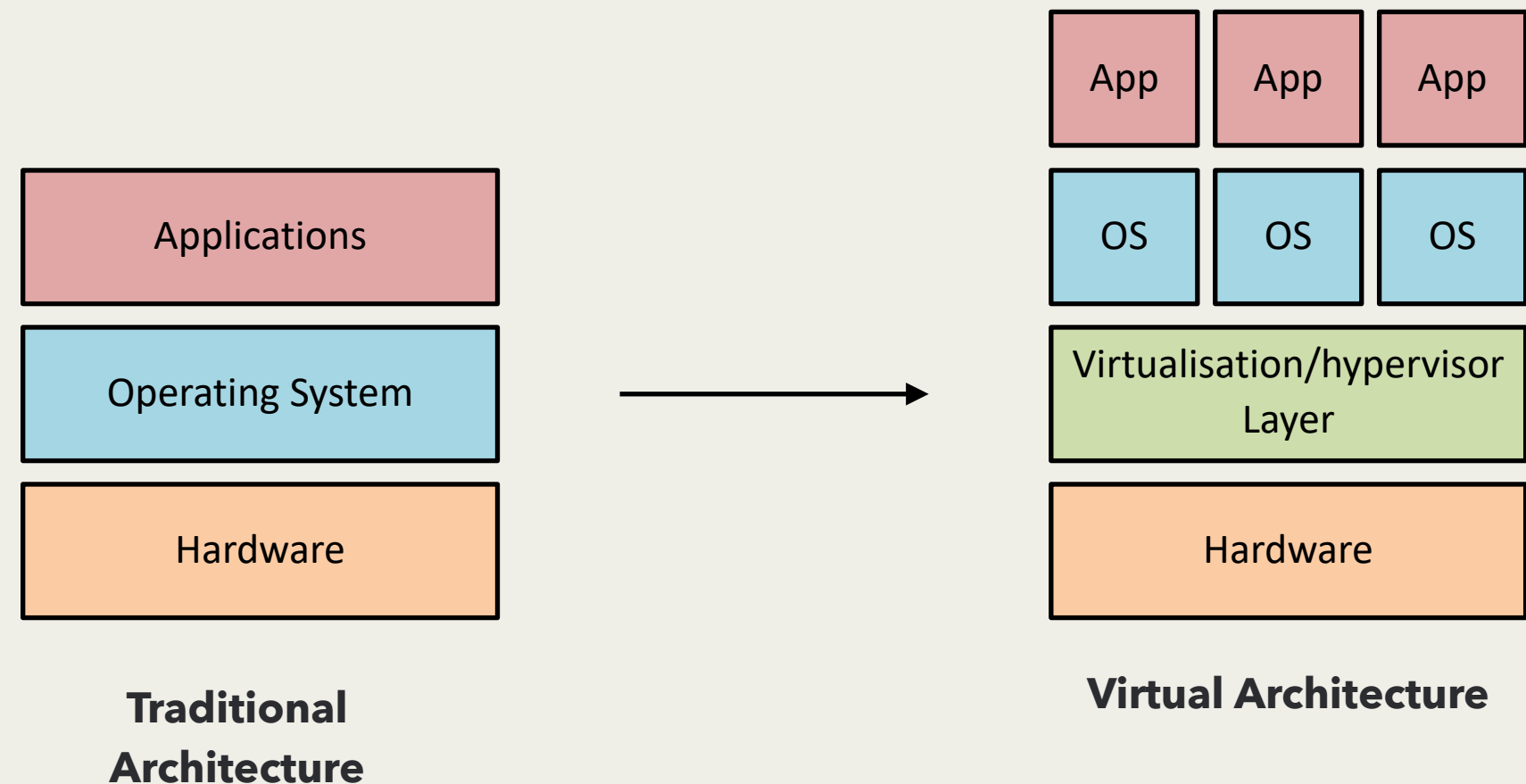
VIRTUALISATION & CLOUD COMPUTING

Virtualisation as the foundation for cloud computing

Resource pooling

Virtualisation allows cloud providers to pool the resources (servers, storage, network) of multiple physical machines into a single virtual pool.

This creates a vast pool of resources that can be dynamically allocated to cloud users on-demand.

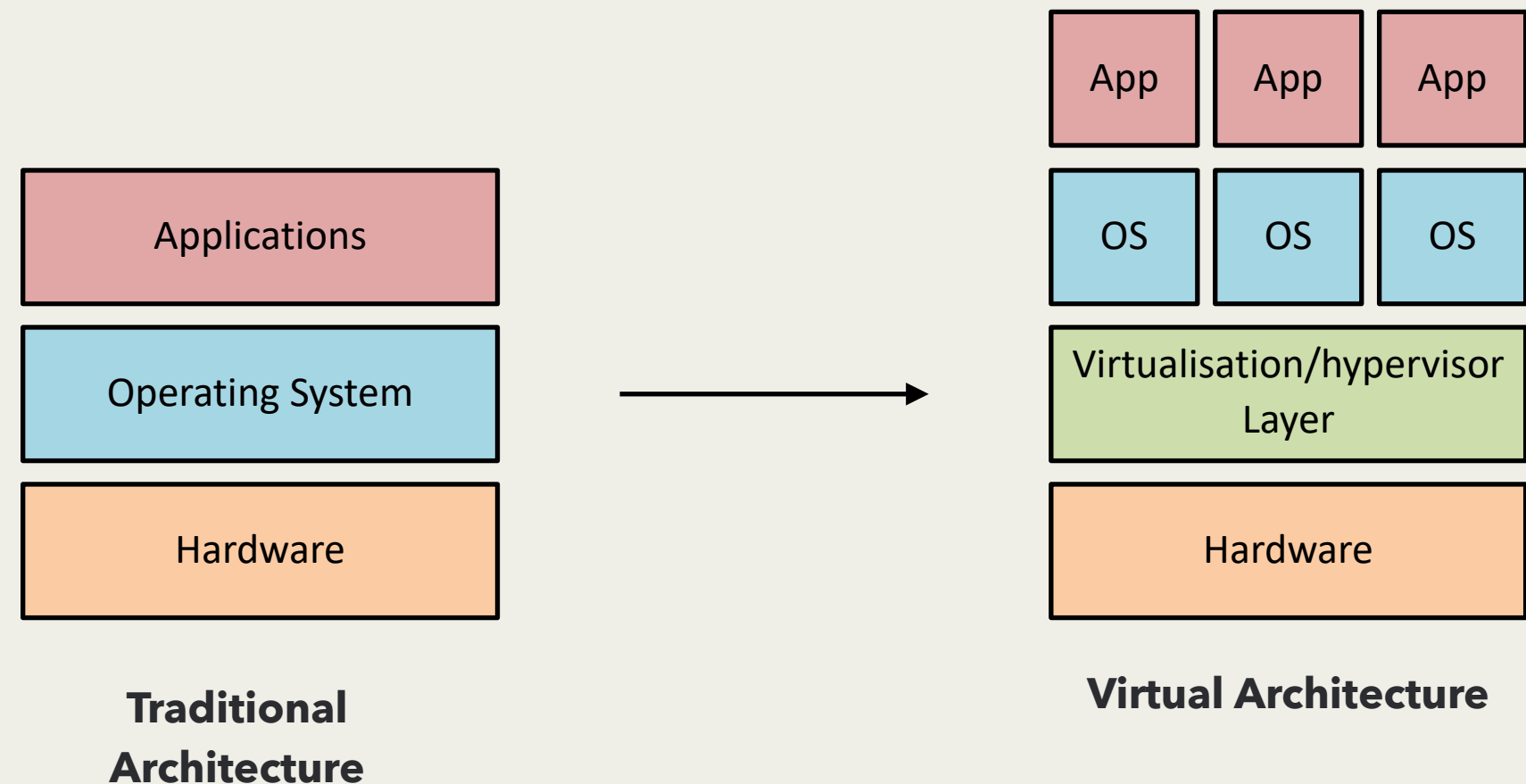


VIRTUALISATION & CLOUD COMPUTING

Virtualisation as the foundation for cloud computing

Multi-tenancy

Virtualisation enables multiple users to share the same physical hardware securely. Each user gets their own virtual environment (VMs, storage, network) isolated from others, providing a secure and independent experience.

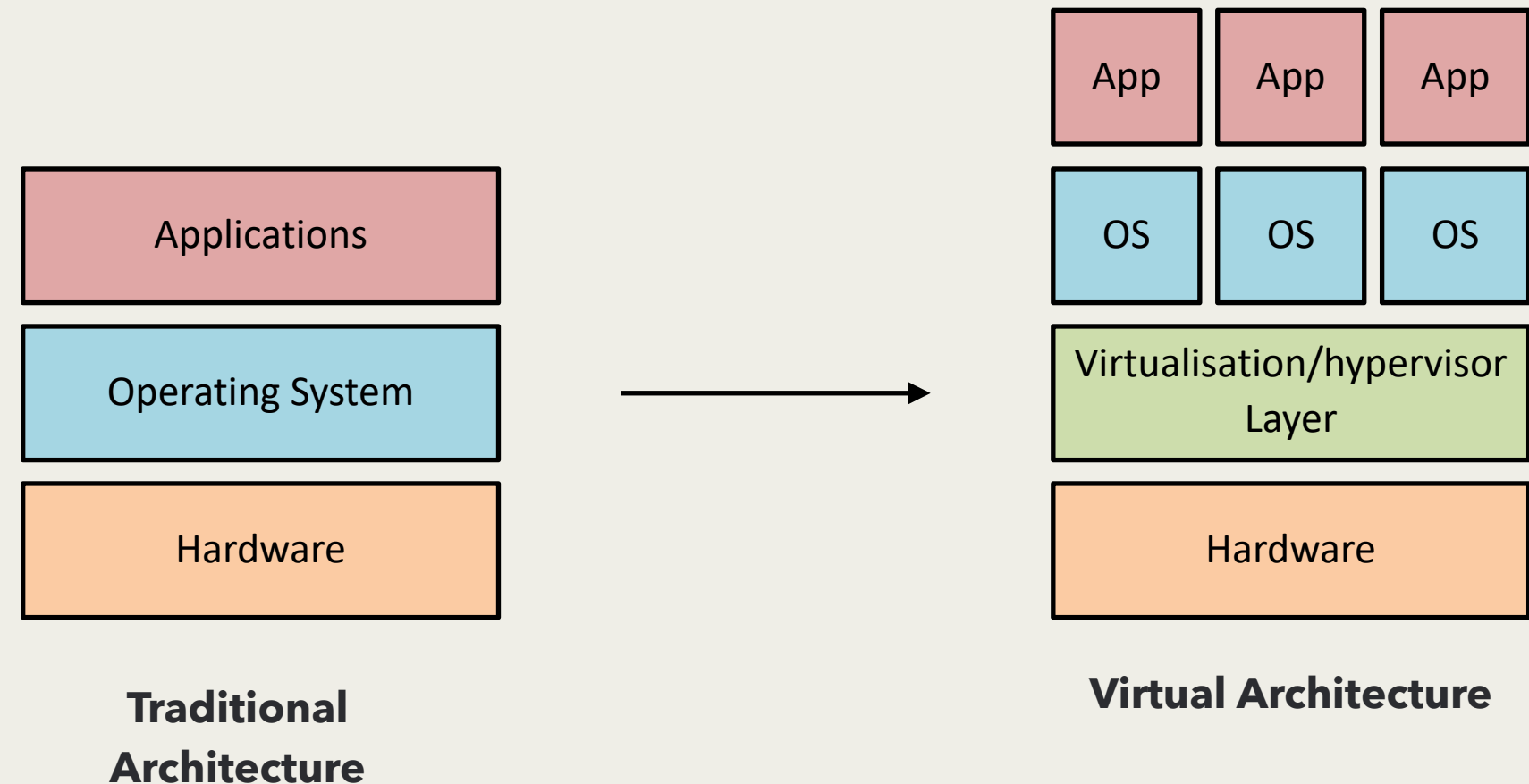


VIRTUALISATION & CLOUD COMPUTING

Virtualisation as the foundation for cloud computing

Elasticity

Cloud providers can dynamically allocate resources from the virtual pool to meet the fluctuating demands of their users.

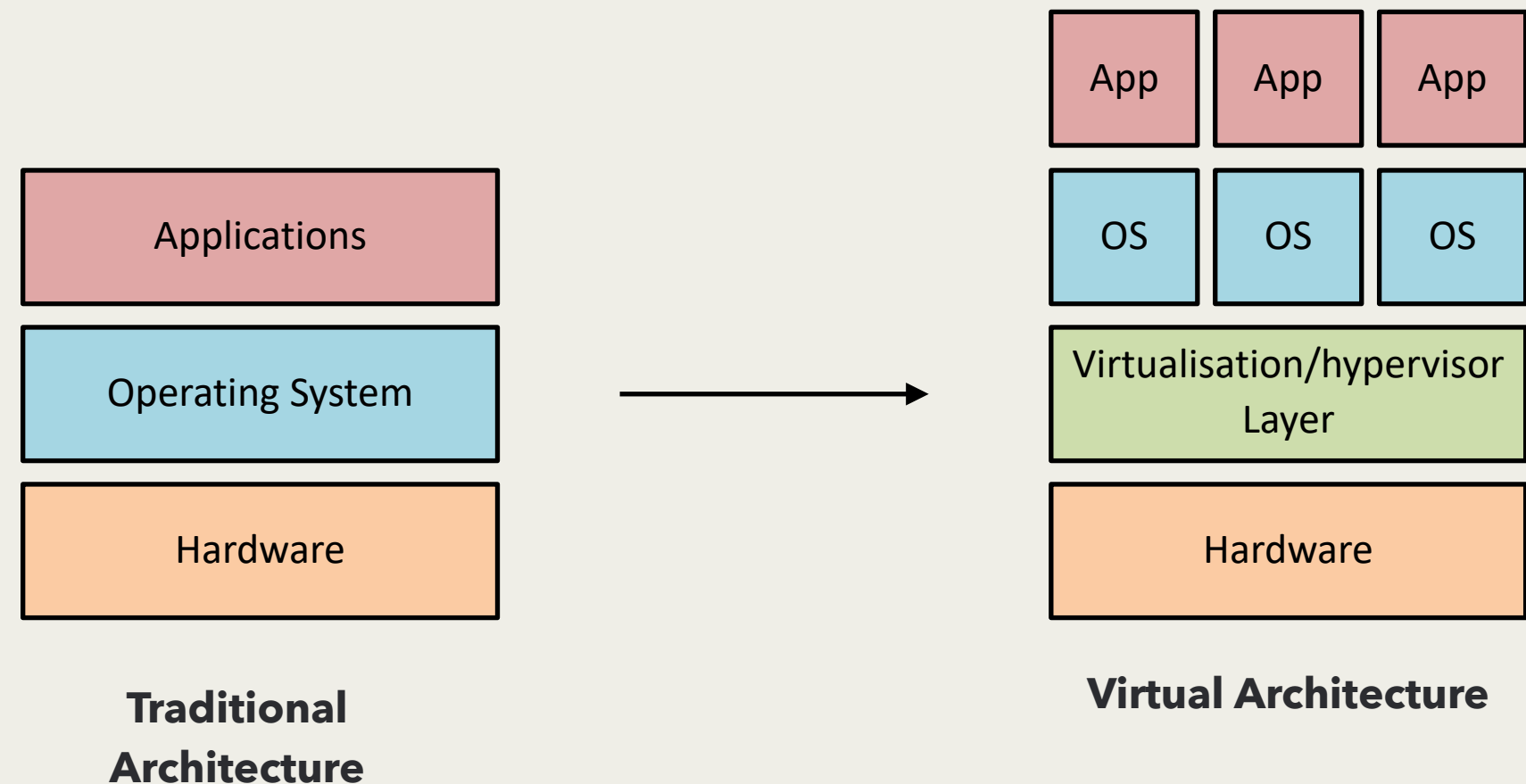


VIRTUALISATION & CLOUD COMPUTING

Virtualisation as the foundation for cloud computing

Self-service provisioning

Users can leverage self-service portals to provision virtual resources like VMs and storage as needed. This eliminates the need for IT intervention and allows users to quickly access the resources they require.

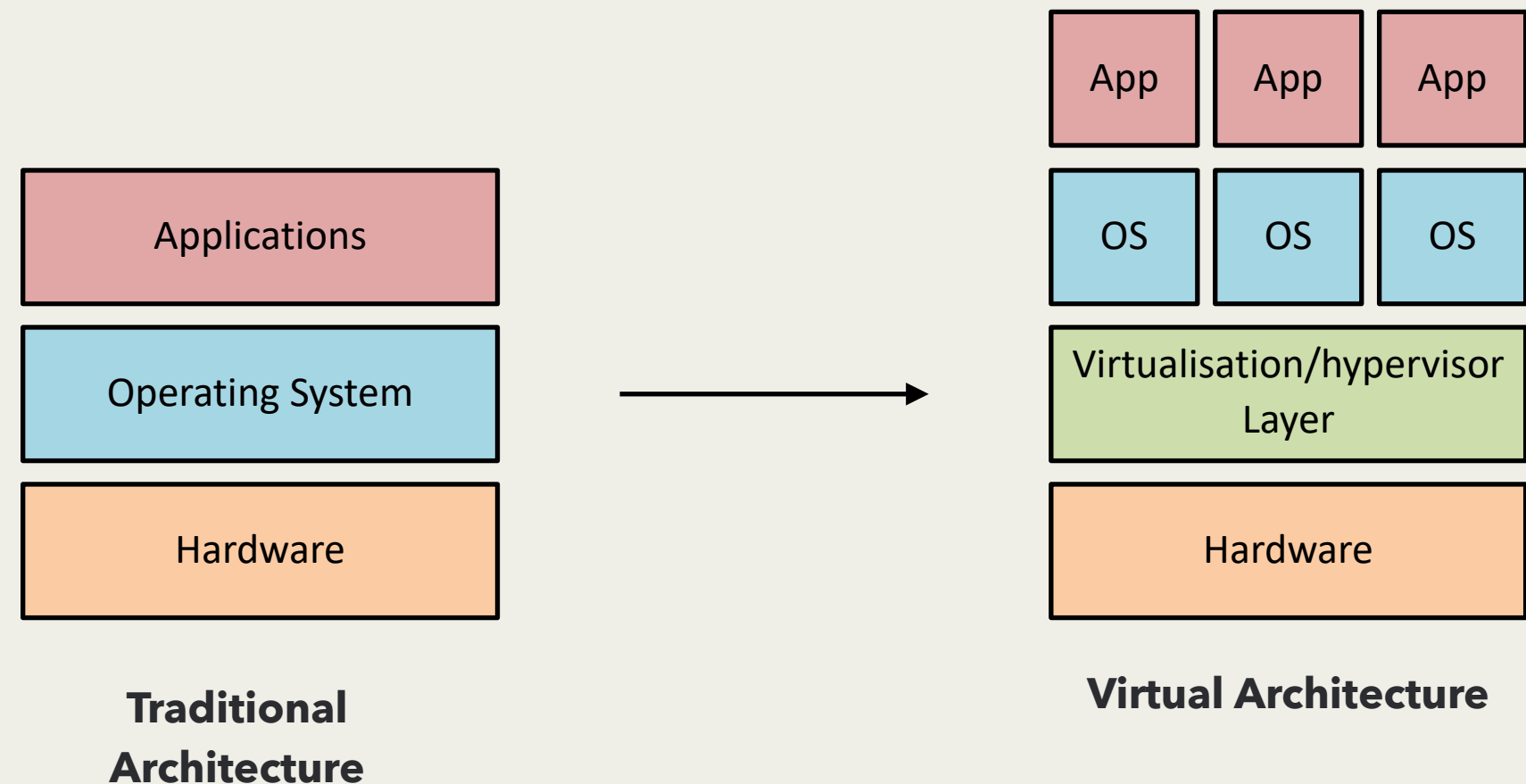


VIRTUALISATION & CLOUD COMPUTING

Types of virtualisation

There are several main types of virtualisation, each focusing on a specific aspect of IT infrastructure

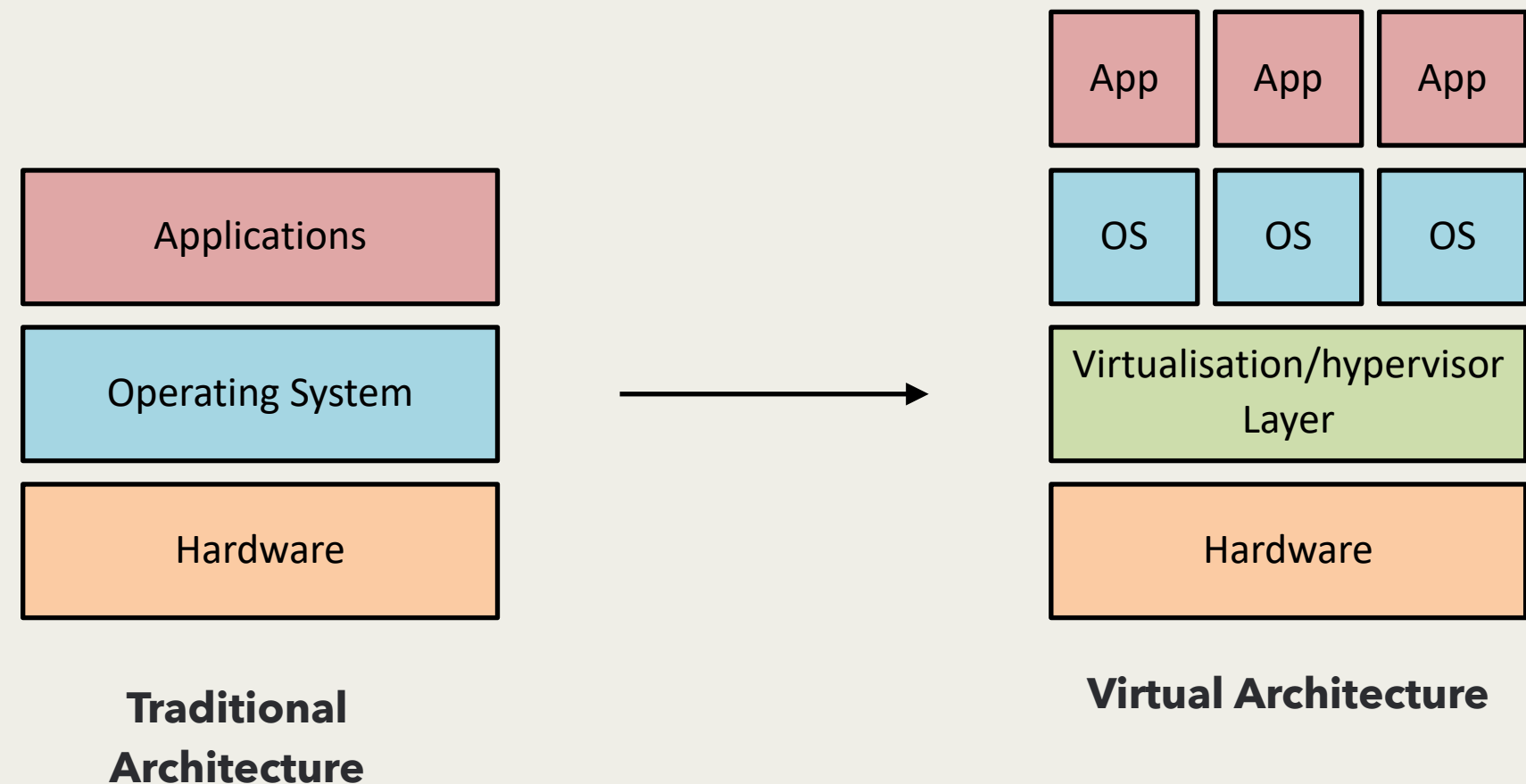
- Server virtualisation
- Desktop virtualisation
- Storage virtualisation
- Network virtualisation
- Application virtualisation



VIRTUALISATION & CLOUD COMPUTING

Server virtualisation

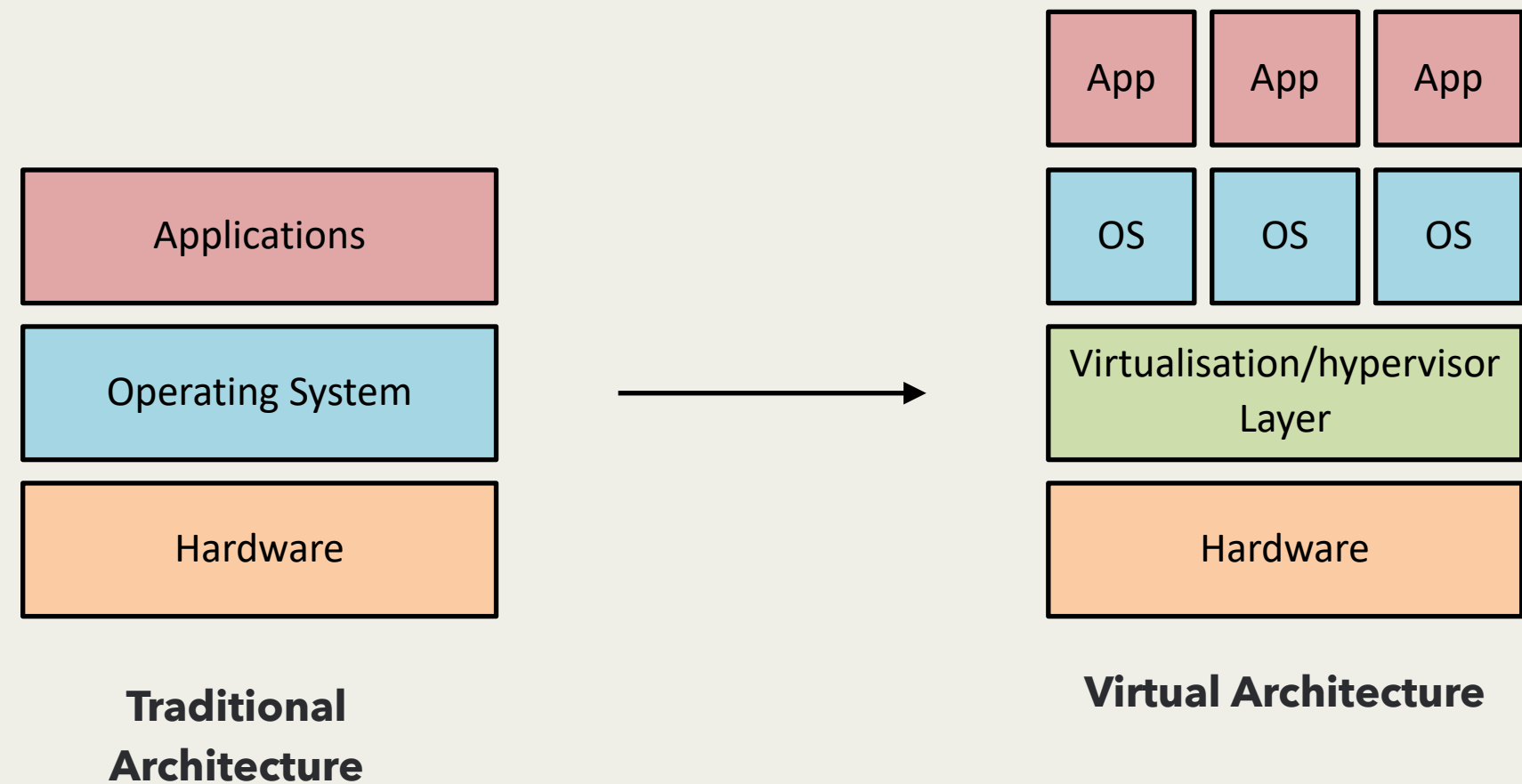
Server virtualisation creates virtual machines (VMs) that emulate physical servers. Each VM runs its own operating system (OS) and applications, isolated from other VMs on the same physical server.



VIRTUALISATION & CLOUD COMPUTING

Server virtualisation

Server virtualisation allows organisations to consolidate multiple physical servers onto fewer machines, improving resource utilisation and reducing costs.



VIRTUALISATION & CLOUD COMPUTING

Desktop virtualisation

It provides users with virtual desktops accessed remotely from any device.

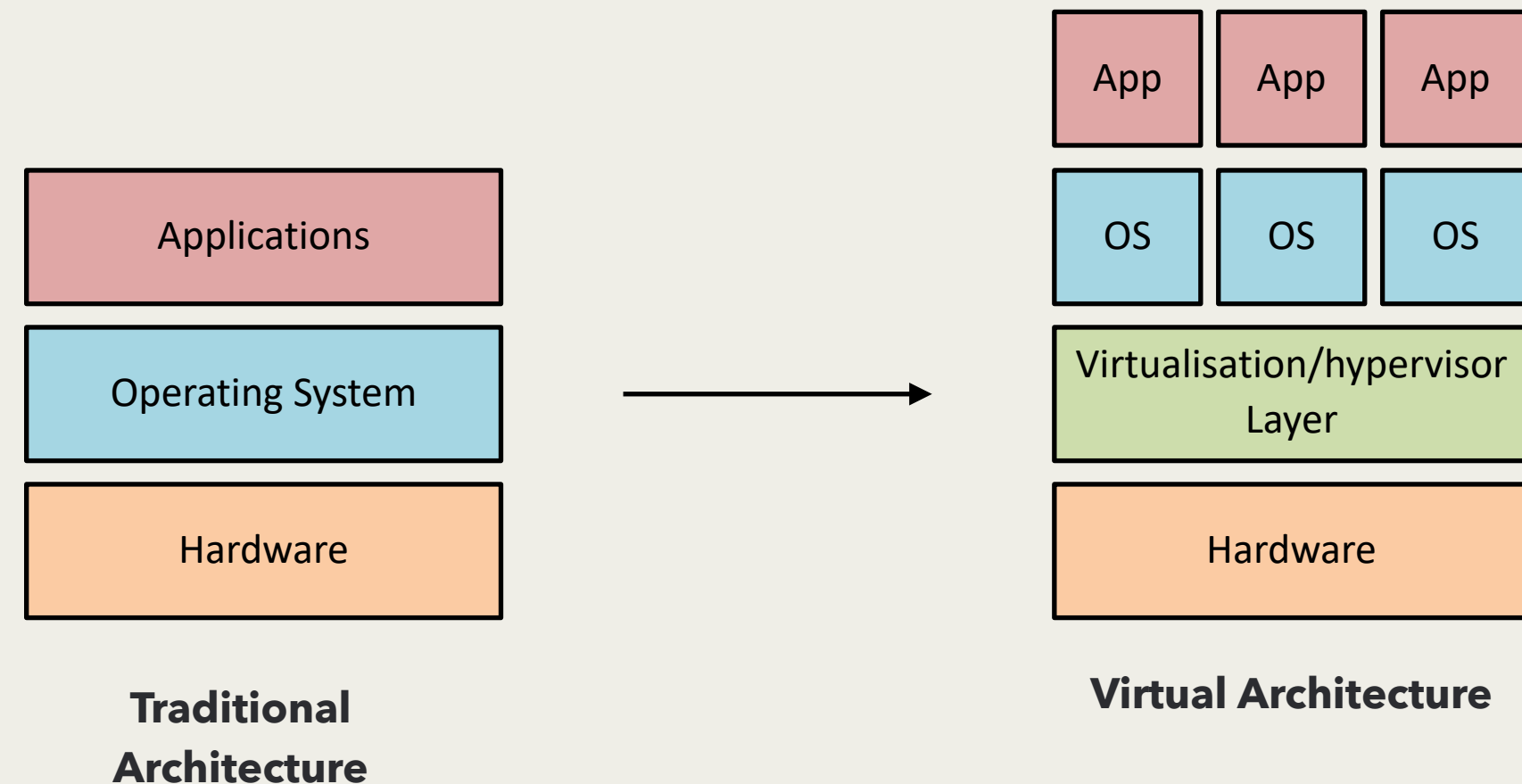
Virtual Desktop Infrastructure

- Provides each user with their own OS and applications.

Session-based Desktop

Virtualisation

- Multiple users share a single server session.



VIRTUALISATION & CLOUD COMPUTING

Desktop virtualisation

It provides users with virtual desktops accessed remotely from any device.

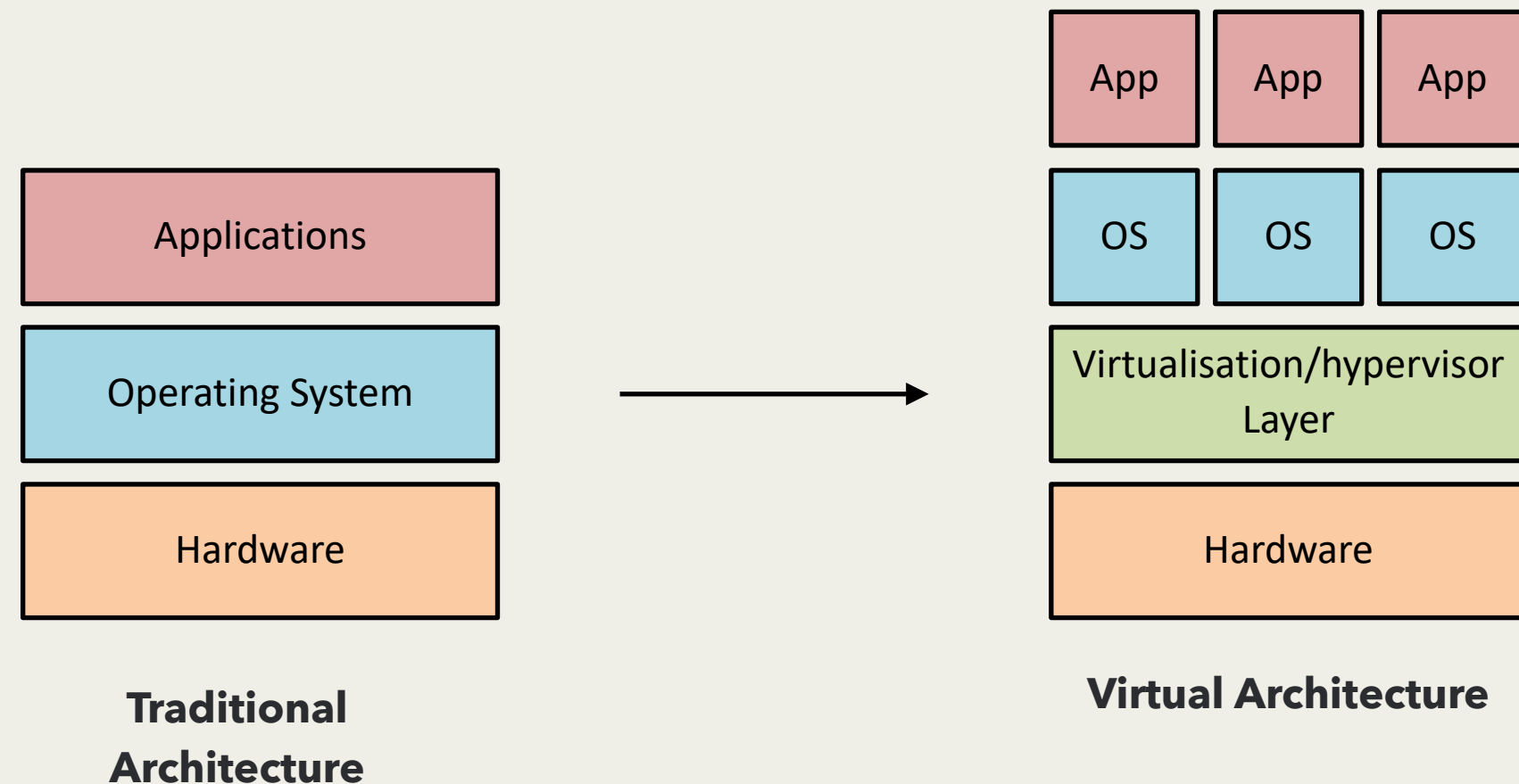
Virtual Desktop Infrastructure

- Provides each user with their own OS and applications.

Session-based Desktop

Virtualisation

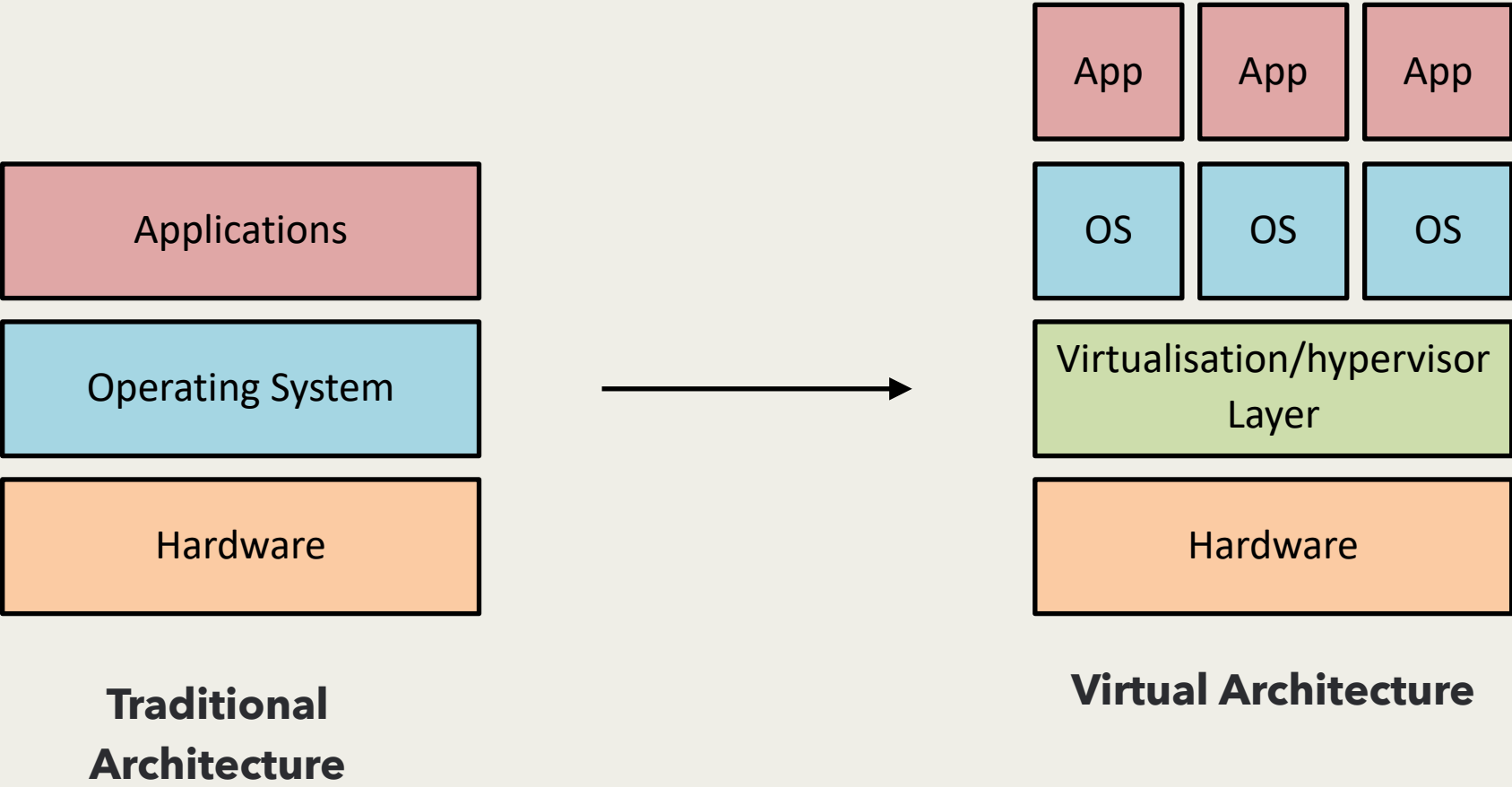
- Multiple users share a single server session.



VIRTUALISATION & CLOUD COMPUTING

Storage virtualisation

Groups physical storage devices from the same or different vendors into a centrally managed storage pool.

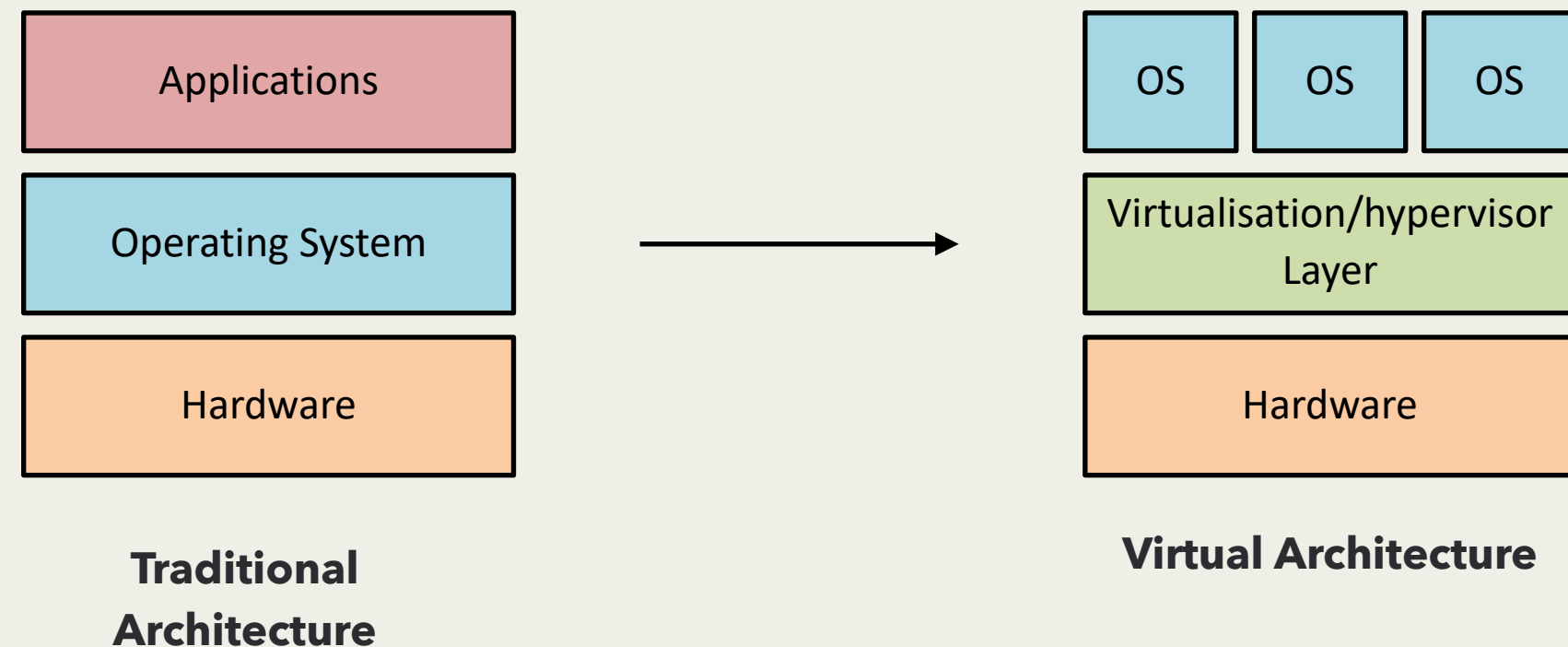


VIRTUALISATION & CLOUD COMPUTING

Storage virtualisation

Storage virtualisation allows for

- Efficient allocation of storage resources
- Improved data availability
- Easier backups

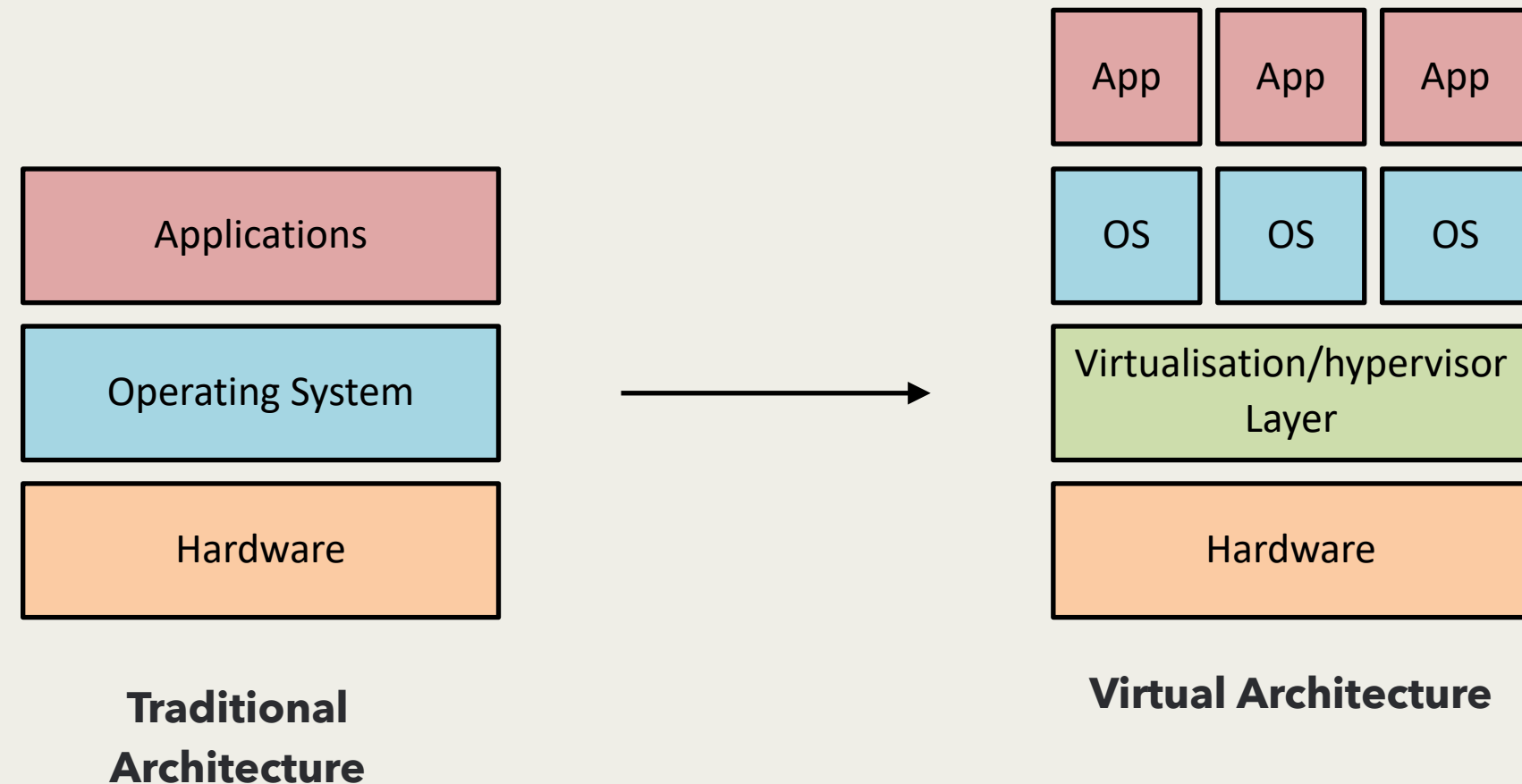


VIRTUALISATION & CLOUD COMPUTING

Network virtualisation

Network virtualisation creates virtual networks that are independent of the underlying physical network infrastructure

- Flexible and scalable network configurations
- Improved security
- Easier network management

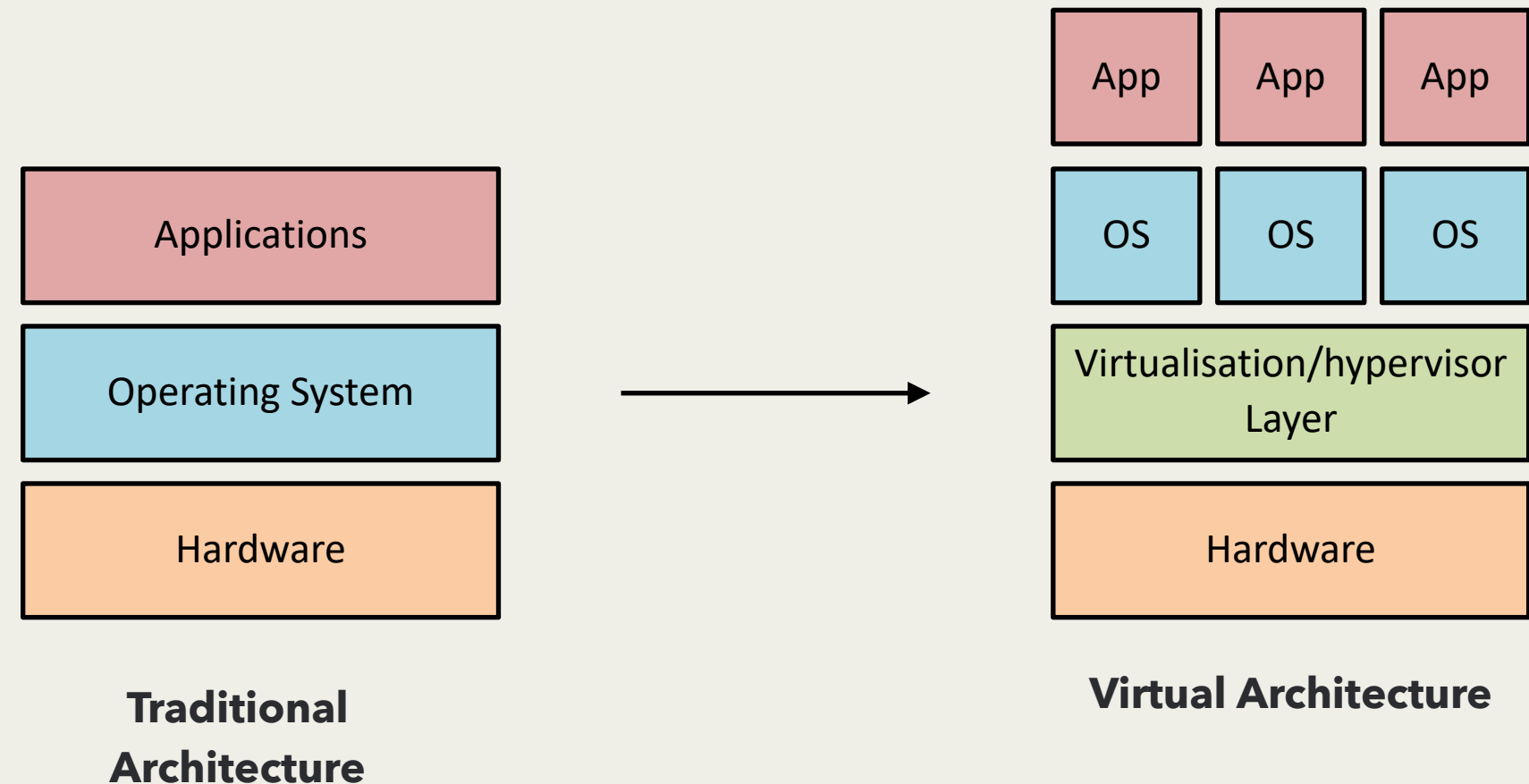


VIRTUALISATION & CLOUD COMPUTING

Application virtualisation

Application virtualisation focuses on applications rather than hardware. It allows applications to be packaged and run independently of the underlying OS.

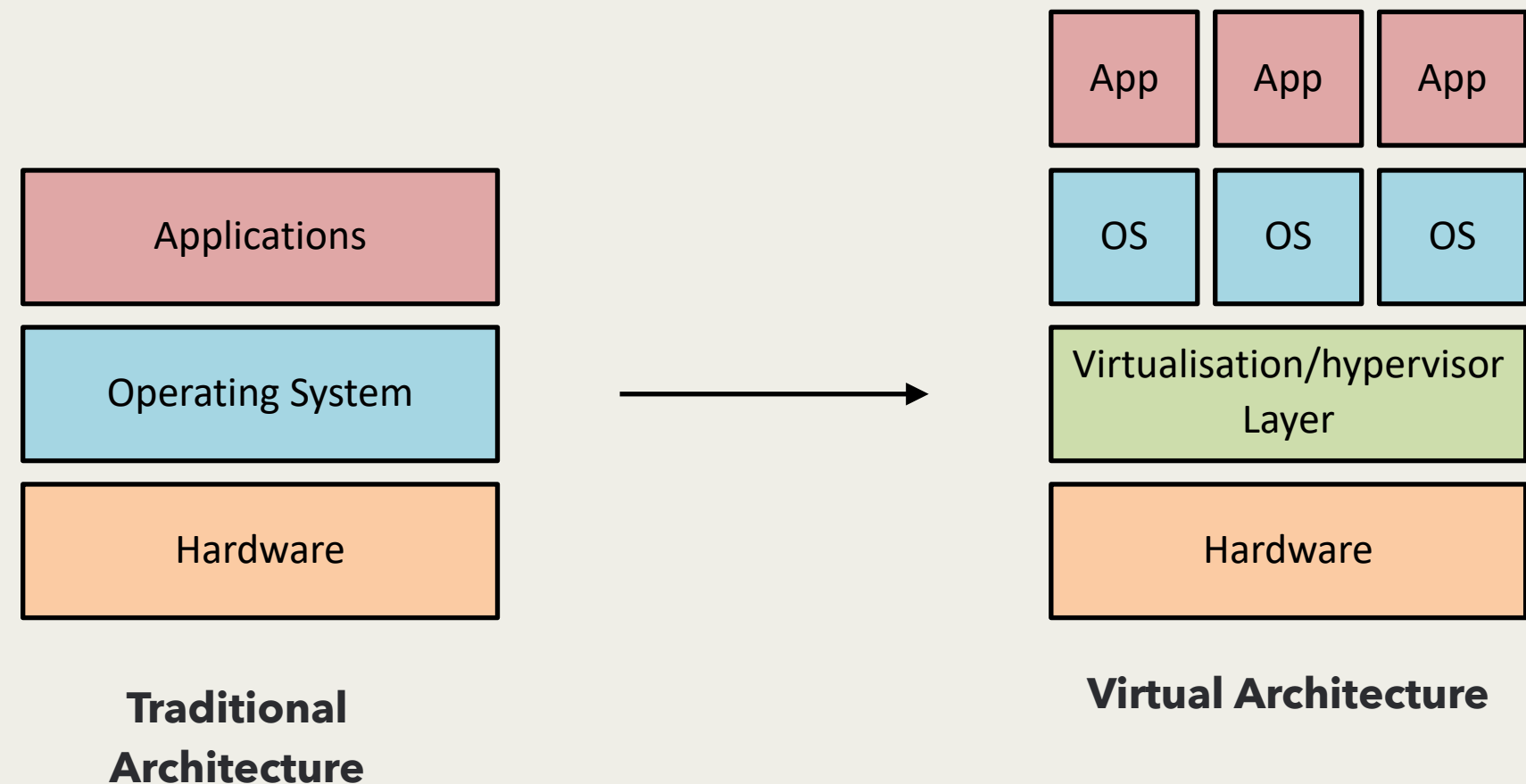
- Simplifies app deployment and management
- Allows applications to run on different OSs



VIRTUALISATION & CLOUD COMPUTING

Types of virtualisation

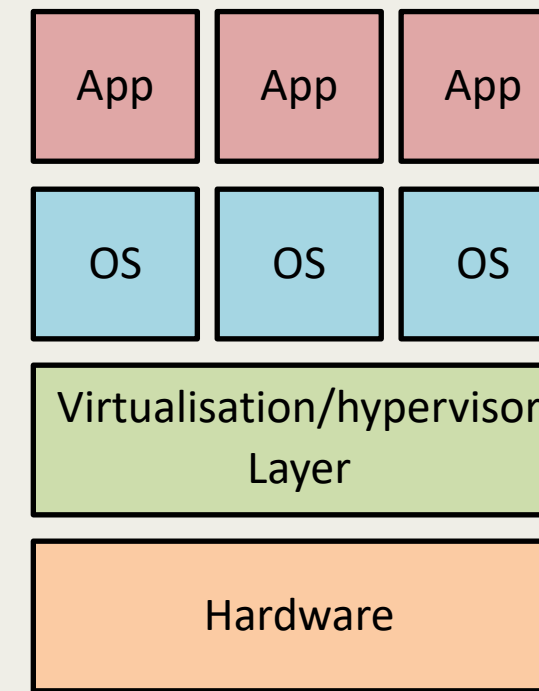
Server, desktop, storage, network, and application virtualisations are the *core types of virtualisation*, but there are additional options like *containerisation* which uses containers for application deployment.



VIRTUALISATION & CLOUD COMPUTING

Hypervisors

Server virtualisation relies on hypervisor technology to create a virtual layer on top of the physical server hardware.

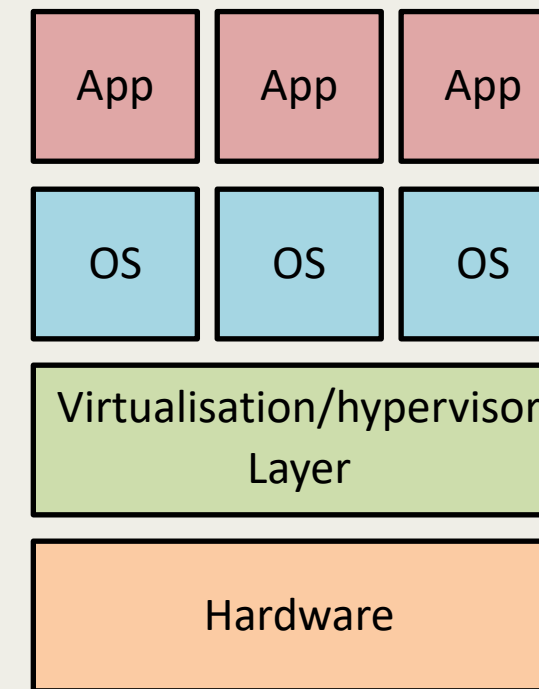


Virtual Architecture

VIRTUALISATION & CLOUD COMPUTING

Hypervisors

The hypervisor manages the allocation of resources (processing power, memory, storage, and networking) among multiple virtual machines (VMs) running on the same physical server.

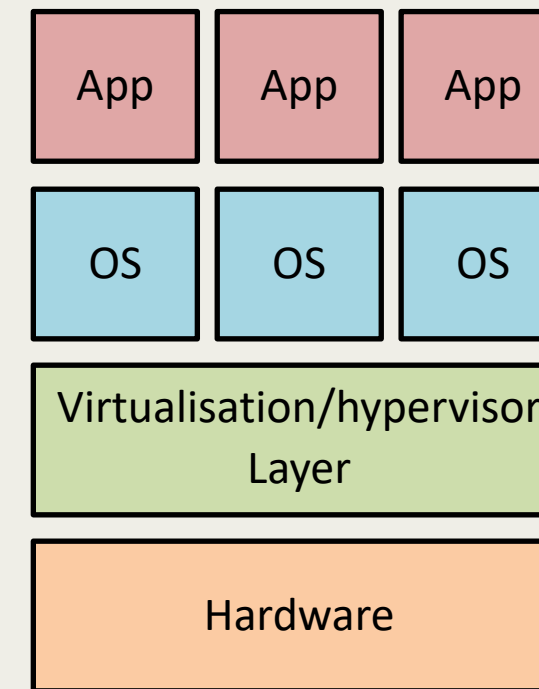


Virtual Architecture

VIRTUALISATION & CLOUD COMPUTING

Hypervisors

Each VM has its own operating system (OS) and applications, completely isolated from other VMs.

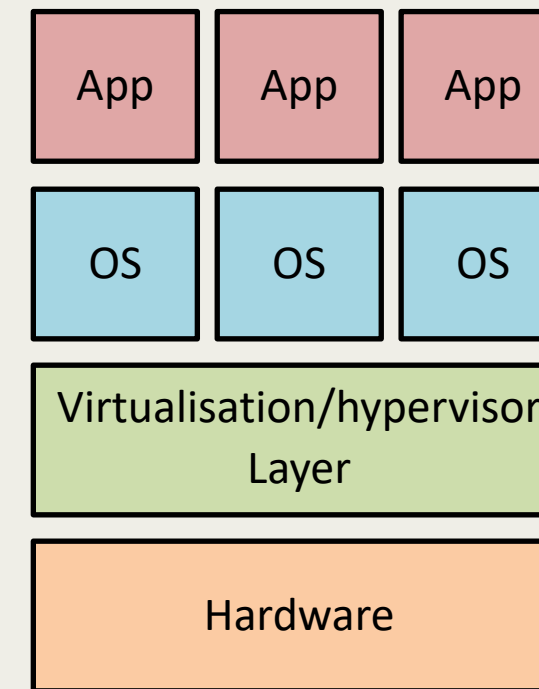


Virtual Architecture

VIRTUALISATION & CLOUD COMPUTING

Hypervisors

The hypervisor ensures each VM has access to the resources it needs, even though they are all sharing the same physical hardware.



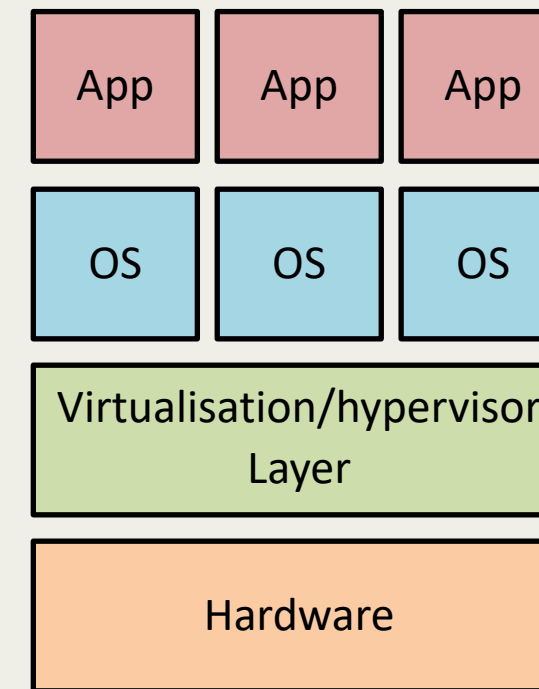
Virtual Architecture

VIRTUALISATION & CLOUD COMPUTING

Hypervisors

Maximise resource utilisation

By consolidating multiple physical servers onto fewer machines with VMs, organisations can significantly increase server utilisation rates.



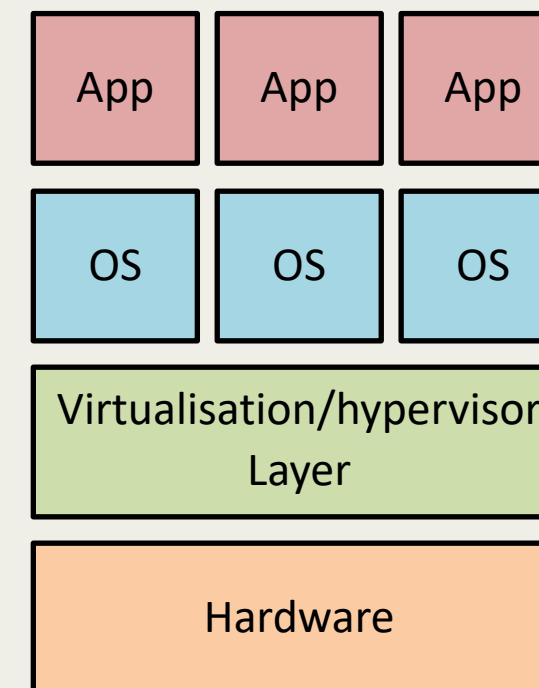
Virtual Architecture

VIRTUALISATION & CLOUD COMPUTING

Hypervisors

Enhanced monitoring and management

Hypervisors provide centralised management tools for monitoring VM performance, resource allocation, and overall system health.



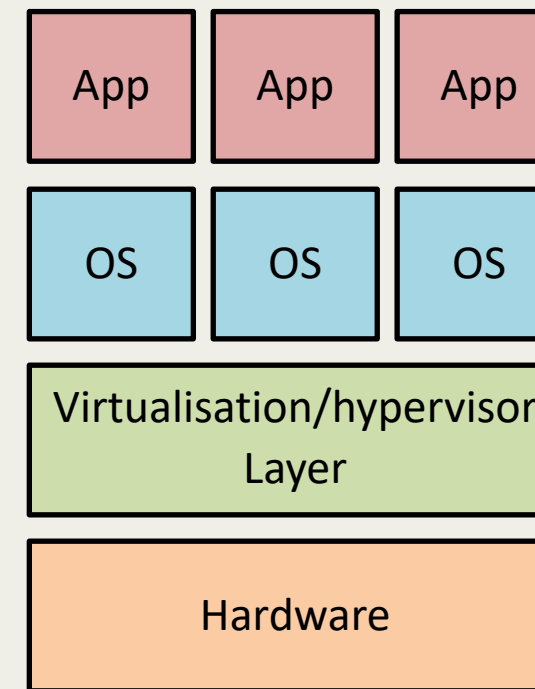
Virtual Architecture

VIRTUALISATION & CLOUD COMPUTING

Hypervisors

Improved reliability and
redundancy

If a physical server fails, VMs can be easily migrated to another server with minimal downtime. Additionally, VMs can be easily backed up and restored for disaster recovery purposes.



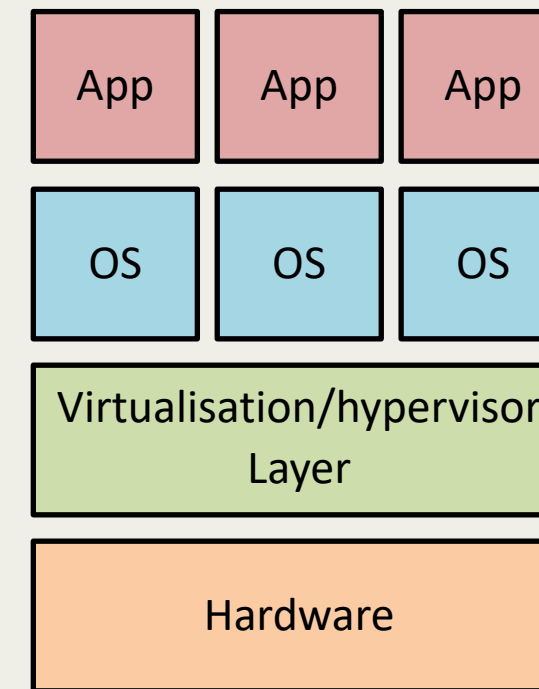
Virtual Architecture

VIRTUALISATION & CLOUD COMPUTING

Hypervisors

On-demand provisioning

Hypervisors enable the creation and deployment of VMs in a matter of minutes, allowing for quicker resource provisioning and greater flexibility in responding to changing needs.



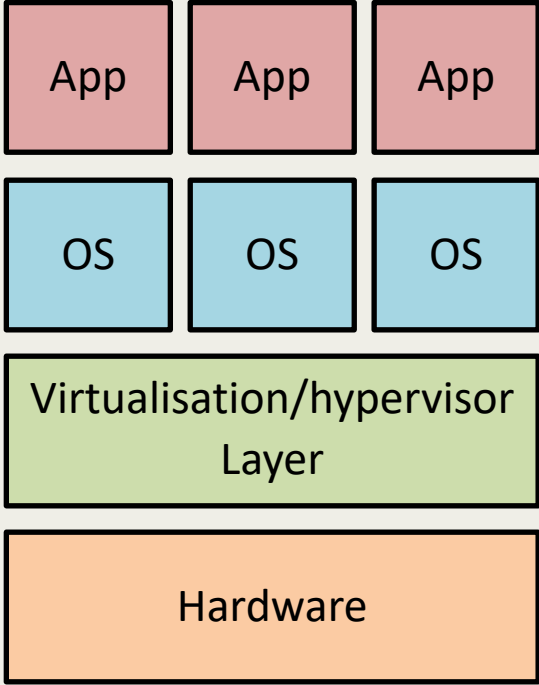
Virtual Architecture

VIRTUALISATION & CLOUD COMPUTING

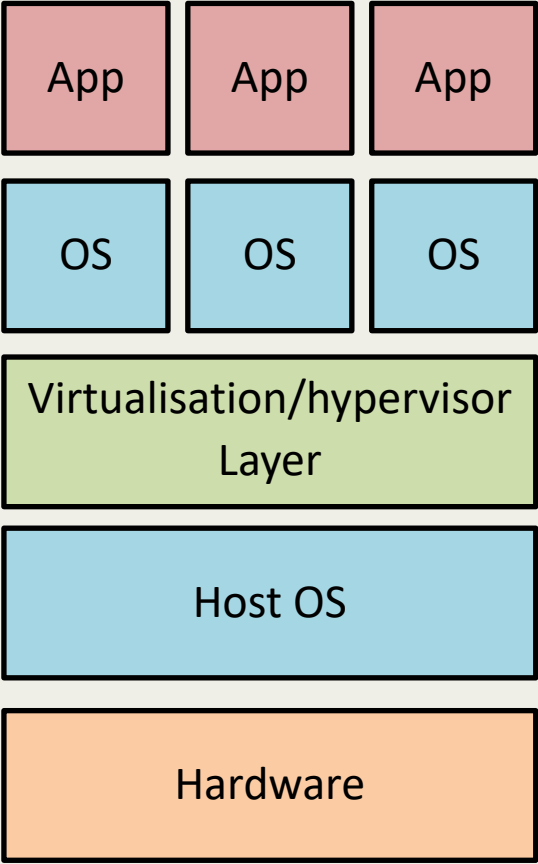
Hypervisors Configurations

In the world of hypervisors, there are two main configurations:

- Bare-metal (Type1) hypervisor
- Hosted (Type 2) hypervisor



**Bare-Metal (Type 1)
Hypervisor**



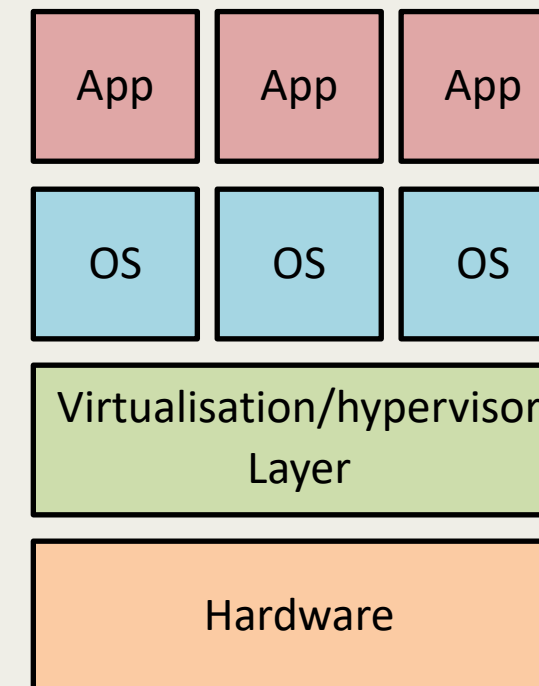
**Hosted (Type 2)
Hypervisor**

VIRTUALISATION & CLOUD COMPUTING

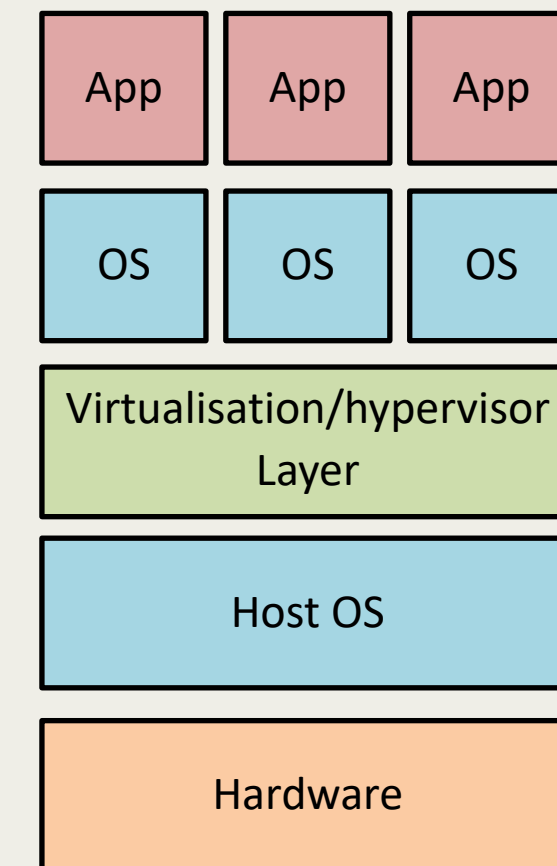
Hypervisors Configurations

Bare-metal (Type 1) hypervisors

Bare-metal hypervisors offer the highest performance and stability in the virtualisation world. They run directly on the physical server hardware, eliminating the need for an underlying OS.



**Bare-Metal (Type 1)
Hypervisor**



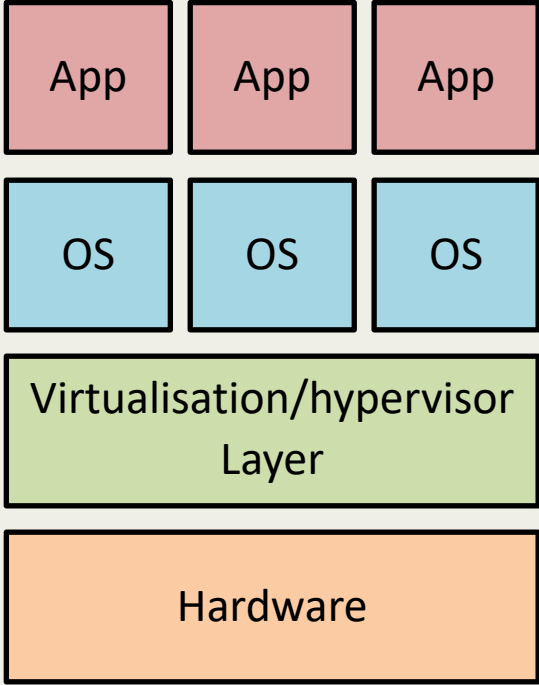
**Hosted (Type 2)
Hypervisor**

VIRTUALISATION & CLOUD COMPUTING

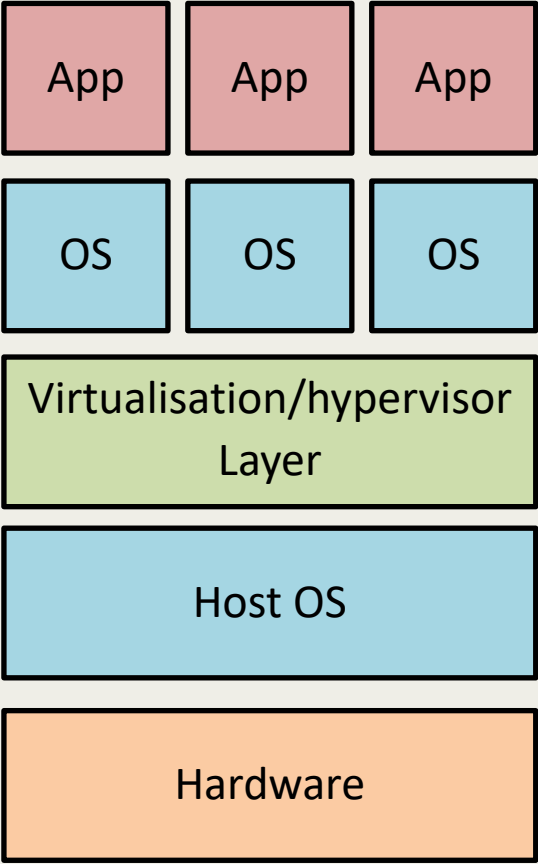
Hypervisors Configurations

Bare-metal (Type 1) hypervisors

Since there's no additional OS layer to mediate hardware access, bare-metal hypervisors deliver the best possible performance for virtual machines (VMs). This is crucial for resource-intensive applications.



**Bare-Metal (Type 1)
Hypervisor**



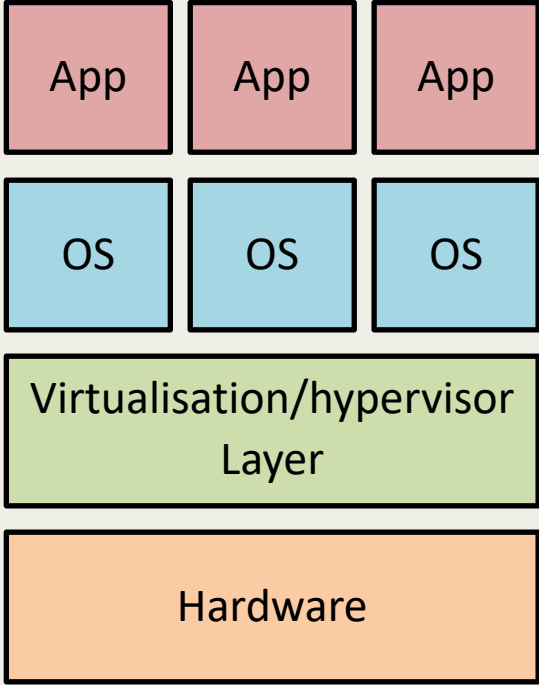
**Hosted (Type 2)
Hypervisor**

VIRTUALISATION & CLOUD COMPUTING

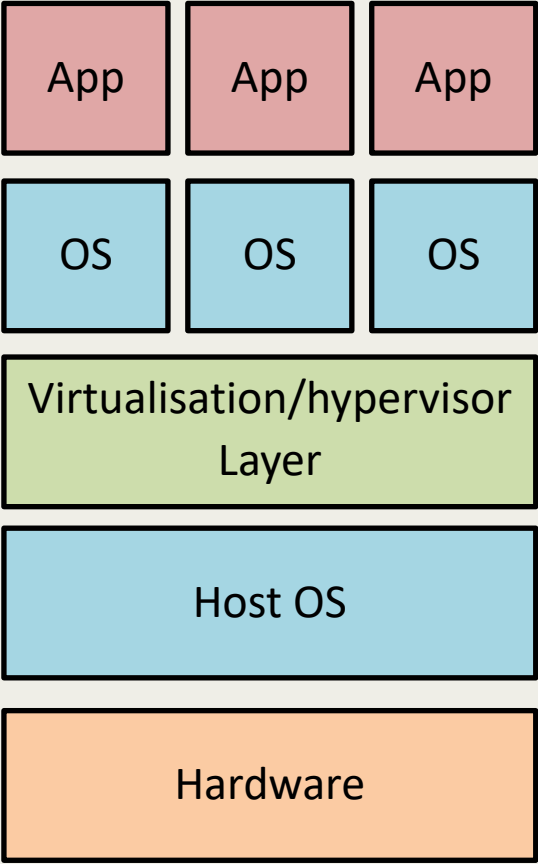
Hypervisors Configurations

Bare-metal (Type 1) hypervisors

Bare-metal hypervisors require minimal resources themselves, leaving more processing power, memory, and storage available for VMs. This translates to better overall system efficiency.



**Bare-Metal (Type 1)
Hypervisor**



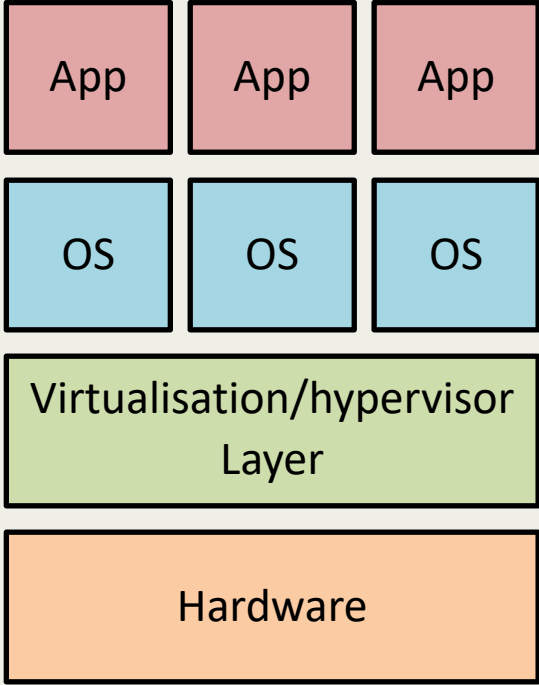
**Hosted (Type 2)
Hypervisor**

VIRTUALISATION & CLOUD COMPUTING

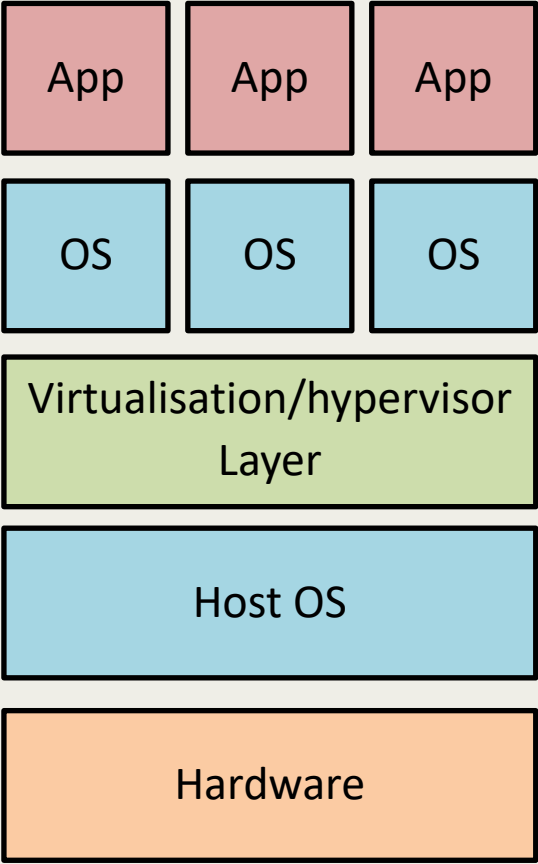
Hypervisors Configurations

Bare-metal (Type 1) hypervisors

Without an underlying OS as a potential vulnerability, bare-metal hypervisors offer a more secure foundation for virtualised environments.



**Bare-Metal (Type 1)
Hypervisor**



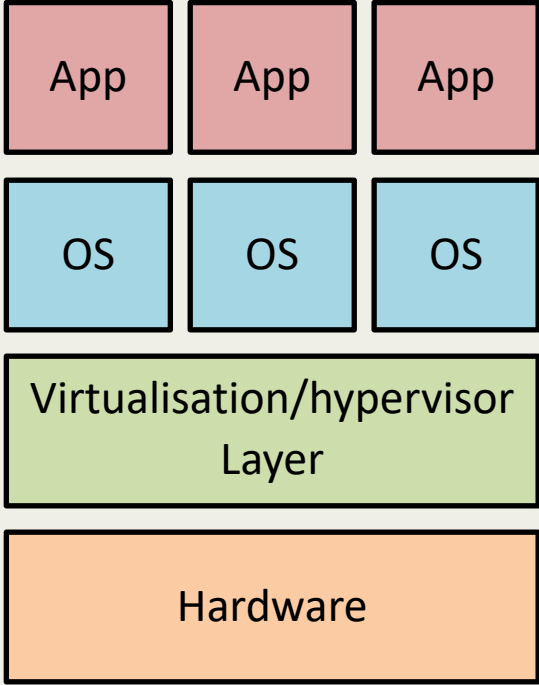
**Hosted (Type 2)
Hypervisor**

VIRTUALISATION & CLOUD COMPUTING

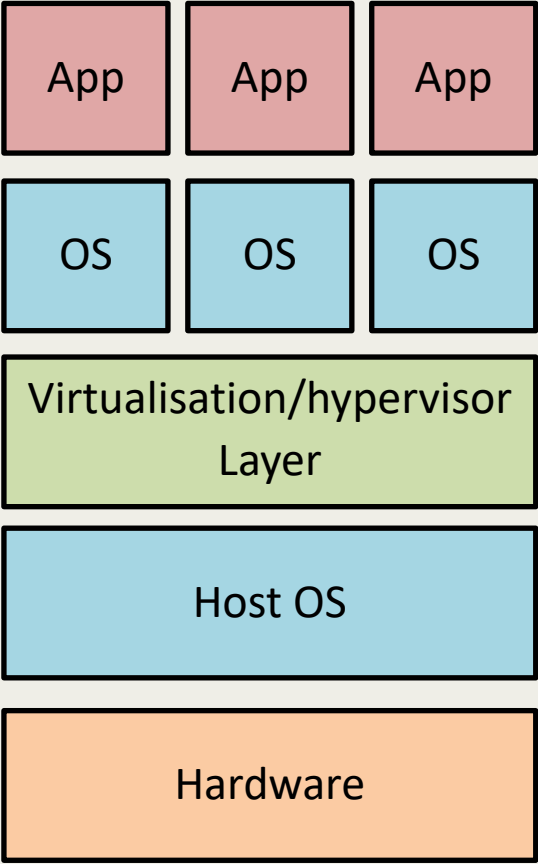
Hypervisors Configurations

Hosted (Type 2) hypervisors

Hosted hypervisors (Type 2) offer a convenient and flexible approach to virtualisation. They run as software applications on top of a traditional operating system (OS) already present on the server.



**Bare-Metal (Type 1)
Hypervisor**



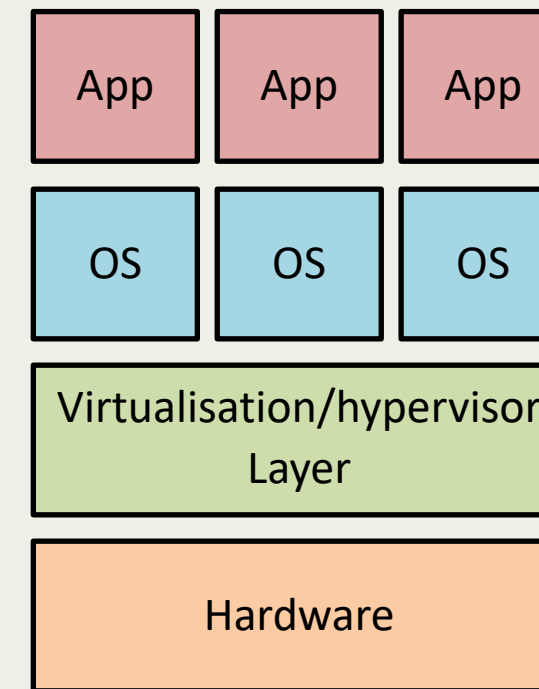
**Hosted (Type 2)
Hypervisor**

VIRTUALISATION & CLOUD COMPUTING

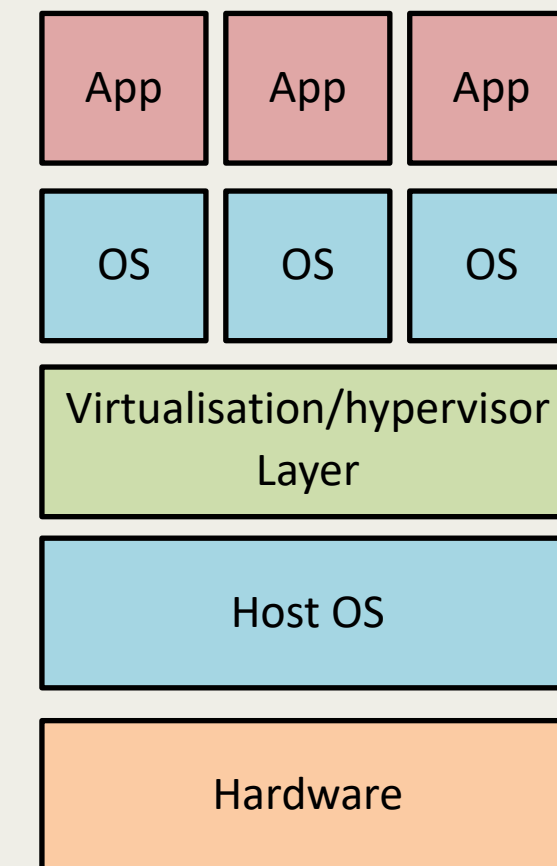
Hypervisors Configurations

Hosted (Type 2) hypervisors

They are generally easier to install and manage compared to bare-metal hypervisors. They leverage the existing OS infrastructure and tools, making them a good fit for environments with less virtualisation expertise.



**Bare-Metal (Type 1)
Hypervisor**



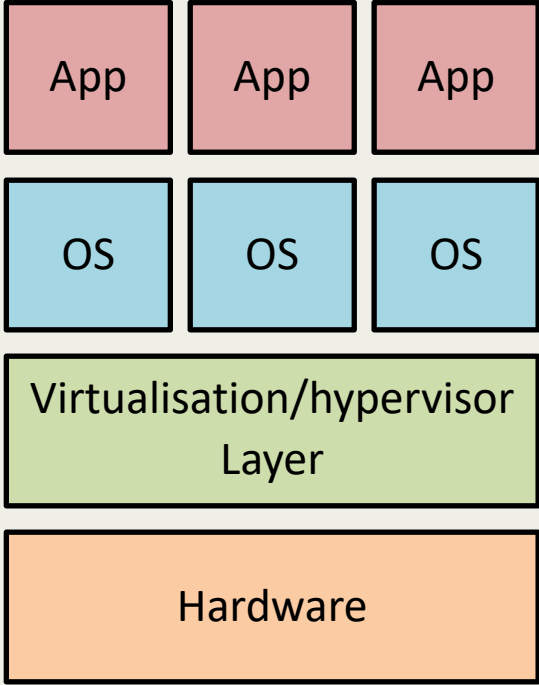
**Hosted (Type 2)
Hypervisor**

VIRTUALISATION & CLOUD COMPUTING

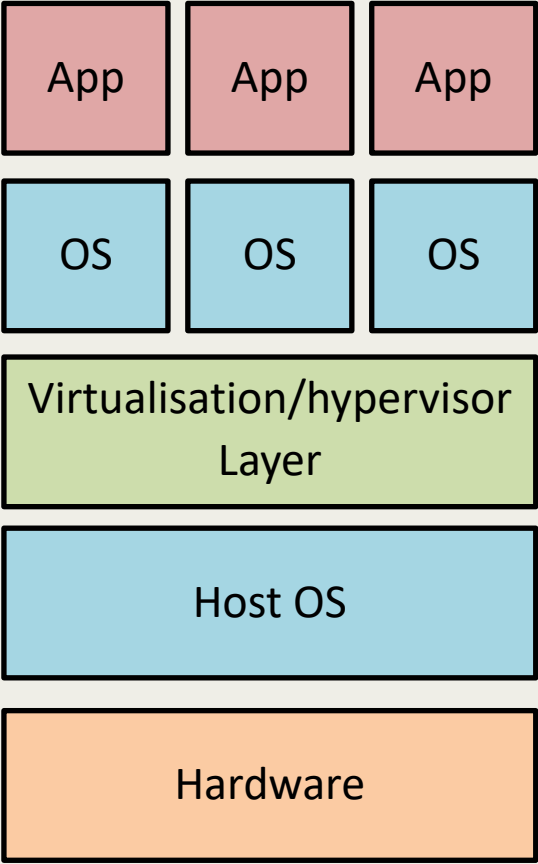
Hypervisors Configurations

Hosted (Type 2) hypervisors

Reliance on the underlying OS security introduces a potential vulnerability. Careful OS hardening and security practices are essential when using hosted hypervisors.



**Bare-Metal (Type 1)
Hypervisor**



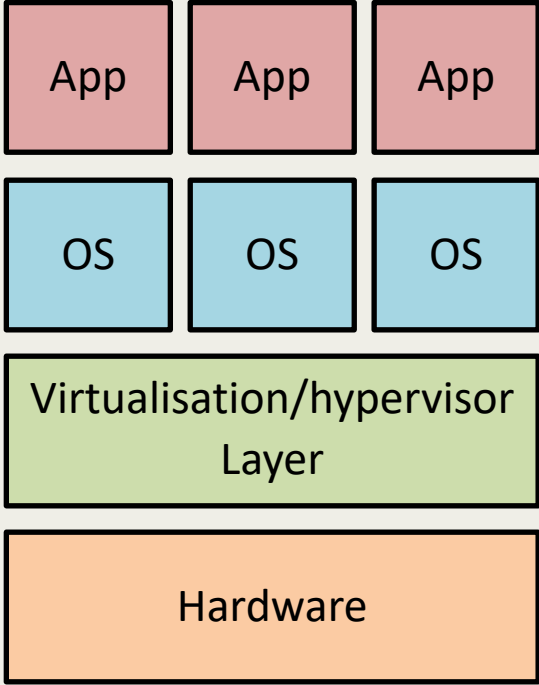
**Hosted (Type 2)
Hypervisor**

VIRTUALISATION & CLOUD COMPUTING

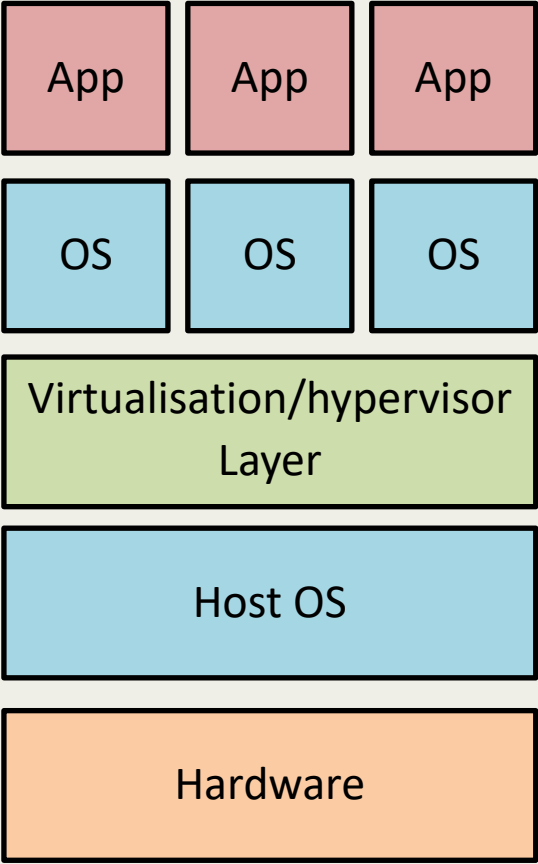
Hypervisors Configurations

Hosted (Type 2) hypervisors

Hosted hypervisors compete with the host OS for system resources. This can impact the overall performance and resource allocation for VMs.



**Bare-Metal (Type 1)
Hypervisor**



**Hosted (Type 2)
Hypervisor**

CLOUD SHARED CONSIDERATIONS

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations

- **Goals:** Confidentiality, integrity, availability, privacy
- **Partner considerations:**
- **Governance,** auditability, regulatory oversight

Operational considerations

- **Availability and performance:** resiliency, performance, SLAs
- **Maintenance and version control**
- **Outsourcing issues:** reversibility, vendor lock-in/portability, interoperability

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations

- **Goals:** Confidentiality, integrity, availability, privacy
- **Partner considerations:**
 - Governance, auditability, regulatory oversight

At a high level, the security and privacy concerns facing cloud computing users are the same as those faced by any cybersecurity professional.

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations

● **Goals:** Confidentiality, integrity, availability, privacy



Confidentiality seeks to protect assets (information and systems) from unauthorised access.

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations

● **Goals:** Confidentiality, integrity, availability, privacy

Integrity seeks to protect assets against unauthorised modification.

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations

● **Goals:** Confidentiality, integrity, availability, privacy



Availability seeks to ensure that assets are available for authorised use when needed without disruption.

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations

● **Goals:** Confidentiality, integrity, availability, privacy

Privacy ensures that we respect the rights of confidentiality not only of our own organisation but also for individuals whose personal information we store, process, and transmit.

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations



Partner considerations:



Governance, auditability,
regulatory oversight

Cloud governance efforts help an organisation work through existing and planned cloud relationships (with CSPs, vendors, etc) to ensure that they comply with security, business, and other constraints.

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations



Partner considerations:



Governance, auditability,
regulatory oversight

Most organisations set up a **cloud governance structure** designed to *assess potential vendors, manage relationships, and oversee cloud operations.*

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations

Auditability is an important component of governance.

Partner considerations:

Governance, auditability,
regulatory oversight

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations



Partner considerations:



Governance, auditability,
regulatory oversight

Cloud computing contracts should specify that the customer has the right to **audit** cloud providers, either directly or through a third party.

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations



Partner considerations:



Governance, auditability,
regulatory oversight

Audits may take place on a scheduled or unplanned basis, allowing customers to gain assurance that the cloud vendor is meeting its security obligations.

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations

Regulatory oversight exists in the cloud world, just as it does in on-premises computing.

Partner considerations:

Governance, auditability,
regulatory oversight

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations



Partner considerations:



Governance, auditability,
regulatory oversight

Organisations subject to cybersecurity regulations (including HIPAA, FERPA, PCI DSS, etc) must ensure that cloud providers support their ability to remain compliant with these obligations.

VIRTUALISATION & CLOUD COMPUTING

Security and privacy considerations

Partner considerations:

Governance, auditability,
regulatory oversight

Some regulatory schemes include provisions about how organisations ensure that third party providers remain compliant, such as using only certified providers or requiring handwritten agreements with providers that the provider's handling of data will be consistent with regulations.

VIRTUALISATION & CLOUD COMPUTING

Operational considerations

Besides being a security issue,
availability is also an operational
consideration.

- **Availability and performance:**
resiliency, performance, SLAs



VIRTUALISATION & CLOUD COMPUTING

Operational considerations

We can increase availability by increasing **resiliency** - the ability of the cloud infrastructure to withstand disruptive events.

- Availability and performance: resiliency, performance, SLAs



VIRTUALISATION & CLOUD COMPUTING

Operational considerations

Performance is a closely related concept to resiliency. It answers the question: *How well can the cloud service stand up to the demands that we place on it?*

- **Availability and performance:**
resiliency, performance, SLAs



VIRTUALISATION & CLOUD COMPUTING

Operational considerations

Customers should negotiate **specific service levels** with vendors during the contracting process and then document those service levels in written agreements called **service-level agreements (SLAs)**.

- **Availability and performance:**
resiliency, performance, SLAs



VIRTUALISATION & CLOUD COMPUTING

Operational considerations

SLAs

- Specify the requirements that the vendor agrees to meet
- Include financial penalties if the vendor fails to live up to operational obligations.

Availability and performance:
resiliency, performance, SLAs

VIRTUALISATION & CLOUD COMPUTING

Operational considerations

Managing change is a difficult issue in enterprise IT, and those issues or concerns do not go away in the cloud.



Maintenance and version control



VIRTUALISATION & CLOUD COMPUTING

Operational considerations

IT teams must now coordinate change

- Internally, with maintenance schedules
- Externally, based on the maintenance schedules of the CSP



Maintenance and version control

VIRTUALISATION & CLOUD COMPUTING

AWS Health provides ongoing visibility into your resource performance and the availability of your AWS services and accounts.

AWS Health Dashboard

▼ Service health

Open and recent issues

Service history

▼ Your account health

Open and recent issues

Scheduled changes

Other notifications

Event log

▼ Your organisation health

Configurations

New

▼ Health integrations

Amazon EventBridge

AWS Health Aware

Time zone settings

Your account health

Stay informed of important events affecting your AWS resources.

Open and recent issues (0)

Scheduled changes (5)

Other notifications (2)

Event log

Scheduled changes (5)

TableCalendar

View upcoming events and ongoing events from the past seven days that might affect your AWS infrastructure, such as scheduled maintenance activities. [View scheduled changes that occurred more than 7 days ago.](#)

Q Add filter

	Event	Status	Region/zone	Start time	End time	Affected resources
	ElasticContainerService task patching retirement	Upcoming	eu-west-1	1 July 2024 at 5:00:00 AM UTC+4	8 July 2024 at 5:00:00 AM UTC+4	11 entities
	ElasticContainerService task patching retirement	Ongoing	us-east-1	21 June 2024 at 1:00:00 PM UTC+4	28 June 2024 at 1:00:00 PM UTC+4	2 entities
	Bedrock planned lifecycle event	Upcoming	us-west-2	30 July 2024 at 11:00:00 AM UTC+4		1 entity

Amazon EventBridge rule

Get notifications for events that might affect your services and resources.

Configure

Find out more

VIRTUALISATION & CLOUD COMPUTING

You can use *AWS Health Events* to learn how service and resource changes might affect your applications running on AWS.

AWS Health Dashboard

▼ Service health

Open and recent issues

Service history

▼ Your account health

Open and recent issues

Scheduled changes

Other notifications

Event log

▼ Your organisation health

Configurations

New

▼ Health integrations

Amazon EventBridge

AWS Health Aware

Time zone settings

Add filter

< 1 >

	Event	Status	Region/zone Info	Start time	End time	Affected resources
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☐	Bedrock planned lifecycle event	Upcoming	us-west-2	30 July 2024 at 11:00:00 AM UTC+4		1 entity

ElasticContainerService task patching retirement

Region/availability zone

eu-west-1

Account specific

Yes

Description

You are receiving this notification because AWS Fargate has deployed a new platform version revision [1] and will retire any tasks running on previous platform version revision(s) starting at Mon, 1 Jul 2024 01:00 GMT as part of routine task maintenance [2]. Please check the "Affected Resources" tab for a list of affected tasks. There is no action required on your part unless you want to replace these tasks before Fargate does. When using the default value of 100% for minimum healthy percent configuration of an ECS service [3], a replacement task will be launched on the most recent platform version revision before the affected task is retired. Any tasks launched after Mon, 24 Jun 2024 01:00 GMT were launched on the new platform version revision.

Category

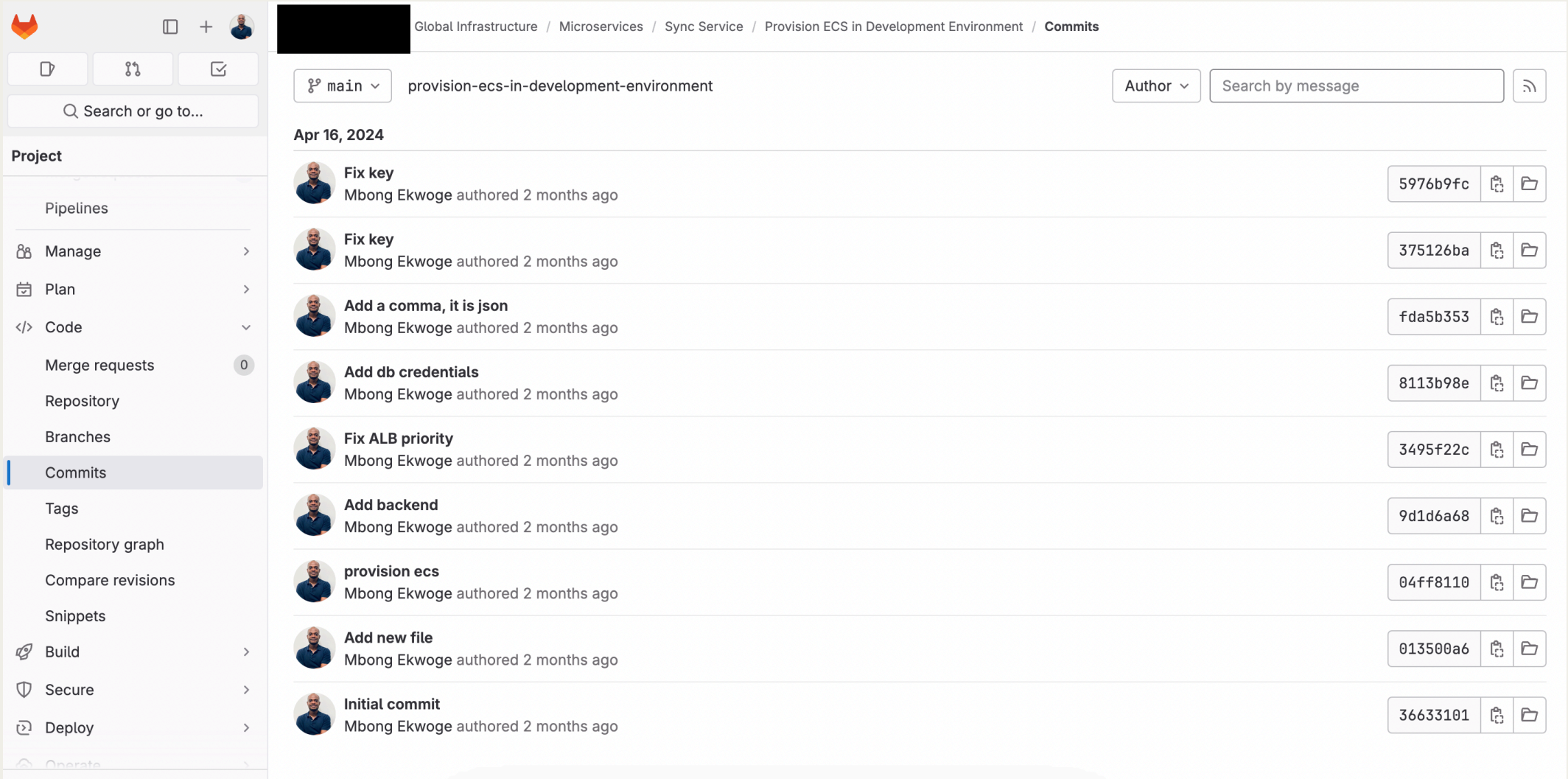
Scheduled change

Affected resources

11

VIRTUALISATION & CLOUD COMPUTING

Version Control allows organisations to manage the development of software by tracking different versions being worked on by different developers.



VIRTUALISATION & CLOUD COMPUTING

Version Control is an essential component of any software development program.

It may also be used to track the configuration of systems and applications.

CloudFormation

Stacks

Stack details

Drifts

StackSets

Exports

Application Composer

laC generator

Registry

Public extensions

Activated extensions

Publisher

Spotlight

Feedback

CloudFormation > Stacks > http-api-private-integrations-tutorial

Stacks (100+)

Filter by stack name

Filter status

Active

View nested

Stacks

http-api-private-integrations-tutorial

2024-05-23 09:38:35 UTC+0400

UPDATE_ROLLBACK_COMPLETE

NESTED

payment-service-CloudWatchLogGroup-10G4UAI65UU6G

2024-04-24 09:45:46 UTC+0400

CREATE_COMPLETE

payment-service

http-api-private-integrations-tutorial

Delete

Update

Stack actions

Create stack

Stack info

Events

Resources

Outputs

Parameters

Template

Changeset

Events (100+)

Detect root cause

Search events

Timestamp	Logical ID	Status	Detailed status
2024-05-23 10:35:38 UTC+0400	http-api-private-integrations-tutorial	UPDATE_ROLLBACK_COMPLETE	-
2024-05-23 10:35:37 UTC+0400	http-api-private-integrations-tutorial	UPDATE_ROLLBACK_COMPLETE_CLEANUP_IN_PROGRESS	-
2024-05-23 10:35:37 UTC+0400	PrivateIntegrationsTutorialService73010F72	UPDATE_COMPLETE	-
2024-05-23 10:35:34 UTC+0400	http-api-private-integrations-tutorial	UPDATE_ROLLBACK_IN_PROGRESS	-

VIRTUALISATION & CLOUD COMPUTING

Operational considerations

Moving to the cloud also introduces some cloud-specific operational considerations that come as a result of **outsourcing** parts of IT operations.



Outsourcing issues: reversibility,
vendor lock-in/portability,
interoperability

VIRTUALISATION & CLOUD COMPUTING

Operational considerations

Organisations moving to the cloud or moving between CSPs, should consider the importance of **reversibility**.



Outsourcing issues: reversibility, vendor lock-in/portability, interoperability

VIRTUALISATION & CLOUD COMPUTING

Reversibility seeks to answer questions like:

Operational considerations

- What if something goes wrong operationally, technically, or financially?

- How difficult would it be to restore the original operations and reverse the move?

- Outsourcing issues: reversibility, vendor lock-in/portability, interoperability

VIRTUALISATION & CLOUD COMPUTING

Operational considerations

Organisations should make **rollback plans** part of their **transition plan**.



Outsourcing issues: reversibility,
vendor lock-in/portability,
interoperability

VIRTUALISATION & CLOUD COMPUTING

The **vendor lock-in** problem in cloud computing occurs when CSCs become reliant (i.e., locked-in) on a single CSP's technological implementation and are unable to switch vendors in the future without incurring significant fees, legal restrictions, or technical incompatibilities.

Operational considerations



Outsourcing issues: reversibility,
vendor lock-in/portability,
interoperability

VIRTUALISATION & CLOUD COMPUTING

Operational considerations

Organisations should strive to avoid
vendor lock-in whenever possible.



Outsourcing issues: reversibility,
vendor lock-in/portability,
interoperability

VIRTUALISATION & CLOUD COMPUTING

Portability is a design principle that says workloads should be designed so that they do not leverage vendor-specific features and may be easily shifted between CSPs.

This isn't always possible but should be considered a good design practice.

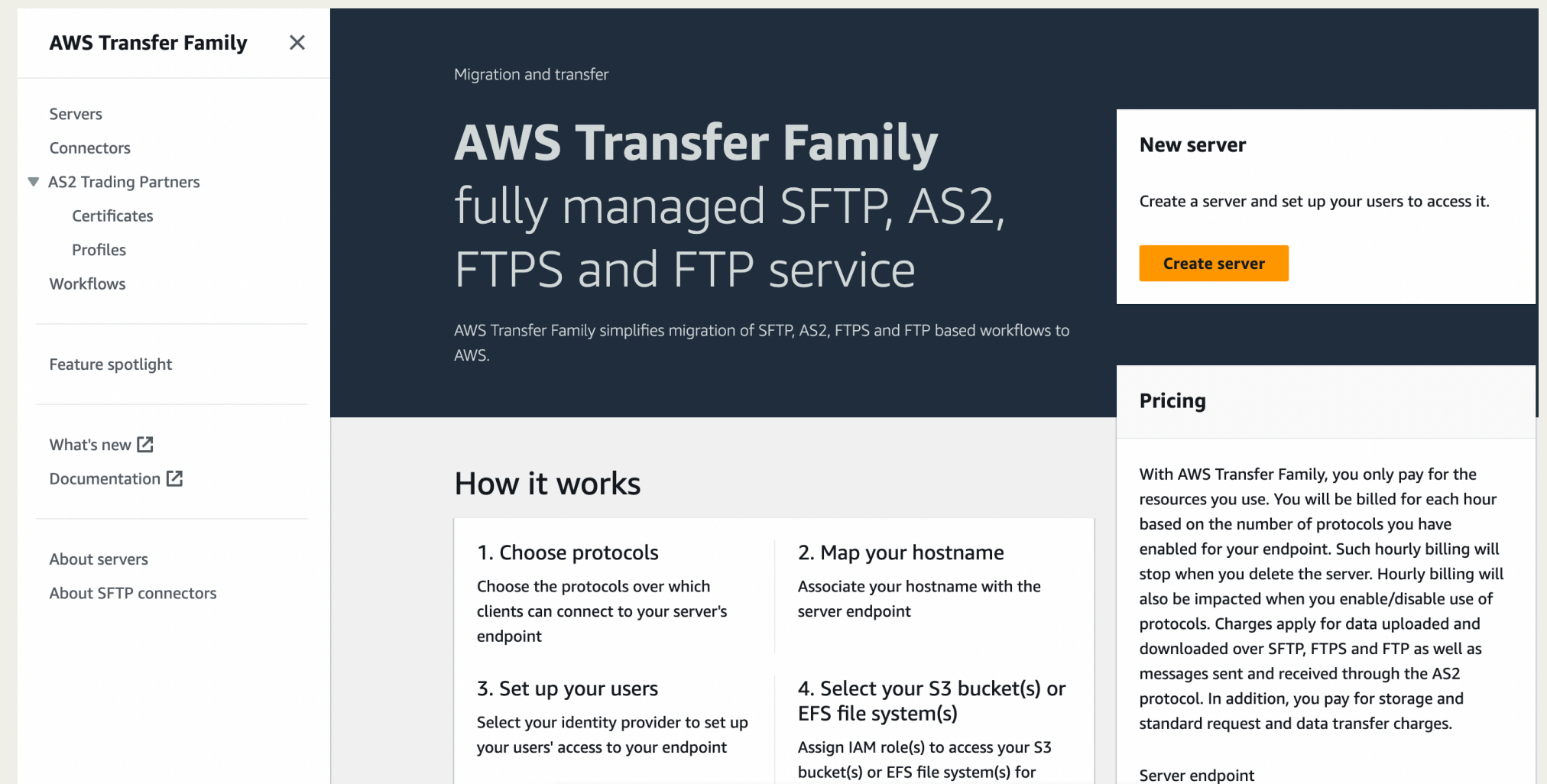
Operational considerations



Outsourcing issues: reversibility,
vendor lock-in/portability,
interoperability

VIRTUALISATION & CLOUD COMPUTING

Each vendor relationship should also provide the ability to export data when required.



VIRTUALISATION & CLOUD COMPUTING

Organisations should consider the interoperability of CSPs whenever adopting a new solution. This is especially important for SaaS and PaaS products.

AWS Marketplace

Manage subscriptions

Private offers

Discover products

Vendor Insights

Private Marketplace

Settings

▼ Refine results

Categories

Infrastructure Software (12879)

DevOps (9288)

Professional Services (8798)

Data Products (4588)

Business Applications (4584)

Machine Learning (3582)

Industries (2482)

IoT (919)

Cloud Operations (705)

▼ Delivery methods

☐ Amazon Machine Image (10994)

☐ Professional Services (8716)

☐ Data Exchange (4590)

☐ SaaS (4176)

☐ SageMaker Model (934)

☐ CloudFormation Template (516)

☐ Container Image (500)

☐ Helm Chart (190)

☐ SageMaker Algorithm (153)

☐ Add-on for Amazon EKS (26)

▼ Publisher

☐ cloudima (1180)

Search AWS Marketplace products

Search for any product via AWS Marketplace...

< 1 ... > ⚙

All products (Over 10000 results) showing 1 - 20

Sort By: Relevance ▼

Fortinet

Fortinet FortiGate Next-Generation Firewall

By Fortinet Inc. | Ver 7.4.4

★★★★☆ 20 AWS reviews | 15 external reviews

Free Trial

Starting from \$0.36/hr or from \$1,920.00/yr (up to 69% savings) for software + AWS usage fees

FortiGate-VM on AWS delivers next-generation firewall and VPN/SD-WAN capabilities for organizations of all sizes. It enables broad network protection and automated security management for consistent enforcement and visibility across your AWS VPCs and hybrid cloud infrastructure. FortiGate natively...

Cloud Storage Security

Antivirus for Amazon S3 - PAYG with 30 DAY FREE TRIAL

By Cloud Storage Security | Ver 7.09.000

★★★★★ 23 AWS reviews | 2 external reviews

Ensure the data in data lakes and application workflows built using AWS storage are free of viruses, ransomware, trojans and other payloads by scanning it with Antivirus for Amazon S3 by Cloud Storage Security. WHY THIS SOLUTION IS DIFFERENT Support for Multiple Data Sources Triple Engine Technol...

Cloud
Brewery



CLOUD SHARED CONSIDERATIONS

CLOUD SHARED CONSIDERATIONS

Interoperability	Performance
Portability	Governance
Reversibility	Maintenance & versioning
Availability	Service levels and SLAs
Security	Auditability
Privacy	Regulatory
Resiliency	Outsourcing

Larger companies may use multiple vendors and two or more CSPs in their delivery of services.

● The business choice (from a customer perspective) is to use the best service for a particular use case, based on *features, cost, or availability*.

CLOUD SHARED CONSIDERATIONS

Interoperability	Performance
Portability	Governance
Reversibility	Maintenance & versioning
Availability	Service levels and SLAs
Security	Auditability
Privacy	Regulatory
Resiliency	Outsourcing

All CSCs and CSPs share a set of concerns or considerations.

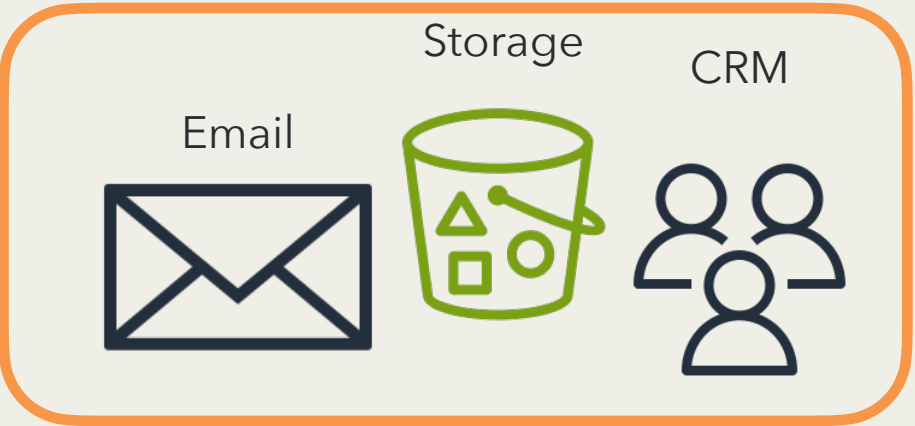
The lectures discuss some major considerations that CSCs consider to decide if cloud computing is appropriate for specific tasks.

The decision should be part of the organisation's cloud computing strategy.

VENDOR LOCK-IN

CLOUD SHARED CONSIDERATIONS

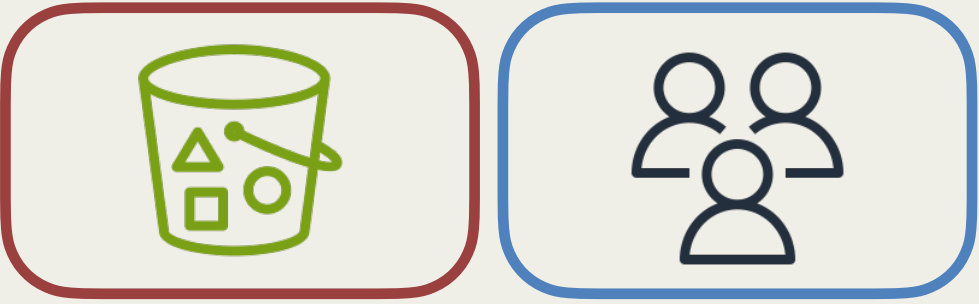
Vendor lock-in refers to a situation where the cost of switching to a different vendor is so high that the customer is essentially stuck with the original vendor.



The company is locked into one cloud vendor

Vendor locked-in

Storage (highest availability) CRM (best price)



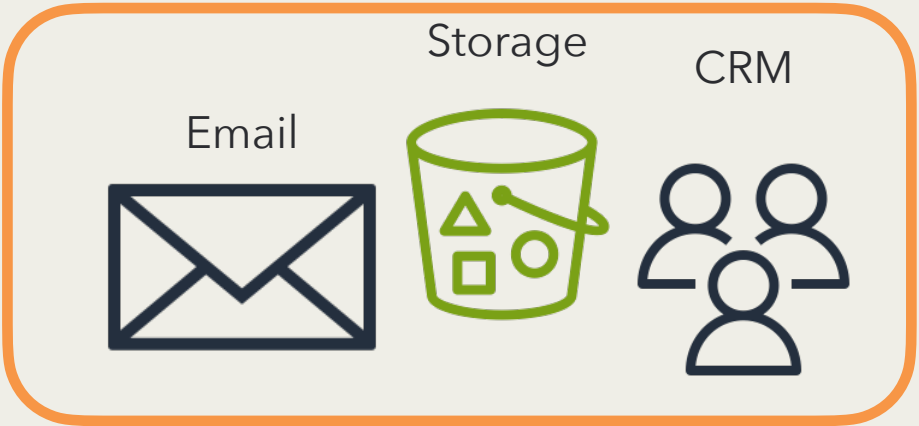
The company has more flexibility with multiple cloud vendors

Avoiding vendor lock-in

CLOUD SHARED CONSIDERATIONS

Why is vendor lock-in a concern?

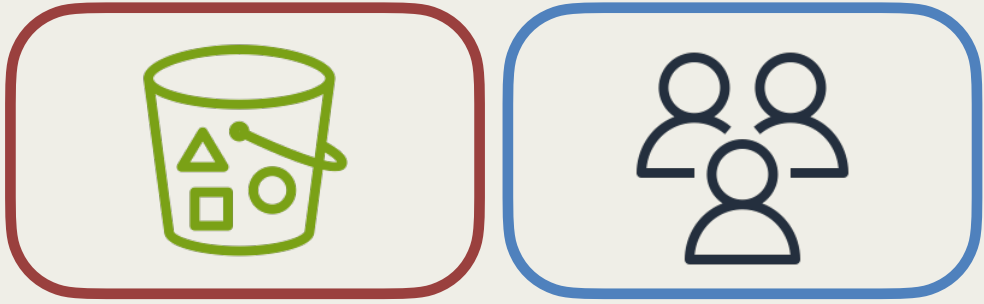
A number of circumstances can negatively impact a business if they are locked-in with a certain CSP.



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Vendor locked-in

Storage (highest availability) CRM (best price)



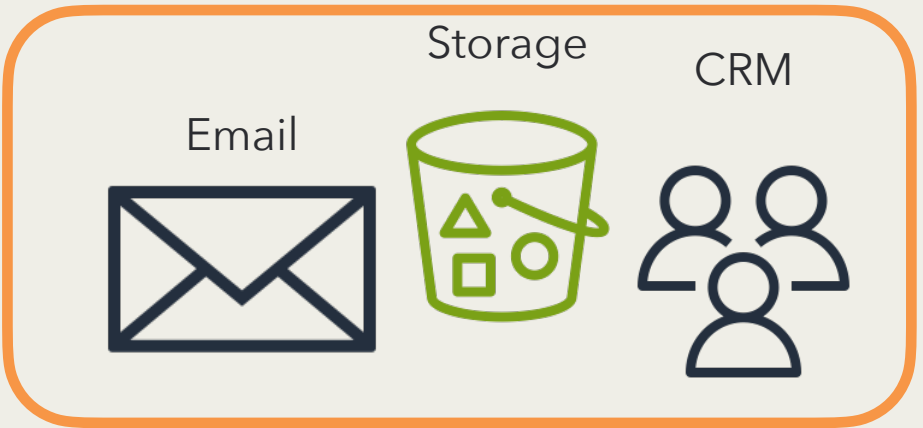
Avoiding vendor lock-in

The company has more flexibility with multiple cloud vendors

CLOUD SHARED CONSIDERATIONS

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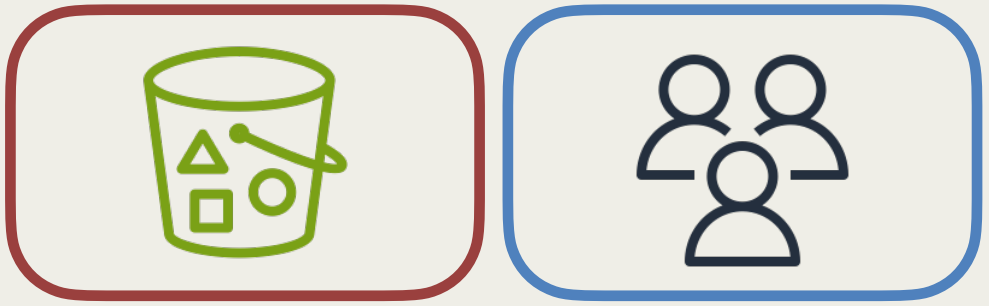
If a CSP's quality of service declines, or never meets a desired threshold, the client will be stuck with it.



The company is locked into one cloud vendor

Vendor locked-in

Storage (highest availability) CRM (best price)



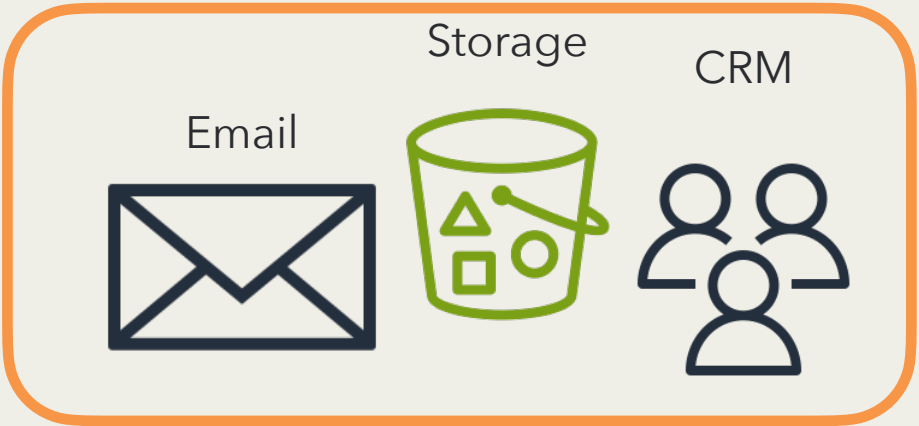
Avoiding vendor lock-in

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CLOUD SHARED CONSIDERATIONS

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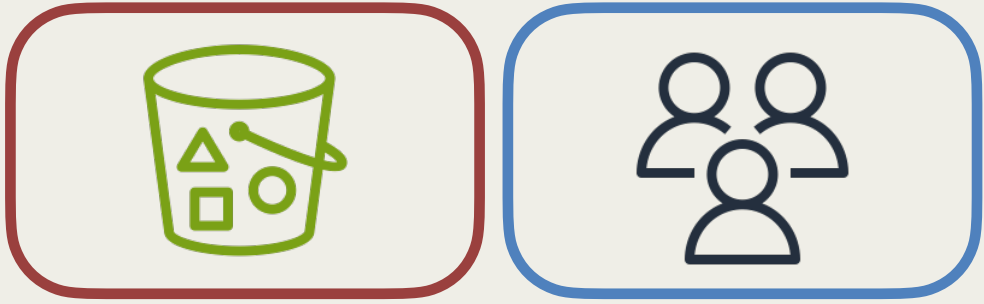
The CSP may drastically change their product offerings in such a way that they no longer meet a business's need.



The company is locked into one cloud vendor

Vendor locked-in

Storage (highest availability) CRM (best price)



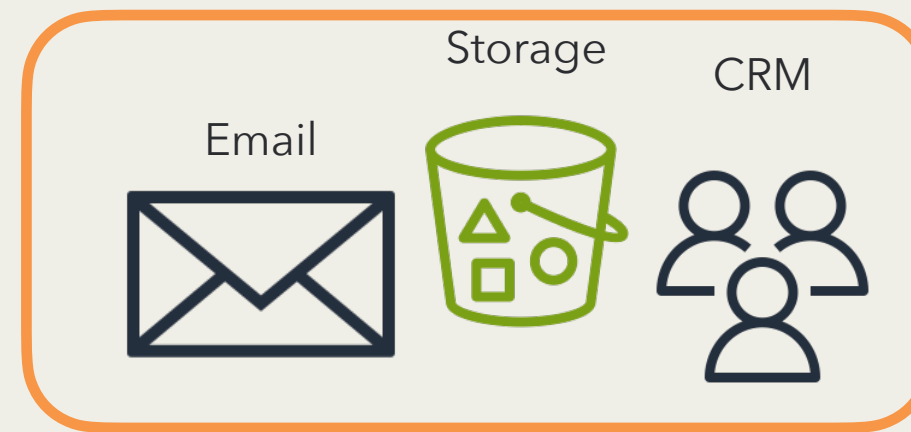
Avoiding vendor lock-in

The company has more flexibility with multiple cloud vendors

CLOUD SHARED CONSIDERATIONS

Why is vendor lock-in a concern?

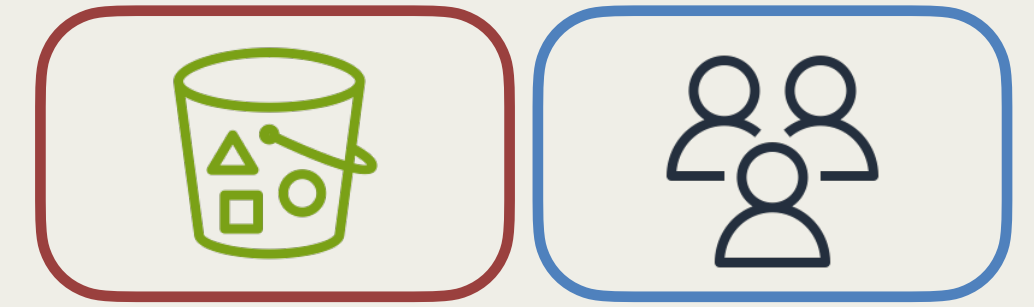
- A CSP may go out of business altogether.
- A CSP may impose massive price increases for the service, knowing that their clients are locked in.



The company is locked into one cloud vendor

Vendor locked-in

Storage (highest availability) CRM (best price)



Avoiding vendor lock-in

The company has more flexibility with multiple cloud vendors

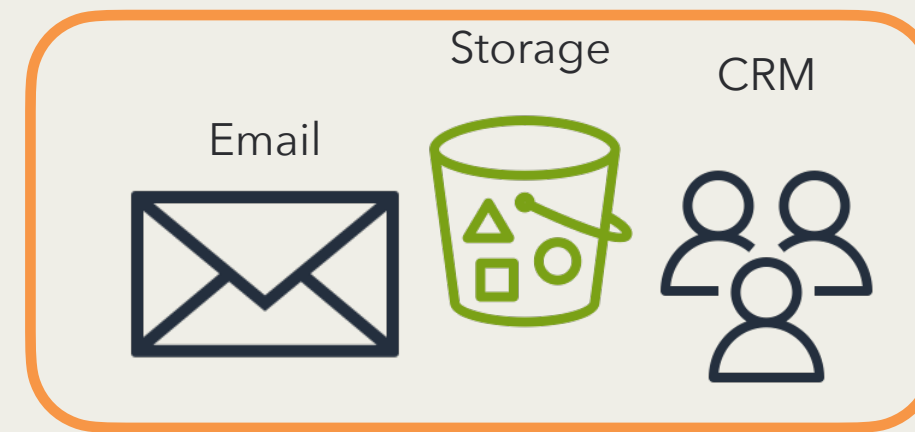
Cloud Brewery 

INTEROPERABILITY

INTEROPERABILITY

Interoperability creates the ability to communicate with and share data

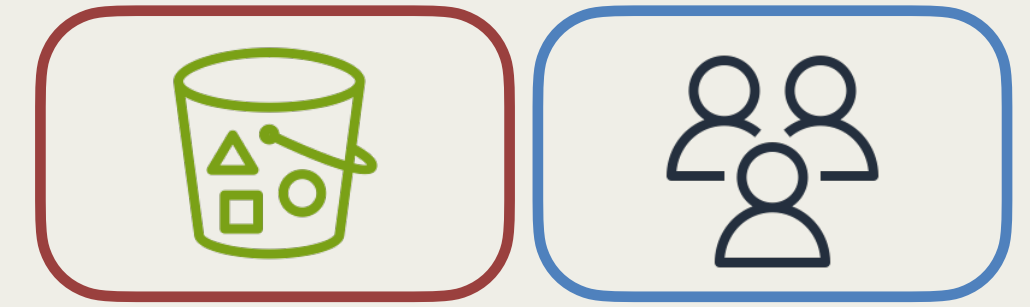
- Across multiple platforms
- Between traditional IT and cloud services provided by different CSPs



The company is locked into one cloud vendor

Vendor locked-in

Storage (highest availability) CRM (best price)



Avoiding vendor lock-in

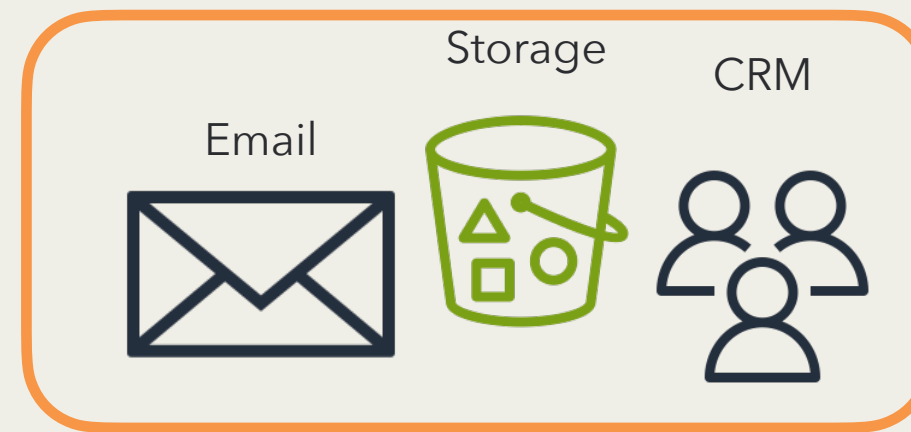
The company has more flexibility with multiple cloud vendors

Cloud Brewery 

INTEROPERABILITY

Avoiding vendor lock-in allows the customer to make decisions on a particular service, regardless of the vendor providing the service, based on

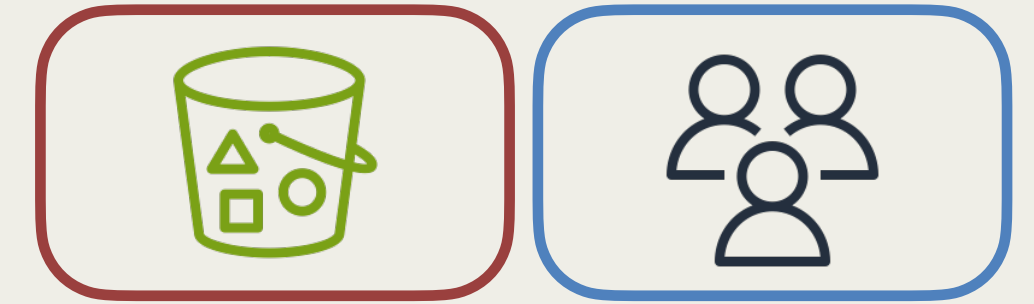
- Cost
- Feature set
- Availability



The company is locked into one cloud vendor

Vendor locked-in

Storage (highest availability) CRM (best price)



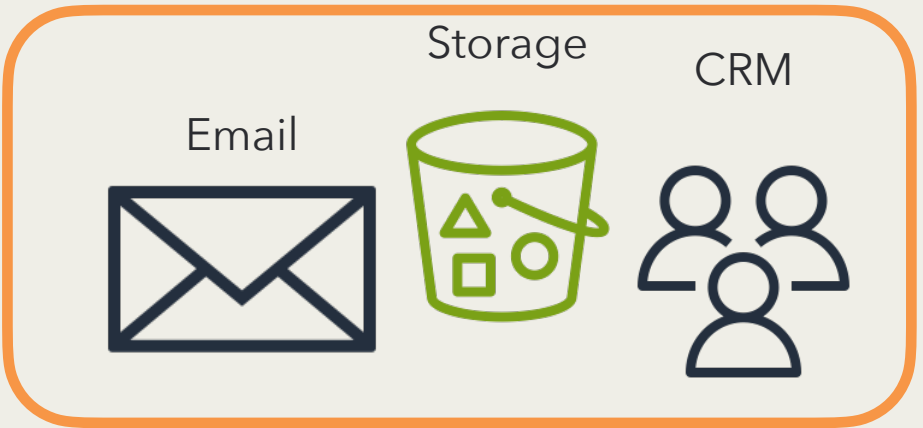
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Avoiding vendor lock-in

Cloud Brewery 

INTEROPERABILITY

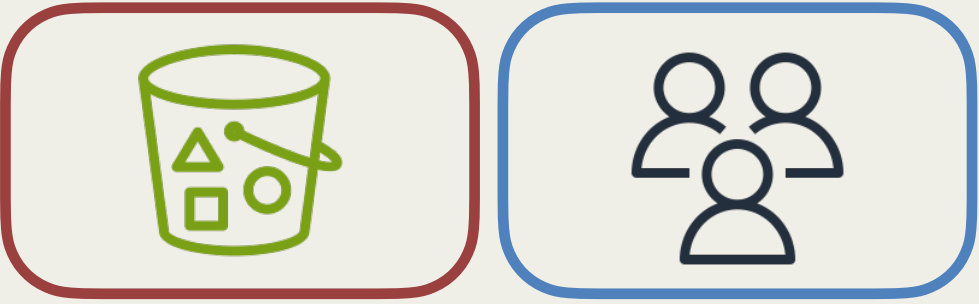
Interoperability leads to a richer set of alternatives and more choices in pricing.



The company is locked into one cloud vendor

Vendor locked-in

Storage (highest availability) CRM (best price)



Avoiding vendor lock-in

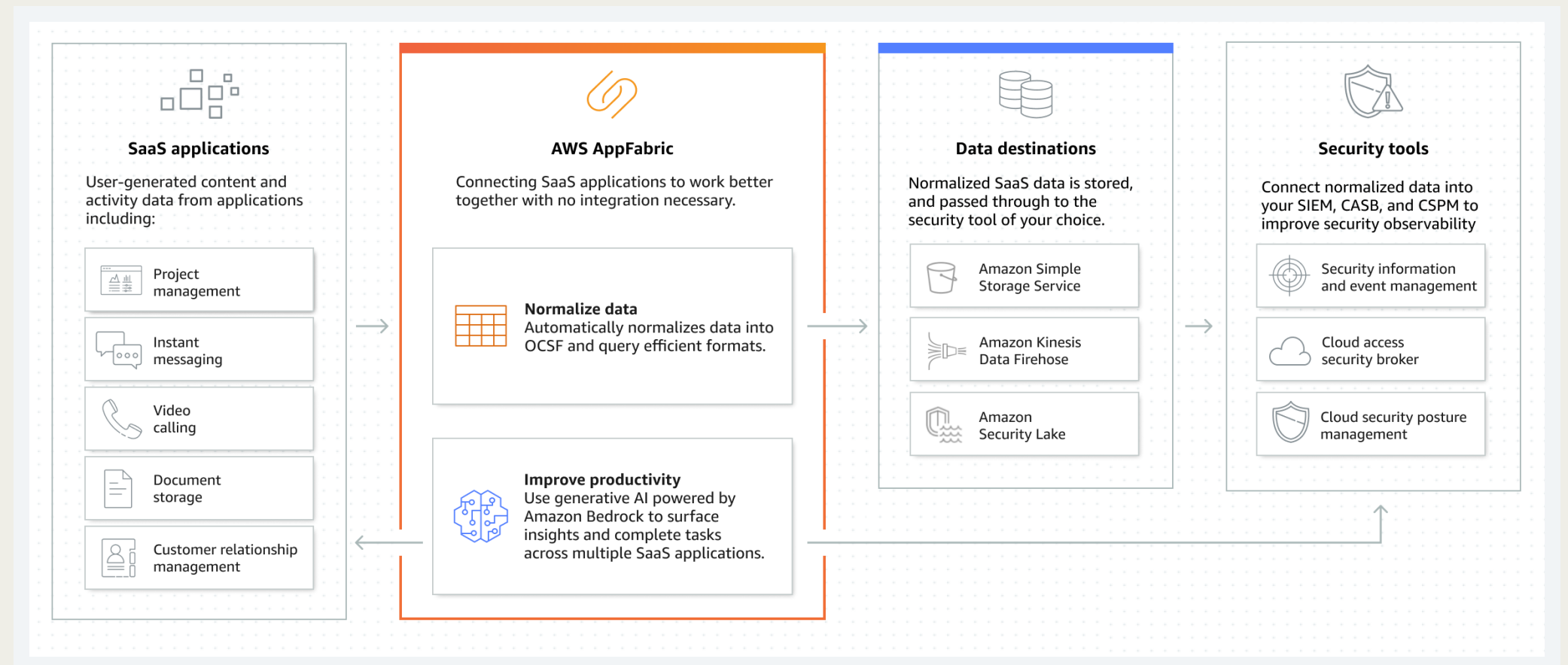
The company has more flexibility with multiple cloud vendors

Cloud Brewery 

INTEROPERABILITY

AWS AppFabric is a managed service that aggregates and analyses data across several SaaS applications.

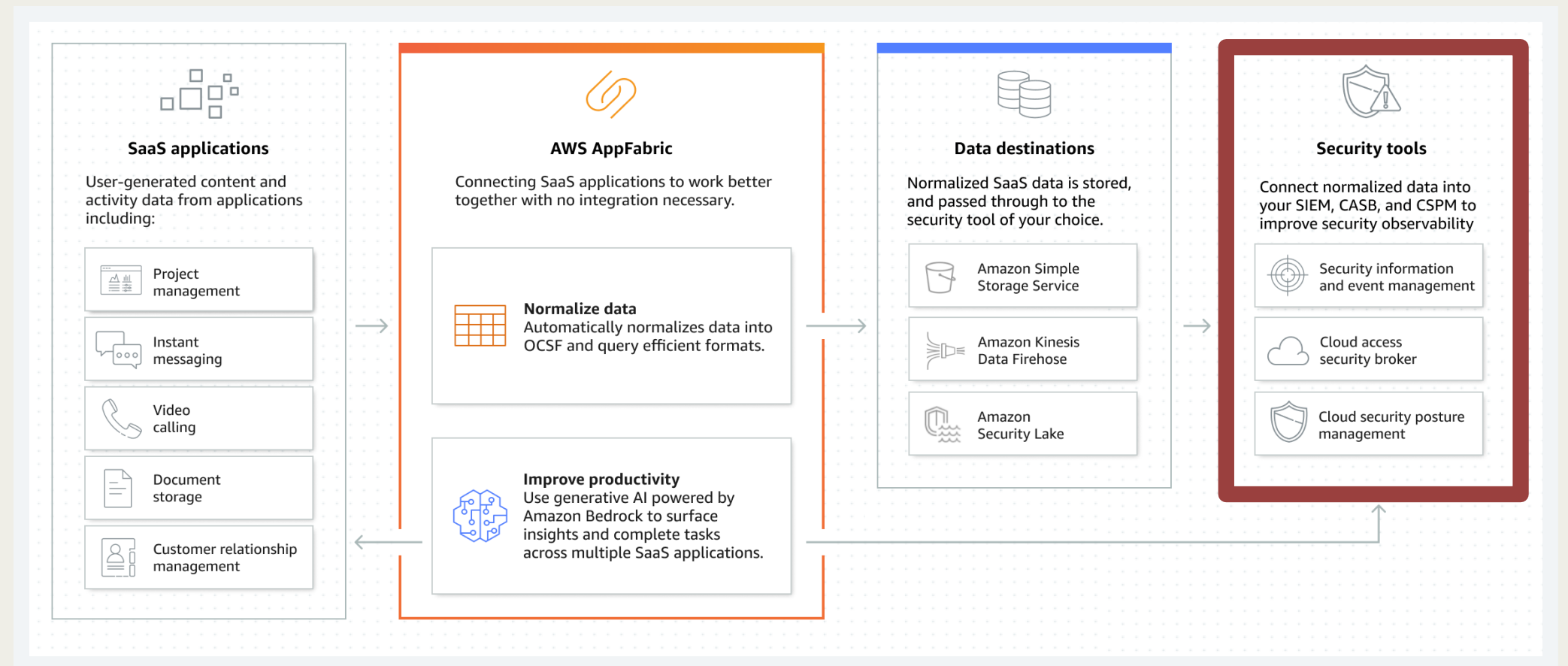
● It breaks down data silos that exist today across various applications.



INTEROPERABILITY

As a security professional, you and your team can use it to

- 1 Analyse user access and event log data
- 2 Output it into a security application of choice



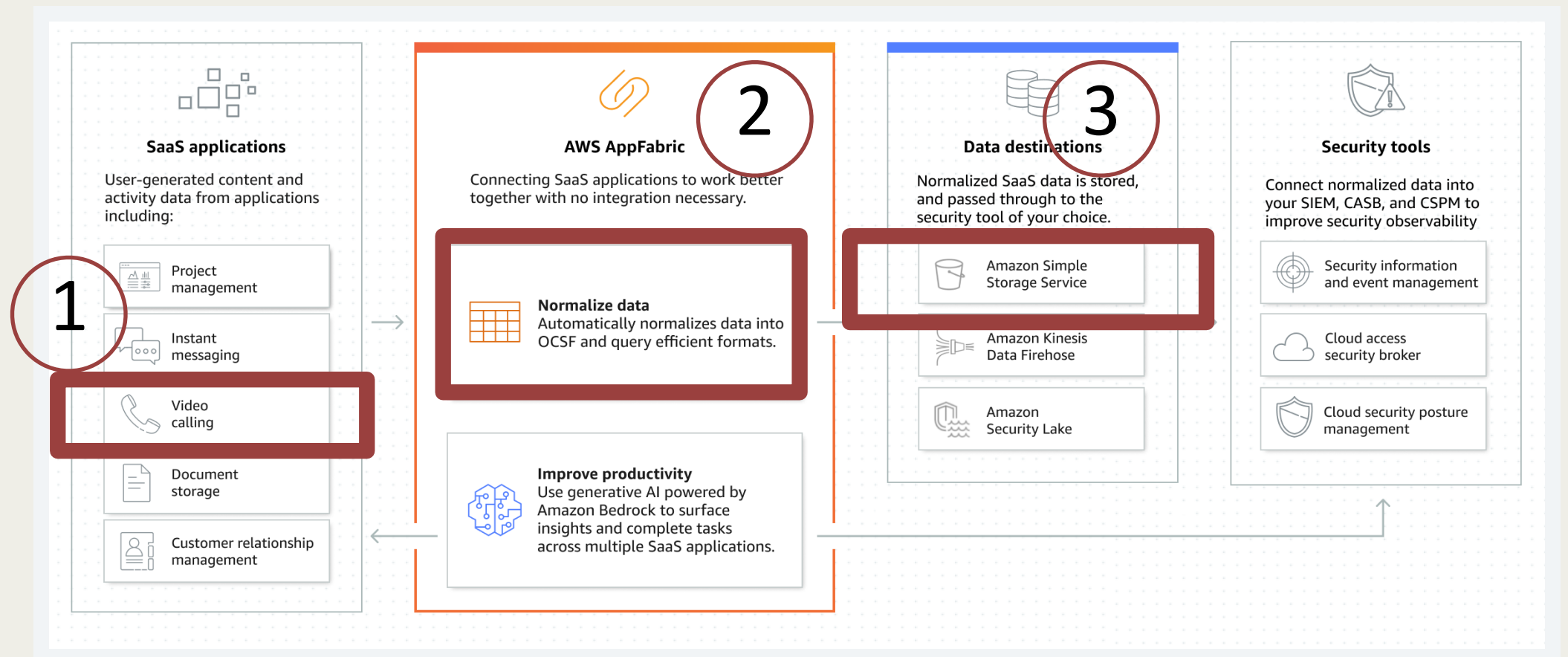
INTEROPERABILITY

AWS AppFabric interoperates with Zoom, for example, to

1 Receive audit logs and user data

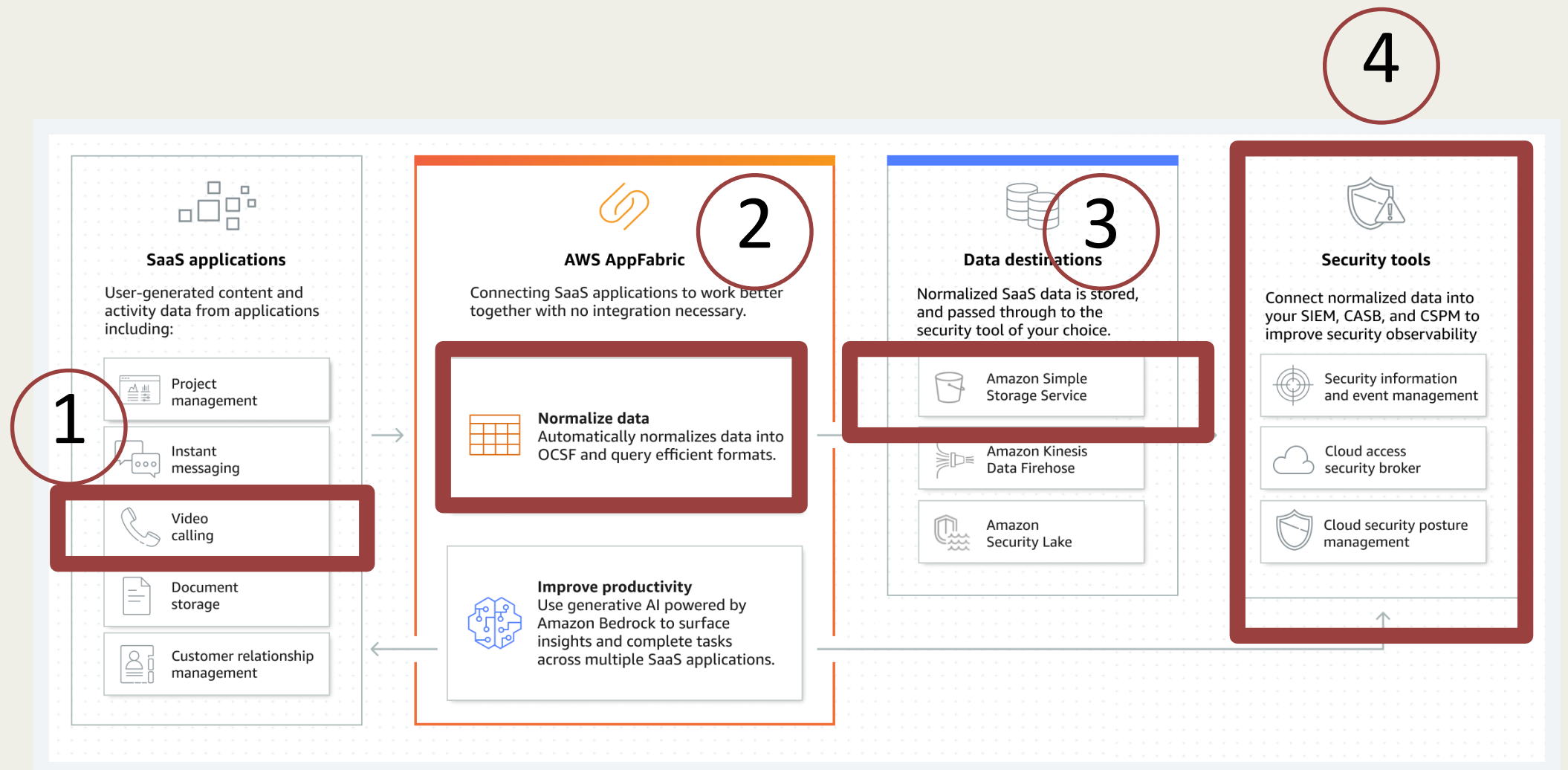
2 Normalise the data into Open Cybersecurity Schema Framework (OCSF) format

3 Output the data to an Amazon Simple Storage Solution (S3) bucket



INTEROPERABILITY

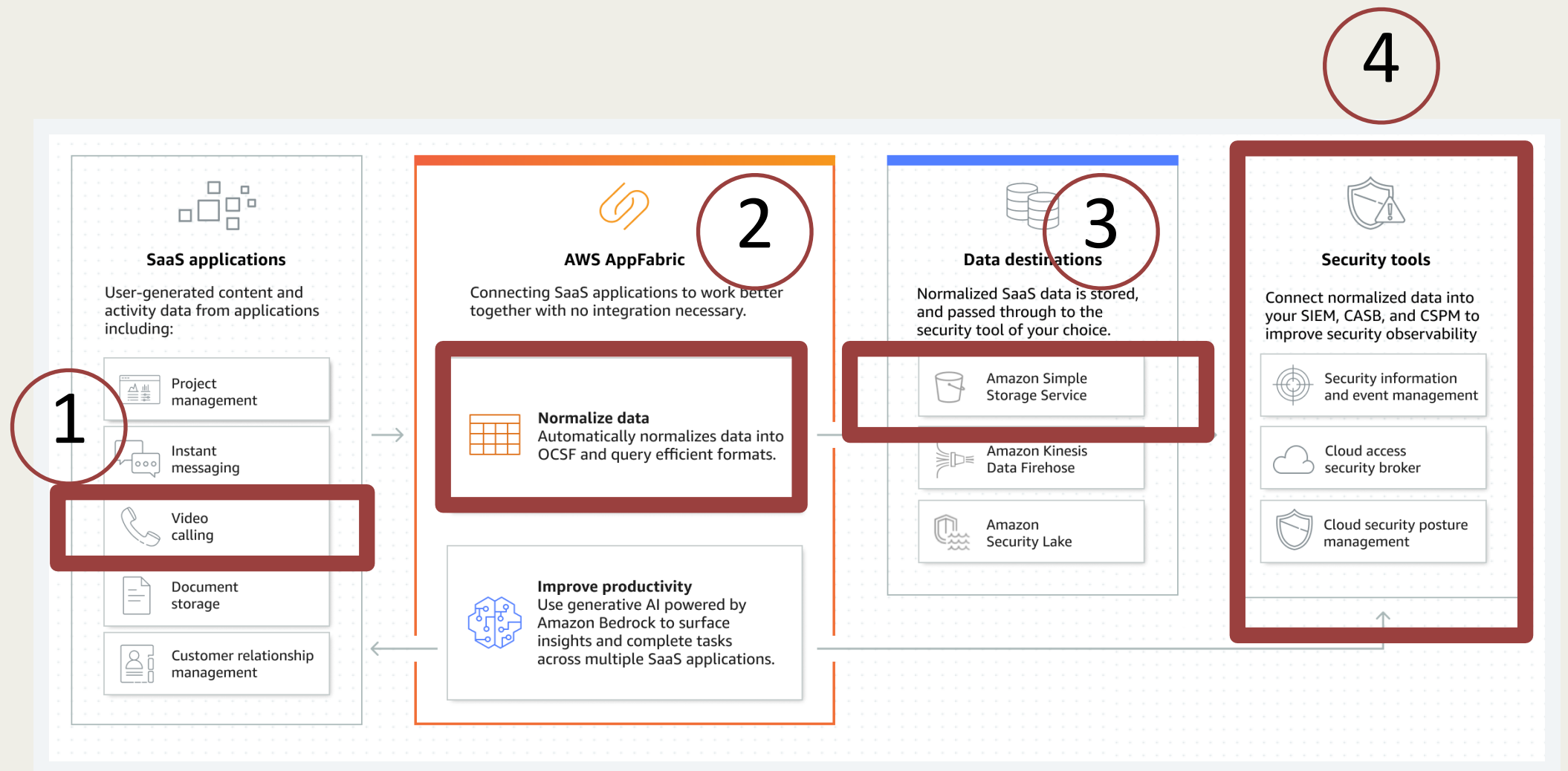
4 The Open Cybersecurity Schema Framework (OCSF) is designed to standardise and streamline the way cybersecurity data is structured and shared across different security tools and platforms, enhancing interoperability and efficiency in threat detection, analysis, and response.



INTEROPERABILITY

4

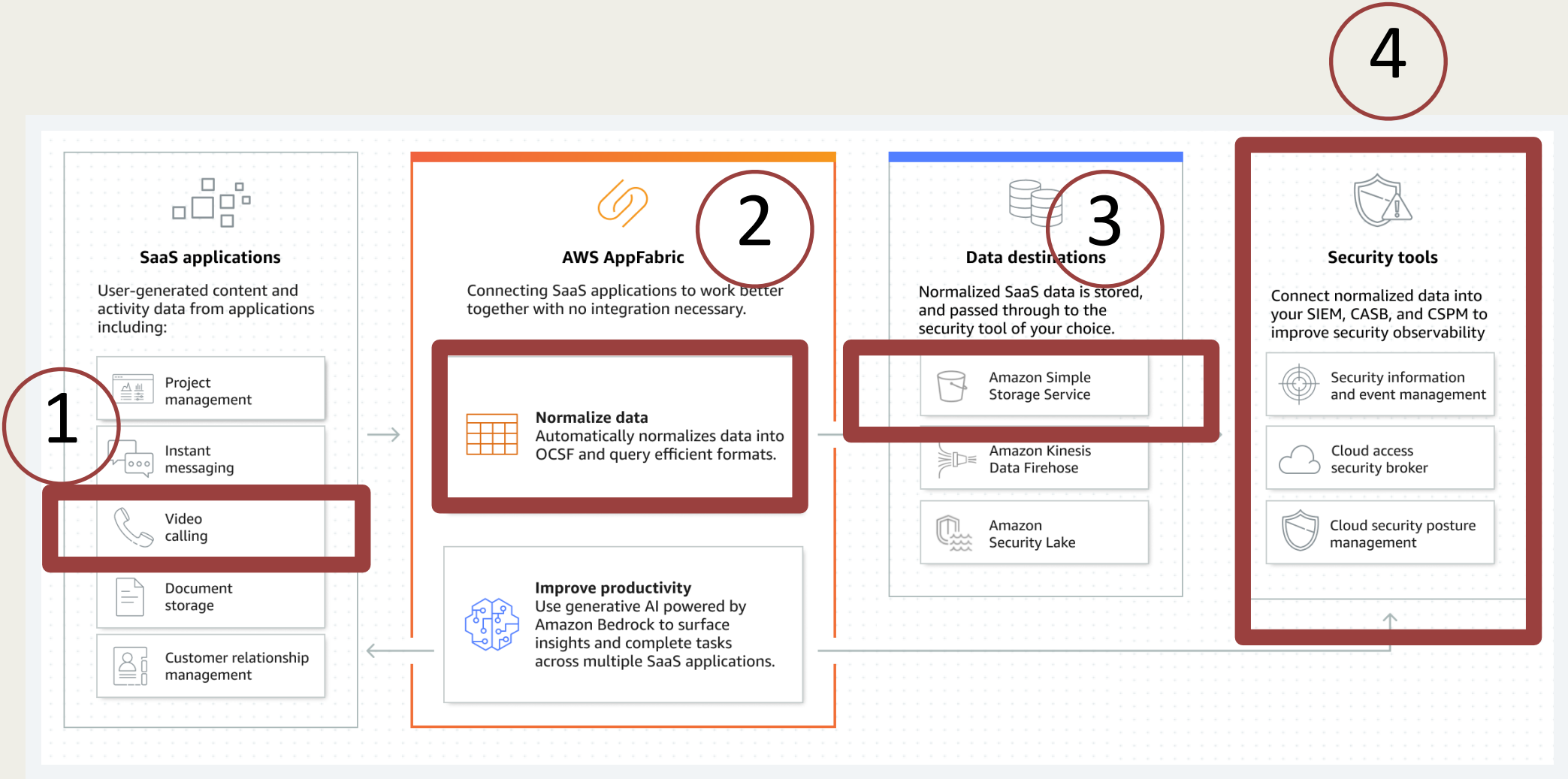
OCSF is tackling a fundamental challenge in the security analytics space: *the absence of a commonly, agreed-upon format and data model for logs and alerts across vendors.*



INTEROPERABILITY

4

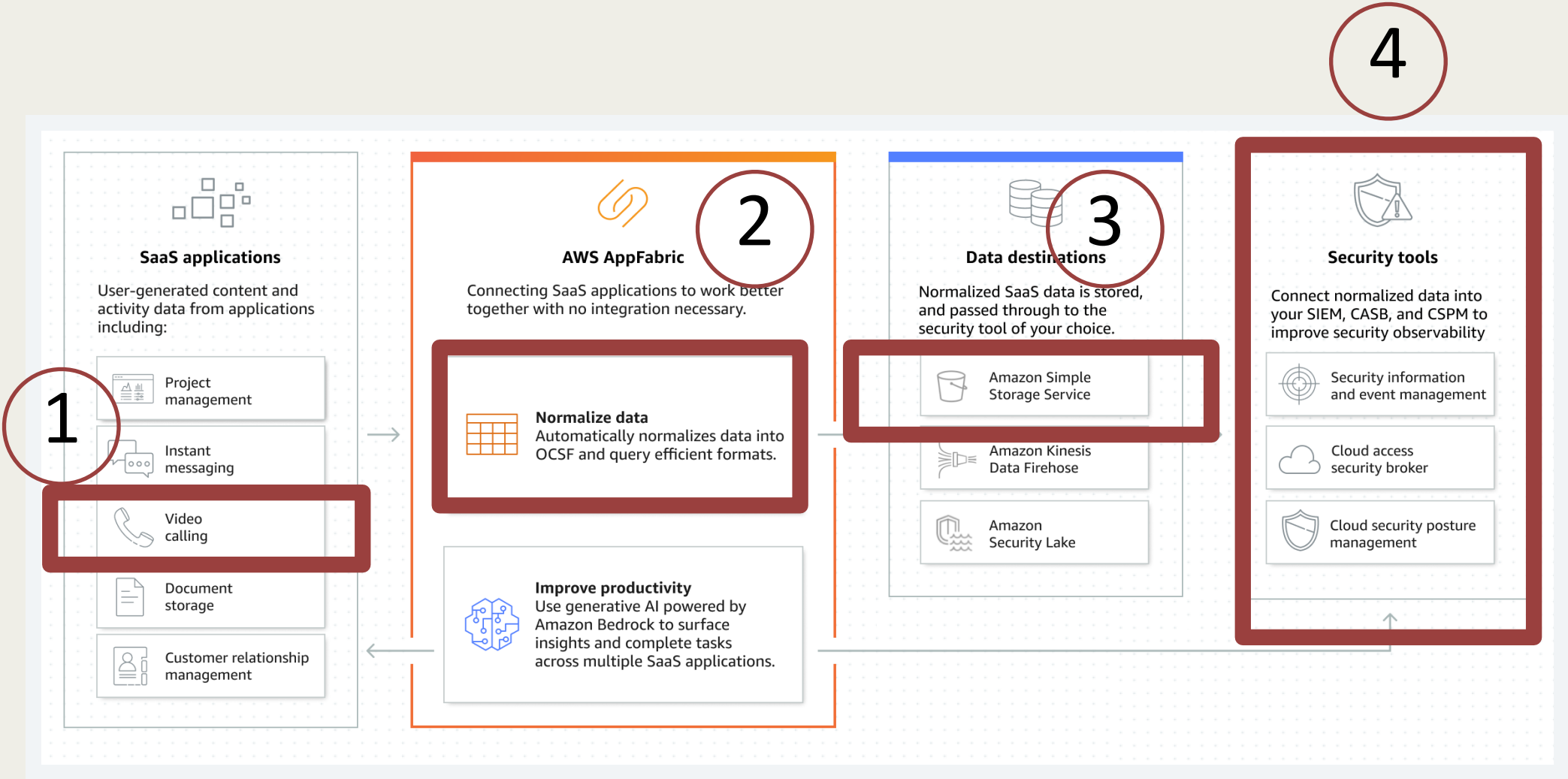
This lack of standard language in cybersecurity tooling and services, creates a challenge for cybersecurity professionals and a lot of manual work.



INTEROPERABILITY

4

The OCSF approach emphasises both **proactive defence** and **reactive response** to cyber threats.



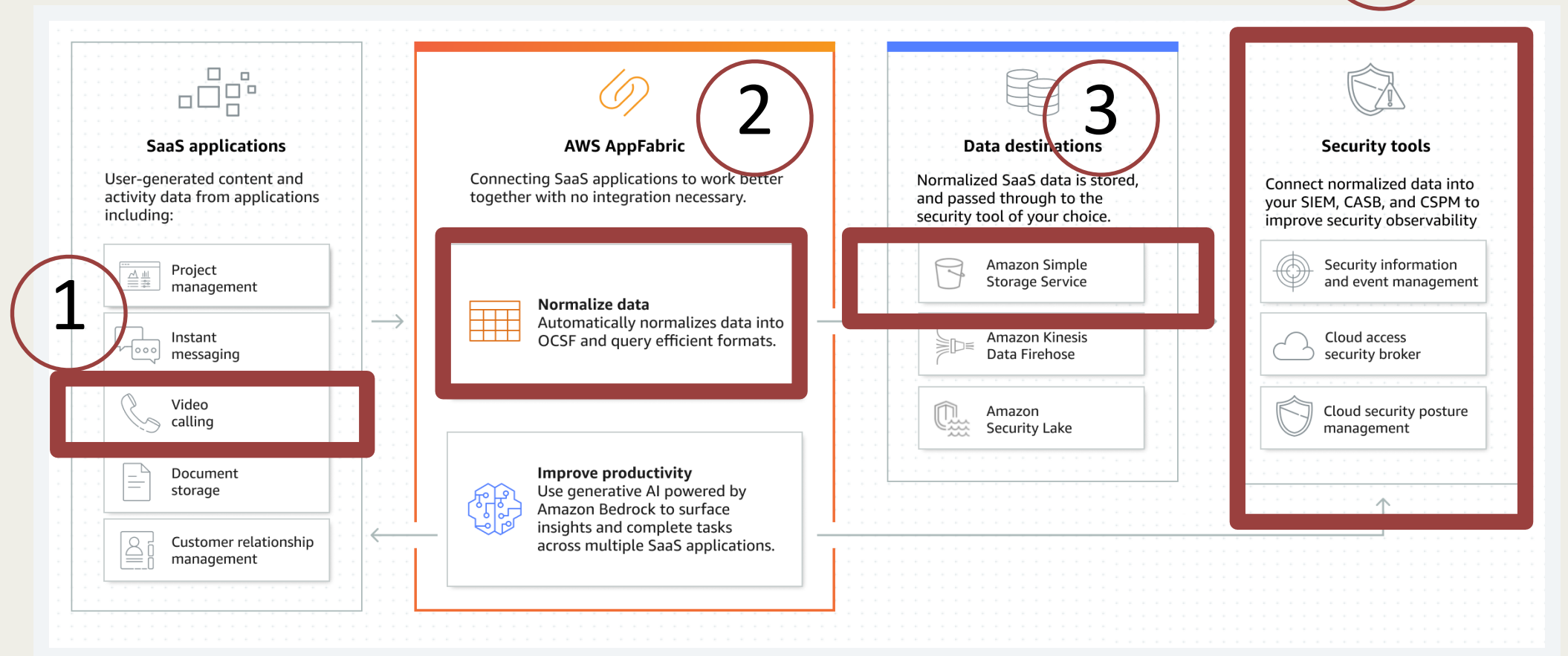
INTEROPERABILITY

4

With a goal of *fostering adaptability and resilience in modern enterprises*, OCSF works by

Integrating open systems for flexibility with closed systems for confidentiality and integrity.

Combining these elements enables enterprises to be *dynamic; adjusting quickly to evolving threats* while *balancing protection with the needs of the business*.



PORTABILITY

PORTABILITY

Data portability

Is focused on the ability to move data between traditional and cloud services or between different cloud services without losing quality.

Architecture portability

Allows for migration of applications or systems without significant changes or loss of service

PORTABILITY

Data portability

Is focused on the ability to move data between traditional and cloud services or between different cloud services without losing quality.

Data portability matters to an organisation that uses a multi-cloud approach, as data will move between CSPs.



Moving data between CSPs should not lead to a loss of system availability or loss of data.

PORTABILITY

Data portability

Is focused on the ability to move data between traditional and cloud services or between different cloud services without losing quality.

Data portability is also important in a **cloud bursting** scenario.

- Peak usage expands from on-premises hosting to include services in a cloud environment in order to meet demand.

- The system must remain fully available both during the cloud burst and when demand returns to normal.

PORTABILITY

Architecture portability is concerned with the ability to access and run a cloud service from a wider variety of devices, running different OSs.

This allows users on a Windows laptop, iPadOS tablet, or Android smartphone, for

● example, to use the same application services, share the same data, and collaborate easily.

Architecture portability

Allows for migration of applications or systems without significant changes or loss of service

REVERSIBILITY

REVERSIBILITY

Reversibility is concerned with a very specific question:

Once an organisation moves an application or workload into a CSP, can it be moved back out again without causing significant impact?

This move-out might entail

- Switching to another CSP
- Deploying a multi-cloud strategy
- Migrating back to traditional on-premises infrastructure

REVERSIBILITY

Reversibility is concerned with a very specific question:

Once an organisation moves an application or workload into a CSP, can it be moved back out again without causing significant impact?

An **element of reversibility** is the **ability to avoid vendor lock-in**. This might take the form of

- Tools that can be used for both importing and exporting data easily.
- Cloud architectures that allow for portability, such as standardised PaaS, which can be replicated to another CSP.

REVERSIBILITY

Reversibility is concerned with a very specific question:

Once an organisation moves an application or workload into a CSP, can it be moved back out again without causing significant impact?

Another **element of reversibility** is the potential **impact on systems operations**.

- If a migration takes days or even weeks, during which the system cannot be used, this is a barrier to reversibility and creates a potential vendor lock-in to a specific CSP.

SERVICE LEVELS AND SLAs

SERVICE LEVELS AND SLAs

Service level agreements (SLAs) specify the required performance parameters of a service.

Depending on the type of CSP, agreeing to a SLA could involve negotiations which can be expensive depending on how strict your requirements are.

● For example, if you need 24-hour support, this will be less expensive than 4-hour support.

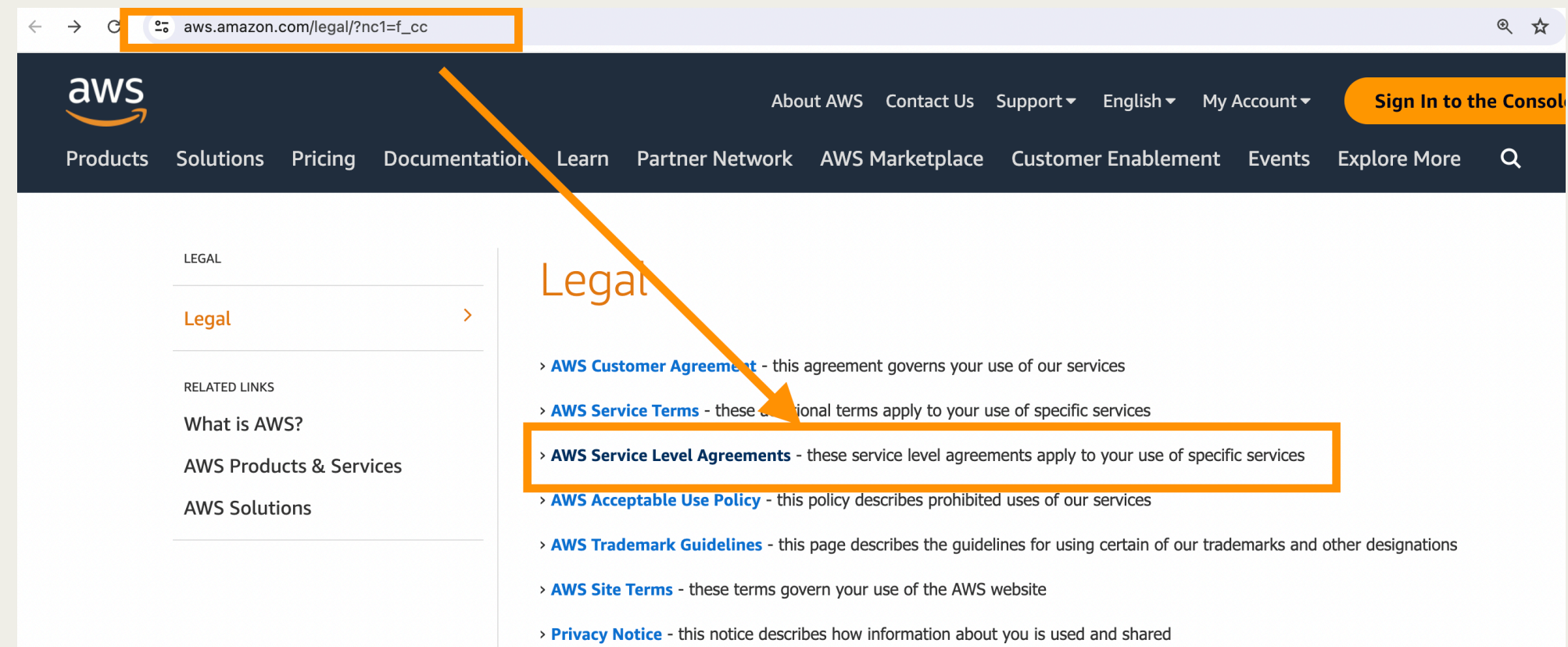
SERVICE LEVELS AND SLAs

Service level agreements (SLAs) specify the required performance parameters of a service.

Some CSPs like AWS will provide a predefined set of SLAs, and customers choose the level of service they need.



Once agreed upon, the SLA becomes part of the contract.



SERVICE LEVELS AND SLAs

Service level agreements (SLAs) specify the required performance parameters of a service.

SLAs commonly include financial penalties if the vendor fails to live up to operational obligations.

Amazon API Gateway Service Level Agreement

Last Updated: May 5, 2022

This Amazon API Gateway Service Level Agreement ("SLA") is a policy governing the use of Amazon API Gateway ("API Gateway") and applies separately to each account using API Gateway. In the event of a conflict between the terms of this SLA and the terms of the [AWS Customer Agreement](#) or other agreement with us governing your use of our Services (the "Agreement"), the terms and conditions of this SLA apply, but only to the extent of such conflict. Capitalized terms used herein but not defined herein shall have the meanings set forth in the Agreement.

Service Commitment

AWS will use commercially reasonable efforts to make API Gateway available with a Monthly Uptime Percentage of at least 99.95% for each AWS region, during any monthly billing cycle (the "Service Commitment"). In the event that a API Gateway does not meet the Service Commitment, you will be eligible to receive a Service Credit as described below.

SERVICE LEVELS AND SLAs

Service level agreements (SLAs) specify the required performance parameters of a service.

AWS, for example, provides **Service Credits**, which are calculated as a percentage of the total charges paid by you for the applicable service.

Service Credits

Service Credits are calculated as a percentage of the total charges paid by you for API Gateway in the affected AWS region for the monthly billing cycle in which the Monthly Uptime Percentage fell within the ranges set forth in the table below:

Monthly Uptime Percentage	Service Credit Percentage
Less than 99.95% but greater than or equal to 99.0%	10%
Less than 99.0% but greater than or equal to 95.0%	25%
Less than 95.0%	100%

AVAILABILITY

AVAILABILITY

Availability is the percentage of time that a workload/resource is available for use.

Within the constraints of the agreed-upon SLAs, the purchased cloud services must be made available by the CSP.

Availability	Maximum Unavailability (per year)	Application Categories
99%	3 days 15 hours	Batch processing, data extraction, transfer, and load jobs
99.9%	8 hours 45 minutes	Internal tools like knowledge management, project tracking
99.95%	4 hours 22 minutes	Online commerce, point of sale
99.99%	52 minutes	Video delivery, broadcast workloads
99.999%	5 minutes	ATM transactions, telecommunications workloads

AVAILABILITY

An important aspect of availability is concerned with the **elasticity** and **scalability** of the cloud service.

● If the CSP has not properly planned for capacity expansion, customers with growing needs will not find the service adequate to their requirements.

Availability	Maximum Unavailability (per year)	Application Categories
99%	3 days 15 hours	Batch processing, data extraction, transfer, and load jobs
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99.999%	5 minutes	ATM transactions, telecommunications workloads

SECURITY - SHARED RESPONSIBILITY MODEL

SECURITY - SHARED RESPONSIBILITY MODEL

SaaS	PaaS	IaaS	On-premises	Responsibility
				Information and data
				Devices (mobile & PCs)
				Accounts & identities
				Identity & directory infrastructure
				Applications
				Network controls
				Operating system
				Physical hosts
				Physical network
				Physical datacenter
Cloud Service Provider			Cloud Service Customer	
			Shared Responsibility	

Larger CSPs spend resources and focus on creating a secure environment.

SECURITY - SHARED RESPONSIBILITY MODEL

SaaS	PaaS	IaaS	On-premises	Responsibility
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				Accounts & identities
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				Physical datacenter
Cloud Service Provider			Cloud Service Customer	
			Shared Responsibility	

The shared nature of the cloud means resources are available to organisations that would otherwise not have access to them.

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				Physical datacenter
Cloud Service Provider			Cloud Service Customer	
			Shared Responsibility	

Large CSPs are very attractive targets for attackers, and there are aspects of cloud computing like multi-tenancy that create new security challenges.

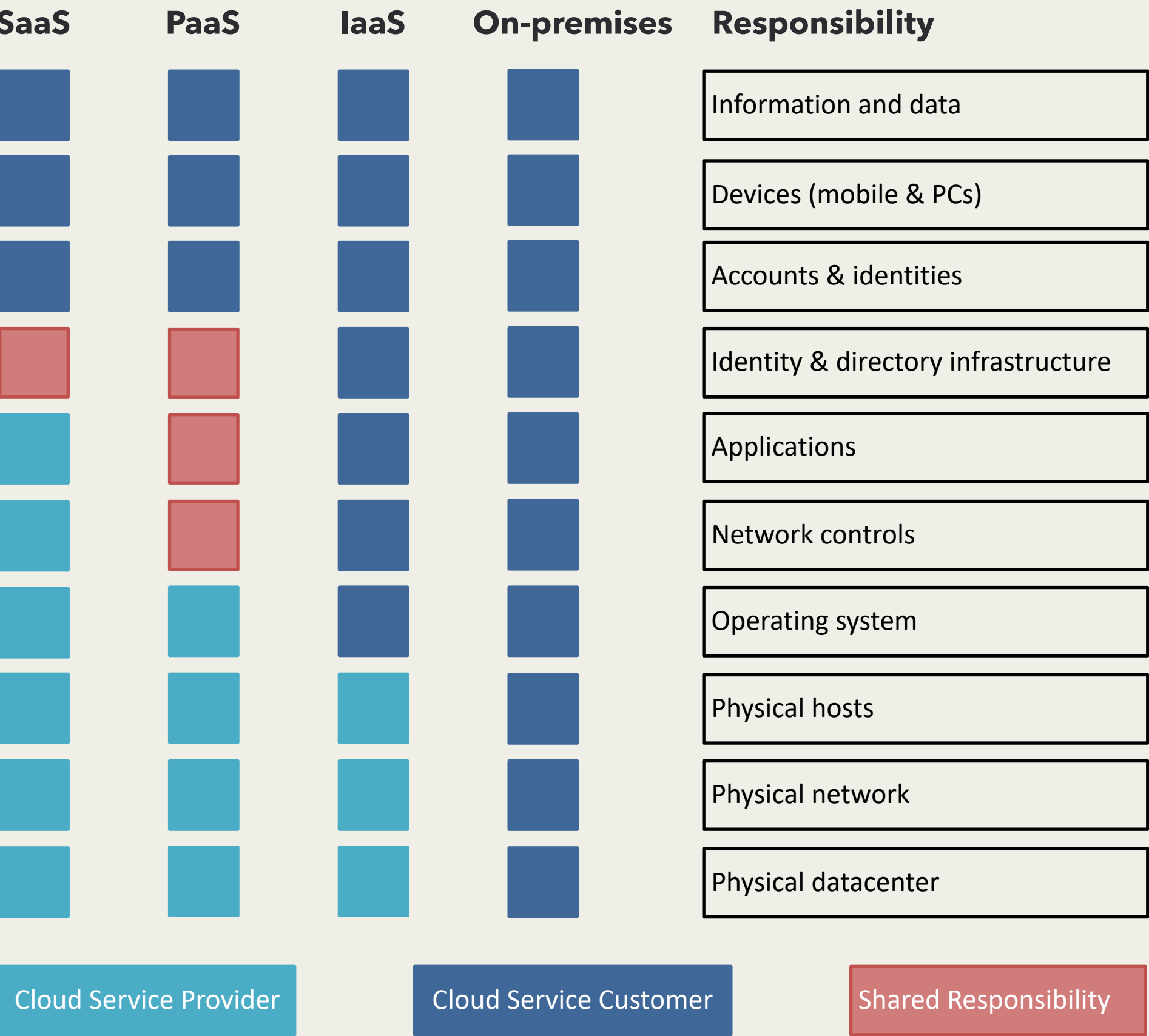
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				Physical hosts
				Physical network
				Physical datacenter
Cloud Service Provider				Cloud Service Customer
				Shared Responsibility

The fundamental architecture of cloud services, which make data and service portability extremely simple, also introduce a new security complexity:

- Laws and regulations that restrict cross-border flow (e.g., GDPR) can make choosing a CSP and architecting infrastructure more challenging.

SECURITY - SHARED RESPONSIBILITY MODEL



The fundamental architecture of cloud services, which make data and service portability extremely simple, also introduce a new security complexity:

- Location of the data may also be a jurisdictional challenge as law enforcement may lack the means to gain access to data outside the jurisdiction.

SECURITY - SHARED RESPONSIBILITY MODEL

SaaS	PaaS	IaaS	On-premises	Responsibility
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Cloud Service Provider			Cloud Service Customer	
			Shared Responsibility	

The owner of data retains ultimate responsibility for the security of the data regardless of what cloud or non-cloud services are used.

SECURITY - SHARED RESPONSIBILITY MODEL

SaaS	PaaS	IaaS	On-premises	Responsibility
				Information and data
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Cloud security involves the protection of

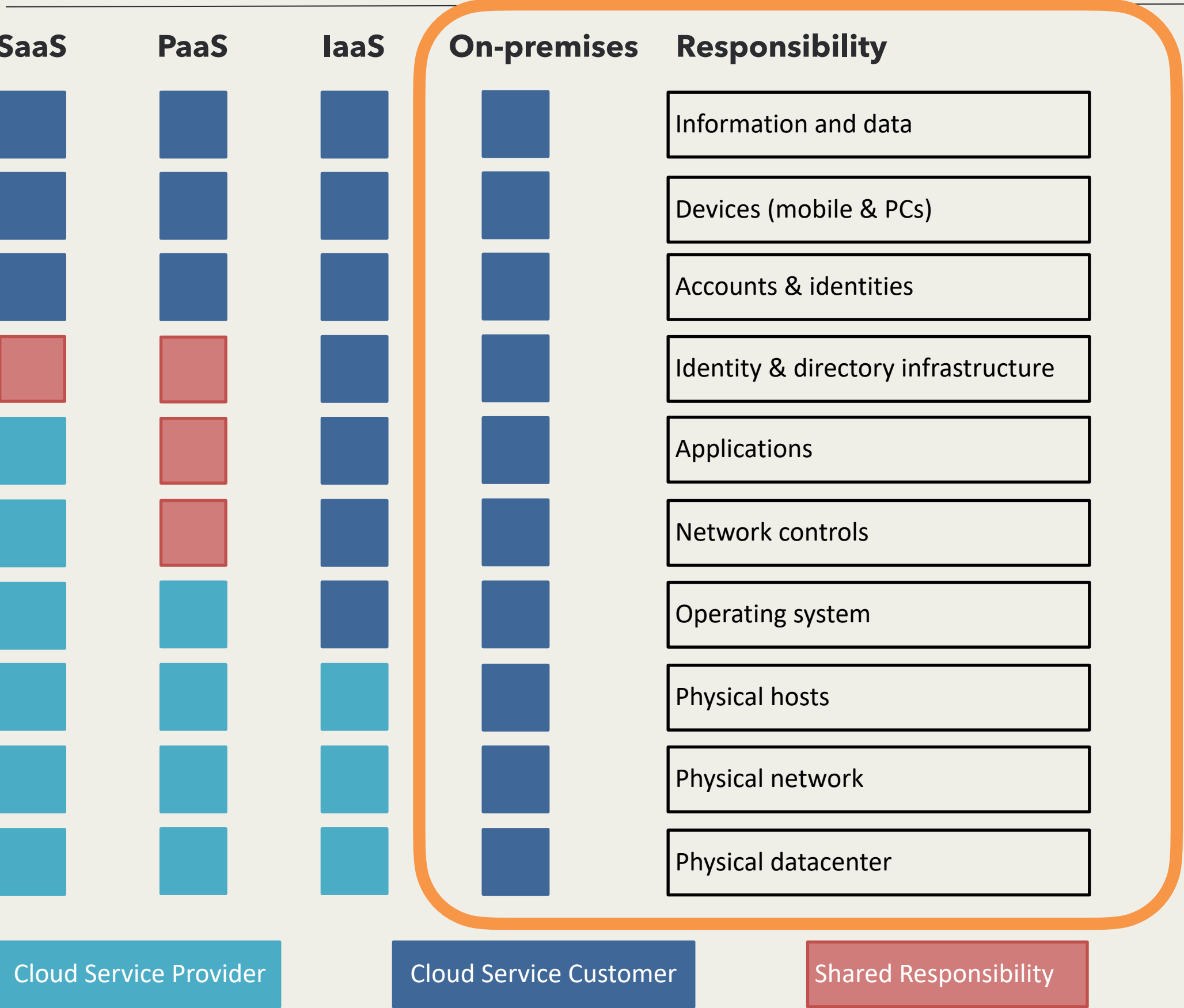
- Data
- Applications
- Infrastructure

SECURITY - SHARED RESPONSIBILITY MODEL

SaaS	PaaS	IaaS	On-premises	Responsibility
■	■	■	■	Information and data
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■	■	■	■	Identity & directory infrastructure
■	■	■	■	Applications
■	■	■	■	Network controls
■	■	■	■	Operating system
■	■	■	■	Physical hosts
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Cloud Service Provider				Cloud Service Customer
				Shared Responsibility

The **Cloud Shared Responsibility Model** is a security framework that defines the division of responsibility between a CSP and its customers for securing the cloud environment.

SECURITY - SHARED RESPONSIBILITY MODEL



In an on-premises datacenter, the customer owns the entire stack and is responsible for its security.

SECURITY - SHARED RESPONSIBILITY MODEL

SaaS	PaaS	IaaS	On-premises	Responsibility
■	■	■	■	Information and data
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Cloud Service Provider				Cloud Service Customer
				Shared Responsibility

For all cloud deployment models, the customer owns the data and identities.

PRIVACY

PRIVACY

Data subject

The data subject is the individual whose personal data is being processed.

Has rights to access, rectify, erase, restrict processing, and object to processing of their data. Must be informed about how their data is being used.

Data controller/owner

The organisation that determines the purposes and means of processing personal data. It can be the same as the data processor.

Responsible for ensuring compliance with privacy principles. Must have a lawful basis for processing data. Must respond to data subject requests.

Data processor

The organisation that processes personal data on behalf of the controller.

Process data to controller's instructions, implement security measures, liable for breaches if acting outside instructions.

PRIVACY

Data privacy concerns include

- Access to data both during a contract and at the end of the contract
- Erasure or destruction of data when requested or as required within the contract
- Monitoring and logging of data access and modification, and storage

RESILIENCY

RESILIENCY

Organisations typically
breakdown their
applications into 4 main
categories:

- Mission-critical (Tier 1)
- Business-critical (Tier 2)
- Business operational (Tier 3)
- Administrative (Tier 4)

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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RESILIENCY

Mission-critical applications

These applications are essential for the core function of an organisation.

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RESILIENCY

Mission-critical applications

Their failure will bring operations to a complete halt and potentially result in significant financial loss or safety hazards.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
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RESILIENCY

Mission-critical applications

Imagine a restaurant. **The mission-critical application is the point-of-sale system** used to take orders and process payments. Without it, they cannot function.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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RESILIENCY

Business-critical
applications

These applications are
crucial for the long-term
success and smooth daily
operations of a business.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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RESILIENCY

Business-critical applications

A **downtime** might not cause immediate disaster, but it **would severely affect productivity and profitability.**

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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RESILIENCY

Business-critical
applications

In our fictional restaurant, the business-critical application is the inventory management system. A downtime might mean they can't take orders, but they can still operate to some extent.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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RESILIENCY

Business operational
applications

These applications support specific workflows and tasks within a business. While important, their downtime may not be catastrophic.

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Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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RESILIENCY

Business operational
applications

They help improve
efficiency, communication,
and collaboration.

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RESILIENCY

Business operational
applications

Using our restaurant analogy, a business operational application could be a reservation system. Downtime means guests cannot book online, but phone calls might still be accepted.

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RESILIENCY

Administrative applications

These applications support basic office tasks and employee needs. Their **downtime may cause minor inconveniences** but **would not significantly impact core operations.**

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RESILIENCY

Administrative applications

In the restaurant, an administrative application could be an employee scheduling tool. Downtime might mean some confusion about shifts, but the restaurant can still operate.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
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RESILIENCY

Cloud architecture

Cloud architecture is the overall design and structure of components and technologies that make up a cloud computing environment.

- Application architecture
- Infrastructure architecture

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RESILIENCY

Application architecture focuses on the internal structure and organisation of your application itself.

- Code organisation
- Data flows
- Communication
- Technology stack

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RESILIENCY

Infrastructure architecture focuses on the underlying cloud resources that power your application.

- Virtual machines (VMs)
- Storage systems
- Networking
- Security configurations

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RESILIENCY

Cloud architecture is the overall design and structure of components and technologies that make up a cloud computing environment.

- Acts as a blueprint for how the application and infrastructure work together.
- It ensures that both the application and infrastructure are designed for **resiliency**.

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RESILIENCY

Resiliency refers to the ability of a cloud system **to withstand and recover** from failures, disruptions, or outages, **while maintaining functionality and acceptable levels** of service - the cornerstone of building reliable and trustworthy cloud applications.

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RESILIENCY

Key concepts that are foundational to cloud resiliency include:

- High availability (HA)
- Fault tolerance
- Disaster recovery (DR)

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RESILIENCY

High Availability (HA)

Focuses on ensuring the continuous operation of a system during failures.

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Monitoring and tooling	Health endpoint monitoring	Service-level indicator monitoring		Functional monitoring	Uptime monitoring	

RESILIENCY

High Availability (HA)

Continuous operation is achieved through **redundancy** by having backups or failover mechanisms in place to automatically take over if a component fails.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
	Multi-region/ multi-AZ	Active/active or warm standby	Warm standby	Pilot light	Backup and restore	Circuit breaker
	Elasticity	Serverless or autoscaling servers				Schedular agent supervisor
	Load-levelling	Competing consumer or queue based		Queue based	Request throttling	Retry
Data management	Durable data storage	Synchronous or asynchronous	Asynchronous			Compensating transaction
Deployment strategy	Deployment pattern	Blue-green or canary		Canary	Rolling	No application resiliency patterns
Monitoring and tooling	Health endpoint monitoring	Service-level indicator monitoring		Functional monitoring	Uptime monitoring	

High Availability (HA)

HA minimises downtime and keeps your applications accessible even during disruptions.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
	Multi-region/ multi-AZ	Active/active or warm standby	Warm standby	Pilot light	Backup and restore	Circuit breaker
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Fault tolerance

Refers to a system’s in-built or inherent ability to **absorb and recover from failures** without complete collapse.

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	Multi-region/ multi-AZ	Active/active or warm standby	Warm standby	Pilot light	Backup and restore	Circuit breaker
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Deployment strategy	Deployment pattern	Blue-green or canary		Canary	Rolling	No application resiliency patterns
Monitoring and tooling	Health endpoint monitoring	Service-level indicator monitoring		Functional monitoring	Uptime monitoring	

Fault tolerance

It is achieved through

- System mirroring - having duplicate systems running
- Implementing logic to handle errors gracefully
- Self-healing systems or systems that restart on encountering issues

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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Deployment strategy	Deployment pattern	Blue-green or canary		Canary	Rolling	No application resiliency patterns
Monitoring and tooling	Health endpoint monitoring	Service-level indicator monitoring		Functional monitoring	Uptime monitoring	

Fault tolerance

It ensures your application can continue functioning with internal errors or component malfunctions.

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Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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Monitoring and tooling	Health endpoint monitoring	Service-level indicator monitoring		Functional monitoring	Uptime monitoring	

Disaster recovery (DR)

It deals with the recovery process itself after a major outage or disaster.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
	Multi-region/ multi-AZ	Active/active or warm standby	Warm standby	Pilot light	Backup and restore	Circuit breaker
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Monitoring and tooling	Health endpoint monitoring	Service-level indicator monitoring		Functional monitoring	Uptime monitoring	

RESILIENCY

Disaster recovery (DR)

It defines the steps, resources, and timeframe needed to restore full functionality after a significant disruption.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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Disaster recovery (DR)

Key metrics associated
with DR include:

Mean Time to Recovery (MTTR)

This is the average time it
takes to restore a system
after a failure.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business- critical	Business operational	Administrative	Application resiliency patterns
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Disaster recovery (DR)

Key metrics associated
with DR include:

Recovery Time Objective (RTO)

It is the maximum
downtime acceptable
before recovery begins.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
Architecture design	Multi-cloud architecture	Multi-cloud or single cloud	Single cloud			Bulkhead design
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Disaster recovery (DR)

Key metrics associated
with DR include:

Recovery Point Objective (RPO)

It is the acceptable
amount of data loss that
can occur during a failure.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
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RESILIENCY

Disaster recovery (DR)

A well-designed cloud architecture incorporates all 3 elements (*MTTR*, *RTO*, *RPO*) to create a robust and resilient system.

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RESILIENCY

Resiliency patterns

In cloud environments, resiliency patterns are established approaches to design infrastructure and applications that can effectively handle and recover from failures, disruptions, or attacks.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
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	Multi-region/ multi-AZ	Active/active or warm standby	Warm standby	Pilot light	Backup and restore	Circuit breaker
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RESILIENCY

Resiliency patterns

By implementing these patterns, you build in mechanisms for your system to:

- Automatically detect and isolate failures.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
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RESILIENCY

Resiliency patterns

By implementing these patterns, you build in mechanisms for your system to:

- The system can automatically repair itself (self-healing) or restart affected components with minimal downtime.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
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RESILIENCY

Resiliency patterns

By implementing these patterns, you build in mechanisms for your system to:

- Maintain functionality despite failures as the system continues to operate at reduced capacity or reroute traffic around the affected area.

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RESILIENCY

Resiliency patterns

CSPs offer geographically separated **availability zones (AZs)** within a region. Distributing infrastructure across AZs protects against outages impacting an entire zone.

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RESILIENCY

Resiliency patterns

With **elasticity**, cloud resources can be automatically scaled up or down based on demand, helping to handle unexpected spikes in traffic or failures without overwhelming the system.

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Resiliency patterns

Microservices architecture is an approach within application architecture that breaks down applications into smaller, independent services that improves isolation.

A failure in one service does not impact the entire application.

Dimension	Infrastructure resiliency patterns	Mission-critical	Business-critical	Business operational	Administrative	Application resiliency patterns
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Resiliency patterns

Microservices architecture is an architecture within application architecture that breaks down applications into smaller, independent services that improves isolation.

Within the microservices architecture are resiliency patterns such as, bulkhead design pattern, which help in achieving resiliency within an application.

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Resiliency patterns

Choosing the right resiliency pattern depends on the specific needs and priorities of your application and business.

Factors like cost, complexity, and desired level of availability play a role in the selection process.

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RESILIENCY

Performance

It is a closely related concept to resilience. Performance answers the question: **how well can the cloud service stand up to the demands that we place on it?**

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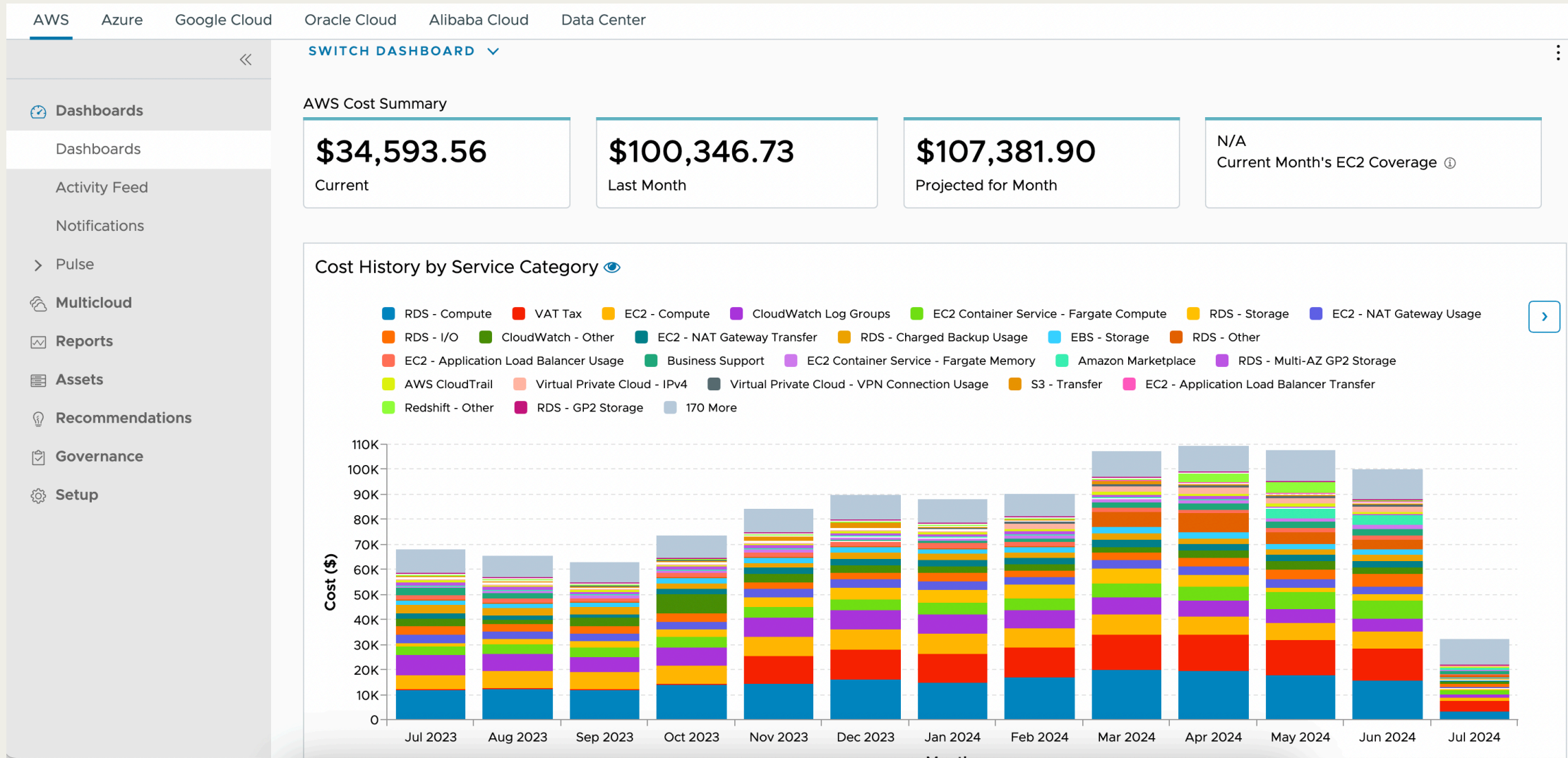
GOVERNANCE

GOVERNANCE



Cloud governance refers to the establishment of policies, processes, and procedures that guide how an organisation uses cloud services.

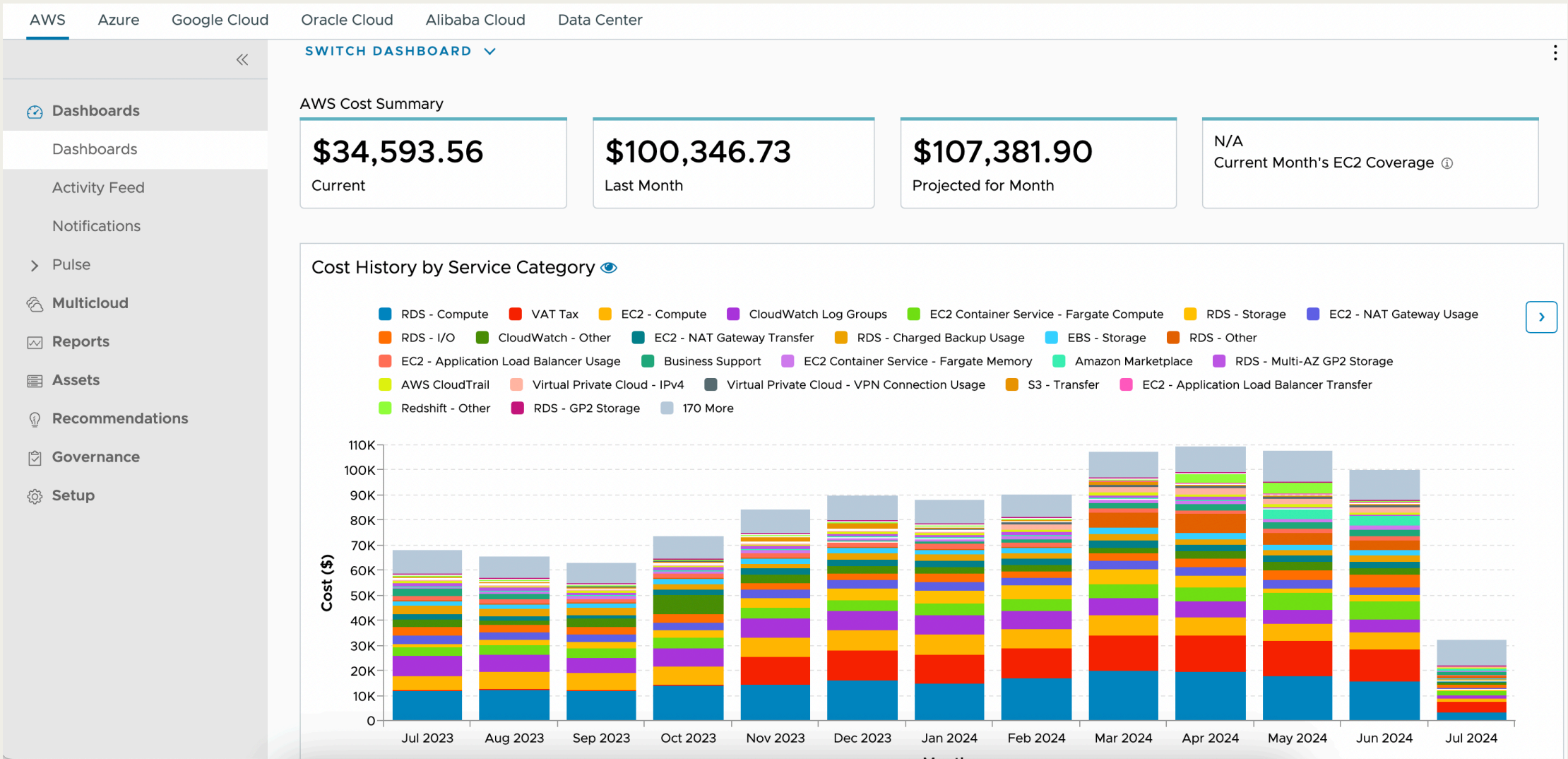
It is essentially a set of rules that ensures efficient and cost-effective cloud adoption.



GOVERNANCE

A company may want to govern aspects of the cloud services used such as

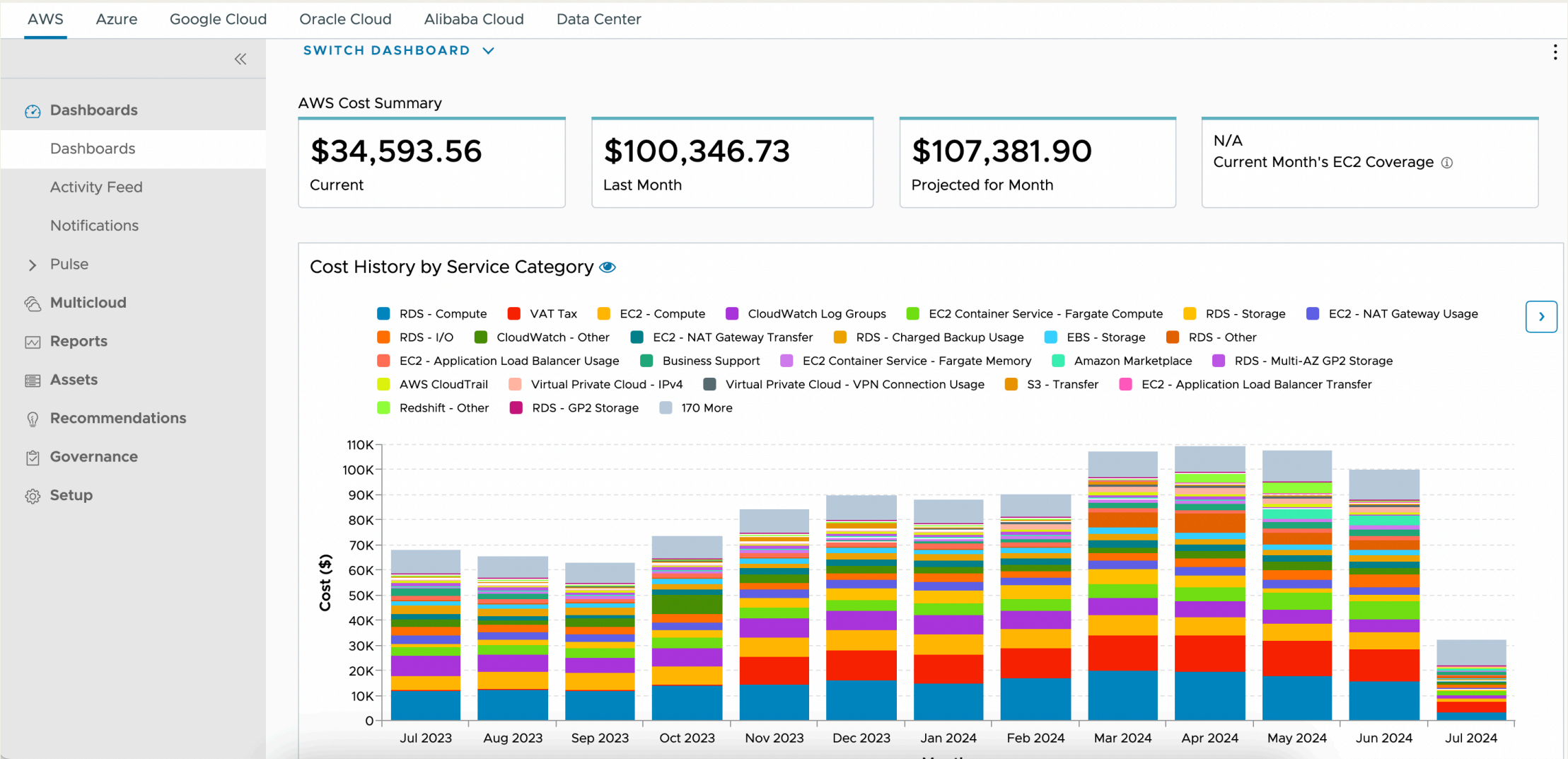
- The number of CSP accounts
- The number of server instances
- The amount of storage utilised
- The size of databases or storage tools



GOVERNANCE

Each of these adds to the cost of cloud computing. Without adequate governance, the cloud bill will continue to grow.

- The number of CSP accounts
- The number of server instances
- The amount of storage utilised
- The size of databases or storage tools



GOVERNANCE

A tool that tracks usage and associated costs will help an organisation use the cloud efficiently and keep its use under budget.

Data View by Cost (\$)

◀ Most Recent Left

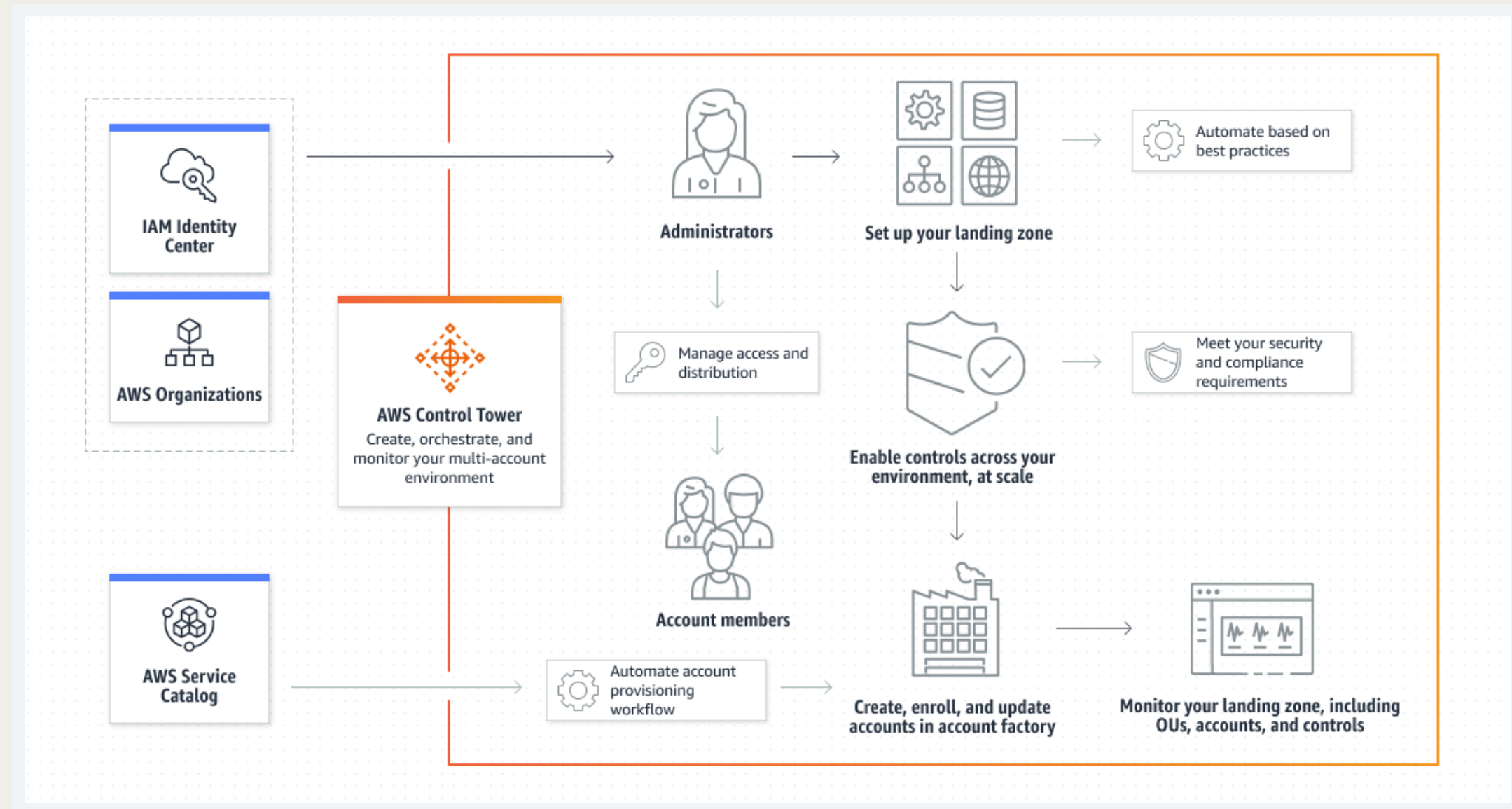
Most Recent Right ▶

rds

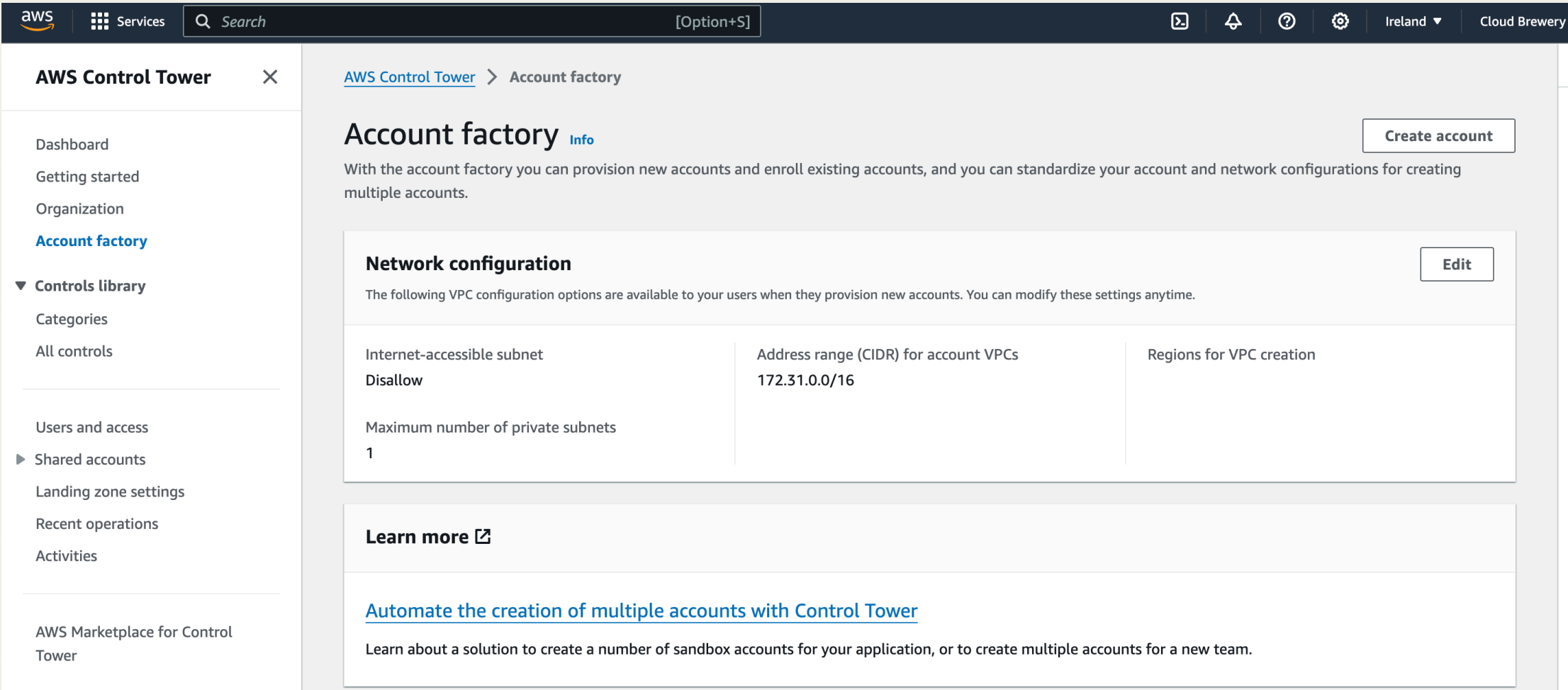
Service Items	2023-07	2023-08	2023-09	2023-10	2023-11	2023-12	2024-01	2024-02	2024-03	2024-04	2024-05	2024-06
RDS - Charged Backup Usage	\$1,256.04	\$1,316.32	\$1,316.63	\$446.31	\$517.32	\$998.80	\$992.56	\$682.74	\$1,503.54	\$699.72	\$340.05	\$370.78
RDS - I/O	\$1,129.10	\$1,054.01	\$957.34	\$1,373.03	\$1,275.93	\$1,436.76	\$1,783.64	\$1,183.73	\$1,365.25	\$1,590.18	\$1,713.51	\$1,839.14
RDS - Storage	\$684.38	\$1,165.47	\$1,642.65	\$2,136.76	\$2,721.81	\$3,366.96	\$4,065.15	\$4,703.16	\$5,326.07	\$3,673.09	\$779.13	\$1,523.20
RDS - Transfer							\$0.00			\$0.00		
RDS - Compute	\$1,596.42	\$1,491.57	\$1,443.46	\$2,647.59	\$1,485.89	\$1,541.41	\$1,548.01	\$4,467.45	\$6,132.94	\$5,935.16	\$6,132.94	\$5,935.10
RDS - Data Transfer							\$0.00			\$0.00		
RDS - Charged Backup Usage	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
RDS - Other							\$1.05			\$1,455.06		

GOVERNANCE

AWS Control Tower enforces cloud governance through a combination of automation and pre-configured **controls**, also known as **guardrails**.



AWS Control Tower uses **Account factory** which acts like a vending machine for AWS Accounts. This automates the creation and enrolment of new accounts within your organisation.



GOVERNANCE

During account creation,
Account factory applies pre-
defined governance polices to
ensure new accounts adhere to
best practices from the get-go.

aws

Services

Search

[Option+S]

AWS Organizations

AWS accounts

Services

Policies

Settings

Get started

Organization ID

o-7y9mv995g8

AWS Organizations > Policies > Service control policies

Disable service control policies

Service control policies (SCPs) enable central administration over the maximum permissions that identities (users and roles) within accounts in your organization can have. This helps ensure that your identities stay within your organization's access control guidelines. [Learn more](#)

Available policies

Actions

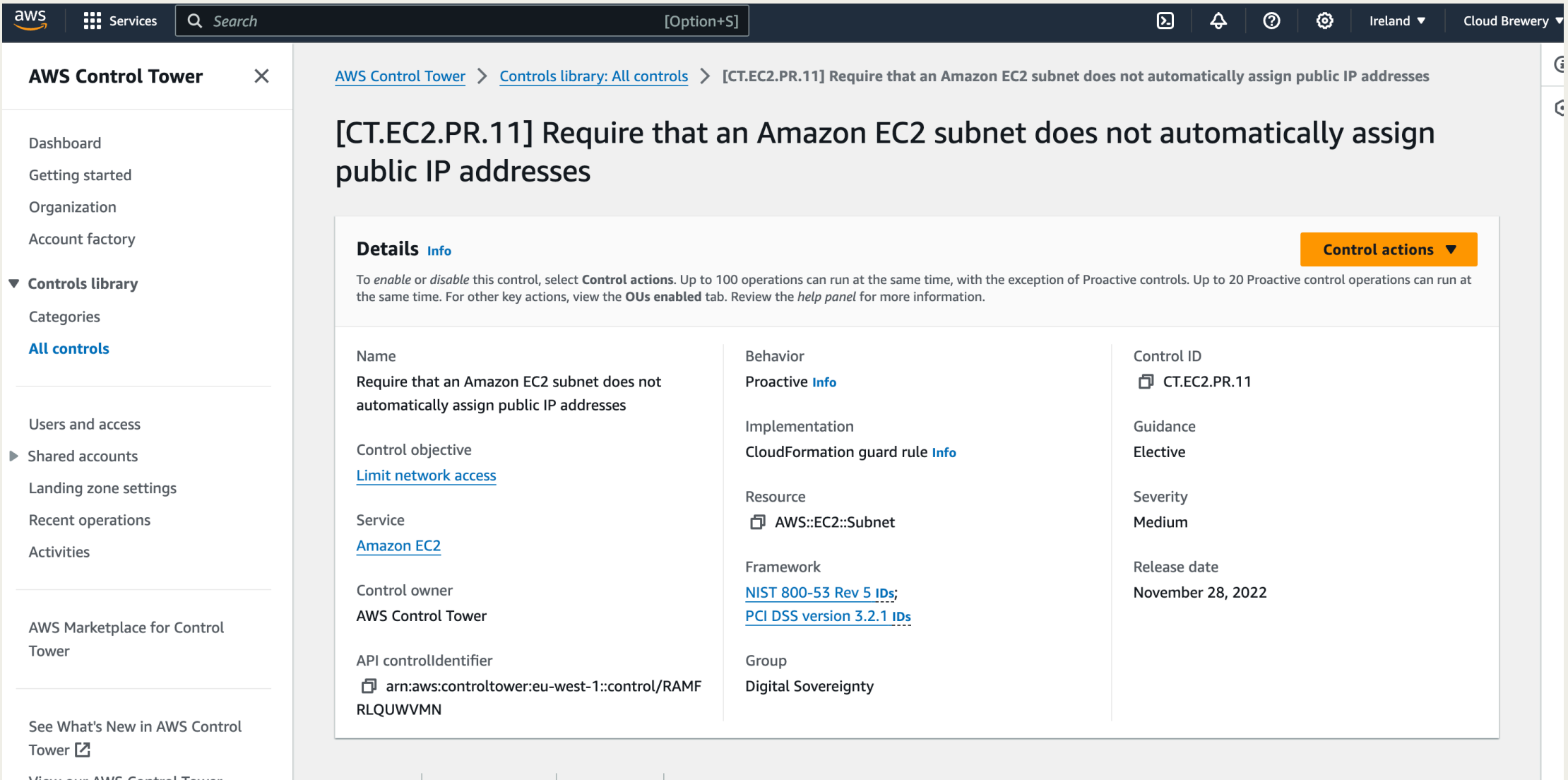
Create policy

	Name	Kind	Description
☐	aws-guardrails-RSRFGl	Customer managed policy	Guardrails applied to an organization
☐	aws-guardrails-tJXLRX	Customer managed policy	Guardrails applied to an organization
☐	aws-guardrails-wyXFIV	Customer managed policy	Guardrails applied to an organization
☐	aws-guardrails-ZRzuxy	Customer managed policy	Guardrails applied to an organization
☐	FullAWSAccess	AWS managed policy	Allows access to every operation

GOVERNANCE

By leveraging these features,
organisations can achieve

- Consistent cloud governance
- Optimise resource utilisation
- Control cloud spending



MAINTENANCE AND VERSIONING

MAINTENANCE AND VERSIONING

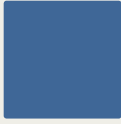


SaaS	PaaS	IaaS	On-premises	Responsibility
				Information and data
				Devices (mobile & PCs)
				Accounts & identities
				Identity & directory infrastructure
				Applications
				Network controls
				Operating system
				Physical hosts
				Physical network
				Physical datacenter
Cloud Service Provider			Cloud Service Customer	
			Shared Responsibility	

Maintenance is the ongoing upkeep and management of the cloud infrastructure. It includes tasks like

- Applying security fixes
- Fixing bugs
- Monitoring performance
- Ensuring everything runs smoothly

MAINTENANCE AND VERSIONING

SaaS	PaaS	IaaS	On-premises	Responsibility
				Information and data
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				Applications
				Network controls
				Operating system
				Physical hosts
				Physical network
				Physical datacenter
Cloud Service Provider			Cloud Service Customer	
			Shared Responsibility	

Versioning is the practice of keeping track of different versions of software, applications, or configurations. This allows you to

- Revert to a previous version
- Understand how things have changed over time.

MAINTENANCE AND VERSIONING

SaaS	PaaS	IaaS	On-premises	Responsibility
				Information and data
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				Applications
				Network controls
				Operating system
				Physical hosts
				Physical network
				Physical datacenter
Cloud Service Provider				
Cloud Service Customer				
				Shared Responsibility

The **division of responsibility** for maintenance and versioning depends on the cloud computing service model.

In **SaaS**, the **CSP** handles all maintenance and versioning. This includes everything from the physical hardware to the software application itself.

MAINTENANCE AND VERSIONING



SaaS	PaaS	IaaS	On-premises	Responsibility
				Information and data
				Devices (mobile & PCs)
				Accounts & identities
				Identity & directory infrastructure
				Applications
				Network controls
				Operating system
				Physical hosts
				Physical network
				Physical datacenter
Cloud Service Provider				Cloud Service Customer
				Shared Responsibility

The **division of responsibility** for maintenance and versioning depends on the cloud computing service model.

In **PaaS**, the **CSP** takes care of the underlying infrastructure, platform tools, and virtualisation software.

The **CSC** is responsible for maintaining and versioning the applications it deploys on the platform.

MAINTENANCE AND VERSIONING



SaaS	PaaS	IaaS	On-premises	Responsibility
				Information and data
				Devices (mobile & PCs)
				Accounts & identities
				Identity & directory infrastructure
				Applications
				Network controls
				Operating system
				Physical hosts
				Physical network
				Physical datacenter
Cloud Service Provider				Cloud Service Customer
				Shared Responsibility

The **division of responsibility** for maintenance and versioning depends on the cloud computing service model.

In **IaaS**, the **CSP** manages the hardware, networking, storage, and virtualisation layer.

The **CSC** is responsible for everything else, including the OS, applications, and any additional software they use.

AUDITABILITY

Auditability

Auditability is the ability for an independent party to examine and verify the controls a CSP has in place, in a process known as a **cloud audit**.

Auditability is crucial because in a cloud environment you are trusting the CSP with your data and resources. An **audit provides assurance** that the CSP is following best practices and adhering to relevant regulations.

Auditability is the ability for an independent party to examine and verify the controls a CSP has in place, in a process known as a **cloud audit**.

Auditability

Audits can identify weaknesses in the CSP's **security posture**, helping prevent data breaches and other security incidents.

Auditability is the ability for an independent party to examine and verify the controls a CSP has in place, in a process known as a **cloud audit**.

Auditability

Many regulations require organisations to implement specific controls for data protection. A cloud audit can help **demonstrate compliance with these regulations**.

Auditability is the ability for an independent party to examine and verify the controls a CSP has in place, in a process known as a **cloud audit**.

Auditability

Audits can help organisations **understand the risks** associated with using a particular CSP and make informed decisions about cloud adoption.

There are different types of cloud audits, each focusing on different aspects.

Auditability

Service Organisation Controls (SOC)
reports provide independent
verification of a CSP's controls
related to security, availability, and
integrity of systems.

There are different types of cloud audits, each focusing on different aspects.

Auditability

Payment Card Industry Data Security
Standard (PCI DSS) audits are
required for organisations that store,
transmit, or process payment card
data.

Auditability is the ability for an independent party to examine and verify the controls a CSP has in place, in a process known as a **cloud audit**.

Auditability

By choosing a CSP that offers a high level of auditability and undergoes regular independent audits, you can gain greater confidence in the security and reliability of your cloud environment.

REGULATORY

A **regulatory environment** is one where a **principle** or **rule** controls or manages the activities of an organisation.

Governance of the regulatory environment involves implementing **policies, procedures, and controls** that assist an organisation in meeting regulatory requirements.

There are several forms of regulations:

- Governmental requirements
- Contractual requirements
- Standards requirements

One form of regulations are
governmental requirements:

Laws implemented through
regulations and have the force
of law.

Governance will put in place a
framework to ensure
compliance with these
regulations.

- Health Insurance Portability and Accountability Act (HIPAA)
- Gramm-Leach-Bliley Act (GLBA)
- Sarbanes-Oxley Act (SOX)
- General Data Protection Regulation (GDPR)

Another form of regulations are those put in place through contractual requirements:

- SLAs regulate business activities.
- Credit and debit card processes are required to implement PCI DSS in order to continue processing payments.

- Enforcement can be through civil courts governing contracts.
- Governance must put in place the framework to ensure compliance.

A third form of regulations (**standard requirements**) is found through standard and non-standard bodies like...

- International Organisation for Standardisation (ISO)
- National Institute of Standards and Technology (NIST)
- Cloud Security Alliance (CSA)
- Center for Internet Security (CIS)

These organisations **make recommendations** and **provide best practices** in the governance of security and risks.

- International Organisation for Standardisation (ISO)
- National Institute of Standards and Technology (NIST)
- Cloud Security Alliance (CSA)
- Center for Internet Security (CIS)

While this form of regulation (**standard requirements**) does not usually have the force of law, an organisation may voluntarily adopt a specific set of guidelines as a framework for implementing security and risk management.

- International Organisation for Standardisation (ISO)
- National Institute of Standards and Technology (NIST)
- Cloud Security Alliance (CSA)
- Center for Internet Security (CIS)

In other cases these may be required based on legal or contractual requirements.

- Contractors to the U.S federal government are often required to implement the NIST control set in order to secure data shared with them.

- Private organisations may have customer-enforced requirements for an ISO 27001 certification over their security program.

OUTSOURCING

OUTSOURCING

In the realm of cloud computing, **outsourcing** refers to contracting with a CSP to manage some or all of your IT infrastructure, applications, or services.

- This approach frees you from the burden of maintaining your own physical servers and software licenses.
- It allows you to leverage the provider's expertise and resources.

OUTSOURCING

Outsourcing your IT infrastructure to a CSP can be a strategic way to improve efficiency, agility, and cost-effectiveness for your organisation.

OUTSOURCING

Two potential drawbacks to outsourcing to a CSP include

- The loss of visibility
- The loss of control

Two potential drawbacks to outsourcing to a CSP include



The loss of **visibility**

In SaaS, the CSP is supposed to provide patching, but there may be little to no visibility other than application interface changes to indicate whether all relevant patches have actually been applied.

OUTSOURCING

Two potential drawbacks to outsourcing to a CSP include



The loss of **control**

Outsourcing firms are often located in different countries, and sharing data across borders can be a violation of privacy or security laws and regulations like the EU GDPR.

OUTSOURCING

Two potential drawbacks to outsourcing to a CSP include



The loss of **control**

Outsourcing services like system administration or even data entry can lead to issues under the U.S International Traffic of Arms Regulation (ITAR), which requires that certain activities be performed by U.S. citizens.

OUTSOURCING

Both CSCs and CSPs need to consider the advantages and risks of outsourcing when developing business strategies

- The cost savings can be significant, but so too can the added risks!

OUTSOURCING

Both CSCs and CSPs need to consider the advantages and risks of outsourcing when developing business strategies



Major CSPs offer specialised cloud services designed to meet major regulations like ITAR, PCI DSS, HIPAA, and GDPR.

HIPAA Resources

- » Architecting for HIPAA Security and Compliance on AWS Whitepaper
- » Healthcare Providers and Insurers in the Cloud
- » AWS Security Assurance Services

 Filter by title

Details of the PCI DSS 3.2.1 Regulatory Compliance built-in initiative

Article • 07/02/2024 • [2 contributors](#) [Feedback](#)

In this article

- [Requirement 1](#)
- [Requirement 10](#)
- [Requirement 11](#)
- [Requirement 3](#)
- [Show 6 more](#)

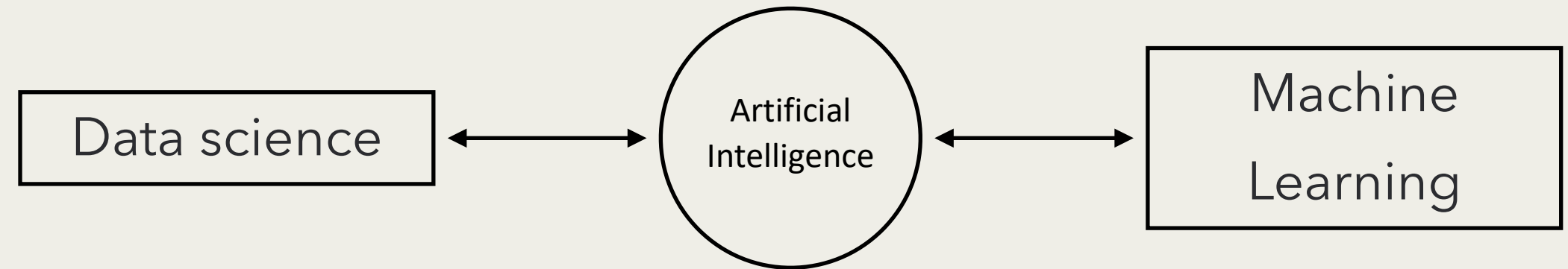
The following article details how the Azure Policy Regulatory Compliance built-in initiative definition maps to **compliance domains** and **controls** in PCI DSS 3.2.1. For more information about this compliance standard, see [PCI DSS 3.2.1](#). To understand *Ownership*, review the [policy type](#) and [Shared responsibility in the cloud](#).

The following mappings are to the **PCI DSS 3.2.1** controls. Many of the controls are implemented with an [Azure Policy](#) initiative definition. To review the complete initiative definition, open **Policy** in the Azure portal and select the **Definitions** page. Then, find and select the **PCI v3.2.1:2018** Regulatory Compliance built-in initiative definition.

EMERGING TECH - DATA SCIENCE, AI & ML

EMERGING TECH: DATA SCIENCE, AI, & ML

Data science, machine learning (ML), and Artificial Intelligence (AI) are all interconnected fields, but serve distinct purposes.



EMERGING TECH: DATA SCIENCE, AI, & ML



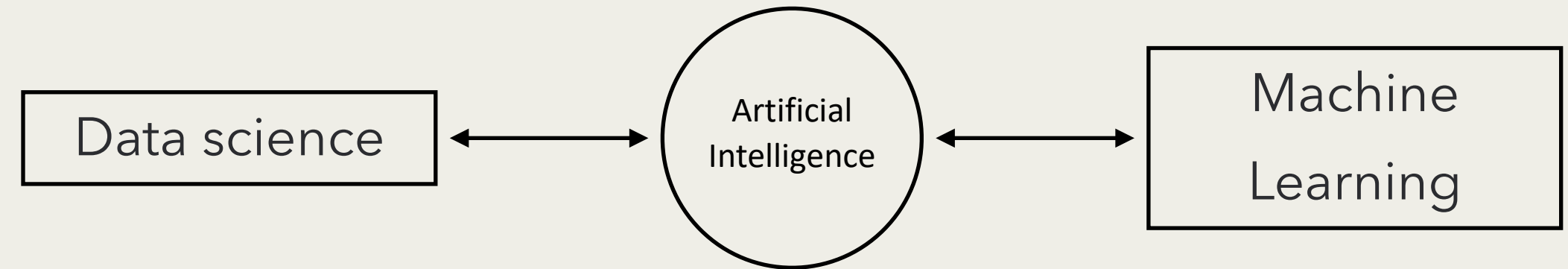
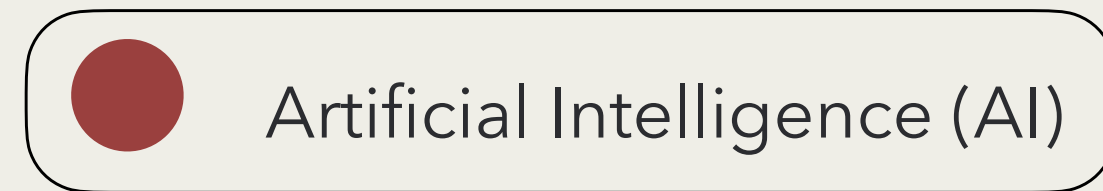
Provides the foundation for both AI and ML.

Data scientists gather, analyse, and interpret data.

Data scientists develop the training data that ML algorithms need to learn and improve.

Data science has applications beyond AI and ML, such as data visualisation and summarisation.

EMERGING TECH: DATA SCIENCE, AI, & ML



AI is the broader concept, aiming to create intelligent machines capable of performing tasks that typically require human intelligence.

AI encapsulates various approaches, including ML and data science.

EMERGING TECH: DATA SCIENCE, AI, & ML



ML is a subfield of AI focused on developing algorithms that can learn and improve from data without explicit programming.

ML algorithms can be used to solve a wide range of problems, including security anomaly detection.

As we conduct ML, we have a few possible goals:

- ☐ Descriptive analytics
- ☐ Predictive analytics
- ☐ Prescriptive analytics

EMERGING TECH: DATA SCIENCE, AI, & ML

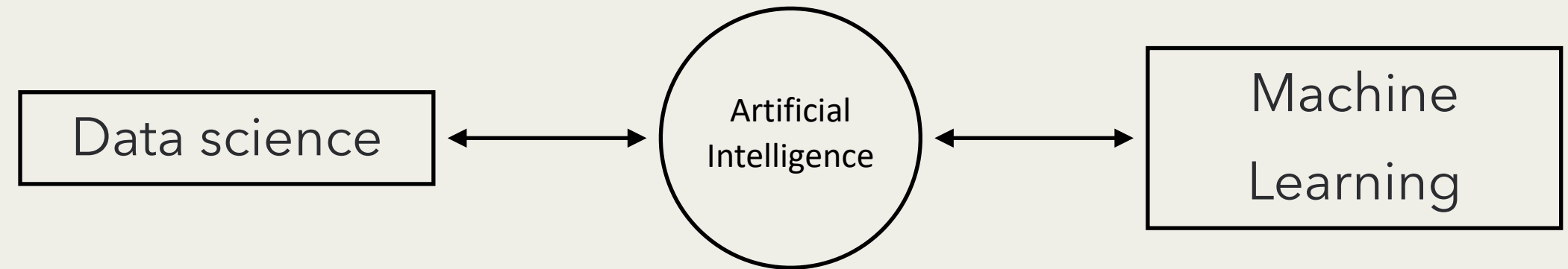
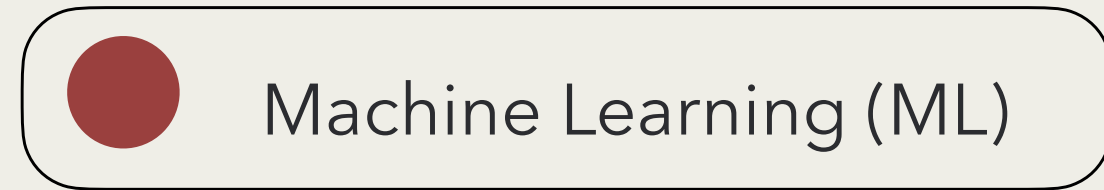


Descriptive analytics simply seeks to describe data.

For example, if we perform descriptive analytics on our customer records, we might ask questions like,

- *What proportion of our customers are female?*
- *How many of them are repeat customers?*

EMERGING TECH: DATA SCIENCE, AI, & ML



Predictive analytics seeks to use our existing data to predict future events.

For example, if we had a dataset on how customers respond to direct mail, we build a model that predicts how individual customers will respond to future mailing.

We might then tweak that mailing to improve the response rate by

- ☐ Changing the day we send emails
- ☐ Altering the content of the message
- ☐ Altering the font size or paper colour

EMERGING TECH: DATA SCIENCE, AI, & ML



Prescriptive analytics seeks to optimise our behaviour by simulating many scenarios.

Imagine we want to determine the best way to allocate our marketing dollars.

- ☐ We might run different simulations of consumer response.
- ☐ Use algorithms to prescribe our behaviour in that context.

EMERGING TECH: DATA SCIENCE, AI, & ML



Prescriptive analytics seeks to optimise our behaviour by simulating many scenarios.

Similarly, we might use prescriptive analytics to optimise the performance of an automated process.

EMERGING TECH: DATA SCIENCE, AI, & ML



Cloud computing has revolutionised the world of ML.

- Many of the applications where we apply AI techniques today simply
- wouldn't have been possible without the scalable, on-demand computing offered by the cloud.

EMERGING TECH: DATA SCIENCE, AI, & ML



Cloud computing has revolutionised the world of ML.

- CSPs now offer very specialised services designed to help organisations design, build, and implement machine learning models.

EMERGING TECH - BLOCKCHAIN

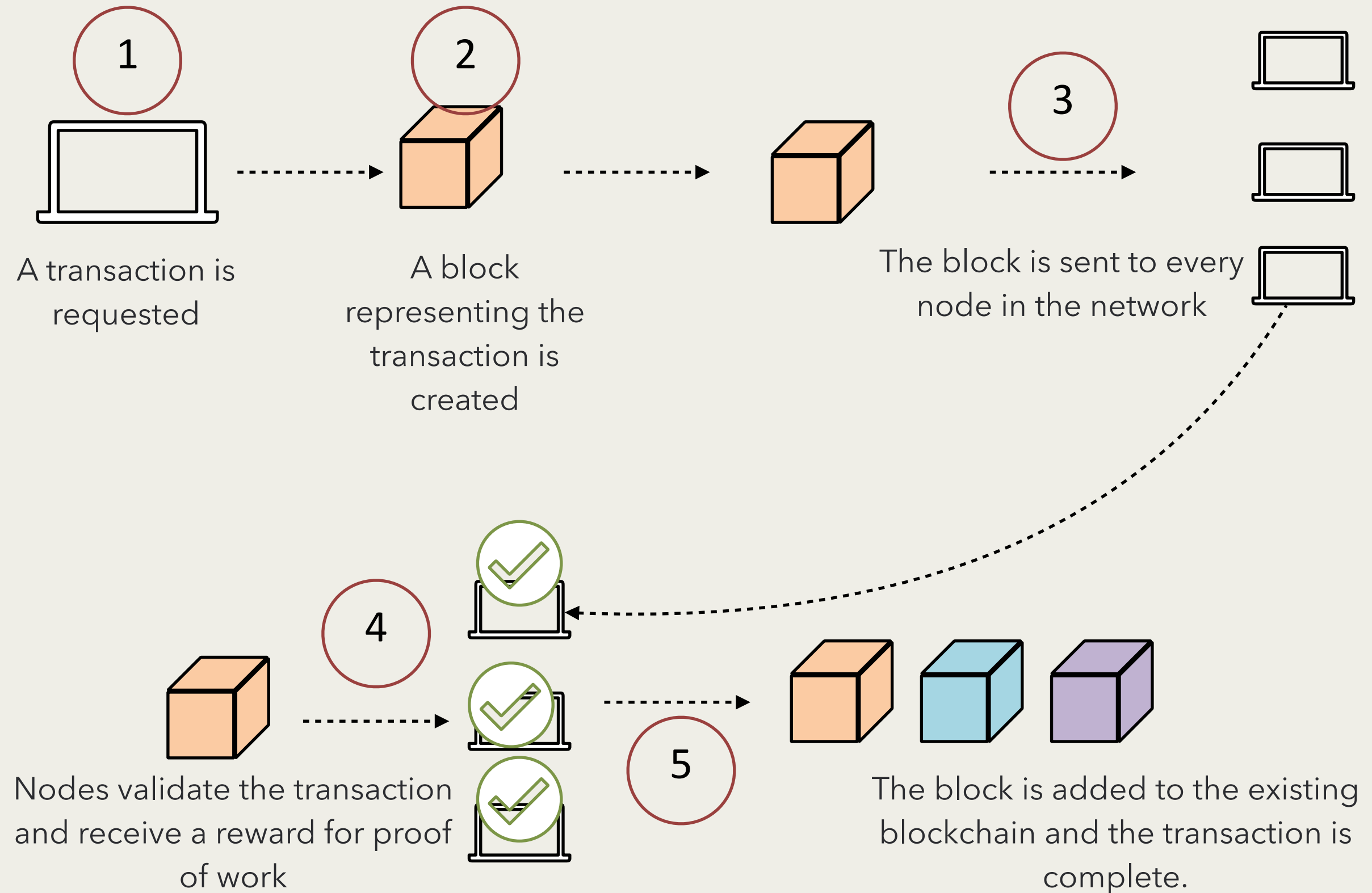
EMERGING TECH: BLOCKCHAIN

Blockchain technology utilises a distributed ledger system for secure and immutable record-keeping.

1 Data is chronologically organised into **blocks**.

2 Blocks are cryptographically linked together to form a tamper-proof chain.

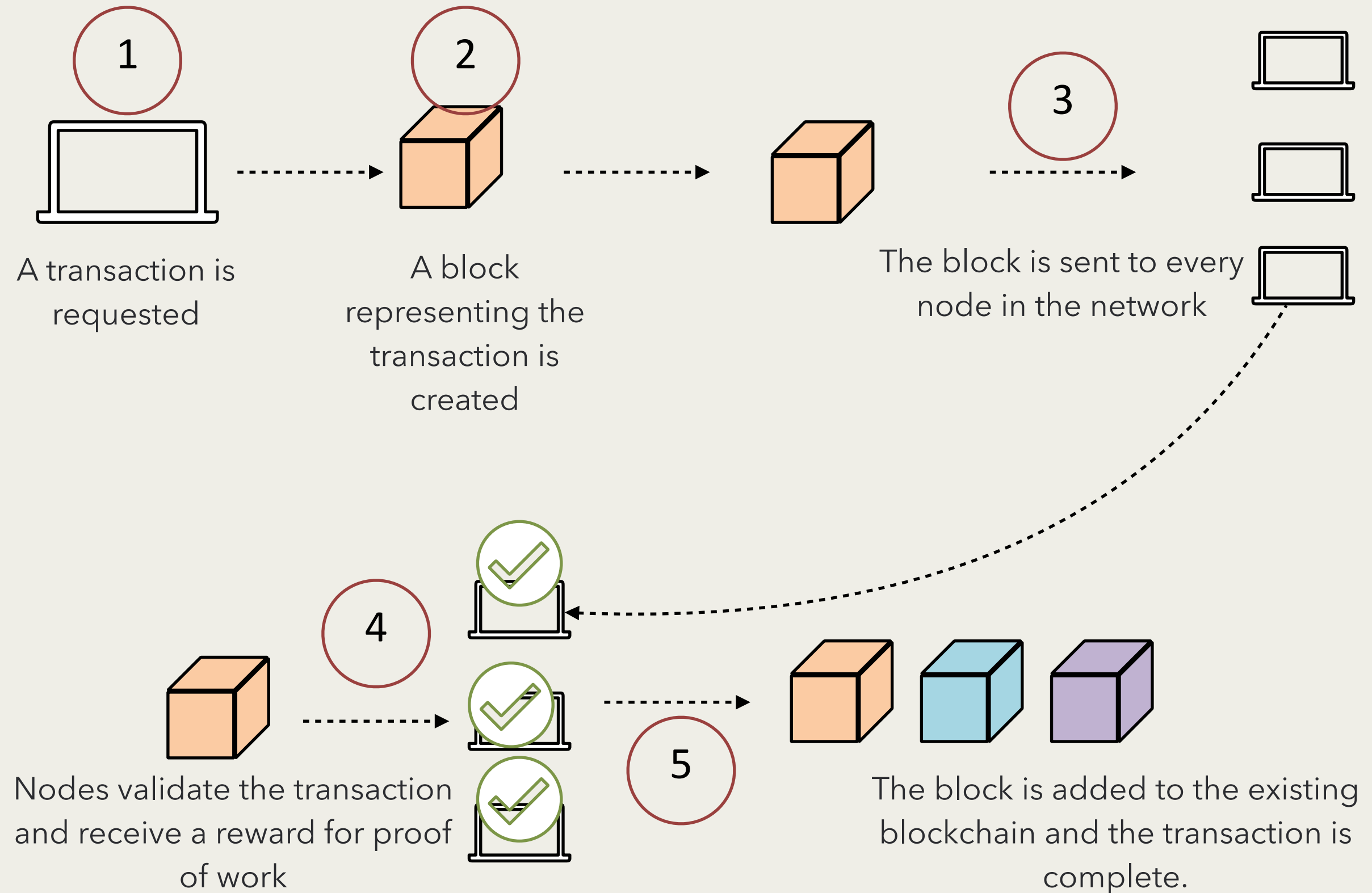
This decentralised network ensures *data integrity and prevents unauthorised modifications.*



EMERGING TECH: BLOCKCHAIN

There are many **use-cases** for a distributed immutable ledger:

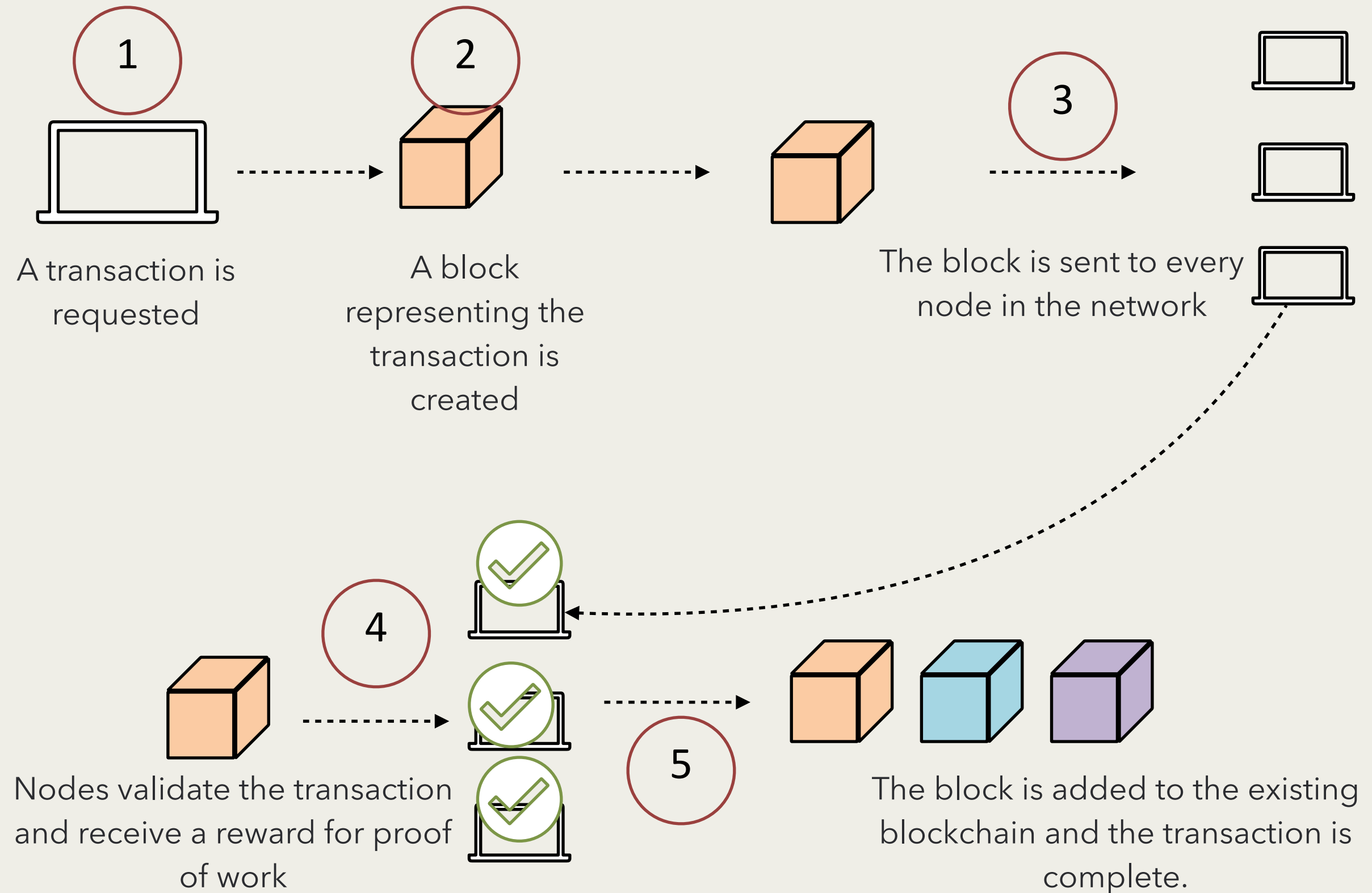
The blockchain was originally invented as a foundational technology for **Bitcoin**, allowing the tracking of Bitcoin transactions without the use of a centralised authority.



EMERGING TECH: BLOCKCHAIN

There are many **use-cases** for a distributed immutable ledger:

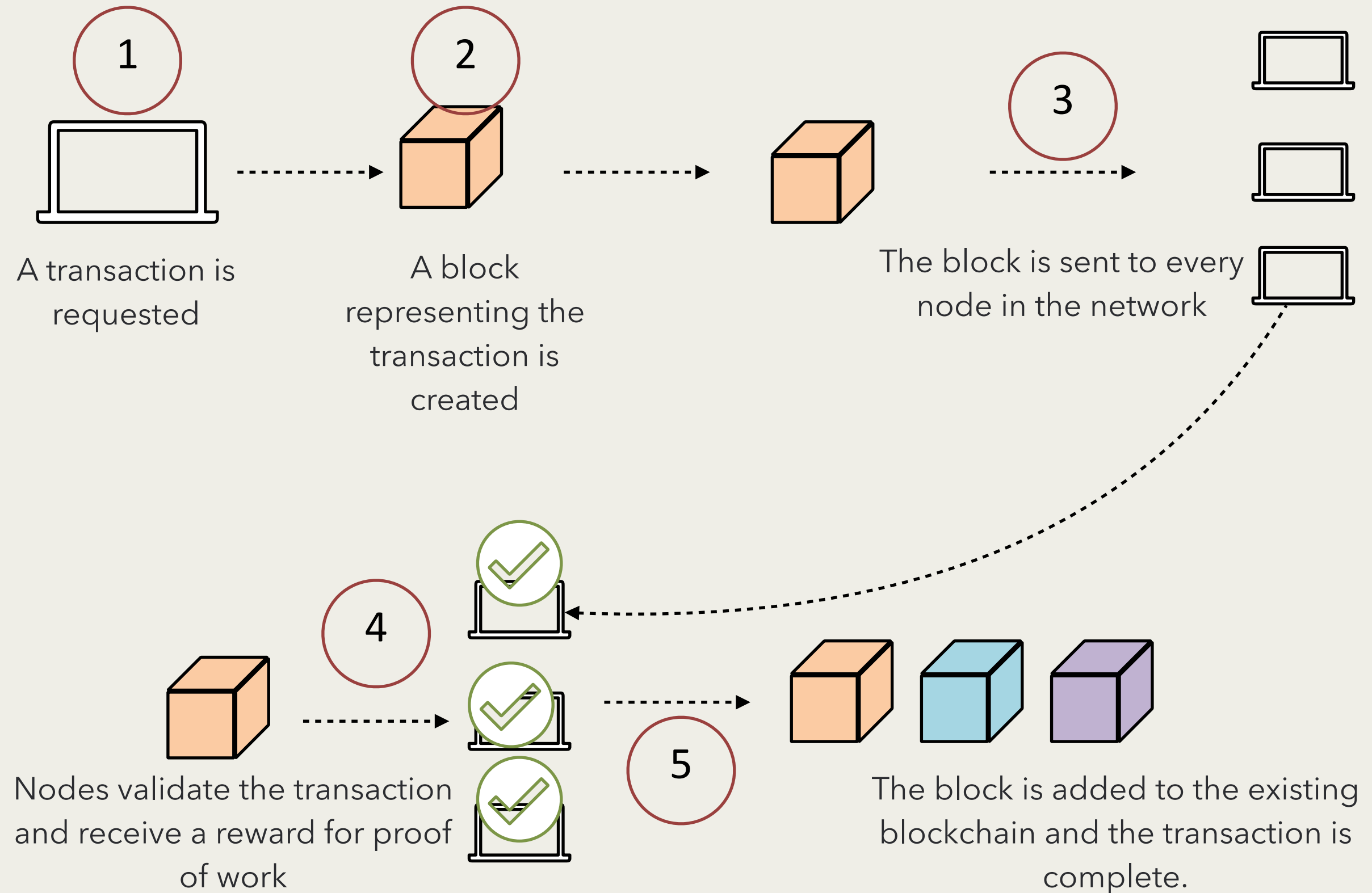
Property ownership records could benefit from a blockchain technology. This approach would place those records in a transparent, public repository that is protected against intentional or accidental damage.



EMERGING TECH: BLOCKCHAIN

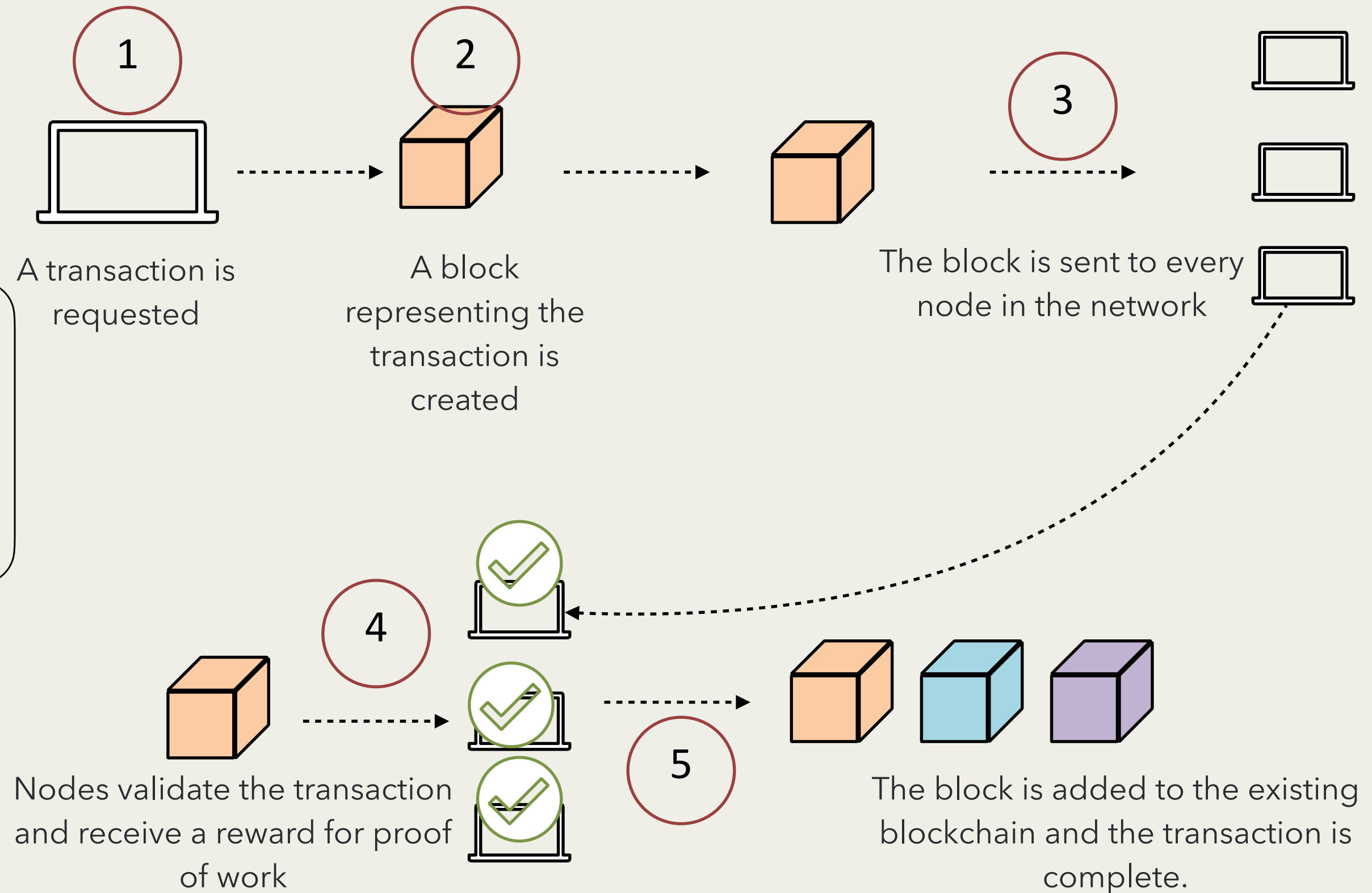
There are many **use-cases** for a distributed immutable ledger:

Blockchain might be used to **track supply chains**, providing customers with confidence that their produce came from reputable sources and allowing regulators to easily track down the origin of recalled produce.



EMERGING TECH: BLOCKCHAIN

Cloud computing enables blockchain by providing computing resources that are scalable, economically efficient, and globally distributed.

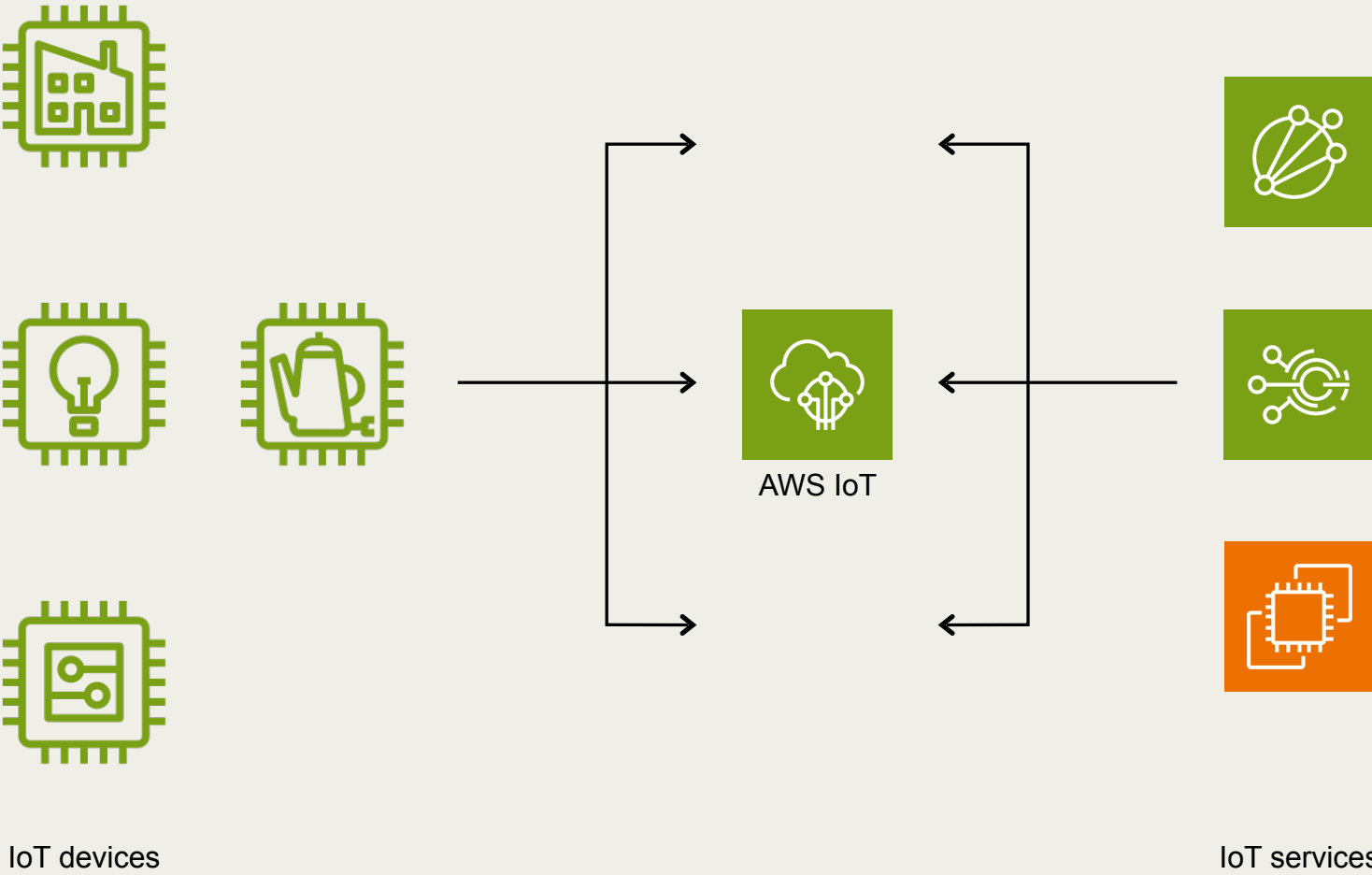


EMERGING TECH - INTERNET OF THINGS (IOT)

EMERGING TECH: INTERNET OF THINGS



IoT (Internet of Things) is a term used to describe connecting non-traditional devices to the Internet for data collection, analysis, and control.

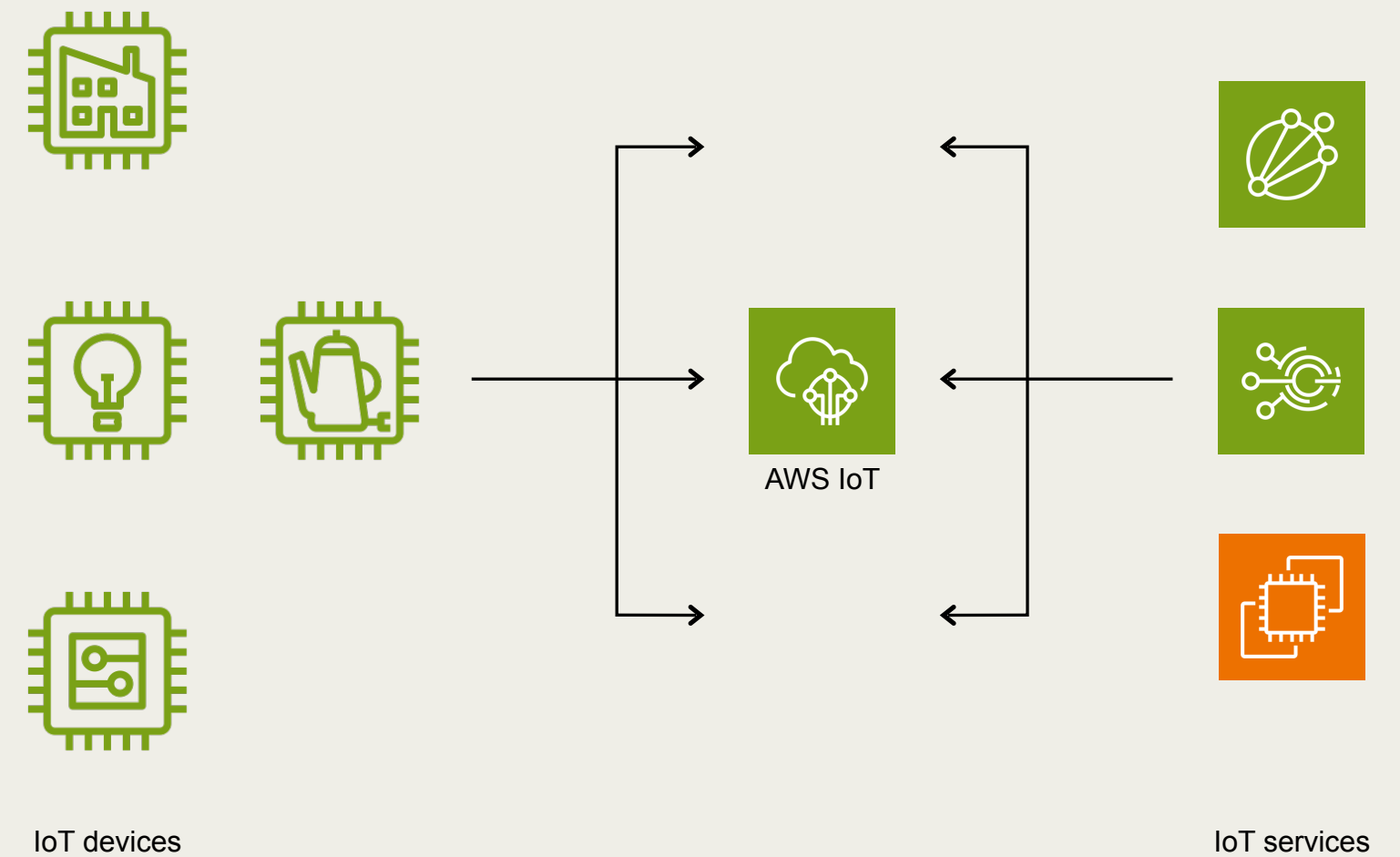


EMERGING TECH: INTERNET OF THINGS

We see IoT applications arising in the home and workplace. All of these devices come with security challenges.



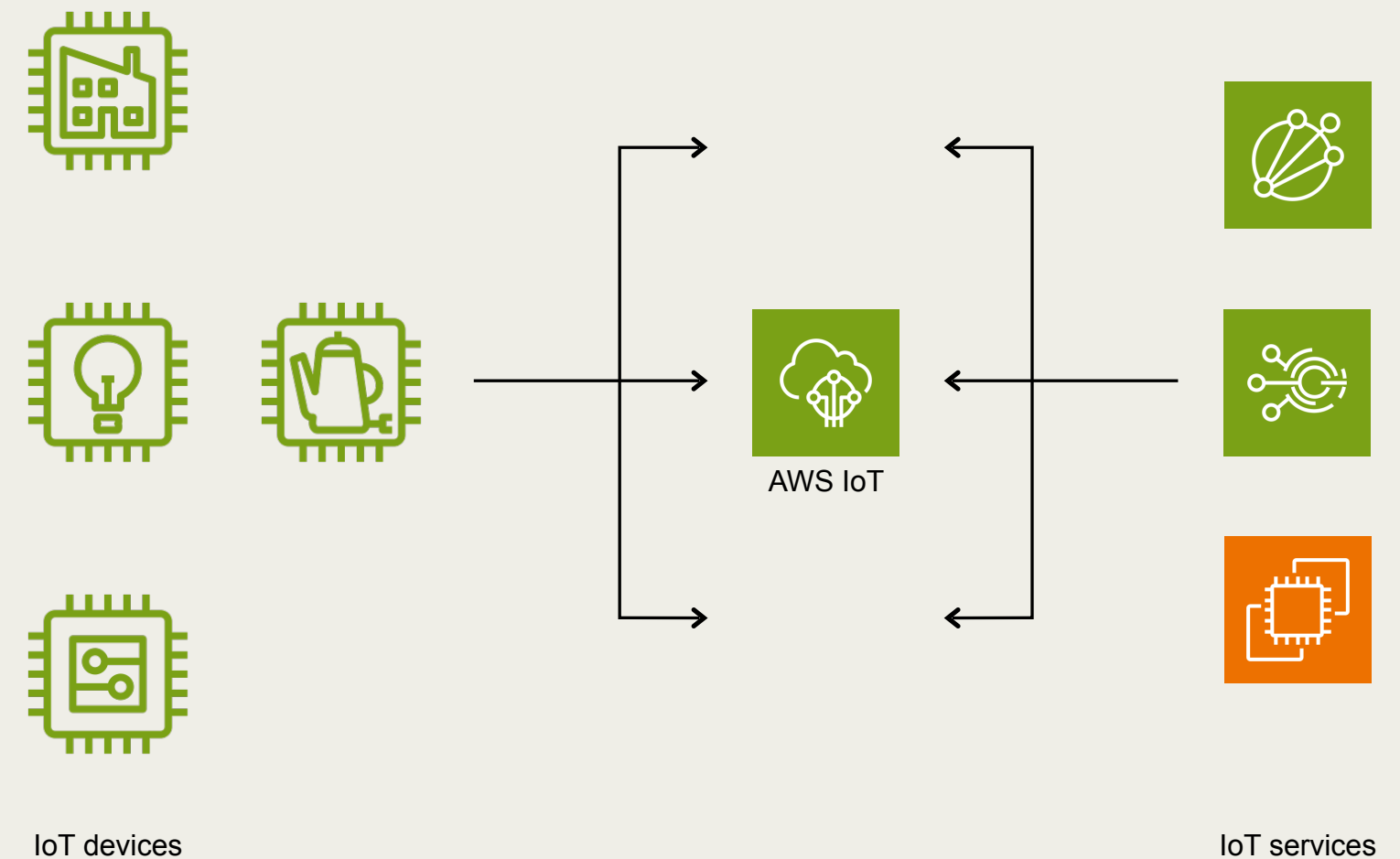
It is often difficult for the consumer to update the software on these devices.



EMERGING TECH: INTERNET OF THINGS

We see IoT applications arising in the home and workplace. All of these devices come with security challenges.

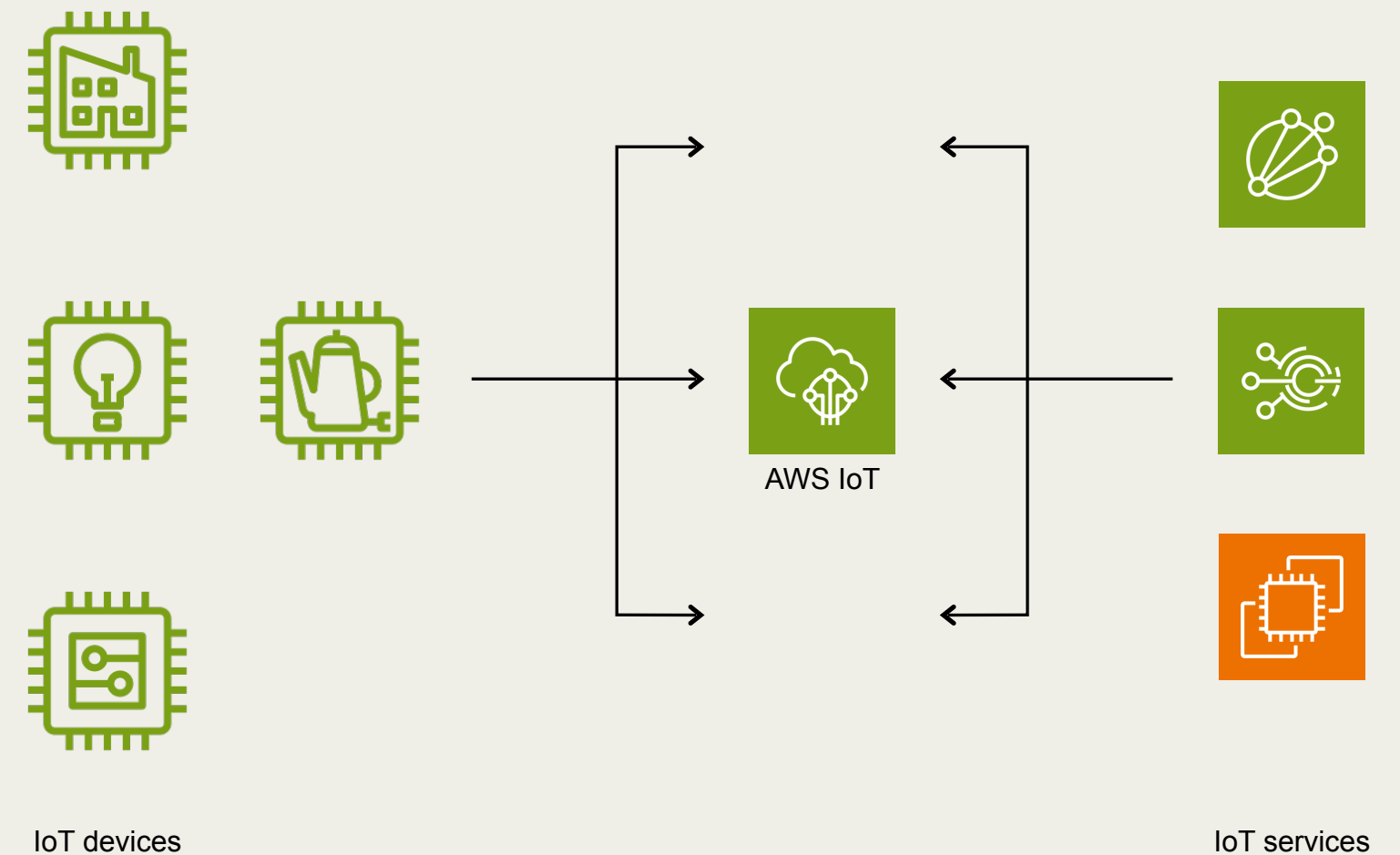
IoT devices connect to the same wireless networks that we use for personal productivity. If compromised, it can be a gateway to the rest of our network.



EMERGING TECH: INTERNET OF THINGS

We see IoT applications arising in the home and workplace. All of these devices come with security challenges.

Smart devices often connect back to cloud services for command-and-control, creating a potential pathway onto the network for external attackers that bypass the firewall.

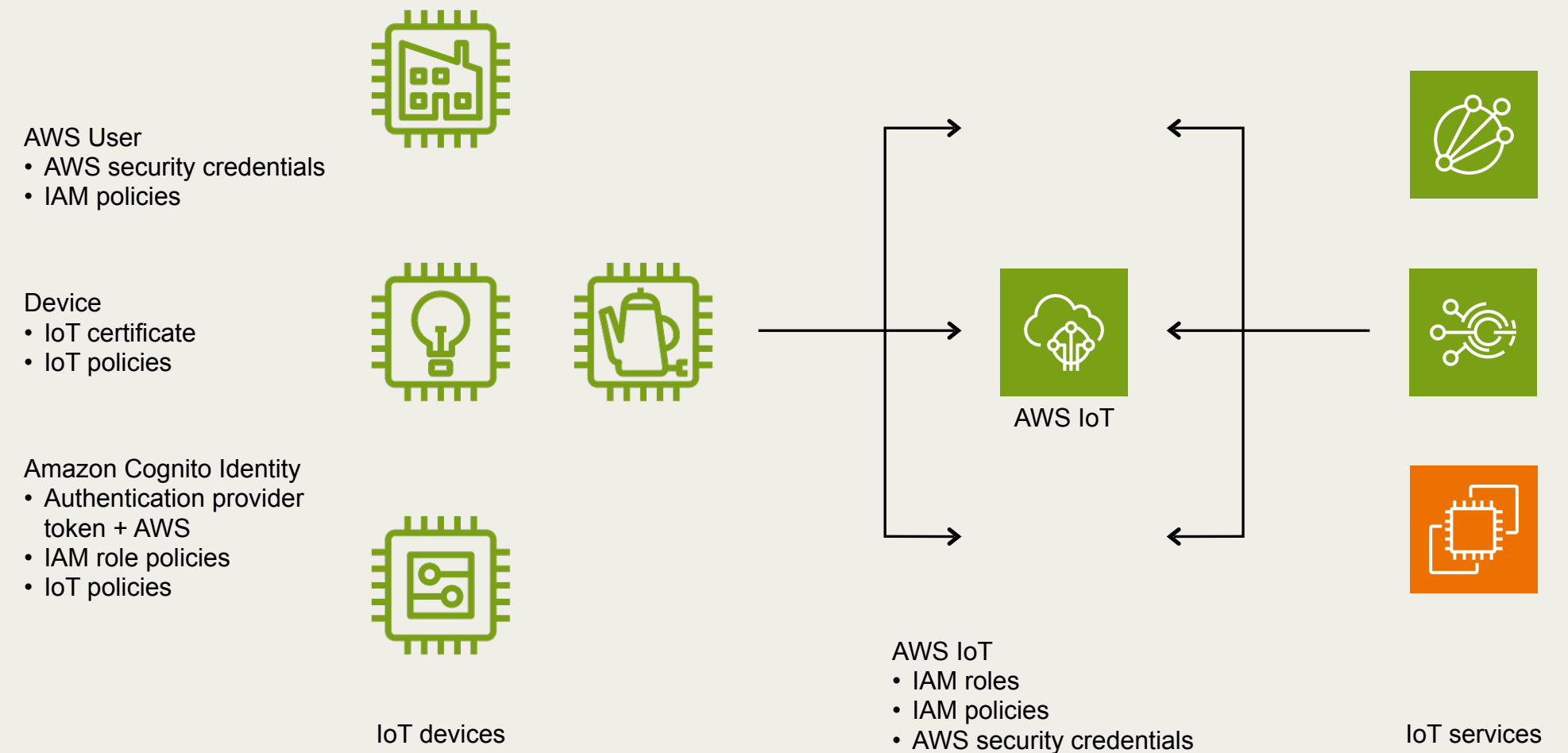


EMERGING TECH: INTERNET OF THINGS

On AWS, each connected device or client must have a credential to interact with AWS IoT.

● All traffic to and from AWS IoT is sent securely over Transport Layer Security (TLS)

● AWS cloud security mechanisms protect data as it moves between AWS IoT and other AWS services.

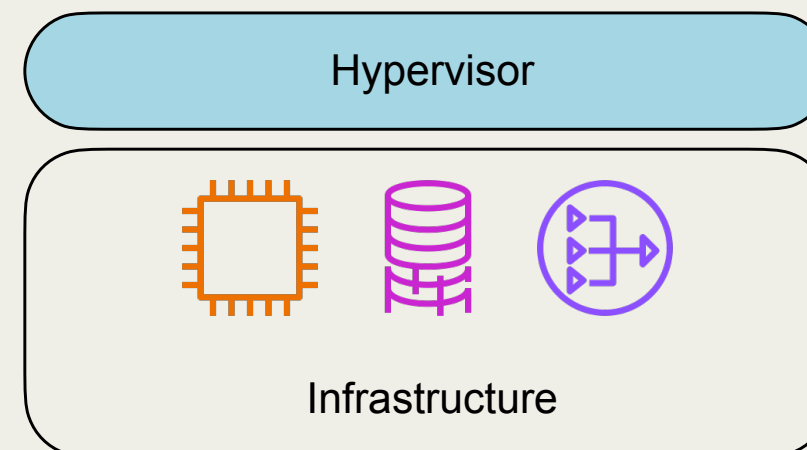


EMERGING TECH - CONTAINERS

EMERGING TECH: CONTAINERS

Containers are an evolution of virtualisation. They are a lightweight approach to package an entire application and make it portable so that it can easily move between hardware platforms.

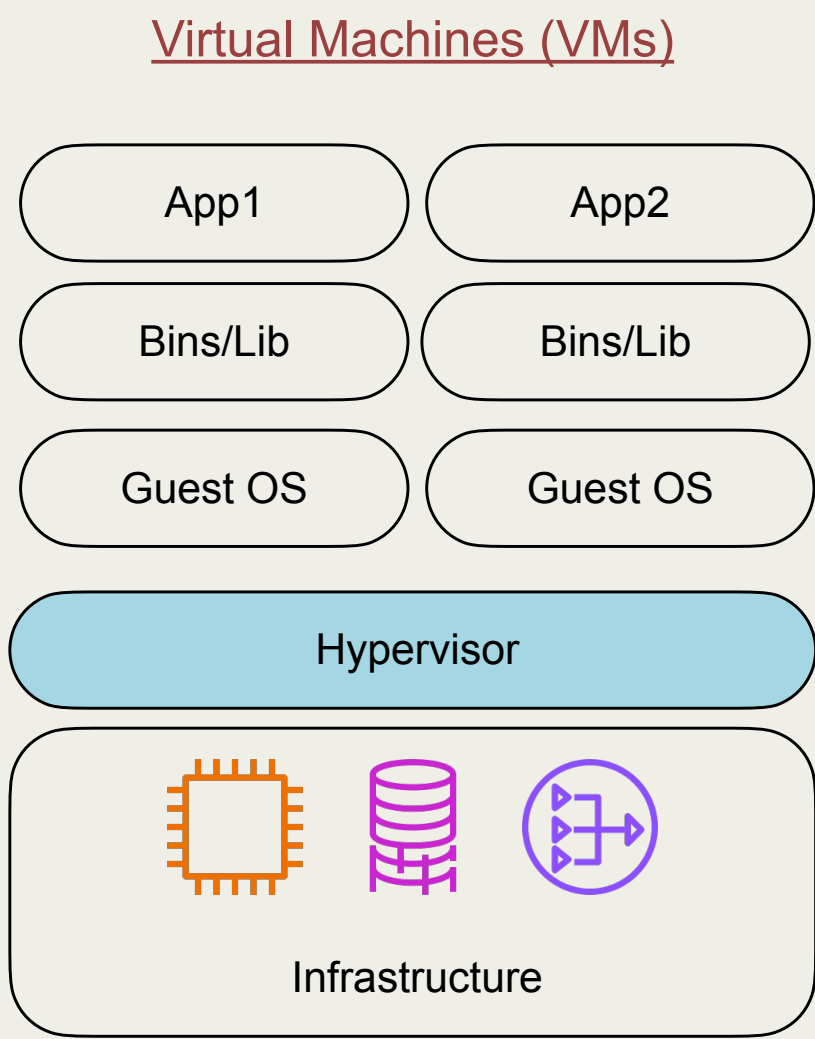
In traditional virtualisation, we have hardware that supports a hypervisor, which then supports guest VMs.



EMERGING TECH: CONTAINERS

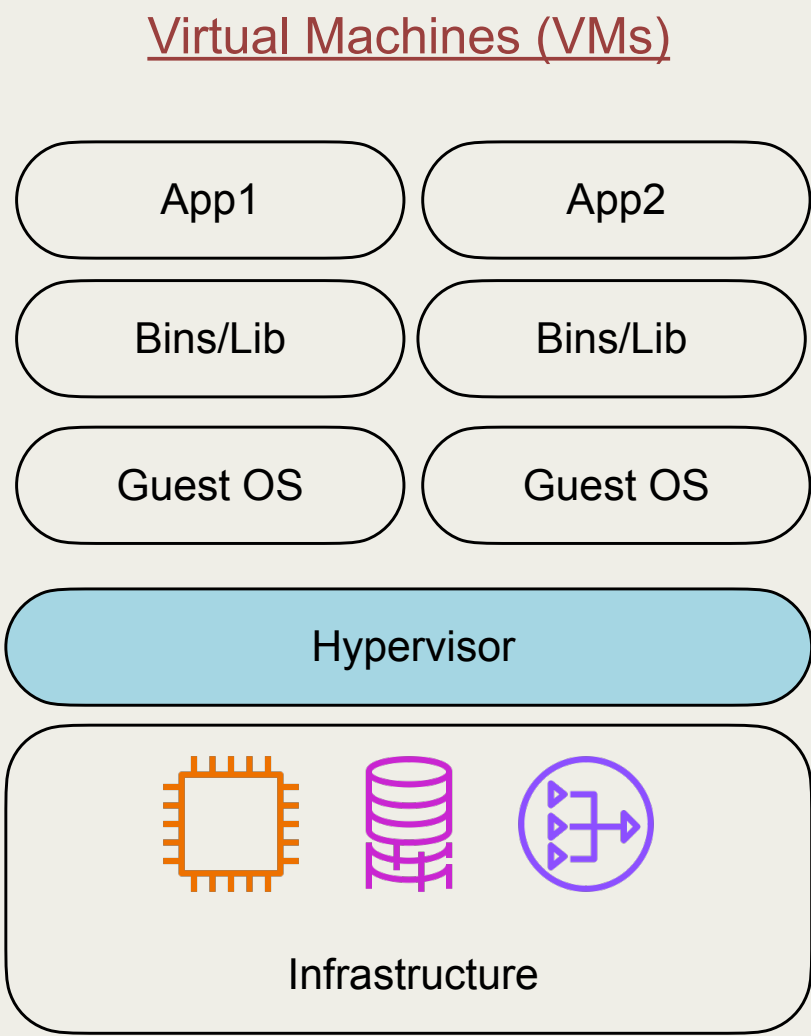


Each of those guest VMs runs its own OS and applications, allowing the applications to function somewhat independently of the hardware.



EMERGING TECH: CONTAINERS

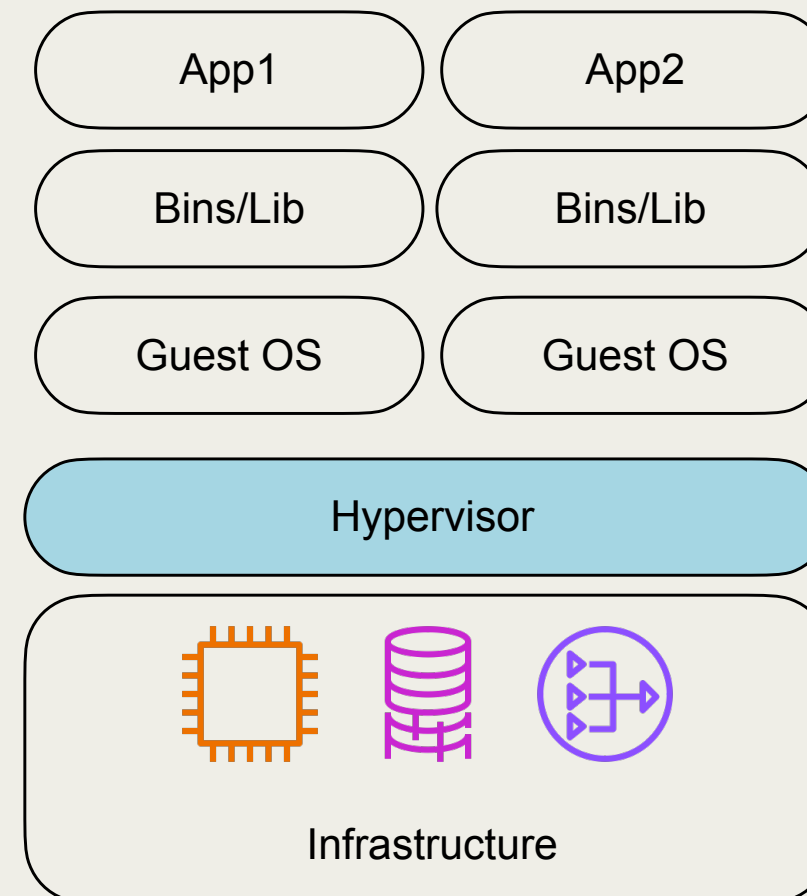
You can move a VM from hardware to hardware, **as long as the machines are running the same hypervisor.**



EMERGING TECH: CONTAINERS

A downside to traditional virtualisation is that **VMs are somewhat "heavy"**. Each VM has to have its own OS and components.

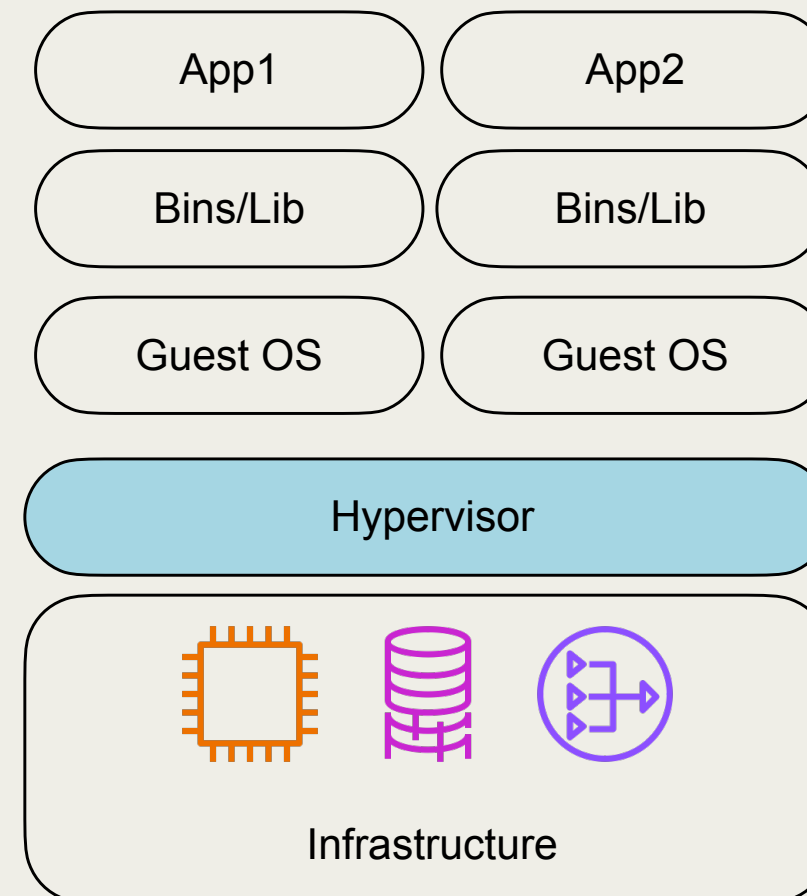
Virtual Machines (VMs)



EMERGING TECH: CONTAINERS

If you are running 10 different Windows VMs on a hypervisor, you have the overhead of running 10 different copies of Windows at the same time.

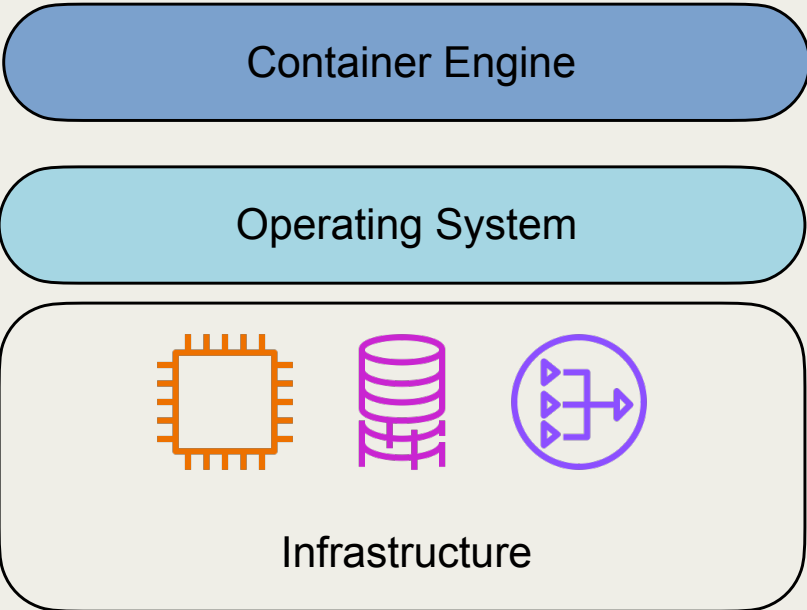
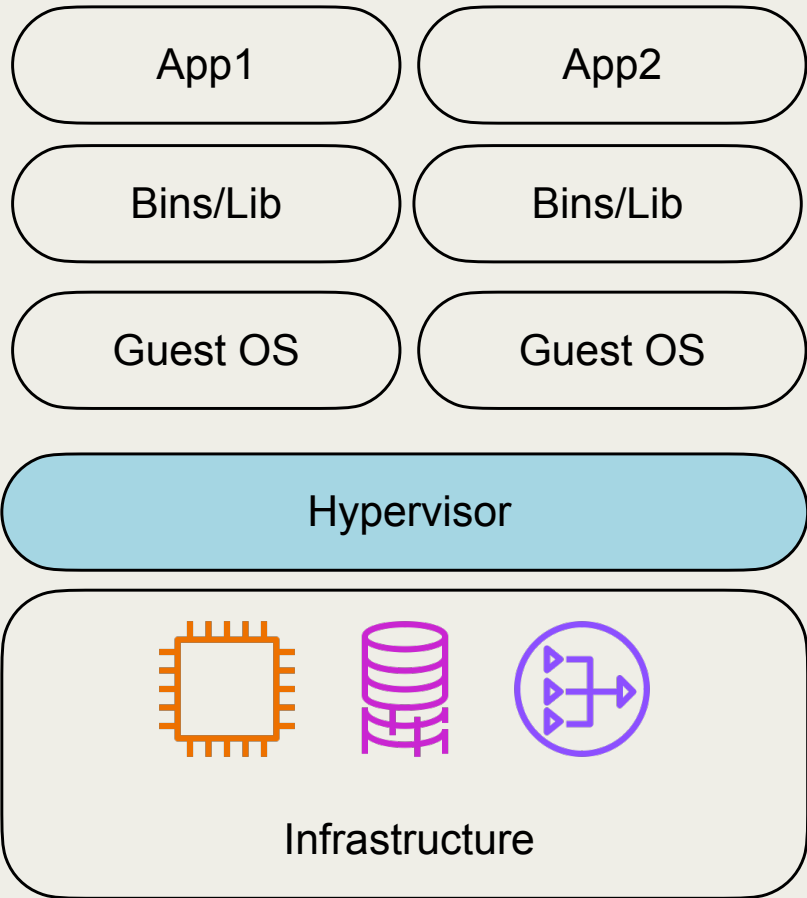
Virtual Machines (VMs)



EMERGING TECH: CONTAINERS

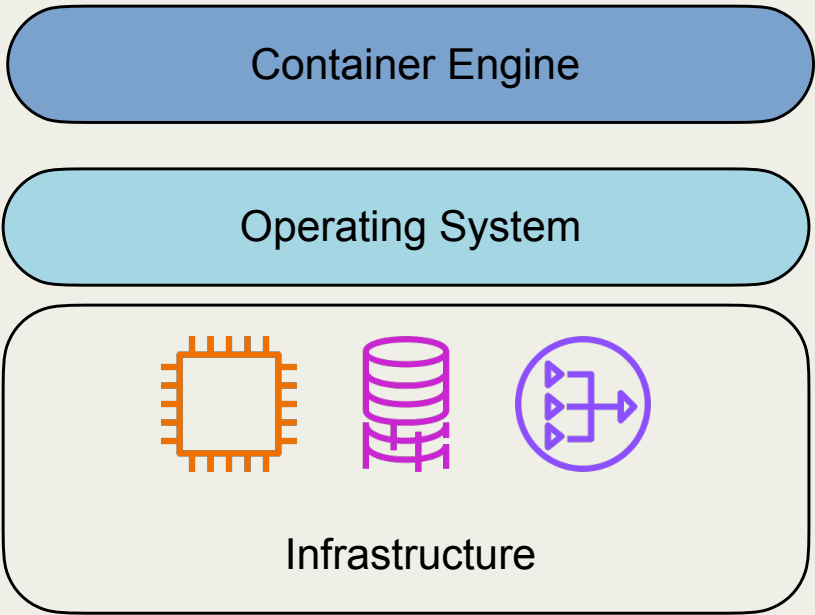
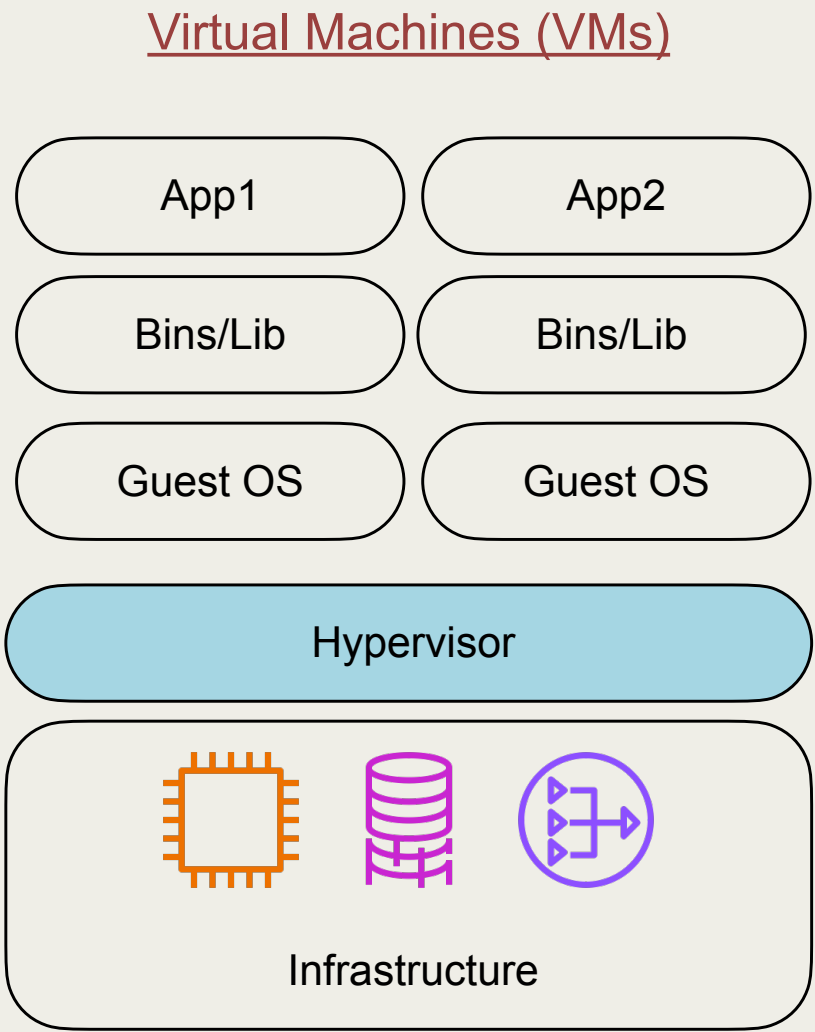
Instead of running a hypervisor, systems supporting containers run a containerisation platform.

Virtual Machines (VMs)



EMERGING TECH: CONTAINERS

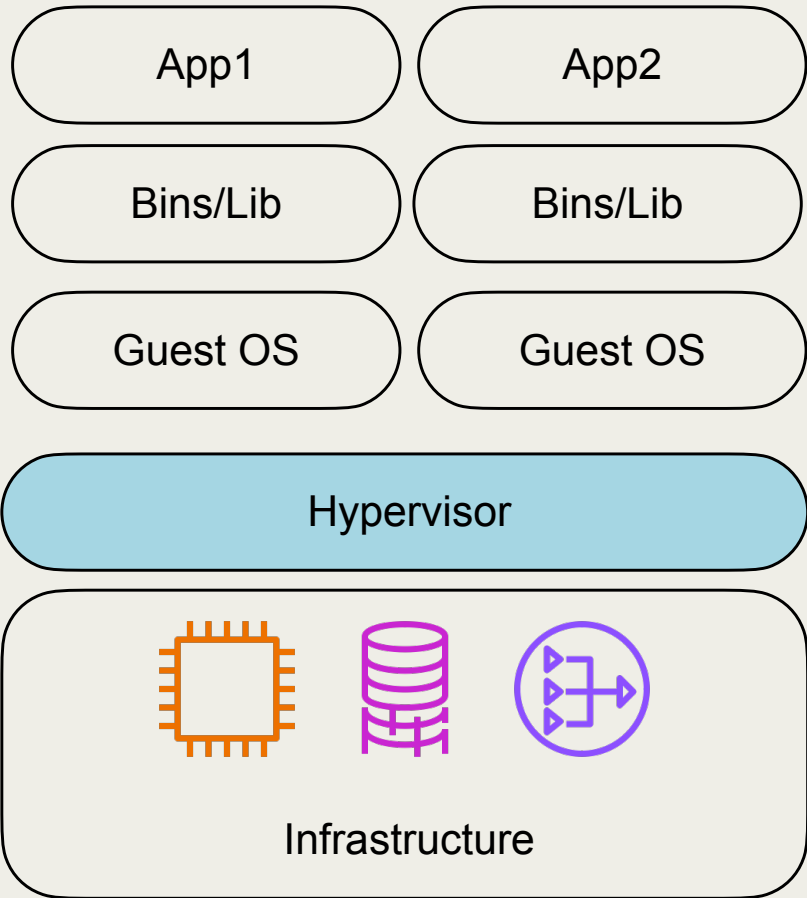
The containerisation platform provides a **standard interface to the OS** that allows containers to function regardless of the OS and hardware.



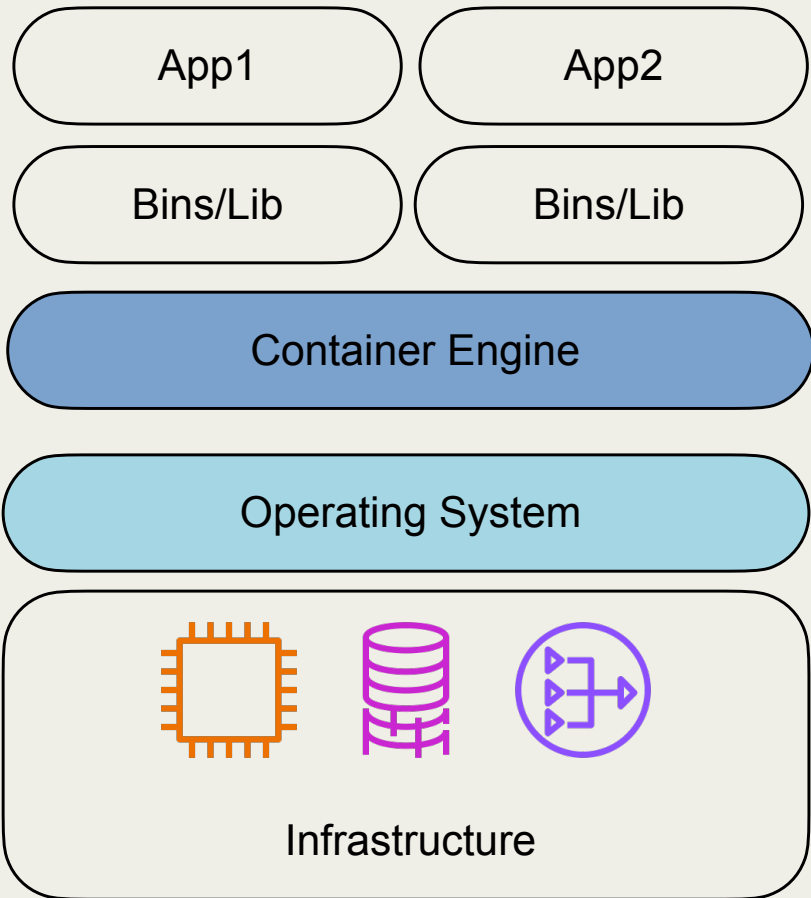
EMERGING TECH: CONTAINERS

Containers package up application code in a standardised format so that it can be easily shifted between systems.

Virtual Machines (VMs)



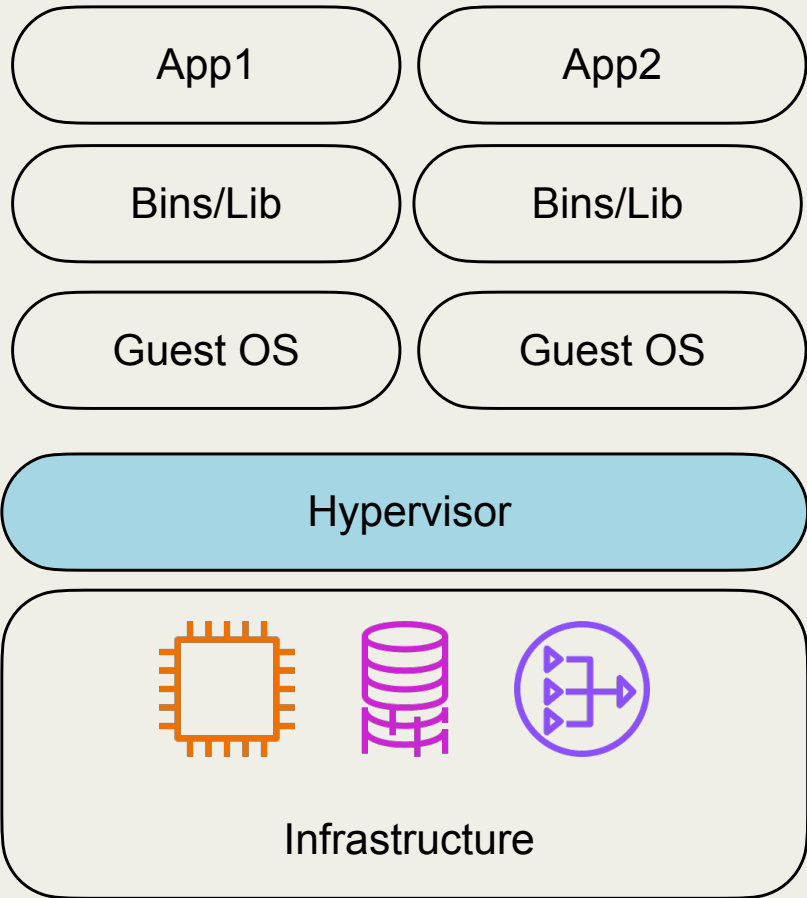
Containers



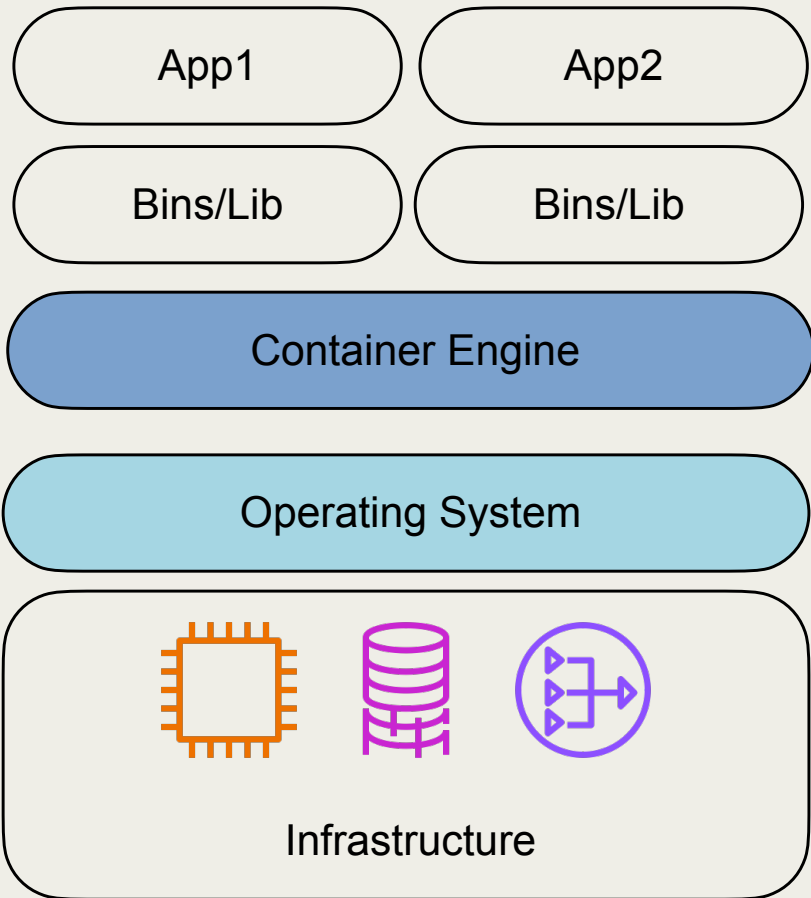
EMERGING TECH: CONTAINERS

The major benefit of containers over VMs is that they do not have their own OS kernel. The containerisation platform allows them to use the host's OS kernel.

Virtual Machines (VMs)



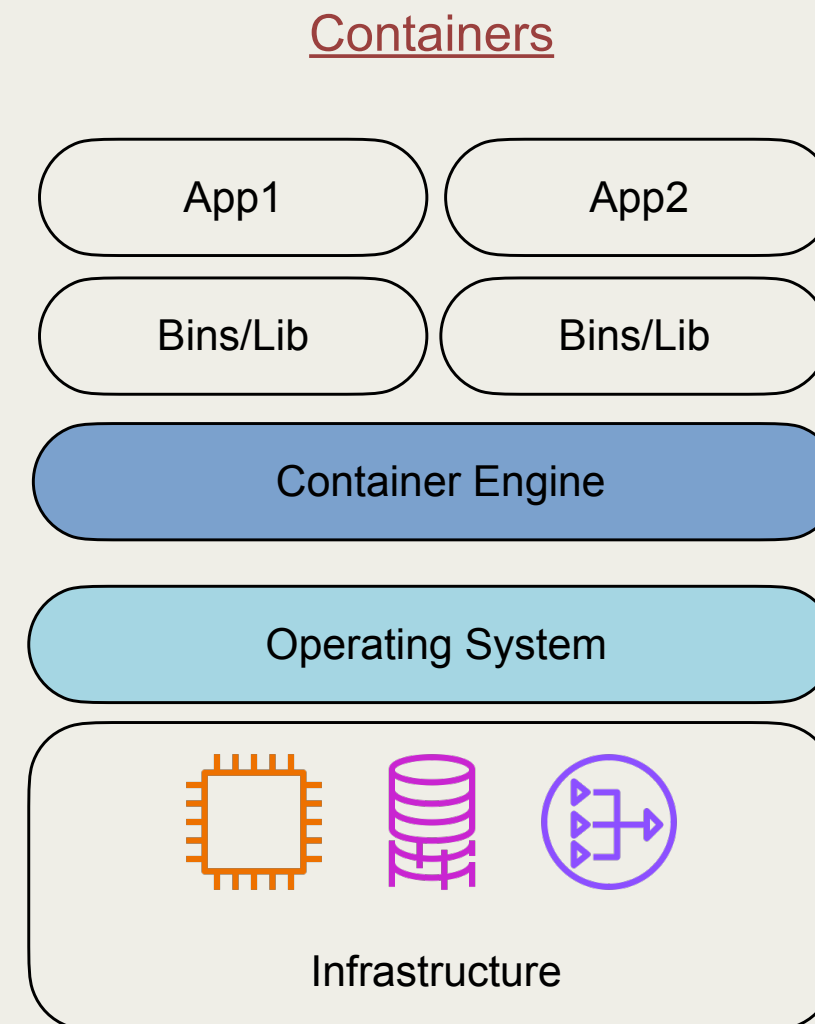
Containers



EMERGING TECH: CONTAINERS

The containerisation platform must **strictly enforce isolation** to ensure that containers cannot access the data or resources allocated to other containers. This is achieved through a combination of Linux features:

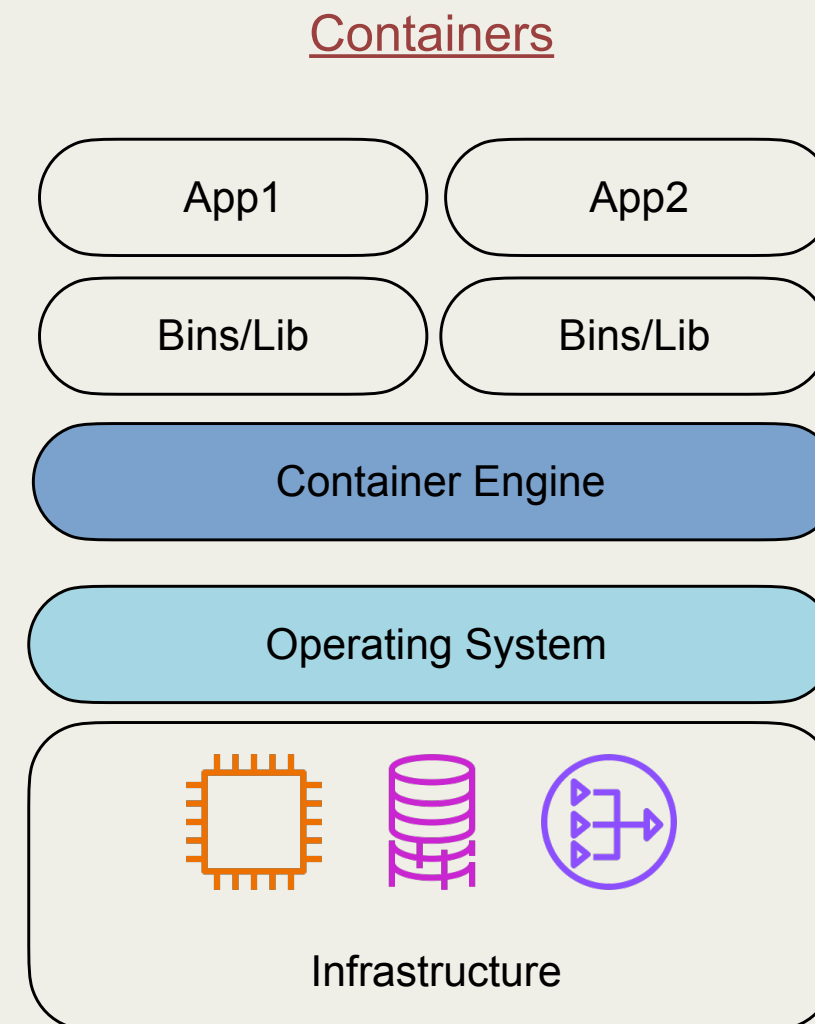
- Namespaces
- Control groups
- Capabilities
- chroot (optional)



EMERGING TECH: CONTAINERS

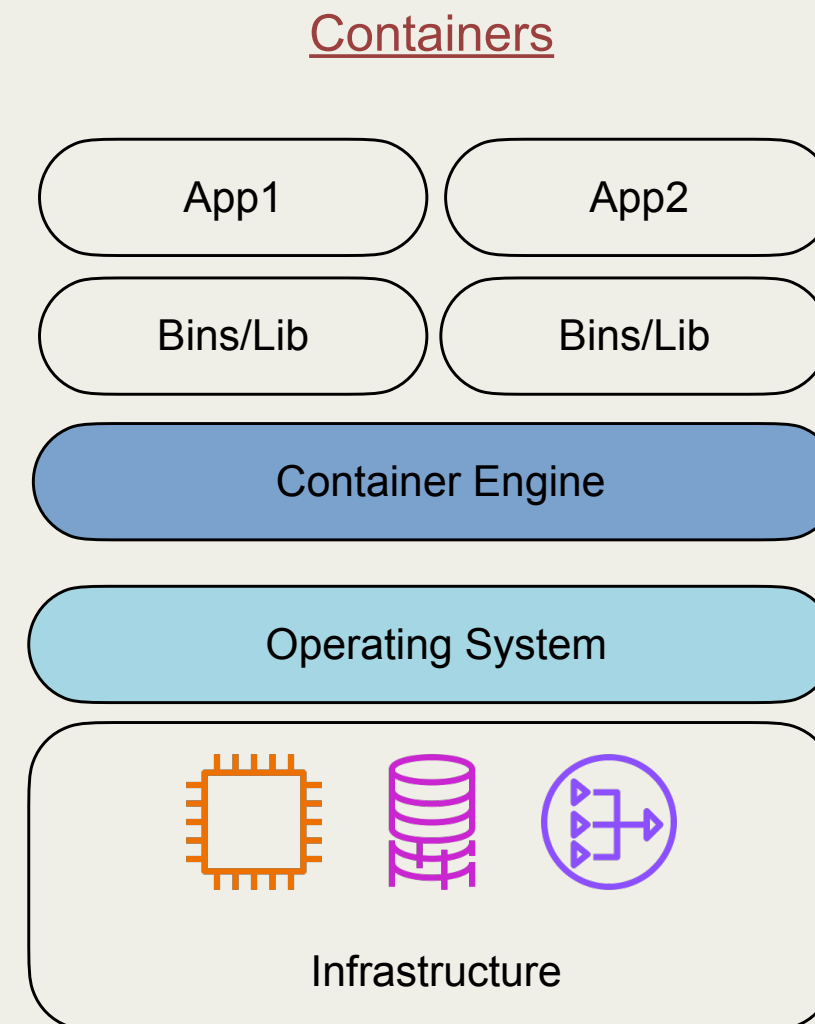
Namespaces create isolated views of system resources for each container. Each container sees its own version of

- Process IDs
- File systems
- Network interfaces/IP addresses
- IPC, User, UTS - isolation for interprocess communication, user IDs, hostname



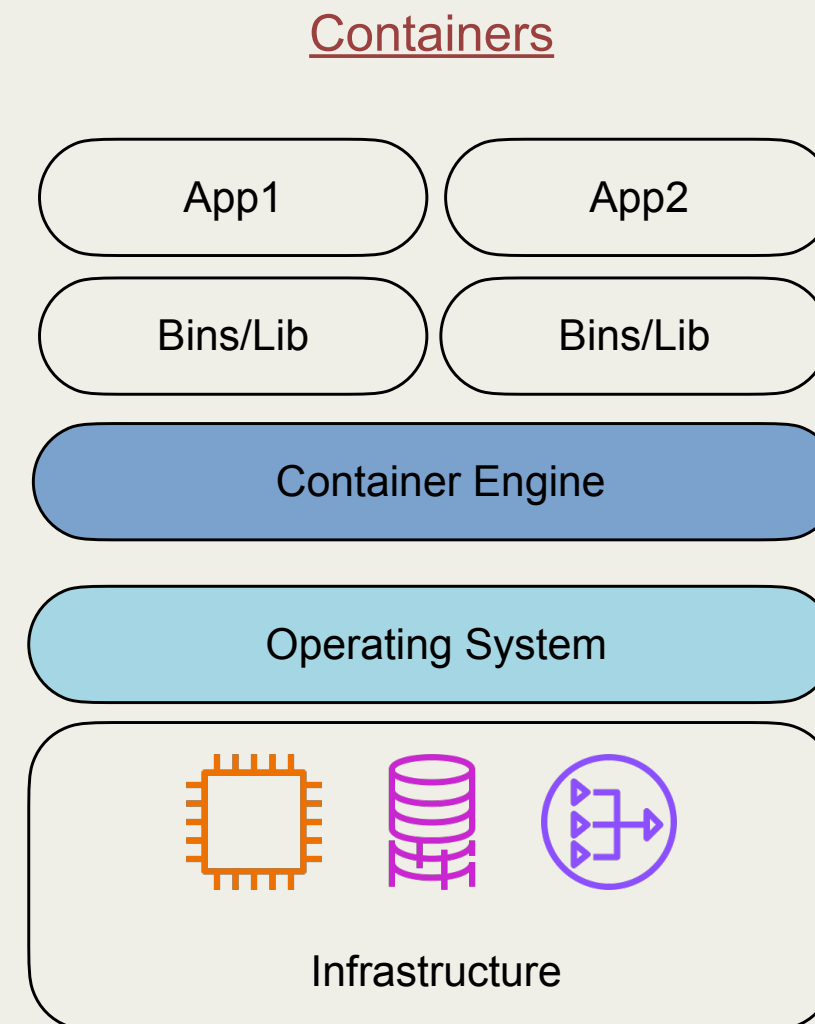
EMERGING TECH: CONTAINERS

Control groups (Cgroups) act like quotas, limiting the amount of memory, CPU, disk I/O, and other resources a container can consume. This ensures fair allocation and prevents one container from hogging resources needed by others.



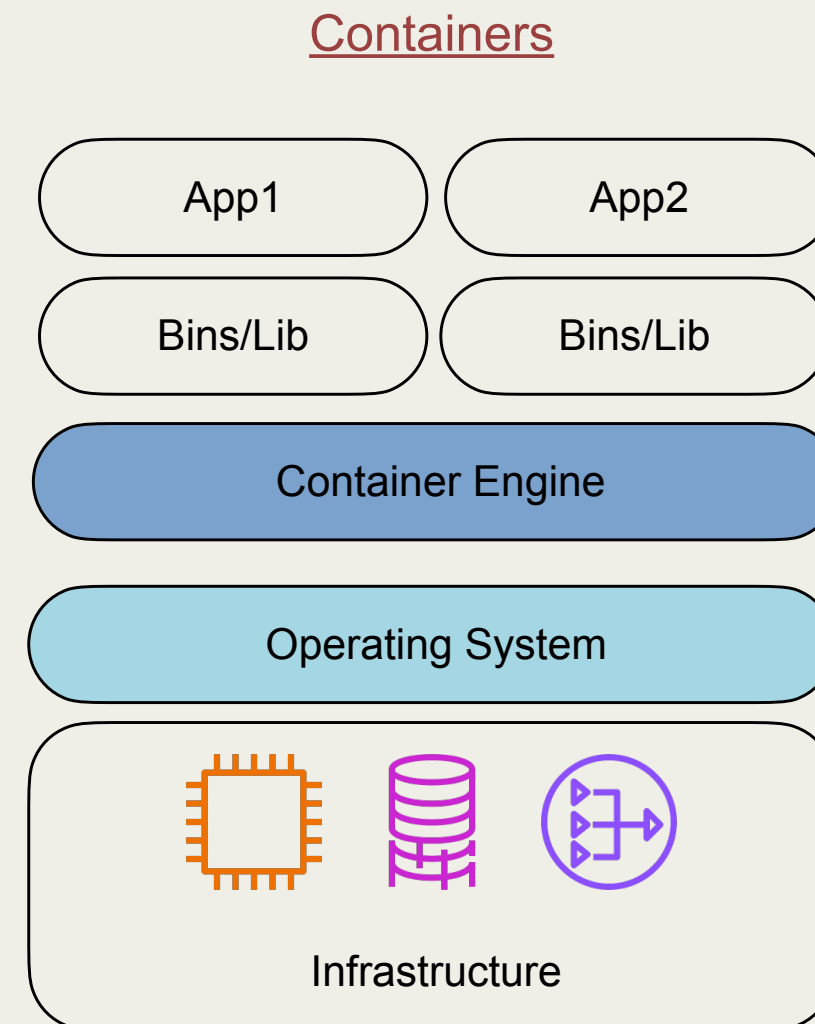
EMERGING TECH: CONTAINERS

Containers inherit a limited set of **capabilities** from the host kernel. This restricts what actions processes inside the container can perform, enhancing security.



EMERGING TECH: CONTAINERS

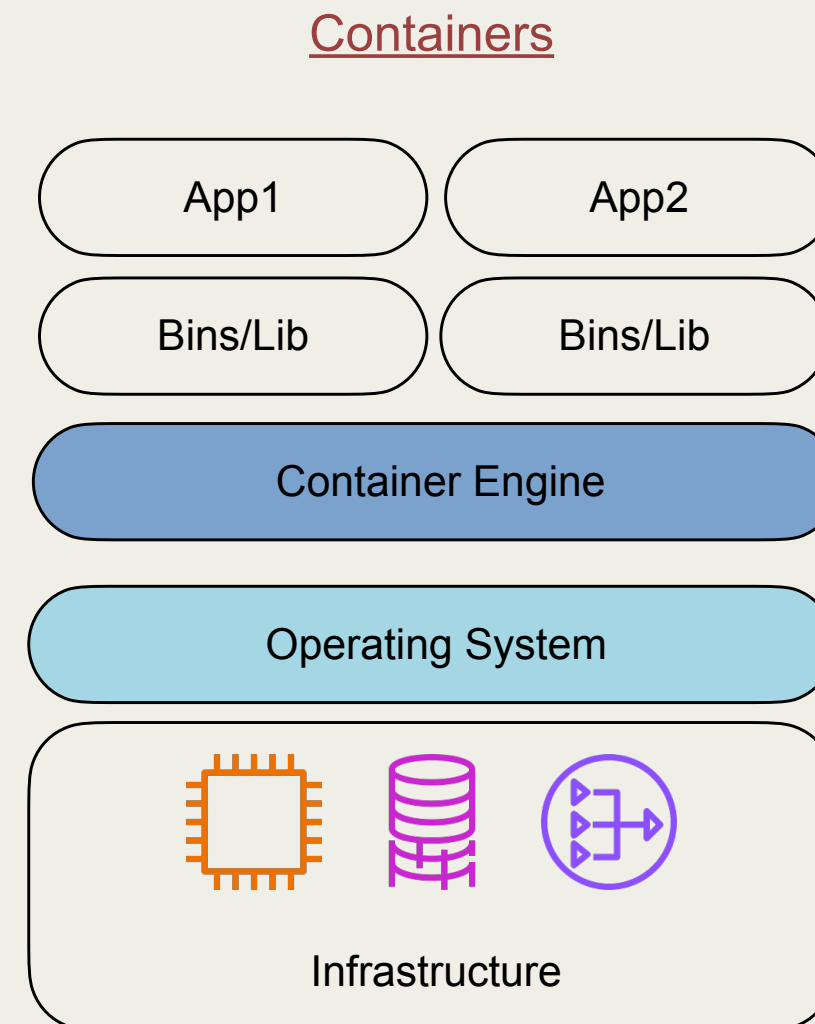
While not always used, **chroot** can further restrict a container's access to the file system by confining its root directory to a specific portion of the host's file system.



EMERGING TECH: CONTAINERS

By combining these techniques, containers achieve a good balance of isolation, resource efficiency, and portability.

- Namespaces
- Control groups
- Capabilities
- chroot (optional)



EMERGING TECH - QUANTUM COMPUTING

EMERGING TECH: QUANTUM COMPUTING

Quantum computing utilises the principles of quantum mechanics (like superposition) to perform calculations on qubits.

These qubits can be 0, 1, or both at the same time, allowing quantum computers to tackle complex problems much faster than classical computers.

Feature	Classical Computing	Quantum Computing
Bit	0 or 1	Qubit (can be a superposition of 0 and 1)
Processing	Sequential	Parallel
Strength	Efficient for well-defined problems	Powerful for complex problems
Applications	General purpose computing, simulations	Drug discovery, materials science, cryptography (both breaking and creating)

EMERGING TECH: QUANTUM COMPUTING

The immense processing power of quantum computing presents both challenges and opportunities for cloud security.

- Factoring and cryptography
- Digital signatures

Feature	Classical Computing	Quantum Computing
Bit	0 or 1	Qubit (can be a superposition of 0 and 1)
Processing	Sequential	Parallel
Strength	Efficient for well-defined problems	Powerful for complex problems
Applications	General purpose computing, simulations	Drug discovery, materials science, cryptography (both breaking and creating)



Factoring and cryptography

Traditional encryption methods rely on the difficulty of factoring large prime numbers. Quantum computers could potentially break these methods, rendering current encryption protocols useless.

Feature	Classical Computing	Quantum Computing
Bit	0 or 1	Qubit (can be a superposition of 0 and 1)
Processing	Sequential	Parallel
Strength	Efficient for well-defined problems	Powerful for complex problems
Applications	General purpose computing, simulations	Drug discovery, materials science, cryptography (both breaking and creating)



Digital signatures

Digital signatures are used to verify the authenticity of data and communications. Quantum computing attacks could forge digital signatures, compromising data integrity and trust within the cloud environment.

Feature	Classical Computing	Quantum Computing
<i>Bit</i>	0 or 1	Qubit (can be a superposition of 0 and 1)
<i>Processing</i>	Sequential	Parallel
<i>Strength</i>	Efficient for well-defined problems	Powerful for complex problems
<i>Applications</i>	General purpose computing, simulations	Drug discovery, materials science, cryptography (both breaking and creating)

EMERGING TECH: QUANTUM COMPUTING

Despite the challenges, cloud computing offers advantages in the quantum computing era:

- Scalability
- Agility

Feature	Classical Computing	Quantum Computing
Bit	0 or 1	Qubit (can be a superposition of 0 and 1)
Processing	Sequential	Parallel
Strength	Efficient for well-defined problems	Powerful for complex problems
Applications	General purpose computing, simulations	Drug discovery, materials science, cryptography (both breaking and creating)



Scalability

CSPs can readily scale their infrastructure to accommodate the ever-increasing processing demands of quantum computing algorithms. This allows for faster development and deployment of quantum computing solutions.

Feature	Classical Computing	Quantum Computing
<i>Bit</i>	0 or 1	Qubit (can be a superposition of 0 and 1)
<i>Processing</i>	Sequential	Parallel
<i>Strength</i>	Efficient for well-defined problems	Powerful for complex problems
<i>Applications</i>	General purpose computing, simulations	Drug discovery, materials science, cryptography (both breaking and creating)



Agility

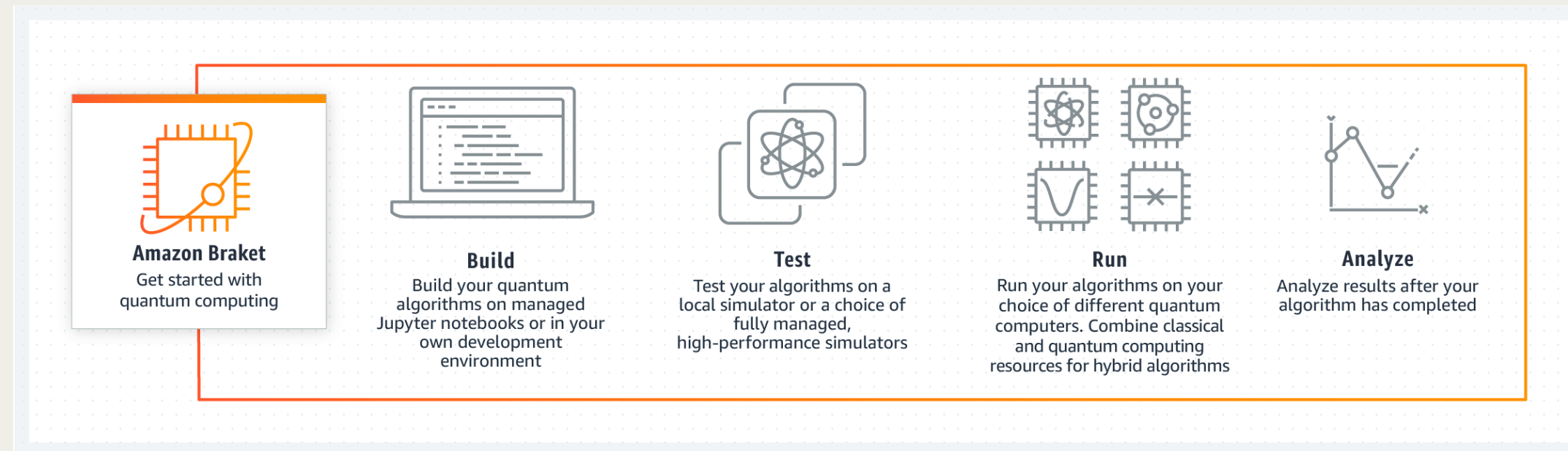
Cloud platforms can facilitate rapid migration of quantum computing algorithms when they become standardised. This agility is crucial for maintaining a robust security posture in a rapidly-evolving landscape.

Feature	Classical Computing	Quantum Computing
<i>Bit</i>	0 or 1	Qubit (can be a superposition of 0 and 1)
<i>Processing</i>	Sequential	Parallel
<i>Strength</i>	Efficient for well-defined problems	Powerful for complex problems
<i>Applications</i>	General purpose computing, simulations	Drug discovery, materials science, cryptography (both breaking and creating)

EMERGING TECH: QUANTUM COMPUTING

Amazon Braket is a fully-managed AWS service that helps researchers, scientists, and developers get started with quantum computing.

It offers a single point of access to a variety of quantum computing technologies.



EMERGING TECH: QUANTUM COMPUTING

Amazon Braket is a fully-managed AWS service that helps researchers, scientists, and developers get started with quantum computing.

Explore and design quantum computing and hybrid algorithms.

Amazon Braket

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Amazon Braket > Algorithm library

Algorithm library

A catalog of pre-built quantum algorithms written in Python. Each quantum algorithm is available as ready-to-run code that can be integrated into more complex algorithms. Open or create a managed JupyterLab Notebook to run the algorithm locally, on a managed simulator, or a quantum computer.

Algorithms (11)

Open notebook

Filter algorithms

Bernstein-Vazirani algorithm

GitHub

The Bernstein-Vazirani algorithm is the first quantum algorithm that solves a problem more efficiently than the best known classical algorithm. It was designed to create an oracle separation between BQP and BPP.

Tags: Textbook

Deutsch-Jozsa algorithm

GitHub

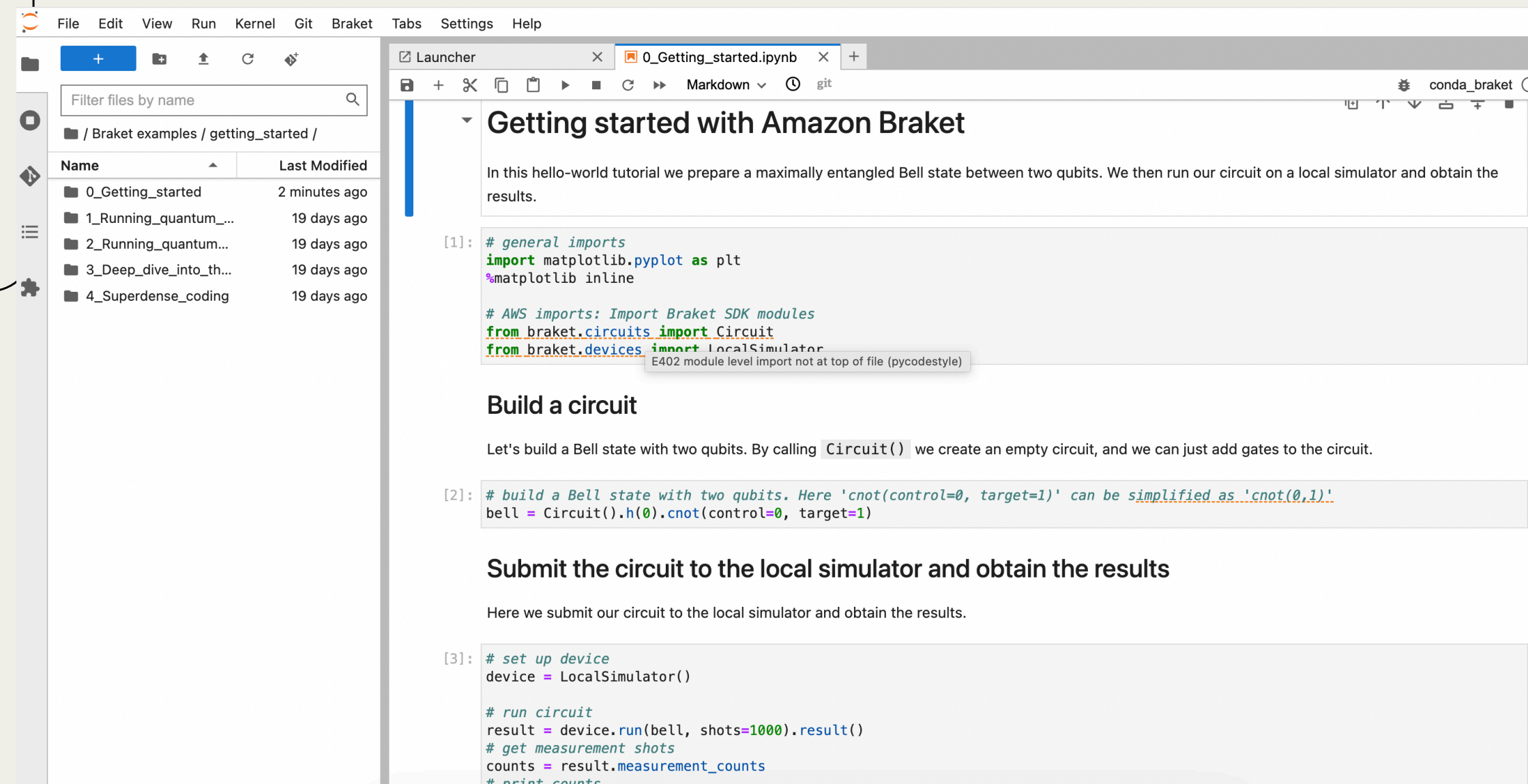
One of the first quantum algorithm's developed by pioneers David Deutsch and Richard Jozsa. This algorithm showcases an efficient quantum solution to a problem that cannot be solved classically but instead can be solved using a quantum device.

Tags: Textbook

EMERGING TECH: QUANTUM COMPUTING

Amazon Braket is a fully-managed AWS service that helps researchers, scientists, and developers get started with quantum computing.

Test algorithms on different quantum circuit simulators.



The screenshot displays the Amazon Braket console interface. On the left, a file explorer shows a directory structure for 'Braket examples / getting_started /' with files like '0_Getting_started', '1_Running_quantum...', '2_Running_quantum...', '3_Deep_dive_into_th...', and '4_Superdense_coding'. The main area shows a Jupyter notebook titled 'Getting started with Amazon Braket'. The notebook content includes:

- Getting started with Amazon Braket**: A text block explaining the tutorial's goal: to prepare a maximally entangled Bell state between two qubits, run the circuit on a local simulator, and obtain the results.
- Build a circuit**: A text block explaining that a Bell state is built with two qubits by calling `Circuit()` to create an empty circuit and adding gates. Below this is a code cell [2] with the following code:

```
[2]: # build a Bell state with two qubits. Here 'cnot(control=0, target=1)' can be simplified as 'cnot(0,1)'
bell = Circuit().h(0).cnot(control=0, target=1)
```
- Submit the circuit to the local simulator and obtain the results**: A text block explaining that the circuit is submitted to the local simulator to obtain results. Below this is a code cell [3] with the following code:

```
[3]: # set up device
device = LocalSimulator()

# run circuit
result = device.run(bell, shots=1000).result()
# get measurement shots
counts = result.measurement_counts
# print counts
```


EMERGING TECH: QUANTUM COMPUTING

Amazon Braket is a fully-managed AWS service that helps researchers, scientists, and developers get started with quantum computing.

Run algorithms on different types of quantum computers.

Amazon Braket

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Permissions and settings

Amazon Braket > Devices

Devices (10)

Filter devices

Show retired devices

	Hardware provider ▲	Device ▼	Availability	Description
☐	Amazon Web Services	SV1	🟢 AVAILABLE NOW	34 qubit Amazon Braket state vector simulator
☐	Amazon Web Services	TN1	🟢 AVAILABLE NOW	50 qubit Amazon Braket tensor network simulator
☐	Amazon Web Services	DM1	🟢 AVAILABLE NOW	17 qubit Amazon Braket density matrix simulator
☐	IonQ	Harmony	🕒 01:08:04	11 qubit universal gate-model ion-trap QPU
☐	IonQ	Aria 1	🔴 OFFLINE	25 qubit universal gate-model ion-trap QPU with error mitigation support
☐	IonQ	Aria 2	🕒 01:08:04	25 qubit universal gate-model ion-trap QPU with error mitigation support
☐	IonQ	Forte 1	🕒 RESERVATION ONLY	36 qubit universal gate-model ion-trap QPU with error mitigation support
☐	IQM	Garnet	🕒 1 day 02:08:04	20 qubit universal gate-model superconducting QPU
☐	QuEra	Aquila	🟢 AVAILABLE NOW	256 qubit analog neutral atom-based QPU
☐	Rigetti	Aspen-M-3	🔴 OFFLINE	79 qubit universal gate-model superconducting QPU

View device

EMERGING TECH: QUANTUM COMPUTING

Amazon Braket is a fully-managed AWS service that helps researchers, scientists, and developers get started with quantum computing.

Create proof of concept applications.

Repositories

Find a repository... Type Language Sort

- amazon-braket-examples** Public
Example notebooks that show how to apply quantum computing with Amazon Braket.
Python 454 Apache-2.0 220 8 (3 issues need help) 23 Updated 12 hours ago
- BraketAHS.jl** Public
Simulating Rydberg atom dynamics with Tensor Networks
Julia 16 Apache-2.0 6 4 (3 issues need help) 6 Updated yesterday
- amazon-braket-containers** Public
Python 13 Apache-2.0 8 0 5 Updated yesterday
- autoqasm** Public
AutoQASM is an experimental module offering a new quantum-imperative programming experience in Python for developing quantum programs.
Python 13 Apache-2.0 8 6 1 Updated yesterday
- amazon-braket-sdk-python** Public
A Python SDK for interacting with quantum devices on Amazon Braket
Python 294 Apache-2.0 118 15 (4 issues need help) 15 Updated 2 days ago

EMERGING TECH - EDGE & FOG COMPUTING

EMERGING TECH: EDGE & FOG COMPUTING

Edge and fog computing are approaches to distributed computing that bring processing closer to the source of data, unlike traditional cloud computing which relies on centralised data centres.

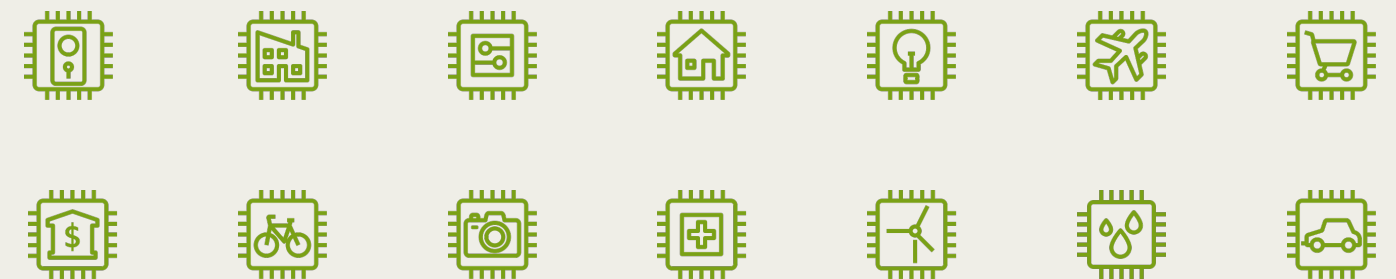
- Faster processing
- Reduced reliance on Internet connectivity
- Better handling of large datasets



Cloud computing - 1000's of data centres



Fog computing - millions of nodes



Edge computing - billions of devices/ sources of data

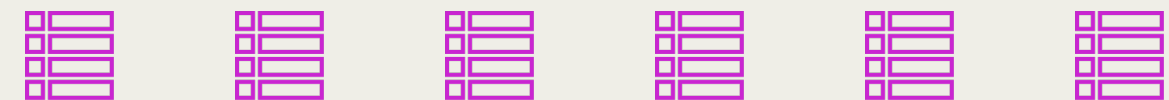
EMERGING TECH: EDGE & FOG COMPUTING

Edge computing processes data directly on the device itself (sensor, smart device) or on a nearby gateway device.

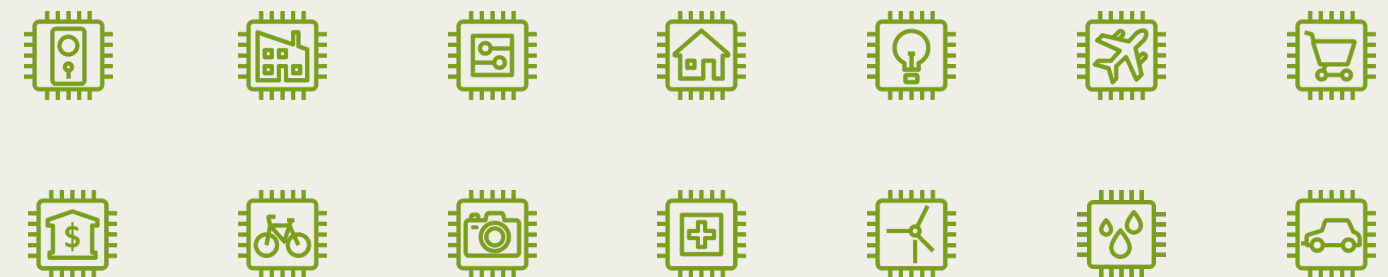
Fog computing acts as an intermediate layer between edge devices and the cloud. It pre-processes, analyses and filters data before sending it to the cloud.



Cloud computing - 1000's of data centres



Fog computing - millions of nodes



Edge computing - billions of devices/ sources of data

Latency: processing data locally reduces the time it takes to analyse and react, crucial for real-time applications like autonomous vehicles or industrial control systems.

Feature	Cloud computing	Fog computing	Edge computing
<i>Location of processing</i>	Centralised data centres	Local servers/gateways	Device itself or gateway
<i>Processing power</i>	High	Medium	Low
<i>Storage capacity</i>	High	Medium	Low
<i>Latency</i>	High	Medium	Low
<i>Reliance on Internet or bandwidth needs</i>	Yes	May vary	No (optional)
<i>Security</i>	Robust security measures	Requires additional measures	Requires additional measures

Bandwidth: large amounts of data from sensors can overwhelm network bandwidth. Edge and fog computing handle some processing locally, reducing the data sent to the cloud.

Offline functionality: edge devices can still operate even without Internet connectivity by processing data locally.

Feature	Cloud computing	Fog computing	Edge computing
<i>Location of processing</i>	Centralised data centres	Local servers/gateways	Device itself or gateway
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<i>Latency</i>	High	Medium	Low
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EMERGING TECH: EDGE & FOG COMPUTING

Both edge and fog computing present certain cybersecurity challenges.

Securing a wider range of devices and locations increases the attack surface.

Feature	Cloud computing	Fog computing	Edge computing
<i>Location of processing</i>	Centralised data centres	Local servers/gateways	Device itself or gateway
<i>Processing power</i>	High	Medium	Low
<i>Storage capacity</i>	High	Medium	Low
<i>Latency</i>	High	Medium	Low
<i>Reliance on Internet or bandwidth needs</i>	Yes	May vary	No (optional)
<i>Security</i>	Robust security measures	Requires additional measures	Requires additional measures

Both edge and fog computing present certain cybersecurity challenges.

Edge devices have less processing power for complex security solutions.

Feature	Cloud computing	Fog computing	Edge computing
<i>Location of processing</i>	Centralised data centres	Local servers/gateways	Device itself or gateway
<i>Processing power</i>	High	Medium	Low
<i>Storage capacity</i>	High	Medium	Low
<i>Latency</i>	High	Medium	Low
<i>Reliance on Internet or bandwidth needs</i>	Yes	May vary	No (optional)
<i>Security</i>	Robust security measures	Requires additional measures	Requires additional measures

Both edge and fog computing present certain cybersecurity challenges.

Disconnected devices might be vulnerable for extended periods.

Feature	Cloud computing	Fog computing	Edge computing
<i>Location of processing</i>	Centralised data centres	Local servers/gateways	Device itself or gateway
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EMERGING TECH: EDGE & FOG COMPUTING

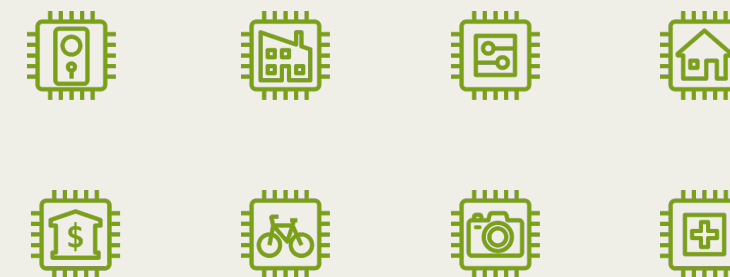
Edge and fog computing can be complementary. An application might use edge computing for real-time tasks and fog computing for preprocessing and filtering data before sending it to the cloud for further analysis.



Cloud computing - 1000's of data centres



Fog computing - millions of nodes



Edge computing - billions of devices/ sources of data

EMERGING TECH - CONFIDENTIAL COMPUTING

CONFIDENTIAL COMPUTING

Confidential computing is a technique for keeping data secure when it is being processed in the cloud or on a remote server.

Trusted Execution Environment (TEE) Security instructions built into the CPU, which define private regions of memory called enclaves, whose contents are protected.	Differential privacy Systematically incorporate random noise yet still be able to compute patterns while withholding information about who is in the dataset.
Synthetic data Computer-generated data: preserves statistical properties of original data, but shields confidential information	Homomorphic encryption Encryption schemes that let users perform computation on encrypted data without first decrypting it.
Federated learning Techniques for training ML models across multiple servers/devices using local data samples without exchanging data.	Secure multi-party computation Cryptographic protocols for parties to jointly compute a function based on their inputs while keeping their inputs private.

CONFIDENTIAL COMPUTING

In a normal client-server computing model, we have a client that might want to request that a server take some action on the client's behalf.

That server might have to access some data that is maintained in a separate storage environment and then perform some processing on that data.

Trusted Execution Environment (TEE)

Security instructions built into the CPU, which define private regions of memory called enclaves, whose contents are protected.

Differential privacy

Systematically incorporate random noise yet still be able to compute patterns while withholding information about who is in the dataset.

Synthetic data

Computer-generated data: preserves statistical properties of original data, but shields confidential information

Homomorphic encryption

Encryption schemes that let users perform computation on encrypted data without first decrypting it.

Federated learning

Techniques for training ML models across multiple servers/devices using local data samples without exchanging data.

Secure multi-party computation

Cryptographic protocols for parties to jointly compute a function based on their inputs while keeping their inputs private.

CONFIDENTIAL COMPUTING

1

In a traditional computing model, we apply encryption to that data that is kept on the storage devices so that unauthorised individuals cannot access it. That is called **protecting data at rest**.

Trusted Execution Environment (TEE) Security instructions built into the CPU, which define private regions of memory called enclaves, whose contents are protected.	Differential privacy Systematically incorporate random noise yet still be able to compute patterns while withholding information about who is in the dataset.
Synthetic data Computer-generated data: preserves statistical properties of original data, but shields confidential information	Homomorphic encryption Encryption schemes that let users perform computation on encrypted data without first decrypting it.
Federated learning Techniques for training ML models across multiple servers/devices using local data samples without exchanging data.	Secure multi-party computation Cryptographic protocols for parties to jointly compute a function based on their inputs while keeping their inputs private.

2

We also add encryption to network communications to prevent eavesdropping attacks between the client and the server and between the server and storage. That is called **protecting data in motion/transit**.

Trusted Execution Environment (TEE) Security instructions built into the CPU, which define private regions of memory called enclaves, whose contents are protected.	Differential privacy Systematically incorporate random noise yet still be able to compute patterns while withholding information about who is in the dataset.
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Federated learning Techniques for training ML models across multiple servers/devices using local data samples without exchanging data.	Secure multi-party computation Cryptographic protocols for parties to jointly compute a function based on their inputs while keeping their inputs private.

CONFIDENTIAL COMPUTING



However, we generally do not worry about the data that is being actively processed by the server. We do not apply encryption to data in memory, which is known as **data in use**.

Trusted Execution Environment (TEE) Security instructions built into the CPU, which define private regions of memory called enclaves, whose contents are protected.	Differential privacy Systematically incorporate random noise yet still be able to compute patterns while withholding information about who is in the dataset.
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3

Confidential computing adds protection for code and data in memory. It does this by offering **Trusted Execution Environments (TEEs)**.

Trusted Execution Environment (TEE) Security instructions built into the CPU, which define private regions of memory called enclaves, whose contents are protected.	Differential privacy Systematically incorporate random noise yet still be able to compute patterns while withholding information about who is in the dataset.
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3

TEEs guarantee that no outside process can view or alter the data being handled within the environment. That provides an added assurance that data is safe through all stages of the computing lifecycle.

Trusted Execution Environment (TEE)

Security instructions built into the CPU, which define private regions of memory called enclaves, whose contents are protected.